The US Treasury Funding Advantage Since 1860*

Clemens Lehner† Jonathan Payne‡ Jack Shurtleff§ Bálint Szőke¶

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Abstract

We provide the first consistent historical estimate of US Treasury funding advantage, as measured by term structures of yield spreads between comparable highest-grade US corporate bonds and Treasurys. We construct a new bond-level dataset with monthly prices and detailed contract features for US highest-grade corporate bonds over 1860-2026. We adjust yield curve estimation techniques to account for tax treatments and embedded options. Existing index-based spreads have mismeasured US funding advantage on long-term bonds in the post-WWI period, particularly during high inflation episodes. A model selection exercise chooses a Treasury wedge that co-varies with the market SDF and is different at short maturities.

JEL classification: E31, E43, G12, N21, N41

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†Columbia University, Department of Finance. Email: clehner28@gsb.columbia.edu

‡Princeton University, Department of Economics. Email: jepayne@princeton.edu

§Federal Reserve Board. Division of Monetary Affairs. Email: jack.k.shurtleff@frb.gov

¶Federal Reserve Board. Division of Monetary Affairs. Email: balint.szoke@frb.gov

1 Introduction

Recent turmoil in the US Treasury market has put the government's rising borrowing costs in the spotlight and prompted the Treasury to buy back its debt on an "emergency" basis. Such interventions to contain borrowing costs are not new. Since the birth of the Republic, policymakers have repeatedly tried to widen the government's *funding advantage*, the interest rate discount at which the Treasury can borrow relative to the safest private borrowers on the same terms. The funding advantage creates real revenue. A government that borrows at a discount sells its debt for more than the present value of the future fiscal surpluses behind it, and this premium, the bond analogue of seigniorage, expands the government budget constraint. But it is hard to measure. Historical Treasurys and safe corporate bonds differ in call options, tax treatment, and other priced characteristics, and the measured spread reflects all of these non-standard features. The funding advantage is only one part of that spread. In this paper, we show how to isolate that part. We then ask how large the US funding advantage has been since 1860, which established "facts" about it survive our correction, and what properties a macro-finance model must have to explain its dynamics.

We estimate zero-coupon yield curves for the safest corporate bonds and for Treasurys, each purged of the price effects of the relevant option and tax features, and define the funding advantage as the corporate-to-Treasury yield spread. We make four contributions. First, we construct a new, largely hand-collected, bond-level dataset of the safest US corporate borrowers, with monthly prices and contract characteristics, including cash-flow streams, call schedules, and tax status, over 1860–2026. The sample contains the highest-grade bonds throughout and, after 1925, only default-free bonds, and we call it the Aaa^{+} sample. Second, we generalize yield curve estimation to control for the price effects of non-standard bond features. Our estimator extracts the yield curve information from these bonds instead of dropping them, which lets us go back in time, reach maturities the existing methods omit, and correct existing Treasury yield curves confounded by these features. Third, we show that the US funding advantage emerges with the National Banking system at 1.5–2.0%, vanishes at US entry into World War I, and settles near 0.5% for most of the twentieth century, with a second structural break at the Global Financial Crisis. Index-based measures exaggerate it in the 1920s–30s and 1970s–80s. Once corrected, the negative unconditional relationship between the funding advantage and government debt-to-GDP in Krishnamurthy and Vissing-Jorgensen (2012) and Choi et al. (2026) is no longer significant. Fourth, we use model selection to identify the features of an affine asset pricing model needed to jointly price the corporate yield curve and the funding advantage and describe government debt issuance by maturity.

The first major task of the paper is to measure the US funding advantage by estimating two yield curves, which is a challenge because historical bond samples are heterogeneous. Many bonds carry embedded options and tax advantages, and it has been recognized for decades that the resulting price effects confound like-for-like comparisons of corporate and government yields. Our bond-level data reveal five such non-standard bond features, which were prevalent and clearly priced by markets. Call options pervade both samples, especially corporate bonds. A capital gains tax advantage on low-coupon bonds distorted yields during 1955–1985. Flower bonds made up all long-term Treasurys during 1955–1971, which is problematic because estates had the right to redeem them at par to pay estate taxes: an effective put option valuable when market prices were below par. Federal income tax exemptions on Treasurys during 1917–1941 made after-tax Treasury yields in those years incomparable with taxable corporate yields and with post-1941 Treasury yields. An exchange option depressed yields on all near-maturity coupon-bearing Treasurys in the 1930s and early 1940s. We cannot drop these bonds, because without them too few remain to estimate a yield curve for much of the 20th century. We cannot ignore their features either, because their price effects bias the estimated curves and misattribute the sources of observed spreads.

Our solution adjusts for the price effects of non-standard features explicitly rather than discarding the bonds that carry them. We generalize the standard bond price equation with feature-specific parametric adjustment functions, which are parsimonious enough to be identified from a date's cross-section of bonds, yet flexible enough to absorb that feature's price effect. Each adjustment is built to separate the feature's price effect from the yield curve common to all bonds in the Treasury or highest-grade corporate market, so non-standard bonds become informative about those curves. The functional forms are also consistent with asset pricing theory, which makes their parameters interpretable and ensures adjustments are made through the relevant mechanisms. The call adjustment shifts a bond's effective duration through the option's moneyness, the flower adjustment prices the embedded put with clientele-specific redemption probabilities, and the tax adjustments recover implicit marginal tax rates that we can check against the historical tax code. When no bond carries a non-standard feature, the adjustments vanish and the estimator reduces to the Kernel Ridge estimator of Filipović et al. (2022).

Our approach delivers the first long-run highest-grade corporate yield curve, an extended Treasury yield curve, and a spread adjusted for tax and option features. Each fills a gap in the literature. First, no long-run zero-coupon yield curve existed for highest-grade corporate bonds, likely because most of these bonds were callable and standard estimation approaches could not use them. We estimate these curves for 1860–2026, with pricing errors comparable to those of modern Treasury estimators. Second, existing Treasury yield curve estimates

had two holes. Excluding flower bonds left their long end beyond seven years missing during 1961–71, and the prevalence of tax exemptions left no taxable yield curve for 1917–1941. Our flower adjustment fills the first hole by pricing the embedded put, allowing us to extract long-term yield information from the distorted flower bonds and deliver an uninterrupted long-maturity Treasury yield series. Our tax adjustments fill the second hole with the first counterfactual fully taxable Treasury curve, which joins our post-1941 estimates into a single series with consistent tax treatment. Third, the resulting spread both extends and corrects the measurement of the US funding advantage. It provides the first measure before World War I, when the advantage was highest, and shows that the commonly used index-based spread has overstated it since World War II.

Our yield curve estimates pass a battery of fit and robustness tests and improve on the leading Treasury estimates wherever those exist. First, pricing errors are small and show no pattern by date, maturity, coupon rate, or option feature. Second, the estimated curves are robust to the composition of the bond sample, even when estimated from callable bonds alone. Third, our interpretable adjustment parameters line up with data we do not target, which validates our specifications. Option-implied volatility tracks realized volatility, flower option values follow inflation, and implied tax rates broadly follow IRS rates. Finally, our Treasury curve improves on the most influential estimates (Gürkaynak et al., 2007; Liu and Wu, 2021; Filipović et al., 2022). Since 1961, it consistently prices bonds with smaller errors despite pricing a far more heterogeneous sample. Tellingly, the gap is widest in 1961–1985, when the capital gains tax advantage distorted yields on low-coupon bonds that escaped these papers' otherwise careful sample restrictions.

The second major task of the paper is to estimate a dynamic affine term structure model that explains movements in the government's funding advantage since 1925.[1] Such models are workhorses of the macro-finance literature because they tie the government's lifetime budget constraint to the prices of its debt and fiscal position (e.g., Jiang et al., 2022; Collin-Dufresne et al., 2023). Technically, we extract a stochastic discount factor (SDF) from the highest-grade corporate yield curve, and a non-pecuniary Treasury wedge from the term structure of funding advantage. Because our wedge allows maturity-specific marginal investors in Treasury markets, our framework nests a wide range of models from the macro-finance and affine term structure literatures.

We start with two problems in the existing models of the Treasury wedge. First and foremost, the factor that Krishnamurthy and Vissing-Jorgensen (2012) and the ensuing literature emphasize, the market value of debt relative to GDP, explains little of the unconditional vari-

[1] We start in 1925 because Section 5 suggests that both Treasurys and highest-grade corporates are default free from then on.

ation in the long-maturity US funding advantage. The apparent fit comes from distortions in the index-based spread these papers use, not from an underlying economic relationship. Once we replace it with our corrected series, the scatter plots and regressions in these papers no longer show a clear negative relationship between the long-maturity funding advantage and the debt-to-GDP ratio. Second, most existing models, such as Choi et al. (2026), take the maturity structure of Treasurys as fixed, which contradicts how the Treasury actually manages its debt and leads to inaccurate conclusions about overall US borrowing costs.

We resolve both problems with the first affine term structure model (ATSM) that prices the highest-grade corporate and Treasury yield curves jointly, with a maturity-specific non-pecuniary Treasury wedge. Its factors are principal components of the highest-grade corporate yield curve, the term structure of funding advantage, and the term structure of government debt. The model explains changes in both curves, while imposing no-arbitrage across time and letting the maturity structure of debt evolve consistent with a parametric issuance policy rule. It brings the rich bond pricing toolkit of the ATSM literature together with the macro-finance models of non-pecuniary Treasury demand and market segmentation, a combination that was out of reach before our yield curves. We use a model selection exercise to identify the properties that macro-finance models of the Treasury wedge should possess. The preferred models price Treasurys differently from highest-grade corporate bonds, price short Treasurys differently from long ones, let the covariance between the SDF and the non-pecuniary wedge help explain the funding advantage, and let the maturity structure of debt evolve dynamically.

Related literature: Our work extends the literature on historical yield spreads (e.g., Fair and Malkiel, 1971; Cook and Hendershott, 1978; Krishnamurthy and Vissing-Jorgensen, 2012; Nagel, 2016; Choi et al., 2026; Cieslak et al., 2024). Relative to it, we collect new primary bond-level data and show how to adjust for tax and option features. We also measure a different object. We do not estimate the spread between Treasury yields and the expectation of the household SDF, the convenience yield, but the funding advantage of the government over the private sector. Reliable convenience yield estimates exist for the modern period (e.g., van Binsbergen et al., 2022; Koijen and Yogo, 2020), but data limitations rule them out for a long sample.

Technically, we build on two literatures. The first estimates zero-coupon yield curves from the law of one price and some restrictions on the shape of the yield curve (e.g., Fama and Bliss, 1987; Nelson and Siegel, 1987; Cecchetti, 1988; Svensson, 1995; Diebold and Li, 2006; Gürkaynak et al., 2007; Liu and Wu, 2021; Filipović et al., 2022). We adapt the Kernel Ridge estimator of Filipović et al. (2022) to heterogeneous bonds because it delivers strong

out-of-sample performance. The second accounts for tax advantages and option features in bond yields (e.g., Robichek and Niebuhr, 1970; Colin and Bayer, 1970; McCulloch, 1975; Cook, 1977; Cook and Hendershott, 1978; Green and Ødegaard, 1997; Duffee, 1996, 1998; Elton et al., 2001). We make three contributions to it. First, we estimate implicit tax effects instead of imposing researcher-chosen tax rates. Second, we value embedded options with option pricing theory, without imposing dynamic no-arbitrage. Finally, we integrate both into modern yield curve estimation.

More broadly, our work joins two literatures, one that connects historical asset prices to government financing costs (e.g., Payne et al., 2025; Jiang et al., 2022; Chen et al., 2022) and one that extends financial time series back in time (e.g., Homer and Sylla, 2004; Goetzmann et al., 2001; Reinhart and Rogoff, 2009; Jordà et al., 2019; Schmelzing, 2020; Officer and Williamson, 2021; Carlson et al., 2022). Like Ghaderi et al. (2025), we expand the available US historical corporate bond data.

Following Krishnamurthy and Vissing-Jorgensen (2012), a literature argues that the non-pecuniary Treasury wedge depends inversely on the market value of government liabilities (e.g., Greenwood et al., 2015; Nagel, 2016; Krishnamurthy and Li, 2023; Choi et al., 2026). These papers disagree on two points. The first is which liabilities matter, all of them (Krishnamurthy and Vissing-Jorgensen, 2012; Krishnamurthy and Li, 2023; Choi et al., 2026) or only short maturities or perhaps only money (Greenwood et al., 2015; Nagel, 2016). The second is how to impose equilibrium. Most papers leave debt supply unmodeled and implicitly assume that it does not react to spreads (Krishnamurthy and Vissing-Jorgensen, 2012; Krishnamurthy and Li, 2023), whereas Choi et al. (2026) has a monopolist government that internalizes how its supply affects the wedge. We show that our corrected funding advantage series no longer supports the negative debt-to-GDP relationship in Krishnamurthy and Vissing-Jorgensen (2012) and Choi et al. (2026), the two "extremes" of this literature. Appendix F replicates and discusses the other papers.

Outline. The paper is structured as follows. Section 2 describes the bond samples and the five non-standard features that markets priced. Section 3 builds the adjustment functions and the estimator for the yield curves and the funding advantage. Section 4 reports the long-run funding advantage, the non-pecuniary revenue it implies, and the fit and robustness of our estimates. Section 5 discusses whether differential default risk and differential state tax treatment explain the measured highest-grade corporate-Treasury spread. Section 6 constructs an asset pricing model for the US funding advantage. Section 7 concludes.

2 Historical Bond Samples: 1860-2026

This section describes the two bond-level datasets underlying our estimation: (i) a newly constructed continuous monthly panel of highest-grade US corporate bond prices and detailed characteristics, such as cash flow stream, call price schedule, and credit rating, covering 1860-2026 and (ii) a corresponding panel of US Treasurys. Our approach requires a private counterpart to Treasurys that is as close to default-free as possible, yet no bond-level record of this segment exists before the 1970s. Constructing one poses two challenges. The first is digitizing a wide collection of historical price records. The harder one is tracking down and understanding the many non-standard characteristics of the bonds in our sample. We close this section by summarizing the five key bond characteristics that complicate estimation: call options, capital gains tax advantage, flower bond provisions, tax exemptions, and exchange privilege. Appendix A documents the data collection, sources, and construction in detail, along with all other data used in the estimation.

2.1 Highest Grade Corporate Bonds

Corporate bond prices. We assemble end-of-month prices from five main sources. Between 1884 and 1973, we hand-collect NYSE closing prices from the "Stock Exchange Quotation/Bond Record" of the *Commercial & Financial Chronicle* (CFC) and similarly from 1934 to 1974 from "Listed Bond Quotations" section of *Barron's*. We do this to fill a gap in the literature that there are no readily available bond-level prices in digitized form. When these sources don't report a price for one of the bonds in our highest-grade sample, we try to fill the gap with closing prices hand-collected from the quotation pages of the *New York Times*. We use actual sale prices, excluding entries that report only bid-ask quotations. For the other periods, we can use existing databases. We use monthly bond prices from *Global Financial Data* (GFD, 1860-1884), the *Lehman Brothers Fixed Income Database* (henceforth Lehman/Warga, 1973-1997) and the *Bank of America Merrill Lynch Bond Index Database* (henceforth BAML, 1998-2026).

Bond characteristics. For each bond in our sample, we assemble the characteristics needed to reconstruct its full contractual cash-flow schedule and identify its embedded options. These include issue and maturity dates, coupon rates and payment schedules, call schedules, sinking fund provisions, convertibility, and denomination. No digitized source records this information before 1973, so we reconstruct it bond by bond back to 1860. For bonds maturing before 1900 we collect them from a wide range of historical sources listed in Table A.6, including railroad manuals, business magazines, exchange listing statements, and corporate

financial statements.[2] For bonds maturing between 1900 and 1973 we take the same characteristics from the sector-specific *Moody's Manuals*, published annually from 1900 onward. From 1973 onward, we obtain detailed bond characteristics from Lehman/Warga and the *Mergent Fixed Income Securities Database* (FISD).

Credit Ratings. From 1973 onward we observe ratings from all major agencies at the bond-month frequency through the Lehman/Warga (1973-1997) and BAML (1998-2026) databases. Before 1973 we hand-collect annual Moody's letter ratings for every bond in our panel from *Moody's Manuals*, which began rating railroads in 1909 and extend to public utilities and industrials in 1914.[3] Before 1909 no agency ratings exist, so we follow Macaulay (1938) and classify as highest grade the issuers whose bonds Macaulay selected for his high-grade railroad bond yield index, a sample he curated for financial strength and pruned before issuers encountered trouble. However, a true highest-grade sample is hard to construct in the 19th century, so the pre-1909 classification should be treated with some caution.

Sample Restrictions. We apply two sets of restrictions. The first is a standard set of instrument filters from the corporate bond literature, following Duffee (1998); Dickerson et al. (2023); Ghaderi et al. (2025), which we apply uniformly across all sources. We keep senior, non-subordinated, fixed-coupon, dollar-denominated bonds of non-financial US issuers and exclude convertible bonds, sinking fund bonds, equity-linked or asset-backed features, Yankee bonds, bonds with floating or otherwise non-standard coupons, and any issue without a fixed maturity. In contrast to other researchers we also keep callable bonds, whose option value our estimation prices explicitly. The second set restricts the sample to highest-grade bonds. Each bond must carry a Moody's Aaa rating at the time we observe its price, and after 1973, when we observe monthly ratings from all major agencies, we require unanimous Aaa ratings from Moody's, S&P, and Fitch.[4] We thus refer to our highest-grade sample as Aaa$^+$ for short.

2.2 US Government Bonds

Our Treasury data combine the comprehensive monthly panel of prices and quantities for all Treasurys over 1790-1925 constructed by Hall et al. (2018) and used in Payne et al. (2025) with the CRSP Treasury Database (1925-2026). We keep bonds with varying tax exemptions

[2]The GFD provides monthly bond prices for 1860-1884 but no detailed bond characteristics.

[3]We rely on Moody's for the historical period because its ratings are the earliest available. Standard & Poor's ratings began in 1922 and Fitch's in 1924.

[4]We apply these restrictions as stringently as the historical record permits.

and/or with embedded options, which our estimation handles explicitly. Appendix A.3 provides further details on the construction of the Treasury bond sample.

2.3 Heterogeneous Bond Samples

It has long been recognized (e.g. Durand, 1942; McCulloch, 1975; Cook and Hendershott, 1978; Cecchetti, 1988; Duffee, 1998; Elton et al., 2001) that non-standard bond characteristics complicate the measurement and comparison of corporate and government yields. Bonds are often treated as securities differing only in coupon, maturity, and default risk. In reality, and especially historically, they are bundles of contractual features, describing how they are taxed, whether the issuer may redeem them early, whether the holder may exchange or surrender them on favorable terms. Many seminal papers have studied individual features in isolation and their implication for bond pricing. However, no systematic account exists of which features mattered most, and when, across a long history. Our bond samples allow us to perform this account for Treasurys and Aaa^+ corporate bonds from 1860 to the present.

We identify five main non-standard bond characteristics in our historical bond sample, summarized in Table 1, that matter most for the measurement of yields and yield spreads. These fall into two broad categories: tax exemptions and embedded options. We describe each characteristic and its period of greatest relevance below, and analyze its price effects in detail in Section 3.4.

Feature	Description	Sample	Period
Callable Bond	Issuer has the right to redeem at the call price	Both	1917–2002
Low Coupon	Income mainly from capital gains, taxed at a lower rate	Both	1955–1985
Flower Bond	Valued at par for estate taxes upon death	Treasury	1918–1976
Tax Exemption	Reduced federal income tax rate on interest income	Treasury	1917–1941
Exchange Privilege	At maturity exchangeable for a new premium bond	Treasury	1933–1942

Table 1: Non-standard bond characteristics and relevant periods

Many of these characteristics share a common institutional backdrop. Before the introduction of US federal income taxation in 1913, neither corporate nor government bonds were

subject to taxes. Since then, income earned by both corporations and individuals from long-term securities holdings has been subject to two types of taxes. Coupon payments are taxed at the relevant (holder-specific) marginal income tax rate τ^{inc}. Capital gains are taxed at the long-term capital gains tax rate τ^{cg} while capital losses can be deducted against ordinary income and are thus valued at the ordinary income tax rate τ^{inc}. Since 1921, the tax code has favored long-term gains over interest income ($\tau^{cg} < \tau^{inc}$) and deferred their taxation until realization, potentially until maturity. In addition, since 1916, all bonds were subject to estate taxes except for the so-called flower bonds.

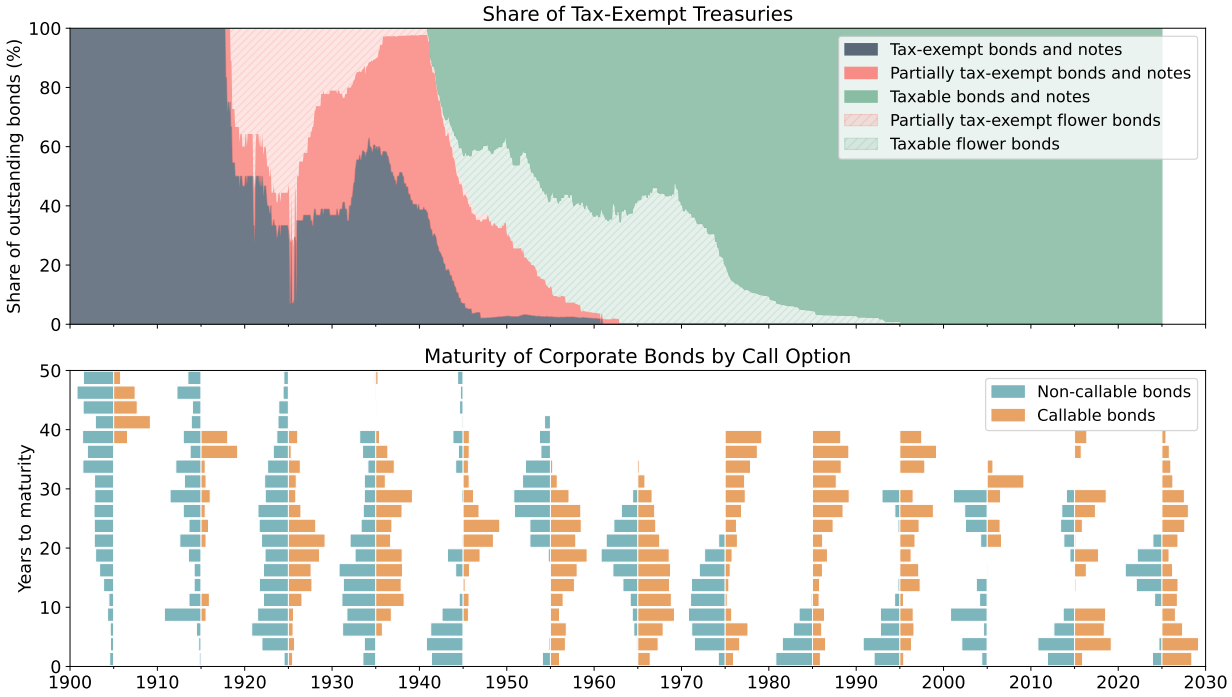


Figure 1: Tax Exemptions on Treasurys and Call Options on Aaa$^+$ Corporate Bonds.

Notes: Both panels plot shares of the number of distinct bonds outstanding in each month. *Top panel:* share of outstanding marketable Treasury bonds and notes by tax treatment, 1900-2026, excluding T-Bills and Certificates of Indebtedness. Different colors represent the tax treatment of each issue. The light hatched areas are flower bonds. *Bottom panel:* histograms of remaining years to maturity of the Aaa$^+$ corporate bonds in our estimation sample, by decade, for non-callable (left) and callable (right) bonds.

Callable bonds: Call provisions let the issuer buy the bond back early at a prespecified call price. Most government bonds were callable until the 1960s. Highest-grade corporate bonds were commonly callable in the 1930s and 1940s and nearly universally so from the mid-1970s to the early 1990s.[5] When callable bonds dominate the corporate sample, they make up most of the long end of the maturity spectrum, while non-callable bonds concentrate

[5]Figure A.21 in Appendix A.4.1 shows the share of callable bonds over time in both markets.

at short maturities (bottom panel of Figure 1). The indenture of a callable bond specifies its call protection along two dimensions: a *call-deferment period*, a window after issuance during which the bond cannot be called, and a *call price schedule*, $\{cp_{i,t}^{(j)}\}_{j=1}^{m_{i,t}}$, where $cp_{i,t}^{(j)}$ is the price the issuer pays to call the bond at time $t+j$. High call prices provide call protection by making it costlier for the issuer to call the bond early.[6]

Capital gains tax advantage on low coupon bonds: A bond's holding-period return has two pieces: the coupon yield, $c_i/p_{i,t}$, and the capital gain (or loss), $p_{i,t+1}/p_{i,t} - 1$. Because equilibrium after-tax (risk-adjusted) returns are equalized across bonds, low-coupon bonds must trade at a discount and earn more of their return as capital gains. This produced a genuine tax advantage only when low-coupon bonds actually traded below par and capital gains were expected, as in 1955–1985. When they instead traded above par, as the majority of bonds did in 1985–2022, capital losses were expected, taxed like ordinary income, and the advantage disappeared.[7]

Flower bonds: Between 1918 and 1971, the US government issued "flower bonds", which were long-term coupon-bearing bonds bundled with an estate tax provision structured like a put option: the right to redeem these bonds *at par value* (not market value) to pay the bondholder's estate taxes upon their death. Before 1976, exercising this option had the added benefit of avoiding long-term capital gains tax on the bond's appreciation to par.[8] Flower bonds were no niche instrument. The top panel of Figure 1 shows they made up a significant share of outstanding Treasurys through the early postwar decades. From 1955 to 1971, almost all Treasurys with maturity beyond 10 years carried the privilege (Appendix A.4.1), and by the late 1960s flower bonds accounted for over 40% of the market for coupon-bearing securities.[9]

Tax exempt Treasurys: From 1917 to 1941, the Treasury issued both fully and partially tax-exempt debt, while corporate bonds were fully taxable. Federal income tax of this era had two layers: a normal tax and progressive surtaxes. Taxable bonds paid both, partially tax-exempt (*pte*) bonds paid only the surtaxes, and fully tax-exempt (*fte*) bonds paid neither. The meaning of partial exemption depended on whether the bond was owned by a

[6]Though the two tools accomplish the same thing in principle, government and corporate bond markets structured call protection differently. Callable Treasurys had long deferment periods, often just a few years short of maturity, paired with a call price fixed at par throughout the call window. Callable corporate bonds tended to do the opposite: short deferment periods (zero to five years) with a step-wise call price schedule that started well above par (e.g., par plus one year's coupon) and declined gradually to par at maturity.

[7]Treasury Bills, always issued at a discount, were taxed as ordinary income starting in March 1941, and were fully tax-exempt before then. They never carry a capital gains tax advantage.

[8]This capital gains tax exemption was repealed by the Tax Reform Act of 1976, after which estates had to pay capital gains tax upon redeeming the bond at par value.

[9]Congress ended flower bond issuance in March 1971. The outstanding stock shrank gradually thereafter as issues matured or were redeemed for estate tax purposes.

corporation (including banks) or an individual.[10] For corporations, federal income tax was mainly the normal tax, with no surtax before 1941, so *pte* Treasurys behaved much more like *fte* Treasurys. For individuals, federal income tax was mainly surtaxes, rising up to 75% with income, on top of a modest flat normal tax rate around 4%, so *pte* Treasurys behaved much more like wholly taxable bonds. The top panel of Figure 1 shows the share of outstanding marketable Treasury bonds and notes by tax treatment, 1900–2026.

Exchange Privilege: During the 1930s and early 1940s, US debt nearly doubled, straining the Treasury's capacity to place new issues with the private sector, and legal restrictions barred issuance below par.[11] As Cecchetti (1988) explains, the Treasury responded by setting coupons high enough for new bonds to trade above par while still selling them at par. The exchange privilege then gave holders of maturing coupon-bearing Treasurys preferential access to these new issues, which let them capture the difference between the issue price and the market value.

3 Yield Curve and Funding Advantage Estimation

> *The yield of any bond may be seriously affected by many extraneous influences [...] tax-exemption, conversion and warrant privileges, an active sinking fund, provision for call prior to maturity, etc. [...] There seems to be no way of analyzing the yield differential between governments and corporates to determine how much is due to tax exemption or other privileges and how much to the quality differential.*
>
> Durand (1942)

The first goal of this paper is to measure the government's borrowing-cost advantage over highest-grade corporations across all maturities, which we refer to as the term structure of government funding advantage. Durand's quote captures why this comparison has long looked intractable: the bond samples described in Section 2 carry non-standard features that get priced by investors and picked up in measured yields. Thus, a change in the measured spread between government and corporate yields need not reflect any change in relative borrowing costs. It can just as easily reflect a change in which non-standard features happen to be priced, and by how much.[12] Durand and the following eighty years of yield curve

[10] WWI Treasury Secretary McAdoo pushed for partially tax-exempt bonds to stabilize interest rates, which were rising with surtaxes, and to keep these bonds in household hands rather than banks, who were absorbing tax-exempt issues to offset their own higher tax burden.

[11] The Second Liberty Bond Act required that new Treasury bonds and certificates of indebtedness be issued at par and new notes at not less than par.

[12] As Cook and Hendershott (1978) write: *"Failure to consider the effects of all of these factors has*

estimation papers were too pessimistic. We propose a resolution.

We estimate Treasury and corporate yield curves purged of the influence of non-standard bond characteristics and, from these, the US funding advantage. In this section, we formally define these objects (Subsection 3.1), discuss the problems with traditional estimation approaches (Subsection 3.2), show how ignoring bond heterogeneity has led to mis-measurement of the US funding advantage (Subsection 3.3), and outline our solution: estimating yield curves from bond price formulas extended with semi-structural wedges whose functional forms are disciplined by the data and by asset pricing theory (Subsection 3.4).

3.1 Defining Yields and Government Funding Advantages

Consider a discrete time, infinite horizon economy with time indexed by $t \in \{0, 1, \ldots\}$. A government and a highest-grade corporate sector both issue liabilities.

Let $q_t^{(j)}$ denote the period-t price the government receives for a zero-coupon bond promising one dollar at time $t+j$ with $q_t^{(0)} = 1$. The sequence $\mathsf{q}_t := \{q_t^{(j)}\}_{j=0}^{\infty}$ is the *Treasury discount function*. Let $y_t^{(j)} := -\frac{1}{j}\log(q_t^{(j)})$ denote the yield of a j-maturity zero-coupon bond, and let $\mathsf{y}_t := \{y_t^{(j)}\}_{j=1}^{\infty}$ denote the *zero-coupon yield curve*. Analogously, let $\tilde{\mathsf{q}}_t := \{\tilde{q}_t^{(j)}\}_{j=0}^{\infty}$ and $\tilde{\mathsf{y}}_t := \{\tilde{y}_t^{(j)}\}_{j=1}^{\infty}$ denote the highest-grade corporate bond discount function and zero-coupon yield curve, respectively, defined analogously for corporate promises with the same default probability as the government.

This paper's premise is simple: the government may borrow more cheaply than the highest-grade private corporations even when both issue the same promises with the same default probabilities. Formally, this means that $\tilde{\mathsf{q}}_t < \mathsf{q}_t$. We refer to the difference $\mathsf{q}_t - \tilde{\mathsf{q}}_t$ as the *premium* on government debt and refer to the *government's funding advantage* as the term structure of corporate to Treasury zero-coupon yield spreads:

$$\chi_t^{(j)} := \frac{1}{j}\log\left(q_t^{(j)}\right) - \frac{1}{j}\log\left(\tilde{q}_t^{(j)}\right), \qquad \forall j \geq 0. \tag{3.1}$$

By definition, $\chi_t^{(j)}$ captures a non-pecuniary benefit, embedded in the prices of hypothetical zero-coupon Treasury bonds, that allows the government to borrow at lower yields and so expands its budget constraint. This non-pecuniary benefit could come from superior Treasury liquidity, regulation, market segmentation, or other features unrelated to the bond's cash flows that advantage Treasurys. Following our definition, it cannot come from differential default risk, tax treatment, or option features. These are bond-specific, clientele-driven

frequently created misunderstanding both in the financial press and in academia about the causes of observed yield spreads." See Section 3.3 for a notable recent example.

distortions underlying borrowing costs which, if issued by the Treasury, do not expand the government budget constraint. A tax exemption on government debt, for instance, opens up a yield spread of its own, but only does so by shrinking future tax revenue. In this sense, it rearranges taxes in the government budget constraint rather than expanding it. Our goal is to measure $\{\chi_t^{(j)}\}_{j\geq 1}$ through the zero-coupon yield spread between highest-grade corporate bonds and Treasury, once it is purged of these other distortions.

3.2 Standard Estimation Approaches and Their Limits

The finance literature offers two broad approaches to estimating yield spreads, such as the US funding advantage. The first constructs indexes of comparable bonds and compares their average yields-to-maturity. The second restricts the sample to bonds simple enough to price with standard yield-curve models, then compares the resulting zero-coupon yields at matched maturities. Both approaches limit historical coverage, particularly in periods when tax and option features dominate the available securities.

Yield Indexes: The first approach (e.g. Homer, 1968; Cook and Hendershott, 1978; Salomon Brothers, 1988) partitions outstanding securities into subclasses based on observable characteristics, x_i, such as issuer, maturity, call features, tax treatment, etc., and computes yield indexes for each subclass by averaging corresponding yields-to-maturity.[13] Comparing subclass indices that differ along a single dimension of x_i isolates that characteristic's price effect. When the dimension is whether the issuer is government or highest-grade corporate, the resulting spread approximates the US funding advantage. This approach lives or dies on how finely *traded* bonds can be partitioned into genuinely like-for-like subclasses, which limits its scope, typically to long maturities.

Commonly used index-based spread: Macrofinance research needs a long-term Aaa-Treasury spread but lacked the bond-level data to compare like-for-like bonds directly, so researchers turned to externally constructed yield indices as a proxy. The most widely used such measure, introduced by Krishnamurthy and Vissing-Jorgensen (2012) and used in many papers since, compares Moody's Seasoned Aaa Corporate Bond index, restricted to maturities beyond 20 years, with the Federal Reserve's long-term government bond yield index, the average yield on *all* outstanding government bonds neither due nor callable within 10 years. We call this the *index-based Aaa-Treasury spread*. Forgoing the full term structure of spreads was defensible for researchers interested only in long-term spreads. But Section 3.3 shows that

[13] The yield to maturity is the *constant* discount rate that equates a bond's price to the present value of its promised cash flows, distinct from the maturity-specific zero-coupon yields y_t.

these indices fail to exclude bonds affected by tax and option features, thereby distorting the profession's understanding of the US funding advantage.

Our index-based spread: Our bond-level data let us build the like-for-like comparison directly. One natural restriction keeps Treasurys and highest-grade corporates with more than 15 years to maturity that are non-callable, non-flower, and non-exchangeable, trading at or above par, and carry no federal income tax exemptions. We call the resulting measure the *restricted index-based Aaa$^+$-Treasury spread*. This restriction minimizes the distortions in Table 1, but it is so stringent that broad coverage exists only from 1990 onward. Moreover, like the standard measure, it covers only the long end of the term structure of spreads.

Yield curve estimation: The second approach, (e.g. Diebold and Li, 2006; Gürkaynak et al., 2007; Liu and Wu, 2021; Filipović et al., 2022) trades rich sample heterogeneity for the ability to obtain estimates for the entire term structure of zero-coupon yields. Commensurate zero-coupon Treasury and corporate bonds are not traded in meaningful numbers, so q_t and $\widetilde{\mathsf{q}}_t$ must be inferred from traded coupon bonds that potentially carry the non-standard features discussed in Section 2.3. The standard solution is to restrict the bond sample to contain only *plain-vanilla* coupon-bearing or zero-coupon bonds, for which price differences can plausibly be attributed to coupon and maturity alone. Identifying $(\mathsf{q}_t, \widetilde{\mathsf{q}}_t)$ from this restricted sample requires Assumption 1*, which says that, within it, all coupon-bearing Treasurys are valued consistently with q_t and all highest-grade corporate bonds are valued consistently with $\widetilde{\mathsf{q}}_t$.

Assumption 1*. The observed price $p_{i,t}$ of government bond i, with promised cash-flow stream $\mathsf{cf}_{i,t} := \{cf_{i,t}^{(j)}\}_{j=0}^{m_{i,t}}$ is given by

$$p_{i,t} = \sum_{j=1}^{m_{i,t}} q_t^{(j)} cf_{i,t}^{(j)} + \varepsilon_{i,t} \tag{3.2}$$

where $\varepsilon_{i,t}$ is a measurement error with mean zero. An analogous expression holds for prices of highest-grade corporate bonds $\tilde{p}_{i,t}$ with discount function $\widetilde{\mathsf{q}}_t$.

This approach does not resolve bond heterogeneity. It requires a completely homogeneous sample of plain vanilla bonds, which makes it untenable for historical Treasury and highest-grade corporate samples, where many subperiods have too few plain vanilla bonds spanning the full maturity spectrum. In fact, the majority of corporate bonds are callable, rendering estimation of a corporate yield curve largely an option-pricing problem. This may explain why no historical zero-coupon yield curve exists for highest-grade corporate bonds.[14]

[14]For corporate bonds, yield curve estimates are scarce (see e.g. Elton et al. (2001), or the Treasury's HQM Corporate Bond Yield Curve series) and none are restricted to highest-grade issues.

Estimating discount functions with (3.2) on a sample where non-standard bond characteristics distort prices can introduce significant bias. This is a version of the *omitted variable bias*: if some bond characteristic is correlated with the bond price, $p_{i,t}$, at date t, leaving it out leads to biased estimate of q_t. In practice, this bias shows up as a systematic correlation between the residual $\varepsilon_{i,t}$ and the omitted characteristic. For these reasons, either one gives up on historical term structures altogether and focuses only on long-maturity yield indexes, which can hardly be created consistently before 1990, or one finds a way to estimate q_t and $\widetilde{\mathsf{q}}_t$ from a heterogeneous sample, i.e., bonds whose non-standard features carry real price effects. We opt for the latter in Subsection 3.4.

3.3 A Case Study: Flower Bonds in the 1970s

Flower bonds illustrate how features that are priced by markets but ignored by researchers can bias estimates of the government's funding advantage. As discussed in Section 2.3, flower bonds could be redeemed at par, rather than sold at market value, to pay estate taxes on the bondholder's death. The value of this option rose with the gap between par and market price at redemption, which meant that the flower provision acted as an effective inflation hedge. During the high-inflation 1970s, rising nominal interest rates pushed bond prices below par, especially for low-coupon, long-maturity bonds, whose duration made them particularly sensitive.[15] As their market prices fell further below par, the estate-tax privilege grew more valuable, raising flower-bond prices and lowering their yields-to-maturity. In that sense, flower bonds traded less like nominal bonds and more like inflation-protected ones.

Figure 2 makes this distortion visible, plotting yields-to-maturity for government and Aaa bonds in January 1976, near the inflation peak, together with three spreads that tell very different stories about the government funding advantage. The spread between long-term non-flower Treasurys and long-term Aaa bonds (20–30 years to maturity), a reasonable approximation to the undistorted spread, is a modest 55 basis points. The spread between long-term Aaa bonds and flower bonds, by contrast, is 297 basis points: more than five times larger. This larger spread embeds the value of the flower-bond privilege and therefore cannot be attributed entirely to the non-pecuniary benefits of holding Treasurys. Finally, the green and gray dashed lines show Moody's Aaa corporate yield index and the Federal Reserve's long-term government bond yield index (LTGOVTBD). The gap between these lines is the index-based Aaa-Treasury spread, the most widely used measure of long-term US funding advantage, which, like the flower bond spread, is very large at 186 basis points. This

[15] Bond price changes are approximately (negative) yield changes scaled by duration. As a result, successive yield increases through the 1960s-70s lowered the prices of long-duration flower bonds the most.

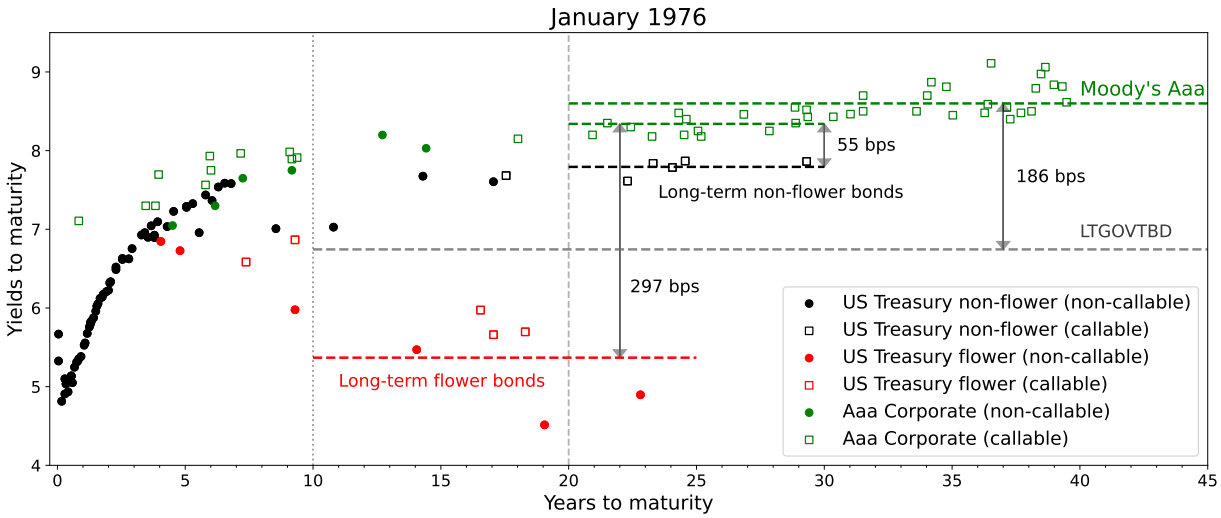


Figure 2: Calculation of Government Funding Advantage

Notes: The figure plots yields-to-maturity (y-axis) against years-to-maturity (x-axis) for every Treasury and Aaa corporate bond outstanding in January 1976. Each marker is one bond: black markers are regular taxable Treasurys, red markers are flower bonds, and green markers are Aaa corporate bonds. Dots denote non-callable bonds and squares denote callable bonds. The dashed green line is the Moody's Seasoned Aaa yield index, and the dashed gray line is the Federal Reserve's long-term government yield index.

is because the LTGOVTBD series averages over flower and non-flower bonds, and enough flower bonds were still outstanding in 1976 to distort that average. By embedding the value of the flower bond privilege, an inflation hedge whose value peaks exactly when inflation is high, the index-based Aaa-Treasury spread substantially overstates the government funding advantage during the 1970s.

3.4 Our Estimation Approach for Heterogeneous Bonds

Our approach prices a bond's non-standard features explicitly rather than discarding bonds that carry them. For each feature k in Table 1, we generalize pricing equation (3.2) by introducing maturity-specific parametric adjustments, $\mathsf{z}_k(\theta_t, \mathsf{q}_t; x_{i,t}) := \{z_k^{(j)}(\theta_t, \mathsf{q}_t; x_{i,t})\}_{j\geq 1}$, to the bond's cash flows that can absorb bond and/or clientele-specific price effects. Each price adjustment function is parsimonious enough to identify from a single date's cross-section of bonds, yet flexible enough to absorb that feature's price effect (if there is any).[16]

The value of each feature depends on both market-wide conditions and the bond's contractual terms. We capture the market-wide component by letting each adjustment func-

[16]Some bonds have multiple non-standard features (e.g. a call option and a tax exemption). For these bonds, the applicable wedge incorporates their interaction. Appendix B.3 provides the joint specifications.

tion depend on a time-varying parameter vector θ_t, collecting tax rates and rate-forecast parameters, and on the period-t discount function q_t, which encodes the market's expectations of future rates. Adjustment functions combine these market-level objects, (θ_t, q_t), with *observable* bond characteristics, such as taxability, call price schedule or bond-specific clienteles, stacked in $x_{i,t}$, to deliver theory-consistent price effects for each characteristic. Given specification of these adjustment functions $\mathsf{z}_{k,t}$, Assumption 1 says that all coupon-bearing Treasurys are priced consistently with q_t and all highest-grade corporate bonds are priced consistently with $\widetilde{\mathsf{q}}_t$, once observed prices are adjusted for these features.

Assumption 1. The observed price $p_{i,t}$ of government bond i belonging to class k with promised cash-flow stream $\mathsf{cf}_{i,t} := \{cf_{i,t}^{(j)}\}_{j=0}^{m_{i,t}}$, is given by:

$$p_{i,t} = \sum_{j=1}^{m_{i,t}} q_t^{(j)} z_k^{(j)}(\theta_t, \mathsf{q}_t; x_{i,t}) cf_{i,t}^{(j)} + \varepsilon_{i,t}. \tag{3.3}$$

where $\varepsilon_{i,t}$ is a measurement error with mean zero. An analogous expression holds for prices of highest-grade corporate bonds $\tilde{p}_{i,t}$ with discount function $\widetilde{\mathsf{q}}_t$.

Moving from Assumption 1* to Assumption 1 is non-trivial, since (3.3) requires decomposing cash-flow valuation into a discount function q_t and adjustments $\mathsf{z}_{k,t}$. What identifies this decomposition is that q_t is the common component across Treasurys, while $\mathsf{z}_{k,t}$ depends on bond-specific factors, capturing special attributes or clientele-driven distortions in the valuation of cash flows. This is because we interpret q_t as a property of the issuer rather than of any single instrument. Once each bond's own distortions are removed, what remains should reflect something common to all of that issuer's obligations. The decomposition is further disciplined by the fact that $\mathsf{z}_{k,t}$ is an explicit function of q_t, rather than an unrestricted residual. Whether this functional form is right is ultimately an empirical question: does (3.3) leave price errors that look like noise rather than systematic bias, and is the estimated q_t robust to the composition of the bond sample? Section 4.2 answers both.

While our primary goal is to fit the discount functions from heterogeneous bonds, each adjustment function is grounded in an internally consistent semi-structural model, so the estimated parameters θ_t have an economic interpretation and, where possible, can be checked against untargeted data (see Section 4.2).[17] The semi-structural interpretation of the price adjustments $\mathsf{z}_{k,t}$ lets us provide the first counterfactual estimate of a *fully taxable* Treasury yield curve between 1917 and 1941.[18]

[17] Following the yield-curve estimation literature we do not impose dynamic no-arbitrage on our samples across estimation dates, so our wedges are not fully "structural".

[18] Many studies, most recently Ghaderi et al. (2025), acknowledge that only tax-exempt Treasurys are observed before 1941, yet do not control for the resulting tax effects.

Our approach nests and supersedes existing modern estimators that use (3.2). By construction, plain vanilla bonds have $z_{k,t}^{(j)} = 1$ for all k, j and t, so in samples with many of them (e.g., 21st century Treasurys) the wedges are irrelevant for estimating q_t, and our approach reduces to off-the-shelf yield curve estimators. However, as plain vanilla bonds grow scarce, the specification of $\mathsf{z}_{k,t}$ becomes relevant and corrects for the resulting bias. Before turning to how we use equation (3.3) for estimation of yield curves and yield spreads in Subsection 3.4.2, we need to specify $\mathsf{z}_{k,t}$ in Subsection 3.4.1.

3.4.1 Adjustment Function Specification

We now specify the parametric form of each price adjustment function. We summarize each $\mathsf{z}_{k,t}$ below and derive them fully in Appendix B.3.

(i). Call Options: Call provisions give the issuer the right to buy their bond back early at a prespecified call price, injecting uncertainty about the timing of cash-flows that bondholders must be compensated for. Whenever an early call looks likely, callable bonds carry higher yields than otherwise identical non-callable bonds, reflecting a positive-valued embedded call option. The left panel of Figure 3 offers a clean example. With market rates around 7% in the mid-1990s, callable Treasurys issued in the 1970s-80s with coupons exceeding 8% (blue squares) were likely to be called at the first available date, so their yields sit consistently above non-callable bonds of similar maturity.

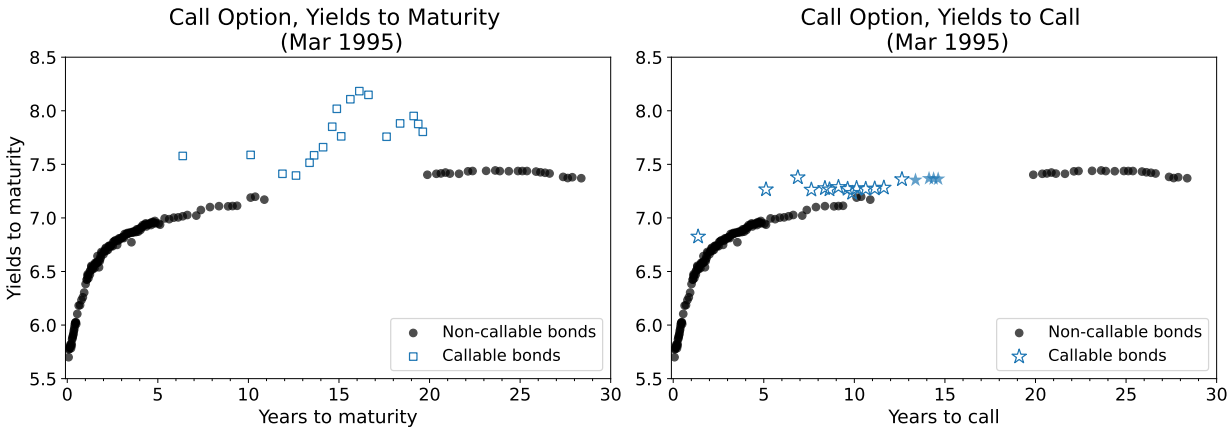


Figure 3: Price Implications of Embedded Call Options.

Notes: Panels depict yields-to-maturity (y-axes) against years-to-maturity (x-axes) for different dates (panel title). Each point corresponds to a separate bond outstanding in the given month. Dots show non-callable bonds, squares represent callable bonds (yields-to-maturity), and stars display callable bonds (yields-to-call). All callable Treasurys in March 1995 were first callable 5 years before maturity.

A bond nominally maturing in $m_{i,t}$ years but likely called at $t_c < m_{i,t}$ is priced, approximately, like a t_c-year bond. Because Treasurys had long deferment periods, the first call date was a good guess at effective maturity, so researchers have long approximated the duration correction by replacing yields-to-maturity with yields-to-call for bonds trading well above par.[19] The stars in the right panel of Figure 3 plot yields-to-call against first call date for the same bonds shown in the left panel. Two features stand out. First, many stars follow the shape of the non-callable yield curve, indicating that once duration is adjusted, callable bonds price approximately off q_t. Second, most stars are still well above the non-callable yield curve, with only the longest maturity callable bonds (shaded) plausibly on the non-callable yield curve. Therefore, fixing the effective maturity to the first available call date is not necessarily sufficient to capture the market's assessment of duration risk for in-the-money bonds. We need a $\mathsf{z}_{call,t}$ that assigns the duration adjustment (the bond's "moneyness") flexibly across the entire call window, not just at one date.

Since issuers are expected to call a bond the first time its market price rises far enough above the call price, the moneyness is pinned down by two forces: (i) the degree of call protection, captured by the call price schedule $\{cp_{i,t}^{(j)}\}_{j=1}^{m_{i,t}}$, and (ii) the market's expectations for future rates, captured by the price path $\{p_{i,t+j}\}_{j=1}^{m_{i,t}}$. For given expectations, more call protection means a lower option value. For given protection, the more likely prices are to rise (rates are to fall), the sooner refinancing becomes attractive, pulling the expected call date forward and raising the option value.

This suggests that $\mathsf{z}_{call,t}$'s role is to adjust the expected duration of the cash-flow stream to reflect bond i's call protection and the bondholder's expectations about future interest rates. Formally, the wedge can be expressed in terms of "call probabilities": the probability that bond i is called at j, evaluated under the risk-neutral measure Q^j appropriate for horizon j.[20] Let $NC_{i,t}^{(j)} := \left\{cp_{i,t}^{(s)} > p_{i,t+s}, \forall s < j\right\}$ be the event that bond i is not called before period $t+j$.[21] Our callable bond price formula is then (see Appendix B.3 for derivation from first principles):

$$p_{i,t} = \sum_{j=1}^{m_{i,t}} q_t^{(j)} \underbrace{\mathbb{P}_t^{Q^j}\left\{NC_{i,t}^{(j)}\right\}}_{i \text{ not called before } j} \left[c_i + \underbrace{\mathbb{P}_t^{Q^j}\left\{cp_{i,t}^{(j)} \leq p_{i,t+j} \mid NC_{i,t}^{(j)}\right\}}_{i \text{ called at } j \text{ cond on not called earlier}} cp_{i,t}^{(j)}\right]$$

[19] The yield to call is the yield to maturity computed as if the bond matured at its first call date, paying the call price as its principal.

[20] The relevant measure for pricing time-$(t+j)$ payoffs is the *forward measure* Q^j. Under Q^j, the discount bond maturing at $t+j$ serves as the numeraire, which lets us remain agnostic about how much the term premium affects the call option value. See Appendix B.3 for a formal definition.

[21] We assume $cp_t^{(s)} \leq p_{i,t+s}$ to be the event that the bond is called at future date $t+s$, given it is callable at $t+s$ and has not yet been called. Bliss and Ronn (1998) discuss why the issuer might find this optimal.

implying

$$z^{(j)}_{\text{call},t} := \mathbb{P}^{Q^j}_t \left\{ NC^{(j)}_{i,t} \right\} \frac{c_i}{cf^{(j)}_{i,t}} + \mathbb{P}^{Q^j}_t \left\{ \{ cp^{(j)}_{i,t} \leq p_{i,t+j} \} \text{ and } NC^{(j)}_{i,t} \right\} \frac{cp^{(j)}_{i,t}}{cf^{(j)}_{i,t}}.$$

To evaluate call probabilities under the appropriate forward measure, we need a model for how bond prices might evolve from period t onward: the call price schedule is data, but the future path of prices, $\{p_{i,t+j}\}_{j=1}^{m_{i,t}}$, is not. We derive these prices from a Hull-White short rate model calibrated period by period so that its implied discount function matches q_t exactly, as in Hull and White (1990). Intuitively, the future short rate fluctuates, AR(1)-style, around a long-run mean pinned down by today's yield curve, with mean-reversion κ_t and volatility σ_t. This calibration is what makes z_{call} depend so critically on q_t: information about expected future rates embedded in the yield curve feeds directly into call probabilities, and hence into the price correction, in a way consistent with how non-callable bonds price in the same cross section. Appendix B.3 gives the full derivation and estimation procedure.

(ii). Capital Gains Tax Advantage on Low Coupon Bonds: Low-coupon discount bonds earn more of their return as capital gains, which are taxed at a lower rate than ordinary income, so they carry a tax advantage when trading below par. Premium bonds carry none, since the premium is amortized against ordinary income. The data bear this out. The left panel of Figure 4, drawn from a period when nearly all bonds traded at a premium, shows essentially no relationship between the portion of income from capital losses and yields. The right panel tells a different story. It is from the late 1960s, when a Congressional ceiling on long-term debt issuance (see Appendix A.4.2) collided with rising interest rates to push the low-coupon Treasurys to deep discounts. There, the smaller a bond's coupon relative to market rates the more its yield is pulled down.

This treatment motivates a price distortion with two properties: it should be zero for bonds expected to realize capital losses ($p_{i,t} \geq 100$), and it should push prices up (yields down) the more a bond's return share comes from capital gains. The second term in the following price formula does both:

$$p_{i,t} = \sum_{j=1}^{m_{t,i}} q^{(j)}_t cf^{(j)}_{t,i} + \underbrace{\eta_t q^{(m_{t,i})}_t \max\{100 - p_{i,t}, 0\}}_{\text{price distortion}} \tag{3.4}$$

where $\eta_t \geq 0$ is the capital gains tax advantage parameter. For discount bonds ($p_{t,i} < 100$), the distortion term is the bond's discount, $(100 - p_{t,i})$, scaled by $q^{(m_{t,i})}_t$. For a given discount, a shorter maturity means that the capital gain makes up a larger share of the

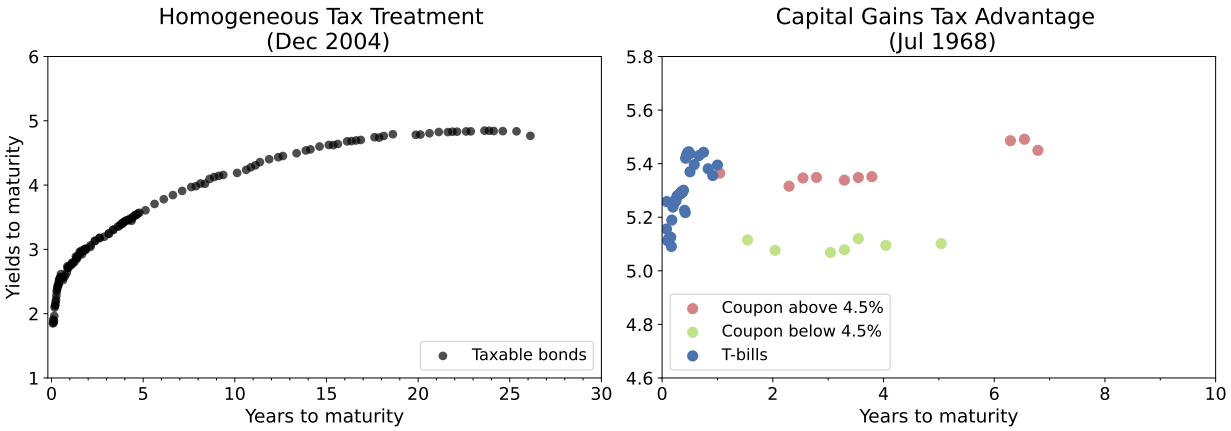


Figure 4: Price Implications of Capital Gains Advantage.

Notes: Panels depict yields-to-maturity (y-axes) against years-to-maturity (x-axes) for different dates (panel title). Each point corresponds to a separate bond outstanding in the given month.

bond's total return, so $q_t^{(m_{t,i})}$ weights the correction accordingly. Formula (3.4) corresponds to the following adjustment function (see Appendix B.3):

$$z_{cg,t}^{(j)} = \begin{cases} 1, & p_{t,i} \geq 100 \\ \frac{1+\eta_t(100/cf_{t,i}^{(j)})\mathbb{1}\{j=m_{t,i}\}}{1+\eta_t q_t^{(m_{t,i})}}, & p_{t,i} < 100 \end{cases} \qquad \forall j$$

In Appendix B.3 we show that, under the assumption that agents believe η_t follows a random walk independent of their discount factor, η_t can be mapped approximately onto the spread between marginal tax rates. In this sense, our formula generalizes the approach used in the micro tax-based asset pricing literature (e.g., Robichek and Niebuhr (1970), McCulloch (1975), Cook and Hendershott (1978)), which uniformly assumes bonds are held to maturity. Relative to that literature, we estimate q_t and η_t jointly rather than imposing pre-chosen IRS tax rates, and we do so without taking a stand on holding periods.

(iii). Flower bonds: Flower bonds could be redeemed at par, rather than market price, to pay the holder's estate tax upon death, making them a put option on the bond struck at par. The option pays off most for low-coupon, long-maturity bonds trading at deep discounts. Figure 5 shows this pattern, especially in the high-inflation left panel: flower bond yields-to-maturity (red dots) sit 1 to 3 percentage points below those of comparable non-flower bonds (black dots) at longer maturities, tracing a separate downward-sloping "flower yield curve."

However, two key features distinguish flower bonds from other option bonds. Only a

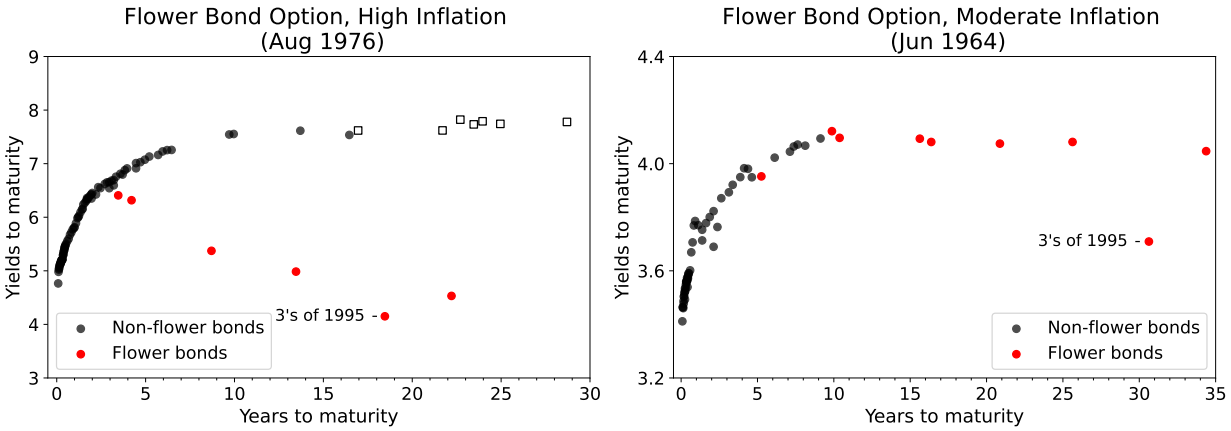


Figure 5: Price Implications of Flower Options.

Notes: Panels depict yields-to-maturity (y-axes) against years-to-maturity (x-axes) for different dates (panel title). Each point corresponds to a separate bond outstanding in the given month. Red color represents flower bonds, black color is for non-flower bonds. Dots show non-callable bonds, squares represent callable bonds (yields-to-maturity).

bondholder who dies holding the bond can exercise the option, and only holders with large nominal estate tax liabilities benefit from it. Aggregate demand for the flower option is therefore satiable. With a no-shorting constraint, that satiable demand may be allocated only to a subset of flower bonds where it is most valuable, those trading furthest below par. This explains the apparent outliers in Figure 5. In the moderate-inflation 1960s (right panel), a single issue, the 3s of 1995, absorbed all demand, while the remaining flower bonds priced as ordinary long bonds. In the high-inflation 1970s (left panel), demand spread across many more bonds for two reasons. Inflation steadily eroded the fixed nominal estate tax exemption, raising the satiation point,[22] and higher nominal rates pushed more low-coupon, long-maturity bonds to deep discounts. Higher demand was then allocated across a much wider, though imperfectly substitutable, set of flower bonds, tracing the more homogeneous curve in the left panel.

The adjustment function must therefore (i) price flower bonds as if they carried a put option on the uncertain time of the holder's death, struck at par, and (ii) let different flower bonds attract different clienteles with heterogeneous willingness to pay for the option. For (i), we reuse the call-option adjustment's interest rate model, with parameters (κ_t, σ_t), to compute the probability that flower bond i trades below par j periods ahead, $\mathbb{P}^{Q^j}\{p_{i,t+j} < 100\}$.[23] Redemption at $t+j < m_{i,t}$ also requires that the holder die and face sizable estate tax

[22]The exemption was fixed at \$60,000 from 1942 until the Tax Reform Act of 1976. With prices roughly 84% higher in 1976 than in 1964, the fixed threshold pulled more estates into taxable status.

[23]In principle (κ_t, σ_t) are identified from callables and flower bonds jointly, but after the early 1960s flower

liabilities. For (ii), we represent clienteles by bond-specific eligibility distributions, flexibly capturing differences in mortality, estate tax liabilities, and anything else relevant to early redemption. Let $e_{i,t}^{(j)}$ be the probability that bond i's clientele is eligible to redeem at $t+j$, independent of whether the price condition happens to make it worthwhile. The probability of actual redemption at $t+j$ is then the product of eligibility and profitability:

$$d_{i,t}^{(j)} := e_{i,t}^{(j)} \, \mathbb{P}^{Q^j}\{p_{i,t+j} \leq 100\}.$$

Investors who expect to become eligible sooner have a higher willingness to pay for the option. To parametrize clientele-specific eligibility probabilities, we start from a reference distribution $\mathsf{e}_t := \{e_t^{(j)}\}_{j\geq 1}$ generated from a hazard rate function $\bar{\alpha}_t \exp(\bar{\beta}_t j)$ with parameters $\overline{\alpha}_t$ and $\overline{\beta}_t$. We then generate alternative clientele distributions by tilting the reference with a logistic function centered at $J_{i,t}$ and elasticity ψ_t:[24]

$$e_{i,t}^{(j)} := A_{i,t}^{(j)} e_t^{(j)} \qquad \text{where} \qquad A_{i,t}^{(j)} = \frac{1}{1+\exp(-\psi_t(j-J_{i,t}))} \tag{3.5}$$

Each flower bond can have its own clientele, with $J_{i,t}$ inferred from traded prices.[25] The deepest-discount flower bond defines the reference clientele, with the lowest expected eligibility time. Every other clientele's cutoff $J_{i,t}$ is the horizon at which bond i's put becomes as attractive as the reference bond's, under the Hull-White distribution with parameters (κ_t, σ_t) and mean function calibrated to q_t (see Appendix B.3.4 for the formal definition of $J_{i,t}$). Flower bonds whose discounts are small relative to the reference's get a large $J_{i,t}$, so only the reference bond is priced for its flower option (as in the 1960s). As other discounts close in on the reference's, the implied $J_{i,t}$ falls and multiple bonds are priced for it (as in the 1970s). Together, these assumptions imply an adjustment function structurally analogous to that of callable bonds. Defining $NR_{i,t}^{(j)} := \left\{\text{not eligible}_{i,t}^{(s)} \cup (100 \leq p_{i,t+s}), \forall s < j\right\}$ the event that bond i is not redeemed before $t+j$, we obtain:

$$z_{\text{flower},t}^{(j)} := \mathbb{P}^{Q^j}\left(NR_{i,t}^{(j)}\right) \frac{c_i}{cf_{i,t}^{(j)}} + d_{i,t}^{(j)} \frac{100}{cf_{i,t}^{(j)}}. \tag{3.6}$$

This formula sets to zero the capital-gains tax advantage $\bar{\eta}_t$ that flower bonds carried on early redemption before 1976. Appendix B.3 derives the full adjustment function, with $\bar{\eta}_t$ restored, that we use in estimation.

bonds traded at such deep discounts that $p_{i,t+j} < 100$ held almost surely for all j irrespective of (κ_t, σ_t).

[24] We do not estimate ψ_t and instead set ψ_t high such that $A_{i,t}^{(j)}$ approximates a heaviside function.

[25] For a flower bond with maturity $m_{i,t}$, (3.5) implies the "probability" that the bond survives to maturity is $1 - \sum_{j\geq 1}^{m_{i,t}} A_{i,t}^{(j)} e_t^{(j)}$. Therefore, if $A_{i,t}^{(j)} = 0$ for all $j \leq m_{i,t}$, i trades like a straight bond.

(iv). Tax Exempt US Treasurys: Interest income on fully tax-exempt (*fte*) Treasurys escaped all federal income tax, for individuals and corporations alike, so *fte* bonds should command a premium over otherwise identical taxable Treasurys. Partially tax-exempt (*pte*) Treasurys escaped only some federal income tax, ranging from a modest fraction of it for individuals to nearly all of it for corporations, so how *pte* Treasurys price relative to *fte* Treasurys is informative about the marginal buyer of these bonds. We can see this directly on the left and middle panels of Figure 6, depicting yields-to-maturity by horizon for every class of government bond outstanding in September 1942 and May 1929, respectively. Two things stand out. First, tax exemption moves yields in a simple pattern. *fte* (green) and *pte* (orange) yields both look like scaled-down versions of the taxable yield curve (black) at every maturity. Second, *pte* bonds trade like *fte* bonds in 1929, but carry a sizable yield spread over *fte* bonds in 1942.

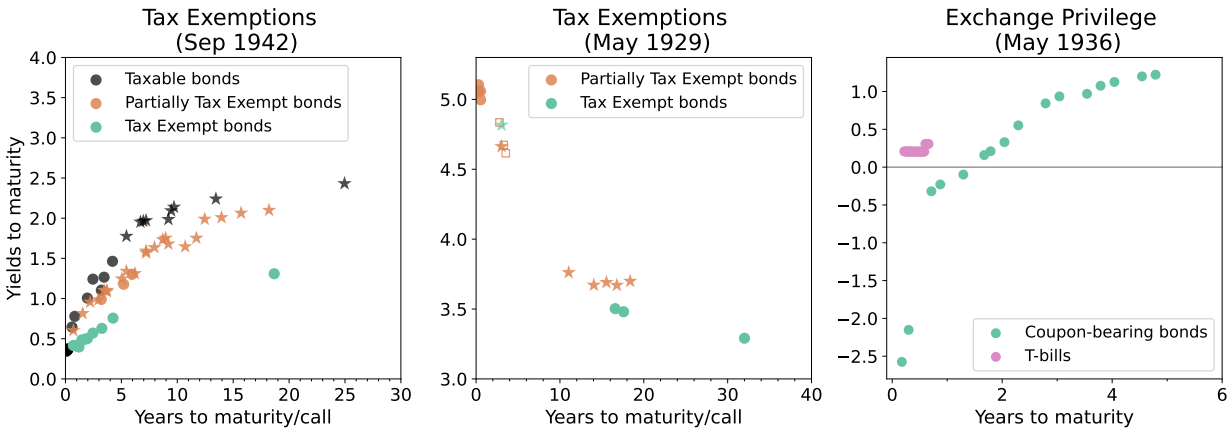


Figure 6: Price Implications of Pre-WWII Treasury Features.

Notes: Panels depict yields-to-maturity (y-axes) against years-to-maturity (x-axes) on different dates (panel title). Each point corresponds to a separate bond outstanding in the given month. Dots show non-callable bonds, squares depict callable bonds (yields-to-maturity), stars represent callable bonds (yields-to-call).

These facts motivate a pair of scaling parameters, $\theta_{fte,t}$ and $\theta_{pte,t}$, linking zero-coupon yields directly: $y_{k,t}^{(j)} = \theta_{k,t} y_t^{(j)}$ for $k \in \{fte,\ pte\}$ and all j, where $y_t^{(j)}$ is the j-period zero-coupon yield implied by q_t. This gives the price adjustment function:

$$z_{k,t}^{(j)} = (q_t^{(j)})^{\theta_{k,t}-1} \qquad k \in \{fte,\ pte\}, \quad \forall j.$$

Conveniently, $(\theta_{fte,t}, \theta_{pte,t})$ carry a structural interpretation. If agents believe taxes follow a random walk independent of their own discount factor, the scaling parameters map directly

onto marginal tax rates.[26] Let τ_t^k denote the marginal tax rate for exemption class k. Appendix B.3 then shows:

$$\theta_{k,t} = \frac{1 - \tau_t^{inc}}{1 - \tau_t^k}, \qquad k \in \{fte,\ pte\}.$$

Since $\tau_t^{fte} = 0$, this maps our fitted scaling parameters directly into the marginal tax rate of a fictitious marginal investor. The ratio $\theta_{fte,t}/\theta_{pte,t}$, which equals the relative yields of *fte* and *pte* Treasurys, reveals the marginal investor's surtax rate τ_t^{pte}: around 35% in 1942 and around 0% in 1929. In addition, after 1941, taxable Treasurys traded alongside *fte* and *pte* bonds, so the relative yields of *fte* and taxable Treasurys, $\theta_{fte,t}$, also reveals the marginal investor's total federal income tax rate τ_t^{inc}.

Before 1941, no fully taxable Treasurys existed, so τ_t^{inc} cannot be inferred from bond prices alone. However, the *after-tax* yield curve $\{y_{fte,t}^{(j)}\}_{j=0}^{\infty}$ can be estimated at each date when *fte* bonds were traded. The challenge is therefore not fitting the curve but pinning down the τ_t^{inc} that scales the after-tax curve into a counterfactual fully taxable curve for 1917 to 1941. In this regard, our estimated τ_t^{pte} is informative, because it maps onto the surtax rate of the marginal investor. The tax code assigned individuals and corporations (banks) very different surtax schedules, so the path of τ_t^{pte} reveals who that investor was. Corporate surtax rates were negligible throughout this period. Surtax rates for wealthy individuals were not, so we bound them using the marginal rates faced by the top 0.1% and 0.01% of earners from Piketty and Saez (2003), for each year since 1913.[27] Whenever our estimated τ_t^{pte} falls inside this individual range, we take individuals to be the marginal investor that year. Whenever it falls below the range, closer to the near-zero *maximum* corporate surtax rate, as in 1929, we take corporations to be marginal instead. Once we infer who is marginal, recovering τ_t^{inc} is immediate. We add the matching statutory normal tax rate to our estimated surtax. The resulting τ_t^{inc} path scales the observed *fte* curve into a counterfactual, fully taxable Treasury yield curve for 1917 to 1941. See the bottom panel of Figure 11 for our estimated τ_t^{pte} path and the marginal investor identification.

(v). Exchange Privilege: The exchange privilege let holders of maturing coupon bearing Treasurys swap into newly issued bonds at par, even though those new bonds immediately traded above par. The value of this exchange option pushed down yields of near-maturity coupon-bearing government bonds throughout the 1930s and early 1940s. In the right panel

[26] An assumption the finance literature imposes uniformly, at least implicitly. See e.g. Robichek and Niebuhr (1970), Colin and Bayer (1970), McCulloch (1975), Cecchetti (1988), Green and Ødegaard (1997).

[27] We choose this range because the surtax schedule is steepest there.

of Figure 6, every bond within 18 months of maturity shows a negative yield-to-maturity. The effect is inversely related to maturity: it fades quickly as maturity lengthens and diverges to $-\infty$ as maturity approaches zero. Since the exchange privilege can only be exercised at maturity, the adjustment $z_{ep,t}^{(j)}$ differs from one only at $j = m_{t,i}$. These patterns motivate the functional form

$$z_{\text{ep},t}^{(m_{i,t})} := \exp\left(\zeta_{1t} \exp(-\zeta_{2t} m_{i,t})\right) \tag{3.7}$$

with level parameter $\zeta_{1t} \geq 0$ and persistence parameter $\zeta_{2t} \geq 0$. Formula (3.7) adjusts for the exchange privilege by scaling up bond i's expected cash flow at maturity. Our formula generalizes Cecchetti (1988)'s calculation, which computes bond-specific realized option values and propagates them across maturities under perfect foresight. We replace perfect foresight with a parametric form fit to the data, though the resulting gain in fit over Cecchetti (1988) turns out to be negligible.

3.4.2 Estimation Implementation

Our tax and option corrections described in Subsection 3.4.1 essentially homogenize our bond samples, making them act like plain-vanilla bond samples. Conditional on the adjustment functions $\mathsf{z}_k(\mathsf{q}_t, \theta_t; x_{i,t})$, one could pair any popular yield-curve estimator, such as Gürkaynak et al. (2007); Liu and Wu (2021); Filipović et al. (2022), with our identifying bond price formula (3.3) to estimate $(\mathsf{q}_t, \widetilde{\mathsf{q}}_t)$. We choose the Kernel Ridge estimator because Filipović et al. (2022) show it has superior out-of-sample properties and because it lets us estimate the corporate-Treasury spread directly via a two-stage procedure described below. We build the estimator here for the Treasury curve. Every object we introduce along the way, the reference curve, the deviation function, the feasibility restrictions, the loss function, has a tilde-marked corporate analog, used in the second stage.

Decomposition: Our approach decomposes the Treasury discount function into two pieces: an anchor curve $\mathsf{a}_t = \{a_t^{(j)}\}_{j=0}^{\infty}$, drawn from a narrow parametric family with $a_t^{(0)} = 1$, and a non-parametric deviation function $\mathsf{h}_t = \{h_t^{(j)}\}_{j=0}^{\infty}$, drawn from a Hilbert space of smooth, twice weakly differentiable functions with $h_t^{(0)} = 0$, such that $\mathsf{q}_t = \mathsf{a}_t + \mathsf{h}_t$.

Anchors: We take the Treasury anchor curve to be flat at level $r_t > 0$ and the term structure of highest-grade corporate-Treasury spread to be flat at level $\widetilde{s}_t$:

$$a_t^{(j)} = \exp(-r_t j) \qquad \widetilde{a}_t^{(j)} = q_t^{(j)} \exp(-\widetilde{s}_t j) \qquad \forall j \geq 0$$

where r_t and $\widetilde{s}_t$ are estimated along with the adjustment function parameters θ_t and the deviation functions h_t and $\widetilde{\mathsf{h}}_t$. These choices are conservative in that the Treasury yield

curve and the term structure of funding advantage default to their simplest shape, without slope or curvature, absent evidence otherwise ($\mathsf{h}_t = 0$ and $\widetilde{\mathsf{h}}_t = 0$).

Feasible discount functions: We impose three restrictions on $(\mathsf{q}_t, \widetilde{\mathsf{q}}_t)$. First, to avoid overfitting, we penalize a weighted combination of the slope and curvature of the deviation from the anchor curve h_t (and analogously $\widetilde{\mathsf{h}}_t$), given by an (α, δ)-indexed norm $||\mathsf{h}_t||_{\alpha,\delta}$ introduced by Filipović et al. (2022), with the effect of discouraging sharp oscillations and kinks. Second, we require that $(\mathsf{q}_t, \widetilde{\mathsf{q}}_t)$ match the observed T-bill rate and short-dated prime commercial paper rates exactly at their respective maturities.[28] This serves a different purpose for each curve. For Treasurys, it lets us absorb whatever curvature Treasury Bill yields require, including their idiosyncratic variation relative to other short-term securities (Duffee, 1996). For highest-grade corporates, where short-maturity data is often unavailable, it grounds the short end of the curve directly, rather than extrapolating it from long-maturity bonds. Third, we require $q_t^{(\infty)} = \widetilde{q}_t^{(\infty)} = 0$, a theoretical restriction imposing the transversality condition on the fitted discount functions. We denote the resulting space of feasible Treasury discount functions as $\mathcal{Q}_{\alpha,\delta}(r_t, \theta_t)$ and the corporate analog as $\widetilde{\mathcal{Q}}_{\alpha,\delta}(\widetilde{s}_t, \widetilde{\theta}_t; \mathsf{q}_t)$.

Two-stage procedure: We estimate each date's Treasury and highest-grade corporate curves sequentially rather than jointly. The first stage fits the Treasury discount function q_t. The second stage takes that estimate as given and fits the corporate-Treasury *spread* directly, applying the same deviation-from-anchor logic to $(\widetilde{s}_t, \widetilde{\mathsf{h}}_t)$ that the first stage applied to (r_t, h_t). This two-stage design pays off practically. Conditioning the spread on the first-stage Treasury curve, rather than estimating $\widetilde{\mathsf{q}}_t$ from scratch, keeps the second stage from picking an arbitrary shape where the corporate sample is sparse at intermediate maturities.

Loss function and estimation: Let θ_t collect the parameters from Subsection 3.4.1:

$$\theta_t = \Big(\underbrace{\tau_t^{inc}, \tau_t^{pte}, \eta_t^{cg}}_{\text{Tax Adjustments}}, \quad \underbrace{\kappa_t, \sigma_t}_{\substack{\text{Option-Implied} \\ \text{Interest Rate Volatility}}} \quad , \quad \underbrace{\bar{\alpha}_t, \bar{\beta}_t, \bar{\eta}_t}_{\substack{\text{Flower Bond} \\ \text{Early Exercise}}} \quad , \quad \underbrace{\zeta_{1t}, \zeta_{2t}}_{\substack{\text{Exchange Privilege} \\ \text{Payout Adjustments}}} \Big),$$

Our loss function is a weighted combination of mean-square price errors and a smoothness penalty, trading off the two via the ridge parameter λ:

$$\mathcal{L}\big(\mathsf{q}_t, r_t, \theta_t\big) = \sum_i \rho_{i,t} \underbrace{\left(p_{i,t} - \sum_{j=1}^{m_{i,t}} q_t^{(j)} z_k^{(j)}\big(\theta_t, \mathsf{q}_t; x_{i,t}\big) cf_{i,t}^{(j)} \right)^2}_{\text{price error}} + \underbrace{\lambda ||\mathsf{q}_t - \mathsf{a}(r_t)||_{\alpha,\delta}^2}_{\text{smoothness penalty}}$$

[28] Appendix A.5.1 provides details on which short dated securities are used.

for exogenous weights $\{\rho_{i,t}\}_i$.[29] Our estimator then solves

$$\min_{r_t,\theta_t}\left\{\min_{\mathsf{q}_t\in\mathcal{Q}_{\alpha,\delta}(r_t,\theta_t)}\mathcal{L}(\mathsf{q}_t,r_t,\theta_t)\right\} \tag{3.8}$$

with the tuning parameters (λ,α,δ) selected via K-fold stratified cross-validation across the maturity spectrum. Conditional on r_t,θ_t, the inner minimization for q_t follows Filipović et al. (2022). We extend them in two ways: an outer layer chooses the anchor and the adjustment-function parameters jointly with the discount function, and the Treasury and Aaa$^+$ corporate curves satisfy transversality and price short-dated bonds without error by construction. Appendix B.1 shows how to collapse the bilevel problem (3.8) into a single set of first-order "moment" conditions, which deliver GMM standard errors for the discount function and for (r_t,θ_t).[30]

4 Results

In this section, we show our estimates of the US funding advantage over 1860-2026, measure what it was worth to the government, and test how well our estimates fit. Three findings stand out. First, the funding advantage moves in long swings that track changes in financial regulation. Second, the standard index-based spread overstates it in the 1920s-30s and 1970s-80s. Third, the advantage was a significant source of revenue in wartime and crises.

4.1 Long-Term Funding Advantage and Non-Pecuniary Revenue

Our estimated funding advantage exhibits three regimes associated with different monetary, fiscal, and financial institutions. Figure 7 plots the 20-year US funding advantage from 1860 to the present (black line). Appendix C.7 reports analogous plots for 1, 5, and 10-year spreads. The funding advantage emerges in the mid-1860s, with the end of the Civil War and the introduction of the National Banking system. National Banks received a "circulation privilege", that is the right to issue bank notes backed by long-term government debt. The funding advantage remains near 1.5–2.0% through this era, then falls to roughly zero at US entry into World War I. Wartime debt issuance coincided with the end of the circulation privilege and the Federal Reserve's new monopoly on money creation. The spread subsequently settles near 0.5% for most of the twentieth century, falls to zero during the inflation surge of

[29] We follow the literature and set $\rho_{i,t}$ proportional to the squared duration times price of bond i.

[30] The loss function for corporates is analogous, given by $\mathcal{L}(\widetilde{\mathsf{q}}_t,\widetilde{s}_t,\widetilde{\theta}_t;\mathsf{q}_t)$. We do not impose that the corporate parameters $\widetilde{\theta}_t$ equal the Treasury parameters θ_t.

the late 1970s, and recovers to 0.5–1.0% by the 2000s. Structural-break tests following Bai and Perron (1998, 2003) and Qu and Perron (2007) confirm breaks during World War I and the 2008 Global Financial Crisis (GFC). These breaks mark shifts in the average level of the funding advantage regardless of the prevailing debt-to-GDP ratio.

The literature's standard measure, the index-based Aaa-Treasury spread of Subsection 3.2 and 3.3, shown in red, tracks our estimate closely, except in two major episodes contaminated by exactly the price distortions our estimates control for.[31] In the 1920s-30s, tax exemptions on government bonds suppress long Treasury yields and inflate the index spread. In the 1970s-80s, high inflation and interest rates inflate both the capital-gains advantage and the option value of flower bonds, pushing the index spread to its historical peak at precisely the moment our highest-grade corporate-Treasury spread is near zero. The gap between the series tells us that most of the variation the index-based measure attributes to Treasury "specialness" in this period is an "inflation risk premium" instead.

As a sense check for our estimation, one that sidesteps modeling distortions entirely by restricting to a homogeneous bond sample, the teal line shows our restricted index-based Aaa$^+$-Treasury spread defined in Subsection 3.2. Reassuringly, our estimate tracks it wherever it can be computed. But we can also see that restricting to homogeneous bonds works only for a narrow range of dates.

How much revenue did the funding advantage raise? We measure it as *non-pecuniary revenue*, the bond analogue of seigniorage. Just as the government earns seigniorage by issuing currency, it earns non-pecuniary revenue by issuing debt at lower interest costs than the private sector. Let $B_{i,t}$ denote the amount of bond i outstanding in period t, including new issues, in units of \$100 par, and let $b_t^{(j)} := \sum_i cf_{i,t}^{(j)} B_{i,t}$ be the dollars due at $t+j$ that the government has promised as of t. The change $b_t^{(j)} - b_{t-1}^{(j+1)}$ is net new promises for date $t+j$, which the Treasury sells at $q_t^{(j)}$ rather than the Aaa$^+$ corporate price $\tilde{q}_t^{(j)}$. Non-pecuniary revenue is the resulting premium, with net retirements entering negatively:

$$R_t := \sum_{j=1}^{M_t} \left(q_t^{(j)} - \tilde{q}_t^{(j)}\right)\left(b_t^{(j)} - b_{t-1}^{(j+1)}\right), \qquad M_t := \max_i \; m_{i,t}.$$

The bottom panel of Figure 7 plots this ex-post measure (black line) as an annual share of federal receipts. Establishing the advantage at the end of the Civil War was costly, as the government absorbed losses when Treasurys began to trade at a premium in the late 1860s. That cost paid off in the long run. In twentieth-century crises (the World Wars, the New Deal, the GFC, COVID), the government drew substantial revenue from its funding

[31] A smaller episode occurred around 2000, when Moody's Aaa index appears to have included in-the-money callable bonds, which inflated the index-based spread. See Figure C.42 in Appendix C.7.

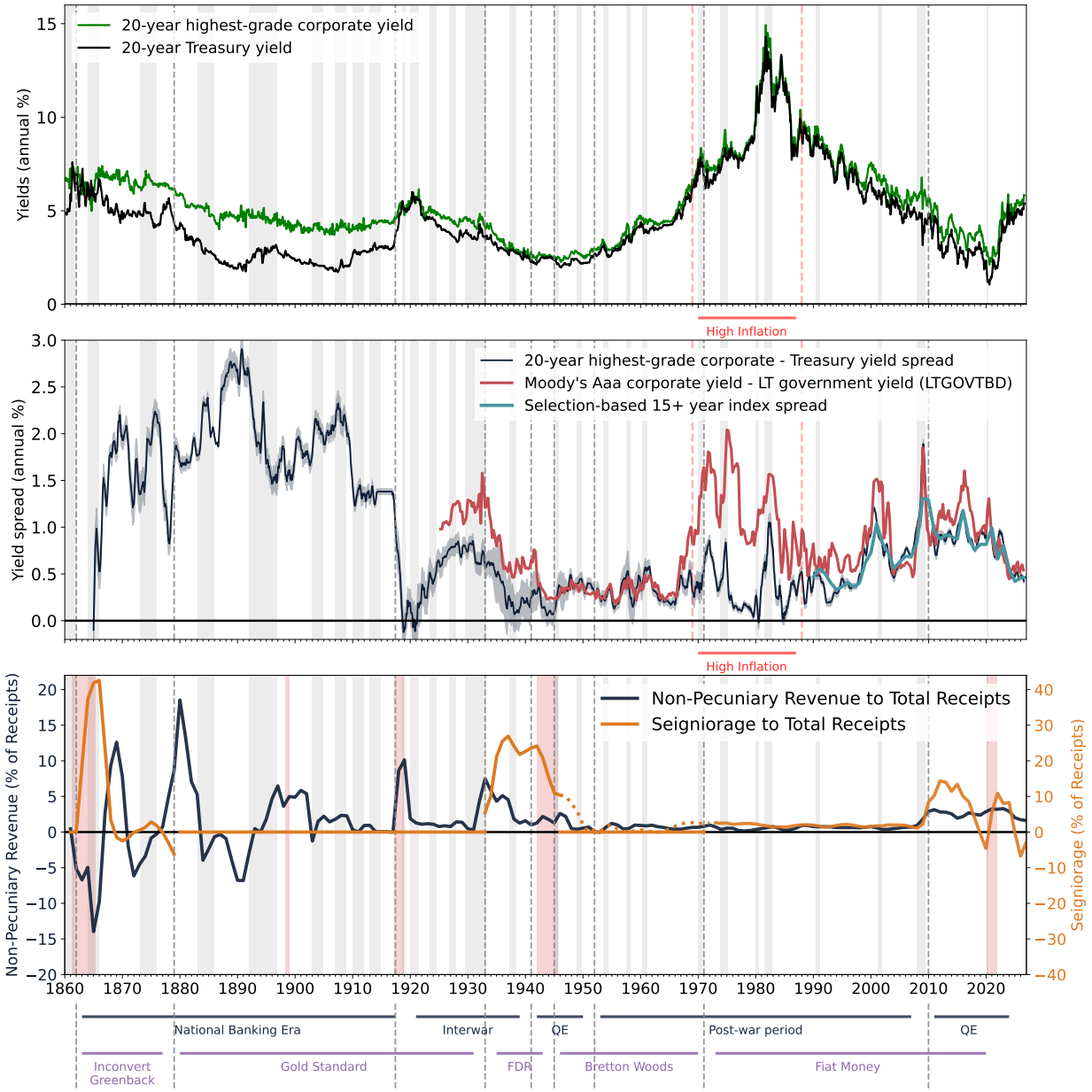


Figure 7: Yields, Funding Advantage, and Non-Pecuniary Revenue, 1860-2026

Notes: Top panel: Our estimates of the 20-year highest-grade corporate and Treasury yields over 1860-2026 in green and black, respectively. *Middle panel:* Our estimates of the 20-year highest-grade corporate-Treasury spread (black solid line). The gray bands show ±1 standard deviation. The teal line depicts our index-based yield index, constructed from a restricted subsample of bonds with maturities exceeding 15 years. The red solid line shows the 12-month centered moving average of the index-based Aaa Corporate-Treasury spread. *Bottom panel:* Seigniorage is defined as new (unbacked) money created by the monetary authority in excess of interest paid on reserves. Series are normalized by total annual federal revenue. Dashed vertical lines denote financial regulatory eras. Bottom labeling shows monetary standards. Light gray intervals are NBER recessions. The light red bands in the bottom panel, from left to right, depict the Civil War, the Spanish-American War, WW1, WW2, and COVID.

advantage. Non-pecuniary revenue reached over 10% of federal receipts in WWI, roughly 8% in the New Deal years, and until recently, roughly 4% post-GFC. These magnitudes are somewhat lower than seigniorage revenue (orange line), which rose sharply during the Civil War, in the decade after 1933, and after the GFC.[32]

4.2 Fit and Robustness

Three considerations frame our assessment of fit. First, data quality, sample size, and the availability of bond-level information improve markedly after WWI, reflected in smaller pricing errors post-1917. Corporate bond prices tend to be noisier than Treasury prices, as many researchers have documented, yet our restricted sample and methodology still deliver corporate pricing errors comparable in magnitude to leading estimators of the modern Treasury curve. Second, we benchmark our Treasury estimates against three prominent yield-curve estimators to assess how our approach compares with existing methods, in increasing order of flexibility: Gürkaynak et al. (2007) (GSW), Filipović et al. (2022) (FPY), and Liu and Wu (2021) (LW). To our knowledge, no comparable benchmark exists for the highest-grade corporate curve.[33] Our functional form is closest to FPY, although we additionally fit the short end exactly with a functional form flexible enough to link it to longer maturities consistently. This distinction matters and has clear benefits during unusual episodes such as the Federal Reserve's yield curve control in the 1940s (see Appendix C). Third, and most important, the price adjustment functions are the paper's real contribution. They let us price bonds that off-the-shelf models discard, extending coverage across historical periods and maturities, while also sharpening the estimates, but only if specified correctly. The rest of this section establishes that they are.

(i). Adjustment functions capture the relevant price effects. The third and fourth panels in Figure 8 show that weighted price errors are small in both samples.[34] More importantly, the bottom panels reveal no systematic pattern in the errors by time, maturity, and option feature. Appendix C shows the same holds for coupon rate and "moneyness". Figure

[32]Seigniorage is 0 during the gold standard (1879-1933). Our baseline calculation is 0 seigniorage during Bretton Woods (1944-1971), though we show a dashed line corresponding to the revenue raised had new issuance not been truly backed by gold.

[33]Appendix C.6 compares our call adjustment with the panel regression of Gilchrist and Zakrajšek (2012), a commonly used technique that controls for callability with a call dummy interacted with default risk and bond characteristics. That regression is designed to purge an index of credit spreads across a large bond universe, and we do not claim it fails for that purpose. As a bond-level call adjustment for yield curve estimation, however, it is not competitive with ours. We show that their panel regression tends to imply negative option values for bonds in our Aaa$^+$ sample, so it adjusts their raw yields in the wrong direction.

[34]Weighted price errors are price errors scaled by duration times price. To a first order, these approximate yield errors and thus we report them in basis points.

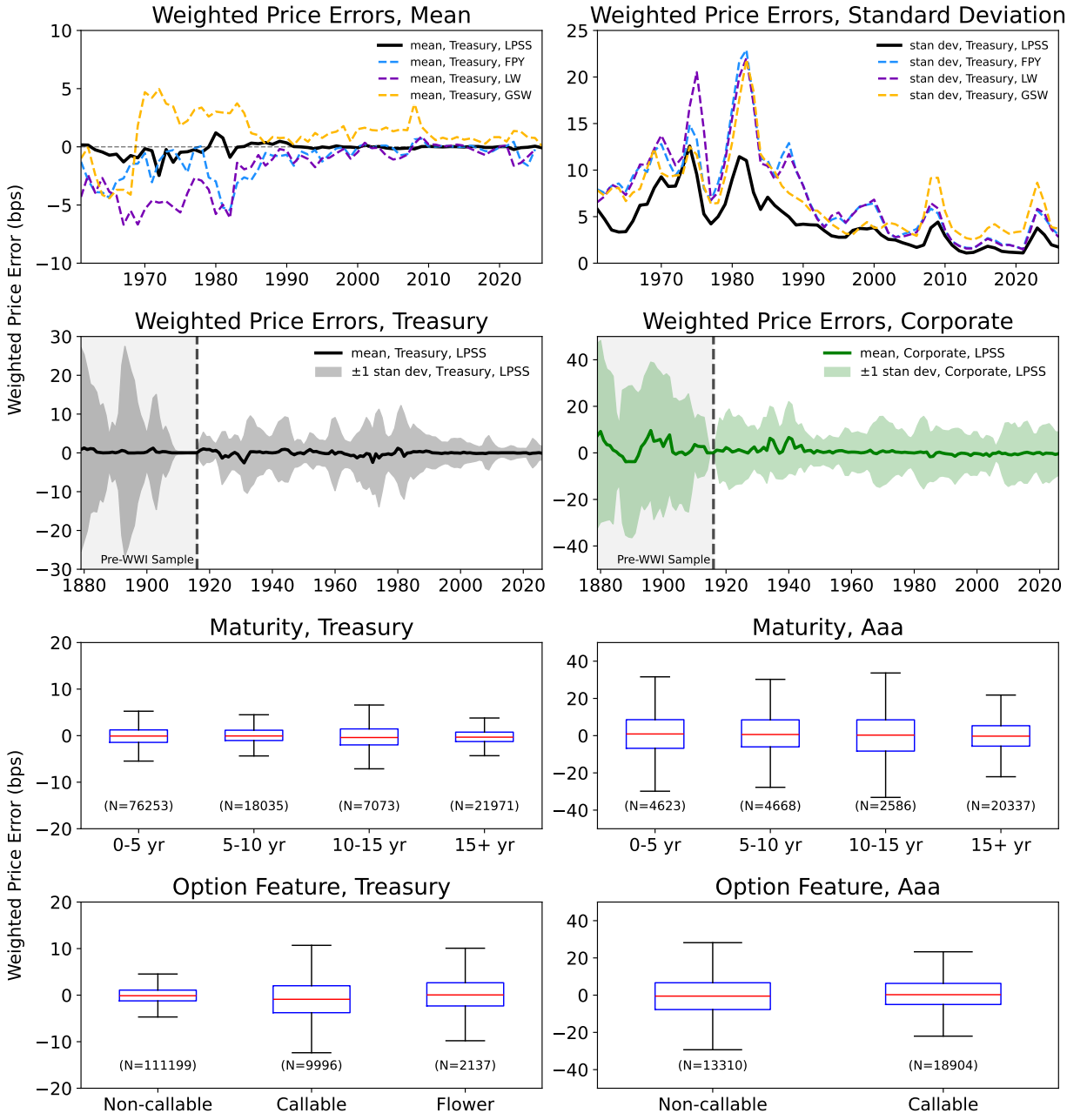


Figure 8: Statistical Fit

Notes: The top row show the time series of means and standard deviations, respectively, of weighted price errors (in basis points) for our (LPSS) baseline Treasury yield curve estimates (black lines) compared to the estimates of Filipović et al. (2022) (FPY), Liu and Wu (2021) (LW), and Gürkaynak et al. (2007) (GSW) since 1961. LPSS measures include errors on option bonds, while FPY, LW, and GSW measures do not. GSW measures exclude errors on Treasury Bills. The second row shows the means and standard deviations of weighted prices for our Treasury (black) and highest-grade corporate (green) samples since 1879 when the greenback period ended. The bottom four panels show weighted price errors within the Treasury and Aaa samples by maturity bins and option features since 1909. Bin sizes are reported in parentheses.

9 offers further evidence by revisiting the dates shown in the figures of Section 3.4, now plotting our tax- and option-adjusted yields in place of raw yields-to-maturity. As desired, the adjusted yields line up cleanly across the full maturity spectrum, tracing a common par yield curve. Together, these results indicate the absence of significant remaining omitted variable bias in our bond samples, i.e., our adjustments homogenize non-standard bonds. The top panels in Figure 8 show that this is no small feat.

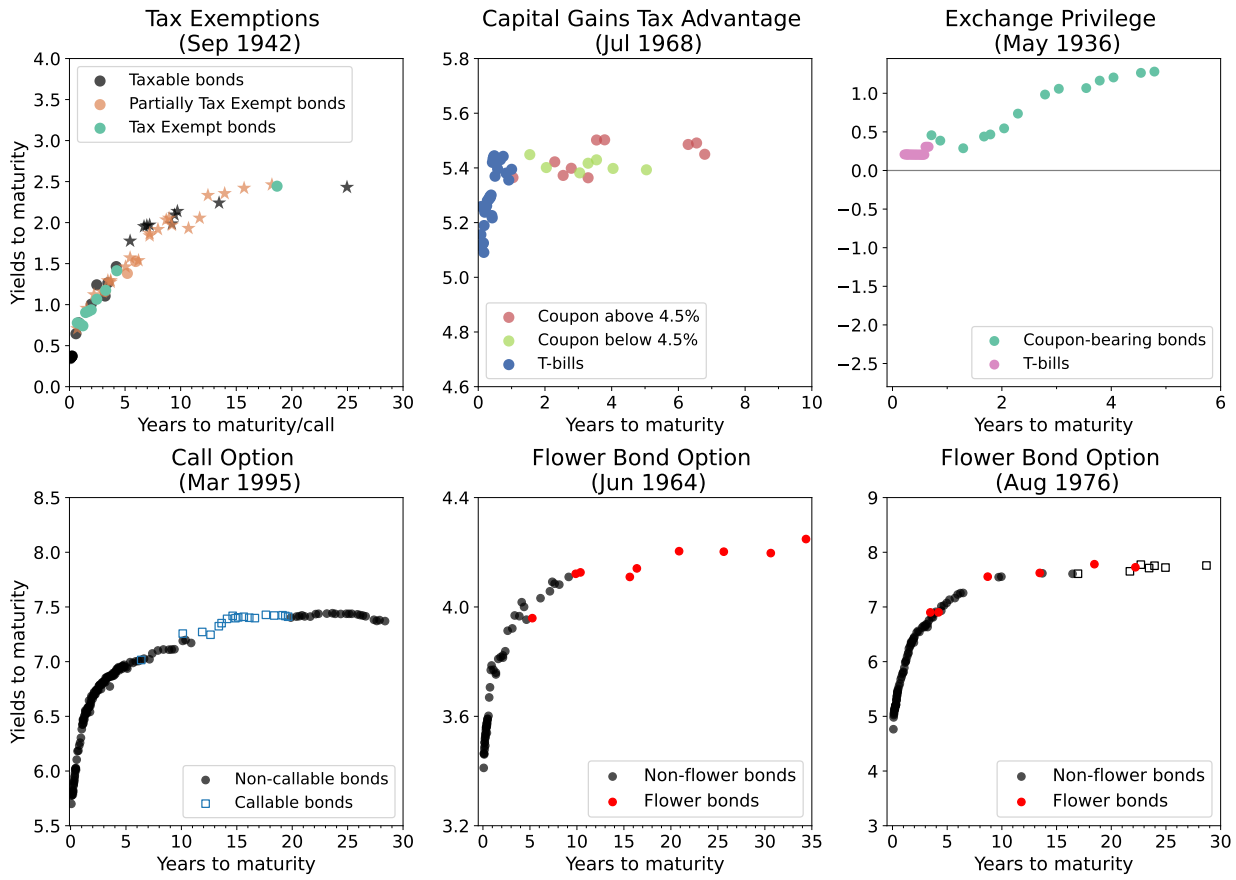


Figure 9: Tax- and Option-Adjusted Yields-to-Maturity

Notes: Tax- and option-adjusted yields-to-maturity are computed from formula (3.3), substituting $z_{k,t}$ with their respective estimated values. Each point corresponds to a separate bond outstanding in the given month. Dots show non-callable bonds, squares represent callable bonds (yields-to-maturity), stars represent callable bonds (yields-to-call).

Our Treasury estimate uniformly outperforms GSW, FPY, and LW despite pricing a far more heterogeneous sample. Tellingly, the gap is widest between 1961 and 1990, when the capital gains tax advantage creates exactly the omitted variable bias our adjustments are designed to absorb. This is the one non-standard bond feature in Table 1 that the strict sample restrictions in GSW, FPY, and LW is unable to control for. Figure 10, for September

1966, makes the mechanism concrete. The left panel shows measured raw Treasury yields, exhibiting a sharp discontinuity at the one-year maturity, where coupon bonds, which carry a capital gains tax advantage, give way to T-bills, which carry none.[35] Our capital gains adjustment absorbs the discontinuity. The right panel of Figure 10 shows how this leads to adjusted yields (hollow dots) that trace a smooth par yield curve (black line), undistorted by the one-year break. For comparison, that panel also shows the three curves considered the literature's gold standard. The tightly parametrized GSW produces a confused shape. The more flexible, non-parametric LW and FPY estimators instead try to fit the discontinuity itself, which produces the large, predictable price errors visible in Figure 8.

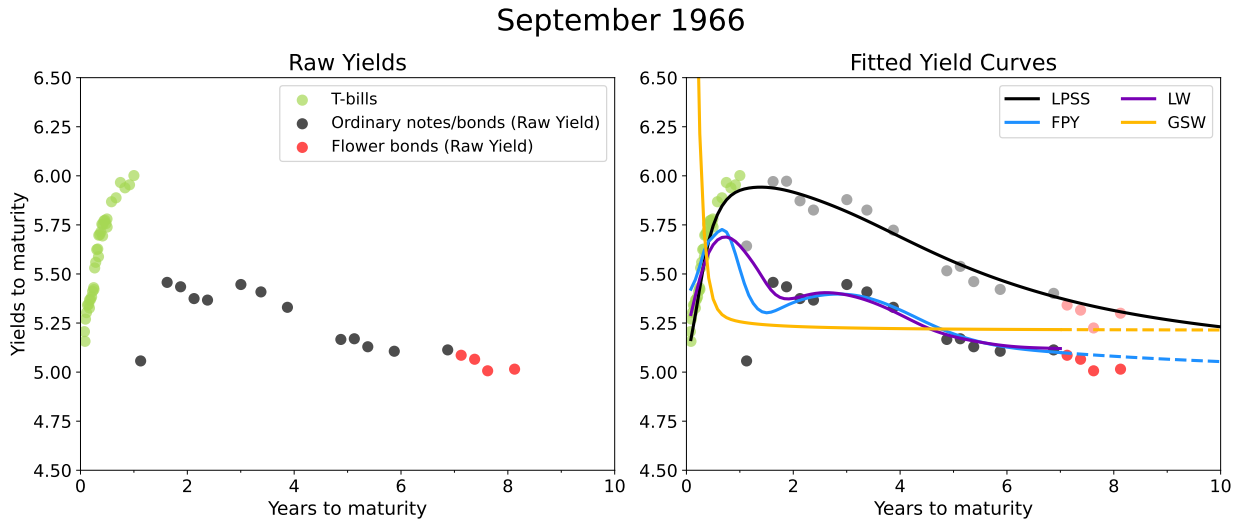


Figure 10: Omitted Variable Bias Correction Over Leading Treasury Estimates

Notes: The left panel shows raw yields-to-maturity for Treasury securities in September 1966 when coupon-bearing securities were subject to capital gains tax advantage. The right panel adds our adjusted yields (shaded dots) and fitted par yield curve compared to three popular yield curve estimates. FPY and GSW are extrapolations beyond 7 years to maturity (dashed lines).

(ii). Adjustment functions are correctly specified. [CL: I find this a bit strong] Our adjustments $z_{k,t}$ are functions of the contemporaneous discount function q_t. If this relationship were misspecified, the inferred q_t would be sensitive to the composition of the underlying sample. It is not. In Appendix C.3.1 we show that estimating q_t on a subsample of callable corporate bonds versus a subsample of non-callable corporate bonds delivers nearly identical curves and the same holds for subsamples with different degrees of moneyness. Appendix C.3.2 shows that swapping long-term flower bonds for long-term non-flower bonds in the

[35] We drop bonds with exchange features, as documented by Garbade (2021, p. 334).

pre-1973 sample barely moves the curve either.[36] This robustness is what lets us trust q_t for counterfactual and out-of-sample exercises, not just for pricing the bonds used to estimate it. We can recover a yield curve for non-callable corporates from a sample of only callable ones in the 1980s, or extend Treasury yields past seven years in the 1960s, when flower bonds are the only long-term government bonds.[37]

(iii). Estimated parameters are plausible. Our adjustment functions are grounded in semi-structural tax and option pricing formulas with economically meaningful parameters. That our estimated parameters line up with their observable counterparts is therefore an external validation of our identification of price effects.

The top panel of Figure 11 compares Aaa$^+$ call option-implied interest rate volatility (solid blue) with one-year-ahead realized volatility (dashed green), from 1920 to 2002.[38] We compute implied volatility period-by-period from estimates of $\tilde{\kappa}_t, \tilde{\sigma}_t$, in the spirit of how the literature backs out implied volatility from the Black and Scholes (1973) formula. Despite being fitted period-by-period to a cross section of Aaa$^+$ bonds, our implied interest rate volatility tracks year-ahead realized volatility well. Periods like the yield curve control show low, steady implied volatility, while the 1970s and 1980s show elevated implied volatility and a high "vol-of-vol," consistent with the realized interest rate process at the time.[39]

The middle panel of Figure 11 plots estimated option prices for two flower bonds, each evidently purchased for estate-tax purposes, as their redemption rates confirm (see Figure A.22 in Appendix A.4). Before 1966, as discussed in Section 3.4.1, the market priced this option almost exclusively on the 3% bond of 1995. Afterward, demand for the flower option grew and spread across multiple bonds as inflation rose. Indeed, the estimated option values track inflation closely, consistent with our "inflation put option" interpretation. Two events in the 1970s then pushed flower option values in opposite directions. When the Treasury stopped issuing flower bonds in March 1971 and interest rates subsequently rose, option values jumped sharply, consistent with growing excess demand. The Tax Reform Act of 1976 had the opposite effect: a sharply higher estate-tax exemption threshold and the elimination of flower bonds' capital gains tax advantage at redemption both cut excess demand, and option values fell dramatically.

[36] We are grateful to Greg Duffee for his insightful suggestions, which inspired Appendix C.3.

[37] Highly valued embedded options shorten flower bonds' effective duration, so their stated maturity can overstate how much they inform the long end. Before 1973, though, this adjustment is modest: a 30-year flower bond behaves like a 20-year bond. Flower bonds therefore still deliver genuine long-end information.

[38] Specifically, the time-t implied volatility of a one-year forward rate is given by the Hull-White forward rate volatility formula $\tilde{\sigma}_{iv,t} = \tilde{\sigma}_t\big((1-\exp(-\tilde{\kappa}_t)\big)/\tilde{\kappa}_t\big)\sqrt{\big(1-\exp(-2\tilde{\kappa}_t)\big)/(2\tilde{\kappa}_t)}$.

[39] A simple forecasting regression of one-year-ahead realized volatility on implied volatility gives a slope coefficient of 0.52, comparable to slope coefficients in VIX-based forecasts of realized S&P 500 volatility.

Finally, the bottom panel of Figure 11 identifies the marginal investor behind our counterfactual taxable Treasury yield curve for 1917–1941. Applying the rule of Section 3.4.1, the surtax rate τ_t^{pte} implied by the relative prices of *fte* and *pte* Treasurys (orange line) sits broadly inside the range of individual surtax rates (gray band) in 1917–1926 and 1932–1941 and near the maximum corporate rate (dashed light blue line) in 1927–1931. We therefore treat individuals as the marginal investor, except in 1927–1931 when the negligible surtax rate suggests that banks played that role.[40] After 1941, we observe fully taxable Treasurys, so τ_t^{inc} can be estimated directly without taking a stand on the marginal buyer. The figure nonetheless suggests that banks became the marginal buyers during WWII. Both the estimated surtax rate (orange) and the total federal income tax rate (black) closely follow their corporate IRS counterparts (dashed light blue and blue, respectively).[41]

5 Default Risk and State Taxes

Our estimates of the US funding advantage compare Treasurys with the highest-grade corporate bonds after adjusting for coupon, maturity, embedded options, and different federal tax treatment. The remaining spread should therefore capture the benefits of holding Treasurys that corporate bonds do not provide, such as liquidity, safe-haven status, collateral value, and favorable regulatory treatment. Two further components, default risk and differential state tax treatment, could in principle remain in the spread and have been studied in the yield curve literature. This section examines each in turn, bounds their contribution, and shows they are of limited importance.

5.1 The Default Risk Premium Puzzle

Realized default risk in our sample is close to zero, and market prices of default risk cannot explain more than a modest fraction of the spread without very high risk aversion (a "default risk premium puzzle") or a believe in a latent risk state that has never realized (a "rare disaster puzzle").

We start with realized default and loss rates.[42] Following Moody's cohort methodology,

[40] Figure C.41 in Appendix C.7 shows the resulting τ_t^{inc}.

[41] We form the corporate IRS rates from corporate normal and surtax rates. During World War II, we additionally assume an effective excess profits tax rate of 5% in 1942, 10% in 1943, and 20% in 1944–1946, which we add to both corporate IRS lines. Bond-implied tax rates fall dramatically following the dissolution of excess profits taxation after June 30, 1946.

[42] We identify defaults and recoveries by matching our bond-level data to the Moody's Default and Recovery Database, which records credit events at the instrument level along with the bond's market price 30 days after default. Following Moody's, we count any credit event, including a single missed or deferred interest

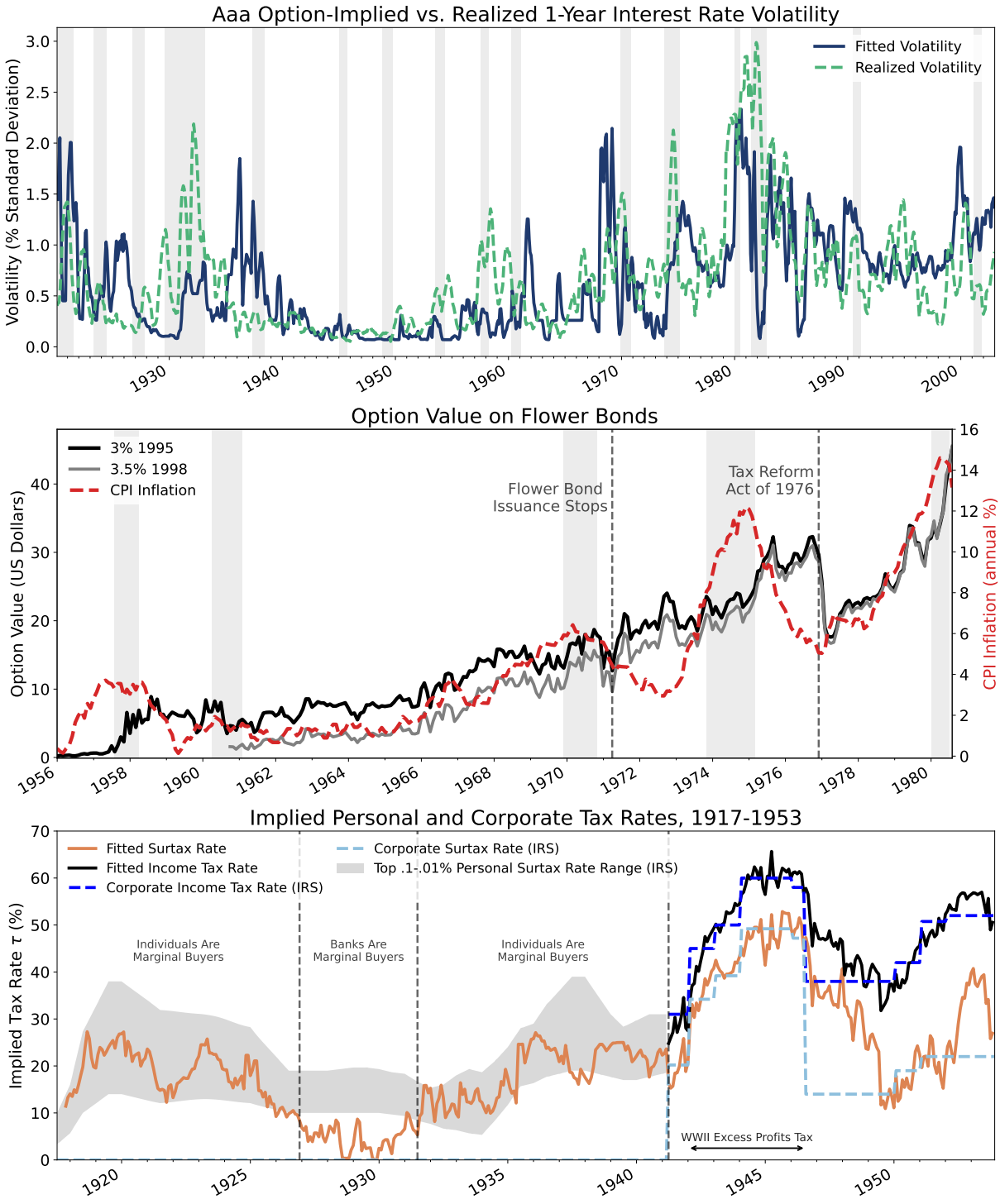


Figure 11: Distortion Parameter-Implied Variables

Notes: Top panel shows Aaa option-implied 1 year interest rate volatility against 1 year ahead realized volatility from 1920-2002. Middle panel shows the fitted option values of the two longest-maturity flower bonds since in 1955. CPI inflation is plotted in red. Bottom panel plots the Treasury bond-implied effective tax rates in black and orange against comparable IRS tax rates.

we track every bond in our estimation panel from the month it first enters the sample as Aaa until its maturity, call, or default. We follow each bond through any downgrade, so defaults that occur long after a bond leaves our Aaa sample still count.[43] Table 2 reports the cumulative five- and twenty-year default and loss rates. Over the long sample, bonds rated Aaa in 1920 or later have a twenty-year cumulative default rate of 0.65 percent and a loss rate of 0.31 percent of par. Moody's published *issuer-level* rate, computed with the same methodology over the same window on its full Aaa universe, is 1.3 percent. The two rates bound the long-run default rate of highest-grade bonds. Moody's rate is an upper bound because it records a default when an issuer misses a payment on *any* rated debt, including the junior and lower rated bonds we exclude, and because it covers every Moody's Aaa issuer. Our rate is a lower bound because we select only the safest bonds of the safest issuers, requiring a unanimous Aaa rating from every agency we observe.

After 1945, default rates are exactly zero for every cohort at every horizon. No bond in our postwar sample defaulted, and no issuer defaulted while its bond was outstanding.[44] Default in the full sample comes mainly from railroad bonds that Moody's rated Aaa in the earliest years of the ratings and that defaulted during the Great Depression or later (see Appendix E). Better rating accuracy, the decline of railroads, or a different policy response to later crises could explain the break. We cannot tell these explanations apart, but each implies that the full-sample rate overstates the default risk of the postwar bonds in our panel.

We next ask how large the default risk premium could be. To remain agnostic about the structural model of default risk, we bound the premium in the spirit of Hansen and Jagannathan (1991). Appendix E.3 gives the environment and derivation. Under any pricing kernel, the default risk premium is a covariance between the kernel and the default event, and a covariance cannot exceed the product of the two volatilities. For a diversified corporate bond portfolio held from time 0 to time t, this implies

$$\frac{\Delta}{\sqrt{p(1-p)}\,\sqrt{\phi}} \leq \frac{\sigma_0(\xi_{0,t})}{\mathbb{E}_0[\xi_{0,t}]}.$$

where p is the physical default probability over t periods (the theoretical counterpart of the estimates in Table 2), Δ is the default risk premium, ϕ is the correlation of defaults across

payment, as a default. See Appendix E.1 for details.

[43] No bond in our sample ever defaulted while rated Aaa. Recorded defaults typically come years after severe downgrades, at an average rating of Baa. Moody's own annual default rate for Aaa issuers is zero in every year from 1920 to 2026.

[44] This is a property of our sample, not of postwar Aaa bonds in general. Some bonds rated Aaa after 1945 did default, but our selection criteria exclude all of them. The criteria require a unanimous Aaa rating across agencies, seniority, no sinking funds, and a non-financial issuer (see Section 2).

	Default rate		Loss rate	
Cohort	5-year	20-year	5-year	20-year
Bonds rated Aaa in 1909 or later	0.00	0.73	0.00	0.36
Bonds rated Aaa in 1920 or later	0.00	0.65	0.00	0.31
Bonds rated Aaa in 1945 or later	0.00	0.00	0.00	0.00
Moody's Aaa issuers, 1920-2026	0.13	1.3		

Table 2: Default and Loss Rates for Aaa-Rated Bonds

Notes: Cumulative rates in percent at the five-year and twenty-year horizons, following Moody's methodology with a withdrawal-adjusted denominator. Every bond is tracked from its first Aaa observation, through any downgrades, to maturity, call, or default. Loss rates value each default at one minus the bond's market price thirty days after default. The Moody's row shows their published issuer-weighted rates. Appendix E.1 reports additional cohorts and lifetime horizons.

corporate bonds, and $\sigma_0[\xi_{0,t}]/\mathbb{E}_0[\xi_{0,t}]$ is the normalized volatility of the t-period pricing kernel $\xi_{0,t}$.[45] We suppress time subscripts on the left-hand side for brevity.

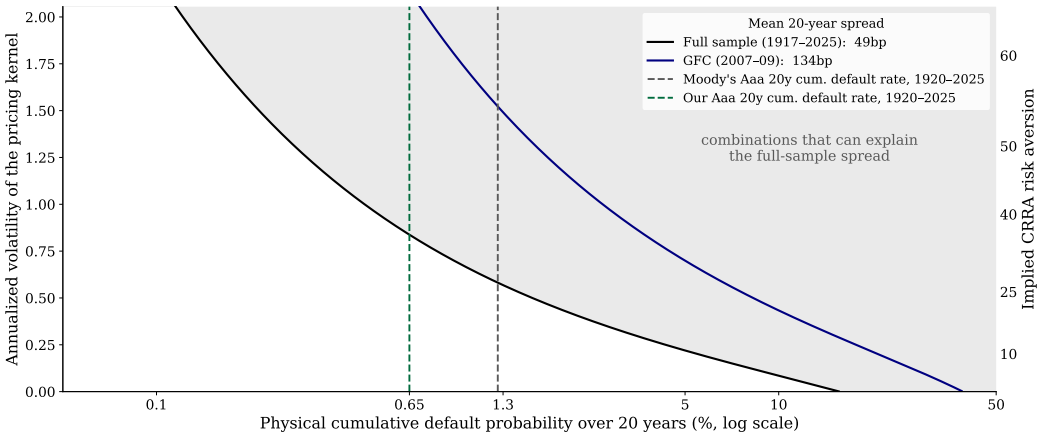


Figure 12: "Hansen-Jagannathan" bound for bond default risk. [CL: Bigger font]

Notes: Each curve traces the minimum annualized volatility of the pricing kernel required to rationalize the period-average Aaa-Treasury spread as compensation for default risk at the twenty-year horizon. The horizontal axis is the physical cumulative default probability p in log scale. The right axis restates the requirement as CRRA risk aversion under power utility with two percent annual consumption growth volatility. Combinations above a curve can rationalize the corresponding spread, combinations below it cannot. Dashed vertical lines mark our sample's realized twenty-year cumulative default rate (0.65 percent) and Moody's rate (1.3 percent).

[45] The bound uses only the existence of a pricing kernel, so it holds for any preferences and any beliefs consistent with observed prices.

Inverting the bound gives, for a hypothesized p, the minimum kernel volatility consistent with the observed spread. Figure 12 traces this minimum at the twenty-year horizon against the assumed physical default probability, with our realized default rate and Moody's rate marked. At either rate, rationalizing the period-average spread requires a kernel more volatile than the equity premium already demands, with constant relative risk aversion of 25 to 65 depending on the spread, several times the level at which the equity premium is deemed a puzzle. Reconciling the spread with default risk therefore requires either extreme risk aversion or a latent disaster that has never realized, one worse than Moody's long-run estimate of 1.3 percent and the 1.8 percent realized default rate in the Great Depression (discussed below). Attributing the spread to default risk implies a default risk premium puzzle or a rare disaster puzzle.

Three further exercises in Appendix E show that default risk cannot account for the bulk of the spread.

1. We subtract CDS premia on Aaa issuers from observed bond-level spreads, following Longstaff et al. (2005). The price of default protection accounts for a quarter to a third of the spread in their own 2001–2002 sample window and about half at the peak of the financial crisis. The default risk premium puzzle persists even for the CDS premium alone. Given Moody's upper bound of 0.13% for the five-year Aaa default probability, rationalizing the median five-year CDS quote of 25 basis points requires risk aversion of roughly 25, which suggests frictions in the CDS market (e.g. Siriwardane, 2019).

2. We measure issuer default risk with the equity-based distance to default, following Gilchrist and Zakrajšek (2012). It accounts for at most a fifth to a quarter of the spread in any decade since 1970. Even this is an upper bound on the default risk premium, because distance to default captures total firm risk, and translating it into a premium would require isolating the common component of default risk.

3. We use the Great Depression as a stress test of realized credit losses. Even in the worst credit episode in our sample, losses on bonds rated Aaa in 1925 amount to 1.6 percent of par by 1940, about 10 basis points per year, against a spread that averaged roughly 100 basis points. A rare disaster large enough to explain the spread would have to be substantially worse than the Great Depression itself.

5.2 State Taxes

State taxes explains at most a modest portion of the spread. For *some* investors, corporate bond interest is subject to both federal and state income taxes, while Treasury interest is

subject only to federal tax. Elton et al. (2001) bound the effect by combining a hypothesized high federal tax rate with a state tax rate of 4–5%, which explains about 20 basis points of the Aa-Treasury spread over 1985–1995. We take this as a comparable bound for our Aaa$^+$-Treasury spread. In practice, the true effect is likely smaller, because the largest holders of corporate bonds pay little or no state tax. For reference, Figure 1 of Koijen and Yogo (2023) shows that pension funds, insurers, and mutual funds held 60–80% of corporate bonds since 1945, which makes them the likely marginal investor in the Aaa bond market. Pension funds are broadly exempt from federal and state taxes. Insurers pay state premium taxes at low rates, if at all, instead of state corporate income tax. Mutual funds pass income through to their investors, and a large share of fund assets sits in tax-advantaged investment accounts.[46] State taxes therefore cannot explain a considerable share of our Aaa$^+$-Treasury spread if the marginal investor is any of these three institutions.[47]

6 US Funding Advantage and Asset Pricing

Recent work connects the market value of US Treasury debt to the government budget constraint (e.g., Cochrane, 2022; Jiang et al., 2022; Collin-Dufresne et al., 2023). This requires going beyond the measurement of the US funding advantage, which has been our focus so far. Instead, we now need to estimate a pricing kernel for Aaa$^+$ corporate bonds and a non-pecuniary benefit wedge for Treasurys, and ultimately build a dynamic asset pricing model of the funding advantage and the maturity structure of government debt. Estimating the kernel and the wedge requires additional structure on the asset pricing equations, which Assumption 2 formalizes.

Assumption 2. Government and highest-grade corporate bonds are default-free. Moreover, there exists a stochastic discount factor (SDF), $\xi_{t,t+1}$ and a Treasury wedge $\Omega^{(j)}_{t,t+1}$ such that, for all $j \geq 1$, the elements of $\tilde{\mathsf{q}}_t$ and q_t solve the Euler equations:

$$\tilde{q}^{(j)}_t = \mathbb{E}_t\left[\xi_{t,t+1}\tilde{q}^{(j-1)}_{t+1}\right], \quad \text{with} \quad \tilde{q}^{(0)}_t = 1 \tag{6.1}$$

$$q^{(j)}_t = \mathbb{E}_t\left[\xi_{t,t+1}\Omega^{(j)}_{t,t+1}q^{(j-1)}_{t+1}\right], \quad \text{with} \quad q^{(0)}_t = 1. \tag{6.2}$$

Conceptually, the Treasury wedge $\Omega^{(j)}_{t,t+1}$ captures the non-pecuniary benefit of holding j-

[46] For example, Bergstresser and Poterba (2002) find that at least 35% of mutual fund assets were held in tax-advantaged accounts at the end of 1998.

[47] Consistent with this, Green and Ødegaard (1997) find that Treasury-implied marginal tax rates fell to zero after the Tax Reform Act of 1986. Because in equilibrium tax-burdened investors sort into tax-advantaged securities of similar default risk, the effective tax rate on Aaa$^+$ bonds is plausibly also small.

maturity government debt for one period. We let the non-pecuniary benefit vary by maturity, which captures any Treasury market frictions that distort pricing across the term structure, such as a liquidity premium on short Treasurys or a regulatory role for long Treasurys, and nests the models in the macro-finance literature. Under Assumption 2, this section jointly estimates the processes for the SDF, $\xi_{t,t+1}$, and the Treasury wedge, $\Omega^{(j)}_{t,t+1}$, from 1925 onward, when Section 5 shows that both curves are effectively default free.

We start in Subsection 6.1 by showing that existing macrofinance models of $\Omega^{(j)}_{t,t+1}$ are inconsistent with the evidence uncovered in the first part of this paper. Contrary to the claims in the literature, $\Omega^{(j)}_{t,t+1}$ is not well approximated by a downward sloping function of the government debt-to-GDP ratio augmented by i.i.d. demand shocks. In addition, current models are inconsistent with observed Treasury maturity management, leading to inaccurate conclusions about what drives overall US government borrowing costs. We conclude that the apparent fit of the models in the macrofinance literature is largely an artifact of funding advantage mismeasurement rather than underlying economic relationships, particularly at long maturities.

To resolve these problems, we estimate the first affine term structure model (ATSM) for $\Omega^{(j)}_{t,t+1}$ (in Subsection 6.2) and reconcile it with the government budget constraint (in Subsection 6.3). This model uses the principal components of the yield curves and the term structure of government debt obligations to jointly explain changes in the discount functions $\{\tilde{\mathsf{q}}_t, \mathsf{q}_t\}$ for highest-grade corporate and government bonds, while respecting no-arbitrage across time and consistency with the government debt process. Being able to fit this model highlights the value of our new yield curve estimates. We are able to build and estimate models of US funding advantage that integrate the large bond pricing toolkit from the ATSM literature with the models of non-pecuniary Treasury demand and market segmentation from the macrofinance literature. We can then determine which classes of models are consistent with the bond pricing and government bond supply data. We conclude that macrofinance models should have that Treasurys are priced differently to corporate bonds, short Treasurys are priced differently to long Treasurys, the covariance between the the SDF and the Treasury wedge is important for explaining US funding advantage, and changes in the US funding advantage relate to government debt maturity changes.

6.1 Challenging The Macro-finance Consensus About $\Omega^{(j)}_{t,t+1}$

Following Krishnamurthy and Vissing-Jorgensen (2012) (henceforth KVJ), a collection of papers have argued that there is an unconditional negative relationship between the US government funding advantage and the market value of outstanding US government liabilities.

We use our new yield curve estimates to reproduce the literature's analysis and challenge their conclusions. In the main text, we focus on KVJ and the more recent Choi et al. (2026) (henceforth CKP), which offer complementary views of how to model $\Omega_{t,t+1}^{(j)}$ for long-term bonds. We replicate and discuss the broader literature (e.g., Greenwood et al., 2015; Nagel, 2016; Krishnamurthy and Li, 2023) in detail in Appendix F.

KVJ and CKP specify a common one-period holding return Treasury wedge at all maturities, j, of the following form:

$$\Omega_{t,t+1}^{(j)} = \exp\Big(\beta_0 + (\beta_1 + \gamma v_t)\log(\mathcal{Q}_t/(P_t Y_t)) + \log(\zeta_t)\Big), \quad \forall j \tag{6.3}$$

where $\mathcal{Q}_t = \sum_{j \leq \mathcal{J}} q_t^{(j)} b_t^{(j)}$ is the nominal market value of government debt promises, P_t is the price level, Y_t is output, $b_t^{(j)}$ is the par value of government cash flow promises in j periods, β_1 is a loading on asset quantities that both papers argue is significantly negative, γ is a rotator for the elasticity,[48] v_t is foreign volatility, and $\log(\zeta_t)$ is a time-t adapted random variable often interpreted as a demand shifter. From an asset pricing perspective, this model restricts deviations from no-arbitrage by imposing that $\mathcal{Q}_t$, v_t, and ζ_t are the only factors that can predict the non-pecuniary component of the pricing kernel for government debt.[49] Equation (6.3) implies that the funding advantage on 1-period Treasurys is given by:

$$\chi_t^{(1)} = \beta_0 + (\beta_1 + \gamma v_t)\log\Big(\mathcal{Q}_t/(P_t Y_t)\Big) + \log(\zeta_t). \tag{6.4}$$

while for $j-$maturity Treasurys, the funding advantage has a similar formula that also includes additional covariance terms.[50] We use our new funding advantage estimates to replicate the literature's estimation of specification (6.3) and challenge its efficacy. The replications lead to two main results, which we discuss below.

Result 1. *The debt-to-GDP factor (β_1) is no longer significant in KVJ's empirical analysis once we use our tax and option corrected funding advantage estimates.*

KVJ test specification (6.3) with $\gamma = 0$ by checking for co-movement (visually and statistically) between the index-based Aaa-Treasury spread and the ratio of the market value of publicly held government debt to GDP. Figure 13 reproduces their scatter plot from Figure 1 in their paper. The top panel reconstructs their scatter plot using their data (the index-based spread from Section 3.3) and time period (1919-2008). The bottom panel plots

[48] KVJ implicitly sets this to zero while CPK includes it in their estimation.

[49] KVJ also allow a collection of privately issued assets, denominated θ_t^P, to generate a non-pecuniary benefit. However, they are not specific about what is in θ_t^P. We discuss this further in Appendix. CKP sets $\theta_t^P = 0$.

[50] See Appendix F.1.1 for a derivation of the full formulas.

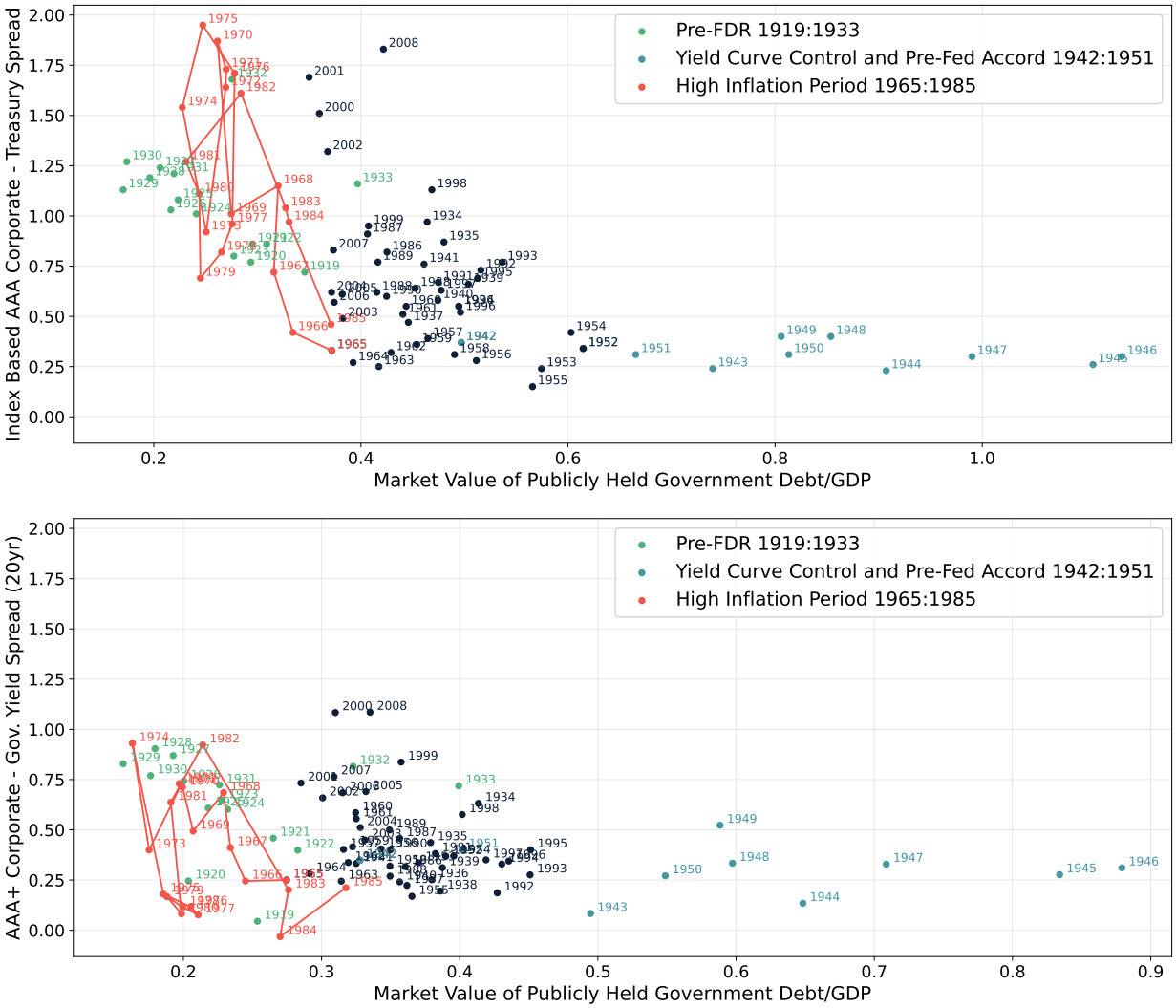


Figure 13: (a) The top panel replicates Krishnamurthy and Vissing-Jorgensen (2012) using their data and time period from 1919-2008. (b) The bottom panel uses our estimate of the 20-year par-yield spread and our market value of government debt/GDP over the same period.

the same time period but uses our estimates of the 20-year funding advantage, corrected for tax advantages and embedded options, together with our measure of the market value of government debt.[51] The stable unconditional negative relationship that KVJ observe using the index-based spread is largely eliminated once we use our new estimates. We highlight, with colors and lines connecting consecutive years, the periods where our estimates diverge most from existing studies: the 1920s, the 1940s, and the 1970s-80s.[52] These are precisely

[51] We follow Hall and Sargent (2022) and compute the Market Value of Marketable Treasury Debt. We do this by using our Treasury zero-coupon discount function to discount all US Treasury promises from outstanding marketable Treasuries at all future maturities. This is different to the KVJ plot, which includes intra-government debt in its value of total market value of government debt.

[52] Our estimates differ in 2000-2002 because Moody's included in-the-money corporate bonds in their index.

the periods driving the shape of the original KVJ plot, and precisely where the distortions from Section 3 are most pronounced. In particular, as Subsection 4.1 shows, correcting for the flower bond distortion moves the red dots for the high-inflation 1960s-70s from the top left of KVJ's plot to the bottom left of ours. Our series therefore no longer implies that the funding advantage expanded when inflation shocks devalued long-term government debt during 1965-85.

	Index-based spread			Our yields		
	(1)	(2)	(3)	(1)	(2)	(3)
	KVJ	KVJ	CKP	KVJ	KVJ	CKP
Sample	1926-2008	1941-2008	1935-2019	1926-2008	1941-2008	1935-2019
Dependent spread	Aaa-Tsy	Aaa-Tsy	S60/L40	20yr par	20yr par	IRR
$\text{Log}(\frac{\text{Debt}}{\text{GDP}})$	-0.800***	-0.779***	-0.233***	-0.153	-0.081	0.088**
	(0.156)	(0.144)	(0.043)	(0.133)	(0.144)	(0.043)
FVD			-0.314***			0.125*
			(0.068)			(0.066)
FVD x $\text{Log}(\frac{\text{Debt}}{\text{GDP}})$			-0.338***			0.062
			(0.062)			(0.060)
Volatility, slope, const	Yes	Yes	Yes	Yes	Yes	Yes
Observations	83 (Y)	68 (Y)	339 (Q)	83 (Y)	68 (Y)	336 (Q)
R^2	0.573	0.637	0.532	0.174	0.095	0.247

Table 3: Selected regression results from KVJ and CKP. The left three columns use the existing published spread series. The right three columns use our yield curves. Specifications (1)/(2) replicate KVJ Table 1.A column 3 using data from 1926-2008 and 1941-2008. Specification (3) replicates CKP Table 1 column 1. FVD is a Foreign Volatility Dummy based on the UK stock market volatility. Log(Debt/GDP) is each paper's own regressor for the same underlying concept. Volatility is stock return volatility in (1) and (2) and the US VIX in (3). The 20yr par spread is our 20-year par-yield spread and IRR is our internal rate of return spread. All logs are natural logs. Standard errors in parentheses: AR(1) for (1)/(2) and conventional OLS for (3). Significance stars are $|t|$-based for (1)/(2) and p-value-based for (3), matching each source paper's own convention: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

Table 3 formalizes our comparison to KVJ. The authors test the spread-quantity relationship in equation (6.4) by regressing their index-based Aaa–Treasury spread on the market value of government debt relative to GDP, controlling for volatility and the slope of the Treasury yield curve. Specifications (1) and (2) in the left panel of Table 3 reconstruct their regressions using their index-based series and the right panels re-estimate them using our series.[53] We summarize the key results in Table 3 and give the full regressions in Appendix

See Figure C.42 in Appendix C.7.

[53]We show regressions for both the original KVJ sample (1926-2008) and a shorter sample (1941-2008) because as discussed in Subsection 3.4.1, there is additional uncertainty about our estimate of the funding advantage between 1917–41 and we want to show our results do not depend on that period.

F.2.1. In both specifications, the left panels reproduce KVJ's finding of a statistically significant negative relationship between the funding advantage and government debt relative to GDP. The right panels, re-estimated with our series, do not. We conclude that KVJ's negative relationship between funding advantage and debt quantity is largely an artifact of index-spread mismeasurement, one that does not survive correcting for tax and option values in the historical bond sample.[54]

Result 2. *Once Treasury maturity dynamics are accounted for, there is no robust unconditional relationship between overall government borrowing costs and debt-to-GDP, whatever the level of foreign volatility, contrary to CKP.*

CKP use a similar empirical specification to KVJ but with two changes that open up additional discussion of Treasury demand and maturity management. First, they use a weighted funding advantage series with fixed 60% weight on the 3-month commercial paper-TBill spread and fixed 40% weight on the long-maturity index-based spread.[55] To replicate and extend their analysis, we use our yield curve estimates to compute a dynamically weighted internal rate of return (IRR) spread. This spread is the difference between the IRR pricing all outstanding Treasury cash flows off the Treasury discount function and the IRR pricing those same cash flows off the corporate discount function.[56] This gives the difference between effective overall private and public borrowing costs, weighted by outstanding government obligations. Second, they add UK stock market volatility as a "demand rotator".

We start by replicating CKP's visual data inspection. The authors argue that their 60/40 short-long weights approximately reflect the fraction of Treasurys maturing in less than and more than 3 years.[57] Figure 14 plots the actual portfolio shares and shows this is not true. CKP's fixed 60/40 short-long weights (the dashed lines) are very different to the historical par-value-weighted shares of government debt maturing in under and over 3 years (the solid lines), particularly before the 1960s.[58] Figure 15 reproduces the scatter plot of the weighted-average spread against the market value of debt/GDP in Figure 2 in

[54]Our analysis does not necessarily rule out that quantity changes explain KVJ's convenience yield, since Aaa corporate bonds may themselves earn non-pecuniary benefits, so the Aaa-Treasury spread need not equal their definition of the convenience yield. Estimating convenience yields directly would require historical equity pricing data and a pricing kernel undistorted by any convenience benefit.

[55]CKP's model has one-period government debt, whereas their empirical spread combines short- and long-maturity series. We replicate their empirical specification as published and extend it to the full term structure with our IRR spread.

[56]See Appendix F.2.3 for a formal definition.

[57]Other papers such as Collin-Dufresne et al. (2023) make a similar assumption of fixed portfolio weights although with a more granular portfolio decomposition.

[58]The weights are particularly volatile during World War I when the Treasury issued large quantities of short-term certificates of indebtedness for cash management purposes before issuing long-term Liberty Bonds. See Garbade (2012) for additional discussion.

CKP, using their weighted-average spread (panel a) and our IRR spread (panel b). Red dots mark periods of high UK equity return volatility, blue dots periods of low volatility. CKP interpret their panel a) as evidence that US debt demand rotates when UK volatility rises. By contrast, we find little evidence of a rotation when using our estimates in panel b).[59] Moreover, our panel b) reinforces our findings from replicating the KVJ regressions. Once we account for dynamic changes in the maturity structure through our IRR spread, the government's overall funding advantage did not change substantially as debt supply expanded. This would be consistent with the government's maturity structure adjusting to maintain a similar funding advantage throughout changes to debt/GDP.

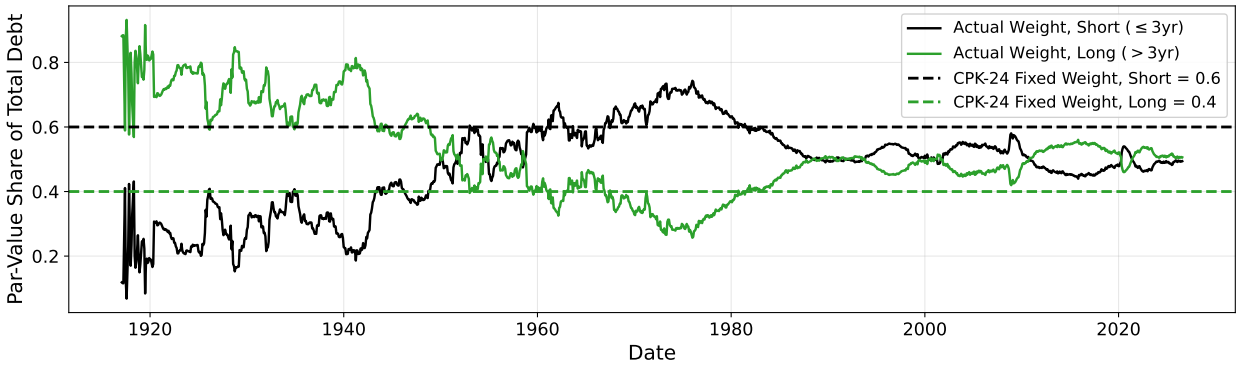
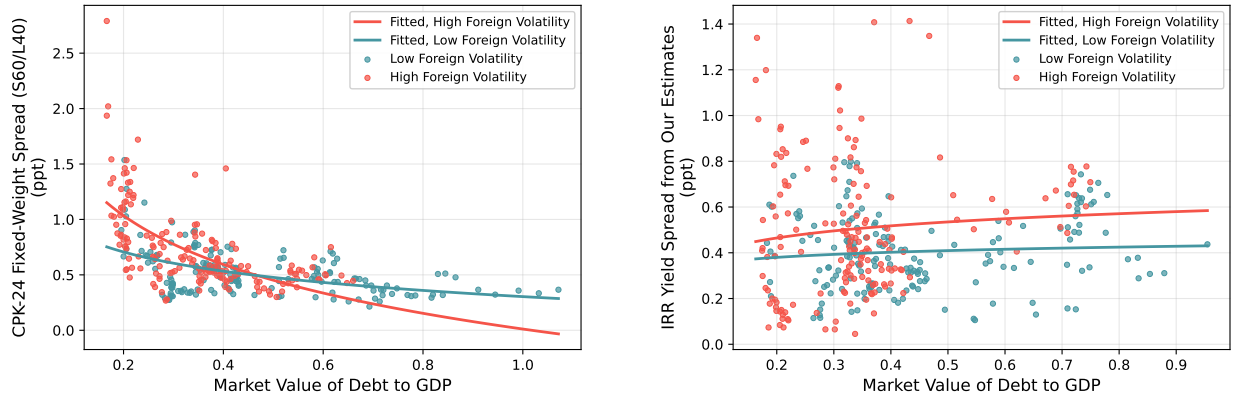


Figure 14: Par-value share of short (≤3yr) vs. long (>3yr) publicly held Treasury debt using our quantity data and Choi et al. (2026)'s fixed weighting.



(a) CKP fixed-weight average spread

(b) Our IRR average yield spread

Figure 15: Spread vs. market value of debt/GDP by foreign-volatility regime for 1935:Q1–2019:Q4. Panel (a) uses CKP's own privately-held par-value debt/GDP measure; panel (b) uses our own market-value-consistent debt/GDP.

[59]When we examine the scatter plots separately for each maturity bin in Appendix F.2.3 we only find evidence for a rotation at maturities less than 1 year.

Specification (3) in Table 3 formalizes these visual observations by replicating CKP's OLS regression with both their data (Column (3) of the left panel) and ours (Column (6) of the right panel). In contrast to CKP, with our IRR spread, the coefficient on debt/GDP becomes weakly positive and its interaction with foreign volatility (FVD) turns insignificant. Appendix F.2.3 replicates their IV analysis with our data, which further extends this conclusion. Our estimates therefore provide no robust evidence that quantities explain average funding advantage spreads, or that foreign volatility rotates aggregate Treasury demand as CKP propose. CKP use the FVD-Debt/GDP interaction coefficient to test and reject the hypothesis that government debt has no market power. However, since our estimate of the interaction coefficient is insignificant, the test in their paper no longer offers support for these conclusions.

Taken together, the results in Table 3 show that the negative spread-quantity relationships emphasized in the literature are not robust features of the historical data, especially after accounting for the maturity structure of government debt.

6.2 An Affine Term Structure Model of Funding Advantage

Subsection 6.1 showed that the quantity-based wedge specifications in the literature do not hold up on our corrected series. We replace these specifications with a collection of affine Gaussian models in which the Treasury wedge $\Omega^{(j)}_{t,t+1}$ depends on the principal components of the yield curves and on the maturity structure of government debt, and we fit and compare these models using our yield curves and data on Treasury debt outstanding by maturity. This delivers a joint estimate of the corporate bond pricing kernel, the Treasury wedge, and government debt issuance by maturity. We first set up the state space, then define the pricing kernel and the Treasury wedge, and finally describe the estimation and the results. We discuss the key points here and defer the details to Appendix G.

6.2.1 State Space

Yield states: Following the affine term structure literature (e.g., see Piazzesi, 2010), we use principal components of the yield curves as states. We let $\tilde{x}_t$ denote the first three principal components of the highest-grade corporate yield curve, x_t denote the first three principal components of the funding advantage curve, and $s_t := (\tilde{x}_t, x_t)$ denote the full set of principal component states. The first three principal components of the Treasury yield curve are commonly identified as the "level", "slope", and "curvature" of the yield curve (e.g., Litterman and Scheinkman, 1991). Appendix G.1 shows that the same interpretation extends to our new estimates. The first three principal components of the corporate yield

curve approximately map to its level, slope, and curvature. The second and third principal components of the funding advantage curve approximately map to its slope and curvature, and the first is a level factor for long maturities.

Debt states: Let Y_t denote nominal GDP and $g_t := \log(Y_t/Y_{t-1})$ nominal GDP growth. Let $b_t^{(j)}$ denote the government's outstanding nominal promised payments due in j periods at time t, $b_t := \{b_t^{(j)}\}_{j=1}^J$ the vector of promised payments, where J is the maximum observed maturity, and $\bar{b}_t := b_t/Y_t$ its GDP-normalized counterpart. Let $\iota_t^{(j)}$ denote new issuance of claims due in j periods, $\iota_t := \{\iota_t^{(j)}\}_{j=1}^J$ the vector of new issuance, and $\bar{\iota}_t := \iota_t/Y_t$ its GDP-normalized counterpart. Let A denote the $J \times J$ aging (shift) matrix, with $(Ab)^{(j)} = b^{(j+1)}$ for $j < J$ and $(Ab)^{(J)} = 0$. Promised payments then evolve according to the accounting identity

$$b_t = A\, b_{t-1} + \iota_t.$$

Following the fiscal literature (e.g., Barrett and Johns, 2024), we approximate the cross section of government promises with three exponentially decaying bonds. Bond i has a decay rate $\varsigma_i \in (0,1)$, so each period a fraction ς_i of its outstanding balance comes due and one unit of the bond promises $H_{ji} := \varsigma_i(1-\varsigma_i)^{j-1}$ in j periods. Let $n_t^{(i)}$ denote the outstanding principal on bond i at the end of period t, $n_t := \{n_t^{(i)}\}_{i=1}^3$ the vector of outstanding principals, and $\bar{n}_t := n_t/Y_t$. The GDP-normalized approximate maturity structure of promised payments is then $H\bar{n}_t$. Replicating the decay rate selection exercise of Barrett and Johns (2024) on our sample, we find that bonds with maturities of 0.45, 2.7, and 8.8 years fit the panel of government promises well, close to their finding. We can therefore interpret the three bonds as approximately the outstanding stocks of short-, medium-, and long-term debt.

State evolution: The model state vector is:

$$X_t := [g_t,\ \bar{n}_t^T,\ s_t^T]^T \in \mathbb{R}^{10}.$$

We impose a homoskedastic VAR(1):

$$X_{t+1} = \mu_X + \Phi_X X_t + V_{t+1}, \qquad V_{t+1} \sim N(0, \Sigma), \tag{6.5}$$

with $\mu_X \in \mathbb{R}^{10}$, $\Phi_X \in \mathbb{R}^{10\times 10}$ and $\Sigma \in \mathbb{R}^{10\times 10}$ symmetric positive definite.

6.2.2 Pricing Kernel and Treasury Wedge

Corporate pricing kernel $\xi_{t,t+1}$: takes an exponential affine form:

$$\begin{aligned}\xi_{t,t+1} = &\exp\left(-\tilde{r}_t - 0.5\lambda_t^T\Sigma\lambda_t - \lambda_t^T V_{t+1}\right), \quad \text{where}\\ \tilde{r}_t = \delta_0 + \delta_1^T X_t, &\qquad \lambda_t = \lambda_0 + \lambda_1^T X_t,\end{aligned}$$

and $\xi_{t,t+1}$ satisfies the corporate bond Euler equation in (6.1). We refer to $\tilde{r}_t$ as the corporate short rate and λ_t as the vector of corporate bond factor exposures.

Treasury wedge for maturity $j \geq 1$ $\Omega_{t,t+1}^{(j)}$: also takes an exponential affine form:[60]

$$\begin{aligned}\Omega_{t,t+1}^{(j)} = &\exp\left(\mu_t^{(j)} - \tfrac{1}{2}(\omega_t^{(j)})^T\Sigma\omega_t^{(j)} + (\omega_t^{(j)})^T(\Sigma\lambda_t + V_{t+1})\right), \quad \text{where}\\ \mu_t^{(j)} := \mu_0^{(j)} + (\mu_1^{(j)})^T X_t, &\qquad \omega_t^{(j)} := \omega_0^{(j)} + (\omega_1^{(j)})^T X_t, \quad \text{with} \quad \omega_t^{(1)} = 0\end{aligned}$$

and the product $\xi_{t,t+1}\Omega_{t,t+1}^{(j)}$ satisfies the Treasury Euler equation in (6.2). To keep the theoretical specification as general as possible, we allow for j-dependence in the level term $\mu_t^{(j)}$ and the risk exposure terms $\omega_t^{(j)}$, which allows the Treasury distortion wedge $\Omega_{t,t+1}^{(j)}$ to potentially vary across maturities.[61] We use model selection to let the data decide the appropriate level of flexibility in the wedge. We compare specifications that vary along three dimensions: the state dependence of the wedge level μ_t, the state dependence of the exposure ω_t, and the number of maturities with their own wedge. Table 4 lists the options along each dimension.

The combinations in Table 4 nest several key models in the literature such as:

1. *Uniform riskless wedge (O0-S0).* The wedge is common to all maturities, but restricted it to be locally risk-free, so it has no conditional covariance with the SDF: $\Omega_{t,t+1}^{(j)} = \exp(\mu_t)$ for all $j \geq 1$, with μ_t known at t and otherwise unrestricted. When μ_t depends on growth and the debt states (M2), this nests KVJ and the bond-in-utility models that follow it (e.g., Jiang et al., 2022).

2. *Riskless wedge with segmentation at the first j maturities (O0-SK for $K > 0$).* This specification allows maturities $j \leq K$ to have their own locally risk-free wedge, $\Omega_{t,t+1}^{(j)} = \exp(\mu_t^{(j)})$, and maturities $j > K$ to share a common one, $\Omega_{t,t+1}^{(j)} = \exp(\mu_t)$, with $\mu_t^{(j)}$ and μ_t unrestricted. The case $K = 1$ nests models that put a liquidity premium

[60] We normalize $\omega_t^{(1)} = 0$ because it is not identified: $q_t^{(1)} = \mathbb{E}_t[\xi_{t,t+1}\Omega_{t+1}^{(1)}] = \exp(-\tilde{r}_t + \mu_t^{(1)})$.

[61] There are limitations on how much flexibility in the j-dependence can actually be estimated. We specify the rank conditions in Appendix G.3.

Wedge level μ_t (M)		Shock Exposure ω_t (O)		Maturity-specific wedge (S)	
M0	μ_0	O0	none (normalization)	S0	none specific; common $\geq$3m
M1	$\mu_0 + \mu_{1,s}^T y_t$	O1	ω_0 constant	S1	3m specific; common $\geq$6m
M2	$\mu_0 + \mu_1^T X_t$	**O2**	$\omega_0 + \omega_{1,s}^T s_t$, rank one	S2	3m, 6m specific; common $\geq$1y
		O3	$\omega_0 + \omega_{1,s}^T s_t$, rank two	**S3**	3m, 6m, 1y specific; common $\geq$2y
		O4	$\omega_0 + \omega_{1,s}^T s_t$, full rank	S4	3m, 6m, 1y, 2y specific; common $\geq$3y
		O5	$\omega_0 + \omega_1^T X_t$, full rank	S5	every maturity specific; no common

Table 4: The three dimensions of a specification: wedge level (M), exposure (O) and maturity-specific wedge (S). In O2, $\omega_{1,s} = ab^T$ for vectors a and b, which has rank one and 11 free parameters, because only the product ab^T is identified. In O3, $\omega_{1,s} = a_1 b_1^T + a_2 b_2^T$, which has rank two and 20 free parameters. In O4, $\omega_{1,s}$ is an unrestricted 6×6 matrix, which has full rank and 36 free parameters. In O5, the exposure also loads on growth and the bond balances, through an additional unrestricted block of ω_1. The selected triple, M1-O2-S3, is shaded.

only on the shortest maturity and stay agnostic about longer maturities (e.g., Nagel, 2016; Krishnamurthy and Li, 2023). The case $K > 1$ nests money search models with different matching probabilities for different Treasurys and CES bond-in-utility models with different weights for different maturities.

3. *Uniform risky wedge (OL-S0 for $L > 0$).* This specification allows a wedge common to all maturities that is exposed to shocks: $(\mu_t^{(j)}, \omega_t^{(j)}) = (\mu_t, \omega_t)$, unrestricted, for all $j \geq 1$. It can be interpreted as one marginal investor across all Treasury securities, who may differ from the marginal investor in corporate bonds. Unlike the riskless wedge, it allows a conditional covariance between the Treasury wedge and the SDF, and so nests models in which the wedge carries a time-varying risk premium that is separate from the corporate risk premium (e.g., Acharya and Laarits, 2023).

4. *Risky wedge with segmentation at the first j maturities (OL-SK for $L, K > 0$).* Maturities $j \leq K$ have their own risky wedge, and maturities $j > K$ share a common wedge exposed to shocks: $(\mu_t^{(j)}, \omega_t^{(j)}) = (\mu_t, \omega_t)$ unrestricted for all $j > K$ and maturity specific for $j \leq K$. It nests models that pair a liquidity premium on the shortest Treasurys with a separate, risky regulatory premium on longer maturities.

6.2.3 Estimation

For each combination across the three columns in Table 4, we jointly estimate the corporate pricing kernel, Treasury wedge, and state space parameters $(\delta_0, \delta_1, \lambda_0, \lambda_1, \mu_0, \mu_1, \omega_0, \omega_1, \mu_X, \Phi_X, \Sigma)$ by maximum likelihood, following Joslin et al. (2011). Pairwise BIC com-

parisons select M1-O2-S3, which allows a covariance between the SDF and the Treasury wedge. The selection is broadly robust to estimating with a maximum Sharpe ratio bound (adapting Duffee, 2010). Appendix G.3 describes the estimation and shows the pairwise BIC comparisons, and Appendix G.1 shows that all approaches fit the yield curves tightly.

The selected model (M1-O2-S3) leads to a number of conclusions about which model features are required to fit the Treasury wedge. First, the data very clearly rejects specifications with no risk exposure (O0), so the Treasury wedge covaries with the SDF. Second, letting short maturities have their own Treasury wedge delivers a smaller but distinct gain over a common wedge, pointing toward models with a liquidity premium. Third, adding the debt and growth states to the wedge level or shock exposure (M2, O5) adds little, in line with Subsection 6.1. Overall, the data are therefore inconsistent with models of the Treasury wedge that lack a liquidity premium on short maturities or a covariance between the SDF and the wedge at longer maturities.

6.3 Debt Issuance, Risk, and the Funding Advantage

We close the paper with impulse response functions (IRFs) that trace how debt supply and the SDF move the funding advantage in the selected model (M1-O2-S3) of Subsection 6.2.

We identify growth and debt shocks recursively. Refunding decisions and auction sizes are announced ahead of price realizations, so we assume that issuance responds to lagged yield curves but not to price shocks within the period, while prices respond to issuance contemporaneously. The Treasury can also react to growth within the period. This timing orders the VAR innovations V_t from equation (6.5) as growth, total issuance, the long-term tilt, the medium-term tilt, and the yield-curve innovations. A Cholesky factorization of the relevant part of the covariance matrix Σ delivers four identified shocks. The growth shock $\epsilon_{g,t}$ is the innovation to nominal GDP growth. The total-issuance shock $\epsilon_{d,t}$ is unexpected total issuance not explained by growth. The two tilt shocks $(\epsilon_{l,t}, \epsilon_{m,t})$ are unexpected issuance of the long and medium bonds in excess of their average shares, net of the shocks before them. Because the yield-curve innovations come last, the responses to these four shocks do not depend on how that block is ordered. Appendix G.4 gives the construction. We do not identify structural shocks within the yield block. Instead, we rotate its residual variation to isolate the direction that raises the level of the SDF. This SDF shock, labeled as "SDF increase", is a statistical rotation, not a structural one. We use it to trace conditional comovement among debt, yields, and the funding advantage, in the spirit of rotations that summarize conditional correlations without a structural meaning.

Figure 16 shows three sets of responses. A rise in total debt issuance (first row) has a small

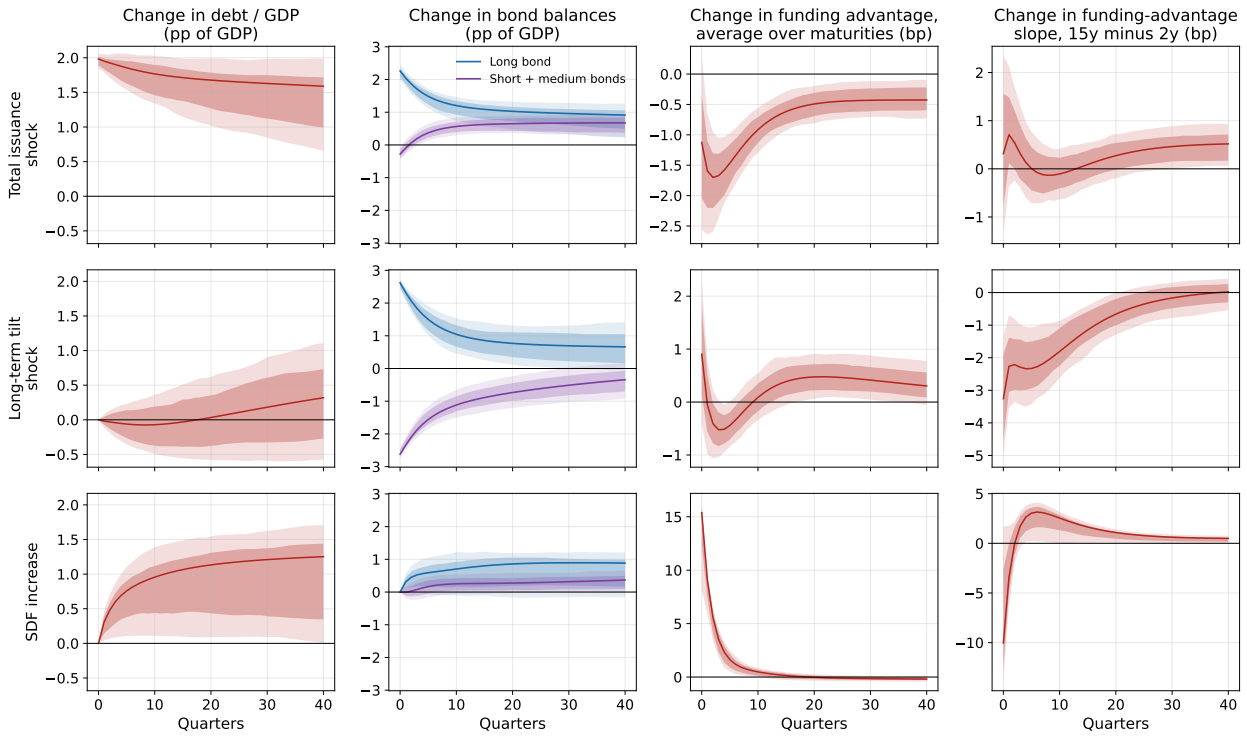


Figure 16: Responses of the Funding Advantage and Debt Issuance

Notes. Impulse responses to a one-standard-deviation shock to total debt issuance (first row), a tilt of issuance toward long-term debt (second row), and a rise in the SDF (third row). Shaded areas show 68% and 90% bands.

but significant effect on the level of funding advantage and none on its slope. This shows that our term structure model retains an element of the inverse quantity-spread relation in KVJ, even though the Treasury wedge does not load on debt/GDP. A tilt of issuance toward long-term debt (second row) lowers the funding advantage at long maturities relative to short maturities, highlighting the importance of maturity-specific supply effects and of modeling the maturity structure of debt dynamically. The effects of these issuance shocks are statistically significant and consistent with downward-sloping Treasury demand, in line with the applied literature that estimates demand elasticities (e.g., Krishnamurthy and Li, 2023; Cavaleri, 2024; Jansen et al., 2025). Their small magnitudes suggest, however, that issuance shocks explain only a limited share of the time variation in the US funding advantage, which dovetails with our findings in Subsection 6.1.

Finally, the third row in Figure 16 shows that a positive shock to the SDF ("bad time") is associated with both a higher funding advantage and higher debt issuance. To understand the SDF response, Figure 17 decomposes the change in the 3-month and 15-year funding advantage into two channels. The level channel is the relative change in the drift μ_t of the

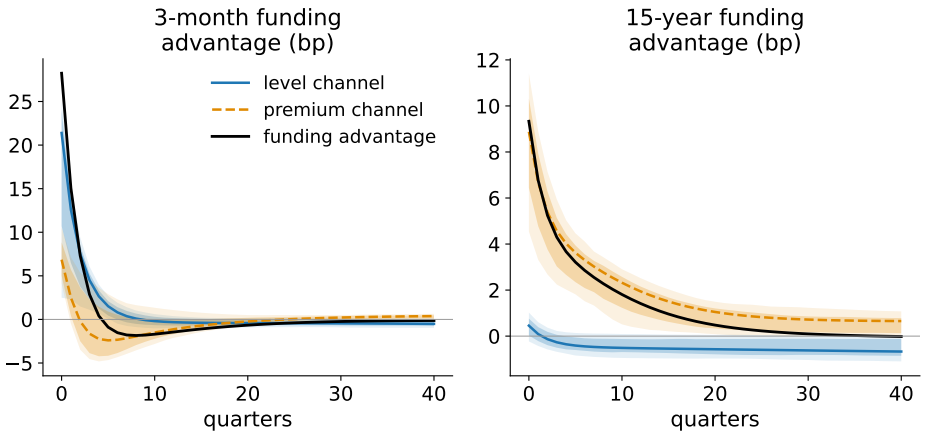


Figure 17: Decomposition of the Funding Advantage Response to an SDF Shock

corporate and Treasury yield curves, and the premium channel is the change in their relative risk premium between Treasurys and safe corporate bonds. Formally, the funding advantage at maturity j can be decomposed as:

$$\chi_t^{(j)} = \underbrace{\frac{1}{j}\sum_{k=0}^{j-1} E_t\,\mu_{t+k}^{(j-k)}}_{\text{level}} + \underbrace{\frac{1}{j}\log\frac{E_t\left[\xi_{t,t+j}\prod_{k=0}^{j-1}\Omega_{t+k,t+k+1}^{(j-k)}\right]}{E_t\left[\xi_{t,t+j}\right]} - \frac{1}{j}\sum_{k=0}^{j-1} E_t\,\mu_{t+k}^{(j-k)}}_{\text{risk-premium}}$$

Figure 17 shows responses of the level term and the risk premium term with respect to an increase in the SDF. At long maturities, the response comes from the relative change in the risk premium. The risk premium rises more on corporate bonds than on Treasurys, consistent with a model in which the shock exposure ω_t in the Treasury wedge makes long-term Treasurys a better hedge against bad times than long-term safe corporate bonds. At short maturities, the response comes largely from the level channel, as the non-pecuniary demand for short-term Treasurys rises in bad times. Together, the responses sharpen our model selection result that the data require a separate wedge on short Treasurys and a covariance between the SDF and the wedge at longer maturities. This is consistent with a flight-to-liquidity premium on short Treasurys and a hedging role for long Treasurys.

7 Conclusion

In this paper, we expand yield curve estimation techniques to deliver the first highest-grade corporate yield curve and the premier Treasury yield curve dating back to 1860. Our estimation techniques adjust for the heterogeneous tax and option features in historically heterogeneous bond samples, allowing us to extract yield curve information that allows for

like-for-like comparison. We use our yield curves to uncover new facts and overturn existing paradigms about the US funding advantage. We build the first affine asset pricing model of the funding advantage to understand its role in Treasury maturity management and US government financing. We show that short Treasurys are priced differently to long Treasurys and the covariance between the SDF and the Treasury wedge is important for explaining US funding advantage dynamics.

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A The Corporate and Government Bond Datasets

This appendix describes the bond-level datasets used to estimate the corporate and Treasury yield curves. We assemble two complementary panels: (i) a new historical panel of highest-grade U.S. corporate bonds spanning 1860-2026, combining existing databases with hand-collected information from historical periodicals and Moody's manuals, and (ii) a comprehensive panel of all U.S. Treasury securities spanning 1776-2026, built from Hall et al. (2018) and the CRSP Treasury database. The corporate panel contains, at the bond–month level, end-of-month trading prices, full coupon and cash-flow schedules, embedded option features (e.g., callability, sinking funds, convertibility), denomination, and Moody's credit ratings. Section A.2 describes the corporate panel; Section A.3 describes the Treasury panel.

A.1 Historical Context

Brief History of the US Bond Market: The US corporate bond market traces its origins to the early 19th century, driven by the need to finance large infrastructure projects. The first corporate bonds were issued by banks and canal companies, but the market truly expanded with the rise of the railroad industry. By the 1850s, railroad companies were expanding into the west at a scale and level of uncertainty that they could no longer raise sufficient capital from the local and fragmented banking sector. The solution was to issue bonded debt to a broader pool of investors, which created what is considered the world's first corporate bond market (Sylla et al., 2006). In the following decades, the corporate bond market expanded rapidly and by the early 2000s was several times larger than the entire UK or US sovereign debt markets.[62] Throughout the early 20th centuries, the market matured, with securities becoming more standardized, and industrial corporations and utilities also began issuing bonds.

Concurrently, throughout the nineteenth century, the federal government issued bonds infrequently, as Congress was responsible for debt management, leading to long-maturity issuances with significant variations in maturities, coupon rates, denominations, and units of account (Payne et al., 2025). The expansion and standardization of federal debt issuance occurred gradually over time, with Congress delegating more autonomy in designing and issuing securities to the Treasury Department between 1917 and 1939. Both markets continued to expand throughout the 20th century and by the mid-20th century, US Treasury securities had become the world's largest and most liquid debt market, with a standardized set of securities at various maturities.

[62]The US actually paid off its entire national debt in 1836.

Both corporate and Treasury bonds traded actively on major exchanges like the New York Stock Exchange (NYSE) and were held by similar investors, including banks, insurers, and wealthy individuals. Both corporate and government bonds shared similar features, such as fixed coupon payments and typically long maturities and were traded more frequently than other asset classes.

Denomination: The denomination of both Treasury and corporate bonds has evolved similarly throughout American financial history. From 1800 to 1933, the US adhered to a gold standard except from 1861 to 1878 when it temporarily suspended gold convertibility and issued a paper currency known as "greenbacks". During this period, both federal and corporate bonds were typically denominated in gold (or greenbacks during the suspension). Following the Gold Reserve Act of 1933, which prohibited private US citizens from holding gold coins, both markets transitioned to nominal dollar denomination. The Bretton Woods Agreement (1944-1971) reintroduced a type of gold standard by establishing an international system of fixed exchange rates with the US dollar convertible to gold until its collapse in 1971 when the dollar was floated. Since then, both Treasury and corporate bonds have been issued exclusively in nominal terms until the introduction of Treasury Inflation-Protected Securities (TIPS) in 1997, which provide explicit inflation indexing.

Credit ratings: The rise of corporate bonds was accompanied by the development of credit ratings. Beginning in 1832, the "American Railroad Journal" published detailed assessments of railroad companies, covering physical descriptions of the railroads, their assets, liabilities, and earnings. In 1868, its former editor Henry V. Poor published the first volume of "Poor's Manual of the Railroads of the United States", a comprehensive resource detailing financial statements, operational statistics, and the capital structure of their securities. In 1909, John Moody in his "Moody's Manual of Railroads and Corporation Securities" first introduced a structured rating system for these securities that established the foundation for modern credit ratings.

Default Risk: In the early 1900s, Moody's Investors Service began assigning credit ratings to bonds and other financial assets, with "Aaa" denoting the highest level of creditworthiness. This rating was based on factors such as physical capital, debt levels relative to assets and revenue, profitability, and liquidity. To qualify for an "Aaa" rating, bonds needed a long-term track record of exceptionally strong interest coverage and substantial physical assets backing the issue, ensuring minimal investment risk. Most bonds were either first mortgages or well protected underlying mortgages. Moody's argued that even in changing economic

conditions, the fundamental strength of these securities would remain intact. As Hickman (1958) found, credit ratings offered investors valuable insights into bond quality and default probabilities.

Policy Interventions: Corporate and government bond markets have historically been subject to different regulatory frameworks, evolving in response to financial and economic pressures. In the decades before the Civil War, only state-chartered banks existed, which were not incentivized to hold Treasurys.[63] This changed with the National Banking Acts of 1863–1866, which established a system of nationally chartered single-branch banks. These banks were permitted to issue banknotes up to 90% of the lower of the par or market value of qualifying US federal bonds, effectively tying their balance sheets to government debt. However, national banks were prohibited from using railroad bonds as backing for their notes and faced strict limitations on the types of loans they could issue.

World War II brought further regulatory intervention, as concerns over war financing led to the government "fixing" the yield curve from 1942 to 1951, with the T-Bill rate set to 3/8% and the long-term bond yield capped at 2.5% (see Garbade (2020) and Rose (2021)). This policy was implemented through coordination between the Treasury and the Federal Reserve, with the Fed agreeing to absorb excess bond supply at the fixed price, and implicit coordination with the banking system, which ended up predominantly holding government debt and reserves. The arrangement ended with the 1951 Treasury-Fed Accord, establishing official Fed independence from the Treasury.

The 2007–2009 financial crisis triggered extensive regulatory reforms, including the Dodd-Frank Wall Street Reform and Consumer Protection Act, which introduced new oversight for financial institutions. Additionally, the Basel III regulations imposed stricter capital requirements and portfolio constraints on banks, penalizing excessive leverage and encouraging the holding of government debt over assets like corporate bonds. In response to the crisis, the Federal Reserve also launched a quantitative easing (QE) program, purchasing long-term government bonds to lower interest rates and stabilize financial markets.

A.2 Corporate Bond Data

The corporate panel combines three types of data that we describe in turn: (i) bond prices, (ii) bond-level characteristics and option features, and (iii) credit ratings. Throughout the paper, the unit of observation is a bond-month and we use end-of-month closing prices on

[63] These banks, chartered by state legislatures, could issue their own banknotes and were subject to diverse balance sheet regulations, often requiring them to hold gold and state bonds. However, no state banks could operate nationally.

the New York Stock Exchange (NYSE).

A.2.1 Bond Prices

We compile end-of-month NYSE trading prices using five primary sources, summarized in Table A.5 and depicted in Figure A.18. The sources are combined along the time dimension and supplemented, where coverage is sparse, by hand-collected end-of-month closing prices from the New York Times.

1860-1884: Global Financial Data (GFD). For the earliest part of the sample we use the corporate bond price file from *Global Financial Data*. GFD covers roughly 800 corporate bonds traded between 1791 and 1884. In our sample window the panel is essentially a railroad panel, reflecting the dominance of railroad financing in the early US corporate bond market. Coverage is thin and irregular before 1860 but becomes much richer between 1870 and 1884, with regular quotations and bond identifiers (bond name, coupon, issuing company) for most observations. GFD does not report issue and maturity dates, denominations, call features, or other bond characteristics so we supplement these characteristics from historical sources described in Section A.2.2.

1886-1964: Commercial & Financial Chronicle (CFC). Our largest hand-collected component are end-of-month NYSE bond prices from the *Commercial & Financial Chronicle*, a weekly business newspaper published from 1865 to 1987. Scanned digital copies of the Chronicle are available from the Federal Reserve Archival System for Economic Research (FRASER) from July 1865 to December 1987. From the issue containing the last week of trading for each month, we transcribe the "Stock Exchange Quotation / Bond Record" table, which lists, for every NYSE-listed bond, the bond's name, interest payment schedule, and the high, low, and closing prices for the relevant period.[64] We collect *actual sale prices only*; bonds that did not trade in the relevant window (and for which the Chronicle reports only bid-ask quotations) are excluded. From 1886-1917 we collect monthly observations only for railroad bonds, since railroads dominate the pre-WWI corporate market and the non-railroad

[64]Bond price quotations on the NYSE switched from a flat-price (price-plus-accrued-interest) basis to a clean-price basis in January 1909, and the CFC's tabulations followed in February 1909. We therefore treat CFC sale prices reported from February 1909 onward as clean prices, and convert them to dirty prices by adding accrued interest when needed for cash-flow valuation. Pre-February 1909 prices are treated as already including accrued interest, except for a small number of entries marked with a †, the CFC's notation for "purchaser also pays accrued interest," which we treat as clean prices. The CFC additionally flags a few individual post-1909 quotations with the prefix "f" wherever a bond still traded flat (which is the standard case for defaulted bonds and for income bonds, whose coupons are payable only out of available earnings) and accordingly we treat all such flagged prices as dirty prices.

tables are sparse. From January 1914 to December 1964 we expand the collection to include all corporate bonds, reflecting the rising importance of utility and industrial securities.

1934-1974: Barron's. For the period 1942-1974 we hand-collect end-of-month NYSE bond closing prices from *Barron's*, a weekly financial newspaper founded in 1921. Prices are taken from the "Listed Bond Quotations" section, which covers actively-traded NYSE-listed corporate bonds and reports closing price, weekly volume, current yield, and the high/low for the year.

1974-1997: Lehman Brothers Fixed Income Database. From April 1974 to December 1997 we use the *Lehman Brothers Fixed Income Database* (Hong and Warga, 2000). The Lehman/Warga database provides month-end bond-level information including price, accrued interest, coupon, maturity, ratings, and embedded option features. We restrict attention to actual trader-quoted prices (as opposed to matrix-priced observations).

1998-2026: Bank of America Merrill Lynch Bond Index Database. After 1997 we use the *Bank of America Merrill Lynch Bond Index Database*, which provides an analogous level of bond-by-month detail (price, accrued interest, coupon, maturity, callability, ratings) for the constituents of U.S. corporate bond indices.

Filling gaps with the New York Times (NYT). Coverage from the five primary sources contains scattered gaps, particularly in the pre-1973 hand-collected period. Where a bond is missing for a given month-end, we collect NYSE closing prices from the bond quotation page of the *New York Times*.

A.2.2 Bond Characteristics

Estimating yield curves requires the full cash-flow schedule of each bond, which in turn requires accurate information about the maturity date, coupon rate and payment dates. In addition, our adjustments require information on embedded options, such as detailed call provisions. For bonds in the post-1972 electronic databases (Lehman/Warga and Bank of America Merrill Lynch), this information is reported directly. For the pre-1973 sample, we hand-collect bond characteristics from a variety of historical sources.

Pre-1900 bonds. For bonds outstanding before 1900, we draw issue and maturity date, coupon, denomination, call provisions, sinking-fund features, and conversion options from the historical sources listed in Table A.6.

Table A.5: Primary sources for corporate bond prices & characteristics, 1860-2026

Source	Period	Frequency	Hand-collected?
Bond prices			
Global Financial Data	1860-1884	End of month	No
Commercial & Financial Chronicle	1886-1964	End of month	Yes
Barron's	1942-1974	End of month	Yes
Lehman Warga	1974-1997	End of month	No
Bank of America Merrill Lynch	1998-2026	End of month	No
New York Times (supplemental)	1860-1974	End of month	Yes
Bond characteristics			
Various	1860-1900	Yearly	Yes
Moody's manuals	1900-1973	Yearly	Yes
Lehman Warga	1974-1997	Monthly	No
Bank of America Merrill Lynch	1998-2026	Monthly	No

"Hand-collected" indicates manual transcription from PDF scans of historical periodicals.

1900-1972 bonds. For bonds outstanding in 1900 or later, our principal source are the *Moody's Manuals*, first published in 1900.[65] For each bond we extract from the relevant Moody's volume:

- issue date, maturity date, denomination, total amount outstanding;
- coupon rate and the payment months in the year;
- full call schedule: call window (first call date and last call date) and call price (typically a step-down schedule);
- any sinking-fund provisions, conversion and other features, where applicable.

We also collect supplementary information on *actual* bond redemptions, that is bonds that were redeemed (called) before maturity. These redemption records are essential for correctly handling callable bonds: we can verify whether the call option was exercised and, if so, on

[65] Initially as *Moody's Manual of Industrial and Miscellaneous Securities*, succeeded by *Moody's Manual of Railroads and Corporation Securities*, and later *Moody's Analyses of Investments*.

Table A.6: Reference sources for pre-1900 corporate bond characteristics

Source	Author / Publisher	Years
Commercial & Financial Chronicle, Investors' Supplement	W. B. Dana & Co., New York	1875-1900
American Railroad Journal	D. K. Minor; H. V. Poor, New York	1832-1886
Listing Statements of the New York Stock Exchange	New York Stock Exchange	1884-1900
Poor's Manual of the Railroads of the United States	H. V. & H. W. Poor, New York	1868-1900
American Railroad Manual	E. Vernon (ed.), New York	1873-1875
The Bond Record	Exchange Printing Company, New York	1893-1896
Gold, Barbour & Swords' Bond-Book	Gold, Barbour & Swords, New York	1886
Catalogue of Railroad Mortgages	Pliny-Fisk Library & Bureau of Railway Economics, Washington D.C.	1919
White & Kemble's Atlas of Railroad Mortgages	White & Kemble, New York	1894
Joseph G. Martin's Boston stock and bond market series (self-published, Boston):		
Twenty-One Years in the Boston Stock Market (1835-1856)		1856
Seventy-Three Years' History of the Boston Stock Market (1798-1871)		1871
Martin's Boston Stock Market: Eighty-Eight Years (1798-1886)		1886
A Century of Finance, 1798-1898		1898
Annual stockholder reports	Major railroad companies	1830-1900

what date and at what price, which we use both for cash-flow construction and for validating our option-pricing assumptions.

Post-1972 bonds. For 1973 onward, bond characteristics (issue date, maturity, coupon rate and schedule, denomination, call schedule, sinking fund, conversion) come directly from Lehman/Warga and Bank of Ameica Merrill Lynch. We cross-validate against the Mergent Fixed Income Securities Database (FISD) where available.

A.2.3 Credit Ratings

To classify bonds by credit quality we rely primarily on Moody's ratings, supplemented for the earliest period by Macaulay's "high-grade railroad bond" classification. From 1973-2026

we take Moody's ratings directly from the Lehman/Warga (1973-1997) and Merrill Lynch (1998-2026) databases at the bond-month frequency. For 1909-1973 we hand-collect annual Moody's letter ratings from the corresponding sector-specific Moody's Manuals (railroads, public utilities, industrials). Moody's first issued credit ratings for railroads in 1909 and extended coverage to public utilities and industrials in 1914.[66] Ratings for each bond are recorded annually as of the manual's publication date.

Before 1909 there are no agency ratings. Following Macaulay (1938), we identify high-grade railroad issuers as those whose bonds he included in his high-grade railroad bond yield index (1857-1935). Macaulay carefully chose companies based on their financial strength and excluded them before they encountered financial trouble, ensuring his index reflected only the most creditworthy issuers. We follow him in both dimensions, taking his selection of companies and the windows over which he includes each of them, so that a company enters our sample only while Macaulay judged it to be truly high grade.[67] We find Macaulay's work convincing because it was assembled with great care from the contemporaneous record by someone with detailed knowledge of the individual railroads and remains the standard benchmark for nineteenth century high-grade yields. However, as emphasized by Homer and Sylla (2004), constructing an index equivalent to a modern Aaa^+ index before 1900 is intrinsically difficult given the small number of true high-grade issuers, and even Macaulay's "high-grade" sample exhibits some heterogeneity in credit quality. Hence, these pre-1909 classifications should be treated with some caution. In addition, the introduction of credit ratings may have changed market risk assessment, as detailed in Bernstein et al. (2025).

A.2.4 Coverage and Sample Restrictions

To create a consistent bond-month panel from our six sources spanning 1860-2026, we first construct a unique bond identifier that tracks each individual bond throughout the entire sample period. We then apply a uniform set of filters across all datasets.[68] The panel is restricted to U.S.-domiciled, USD-denominated, non-financial, senior, non-subordinated, regular corporate bonds with fixed semi-annual coupons. We exclude putable and convertible issues, sinking fund bonds, mortgage-backed, asset-backed, agency-backed, equity-linked and structured notes, as well as Yankee bonds, bonds with floating or otherwise non-standard coupons, and any issue without a fixed maturity. Callable bonds are kept, but only when

[66] We focus on Moody's ratings because they are the earliest available; Standard & Poor's (then Poor's) ratings began in 1922 and Fitch's in 1924.

[67] The only change we make is that we don't restrict to only long maturity bonds.

[68] These restrictions follow Duffee (1998) for Lehman/Warga, but we apply them uniformly across the Bank of America Merrill Lynch and historical samples as well, in line with Ghaderi et al. (2025) and Dickerson et al. (2023).

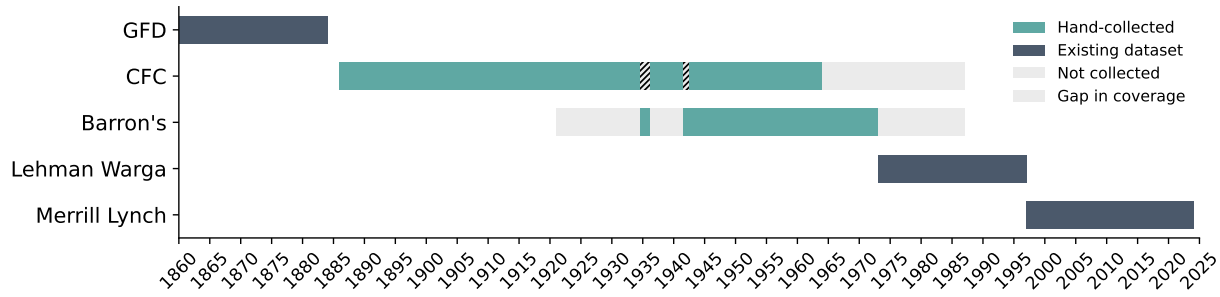


Figure A.18: Corporate Bond Price Data Sources And Coverage

Data sources for bond prices from 1860-2026. GFD, Lehman Warga, and Merrill Lynch are existing datasets, while bond price data from the CFC and Barron's was manually collected using scans from digital archives. Light gray areas indicate periods of the publications that are available as prints and scans but have not yet been digitized.

their full call schedule is known. We exclude bonds whose nominal life (maturity date minus issue date) exceeds 50 years, which removes a small number of perpetual-like issues. We also require that each bond carries a unanimous Aaa rating from every agency rating at the time we observe its price. From 1973 onward this includes Moody's, S&P, and Fitch, before 1973 we only rely on Moody's ratings. For prices, we use only *actual transaction prices*: end-of-month NYSE sale prices for the hand-collected sources (CFC, Barron's, NYT) and trader-quoted dealer prices for the modern electronic databases (Lehman/Warga and Bank of America Merrill Lynch).

Figure A.19 reports the number of bonds in the resulting estimation sample by month, split by maturity bucket and by callable versus non-callable.

A.3 Treasury Data

We use a comprehensive panel of prices and quantities of all U.S. Treasury securities from 1776 to 1925, compiled by Hall et al. (2018) and used in Payne et al. (2025). We complement this historical panel with the CRSP U.S. Treasury database from 1925 to 2026.

Coverage and frequency. Quantities outstanding are quarterly from 1776 to 1871 and monthly thereafter. Prices are monthly: where available we use the end-of-month closing price; otherwise we use the average of the high and low prices for the month, or failing that, the average of the bid and ask quotes.

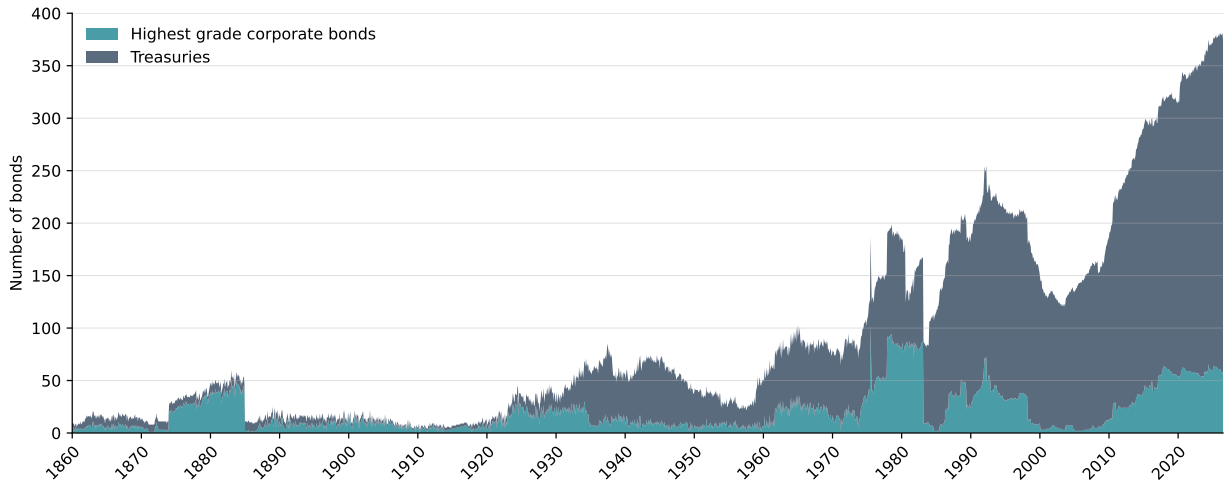


Figure A.19: Number of Bonds in the Corporate Estimation Sample

Notes: The figure plots the number of bonds with an observed end-of-month price in each month from 1860 to 2026, after applying the sample restrictions in Appendix A.2.4 for highest-grade corporate bonds (teal) and in Appendix A.3 for Treasurys (gray).

Sample restrictions. For the historical (pre-1925) sample we apply two filters: (i) we require remaining maturity greater than one year, to avoid the liquidity premia that characterize short-term Treasury debt and (ii) we drop bonds whose currency denomination (gold versus currency) is ambiguous. In the modern sample, we follow Gürkaynak et al. (2007) and drop on-the-run notes and bonds beginning in 1984. Additionally, beginning in 1973 once the sample is densely populated, we drop stale long-term bonds with <10 years to maturity that were 20+ year bonds at original issue. During the peak interest rate period 1981-1984, we drop all old bonds and notes less than 20 years to maturity with coupons below 11 percent due to noise and that their coupons were well below prevailing government borrowing costs. We exclude Treasury Inflation-Protected Securities (TIPS) in our baseline estimation, though we fit real yield curves in order to back out their contribution to the term structure of the government's promised debt repayment in Section 6.

A.4 Bond Characteristics and Institutional Treatment

This subsection provides further details on differences in bond characteristics and institutional treatment between US government and corporate debt that present challenges for measuring the funding advantage of the US government.

A.4.1 Additional Composition Plots

Figure A.20 shows the share of government and corporate debt trading at a discount (as measured by whether the price is less than 98 dollars). Figure A.21 shows the composition of callable and non-callable bonds among outstanding US government and corporate debt. Figure A.22 shows the declining quantities of the four most attractive flower bonds during the 1960s and 70s. The lowest-coupon and deepest-discount flower bond, the 3% bond of 1995, was the first bond to be actively redeemed during the 1960s as inflation increased, with other flower bonds following suit in the late 1960s. Figure A.23 shows the number of flower and non flower bonds for maturity bins less than 5 years, between 5 and 10 years, and more than 10 years. Evidently, flower bonds made up most of the long-term debt during the period from 1960-1974. Figure A.24 shows which investors held different types and maturities of bonds in October 1969, taken from United States Department of Treasury (1969). All bonds of final maturities over 20 years were valuable flower bonds. Estate owners were contained in the final "held by all other investors" column, placing an upper bound on the share of publicly held long-term flower bonds held by estates in this month of $5511/(16107 - 3891) \approx 45\%$. The largest institutional holders of these bonds were pension funds and life insurance companies. Taken in tandem with the fact that they accounted for a large share of outstanding Treasurys in the 1960s, we find evidence that the market for flower bonds was not segmented from other Treasury securities during this time. Rather, they were an integral part of the Treasury market priced by similar investors holding non-flower securities.

A.4.2 Interest Rate Ceiling on Government Bonds

A key legislation that unexpectedly gained prominence in the late 1960s and early 1970s was the Congressional mandate, established in 1917, which imposed a 4-1/4 percent interest rate ceiling on new long-term Treasury bonds.[69] Consequently, when interest rates surpassed the 4-1/4 percent ceiling in the 1960s, the US Treasury was unable to issue new bonds with maturities exceeding 5 years (extended to 7 years in 1967) and so the average maturity of outstanding US debt declined significantly, and the government bond portfolio became heavily concentrated in seasoned discount bonds. The ceiling on long-term US Treasury bonds was effectively lifted in 1971, when $10 billion worth of bonds were authorized without

[69] As explained in Department of the Treasury (1976), the primary rationale for setting a ceiling rate of 4-1/4 percent on long-term government bonds was to minimize borrowing costs tied to the United States' involvement in WWI. This ceiling was intentionally set 25 basis points below prevailing market yields, reflecting the belief that the American public would buy Liberty Bonds for reasons beyond comparative yield considerations.

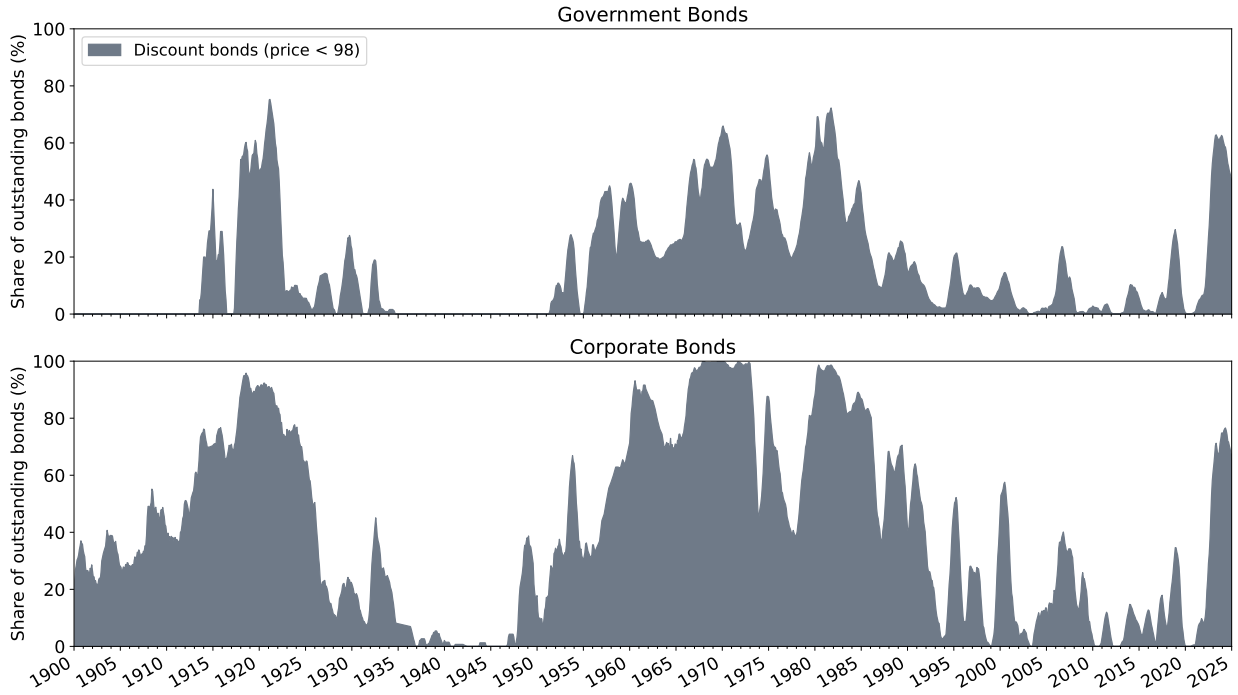


Figure A.20: Composition of Government & Corporate Debt by Discount: This figure shows the share of bonds trading below par among outstanding US government and corporate debt from 1900 to 2025.

regard to the ceiling. Since then, the bond authorization limit has been raised multiple times, and the issuance of long-term bonds has become a regular component of the Treasury's refunding operations.

A.4.3 Circulation Privilege

During the National Banking Era (1863–1913), US federal bonds held a special regulatory status known as *circulation privilege*, which allowed federally chartered banks to issue national bank notes backed by eligible US Treasury bonds. This privilege created a strong institutional demand for long-term government securities, as banks could profitably convert them into currency liabilities. Crucially, corporate bonds were not eligible for this privilege, so the resulting yield suppression applied exclusively to US government debt. As a result, the circulation privilege helped support bond prices and depress long-term yields of government bonds, contributing to a widening of the Aaa–Treasury spread during this period.

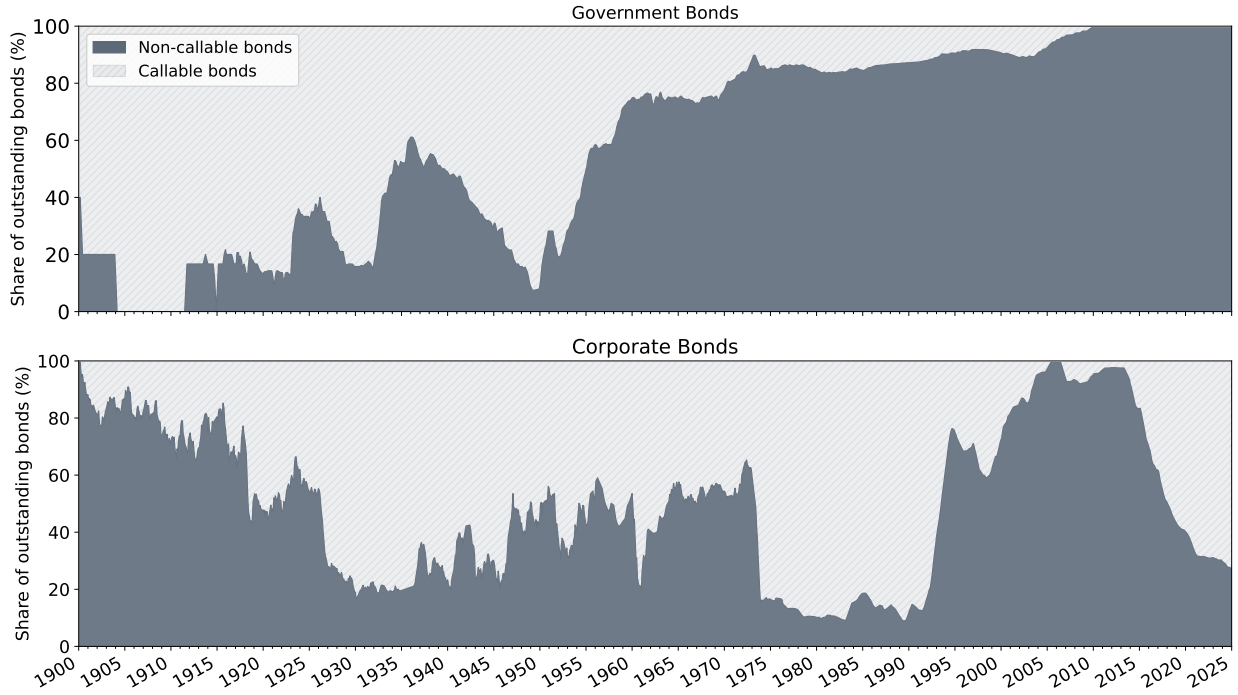


Figure A.21: Composition of Government & Corporate Debt by Callability: This figure shows the share of callable and non-callable bonds among outstanding US government and corporate debt from 1900 to 2026.

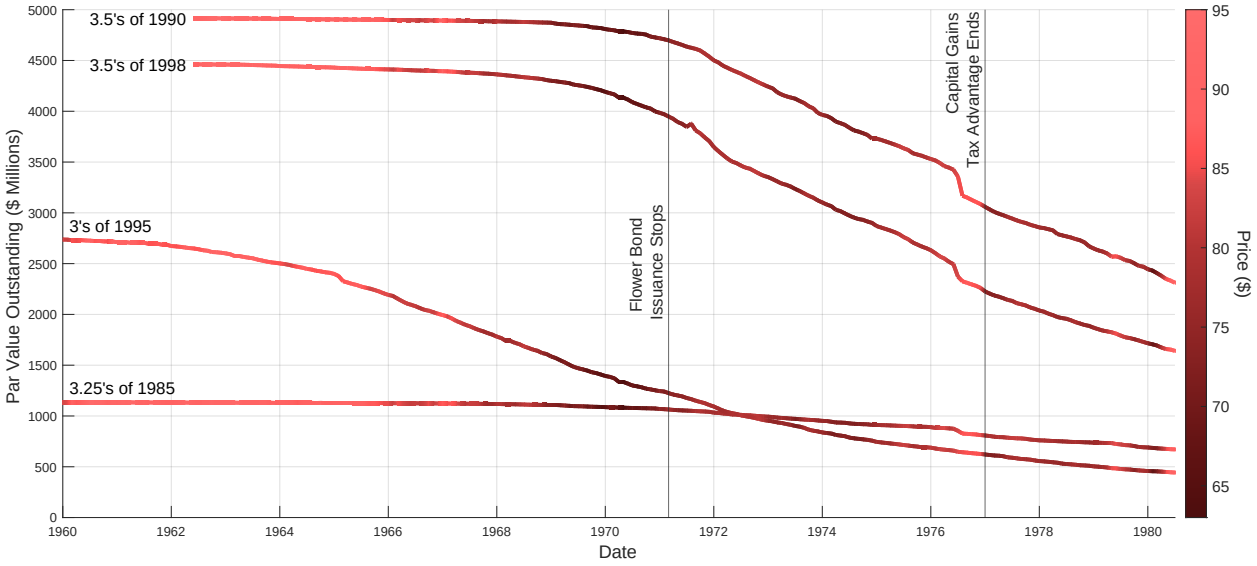


Figure A.22: Quantities Outstanding of Flower Bonds

Notes: Figure shows the par value outstanding, in millions of dollars, of the four longest maturity non-callable flower bonds. Color varies with the corresponding bond price. Darker colors represent lower prices, lighter colors represent higher prices.

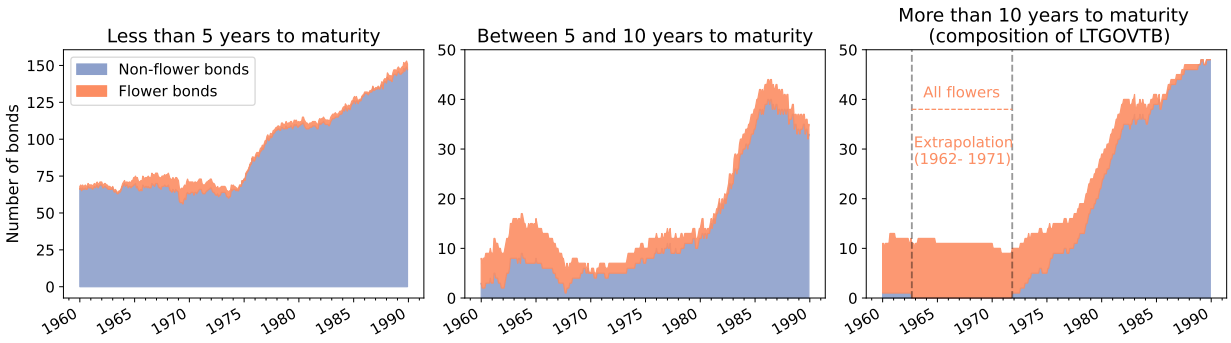


Figure A.23: Composition of Treasurys: 1960-1990. The left subplot shows the number of bonds with less than 5 years to maturity. The middle subplot shows the number of bonds with 5-10 years to maturity. The right subplot shows the number of bonds with more than 10 years to maturity.

December 1969 75

TREASURY SURVEY OF OWNERSHIP, OCTOBER 31, 1969

Table TSO-2. - Interest-Bearing Marketable Public Debt Securities by Type and Maturity Distribution

(Par values - in millions of dollars)

Classification	Total amount outstanding 1/	Held by investors covered in Treasury Survey									Held by all other investors 3/
		U.S. Government accounts and Federal Reserve banks	5,791 commercial banks 2/	497 mutual savings banks 2/	Insurance companies		488 savings and loan associations	469 corporations	State and local governments		
					293 life	460 fire, casualty, and marine			316 general funds	187 pension and retirement funds	
By type security:											
Issued by U.S. Government:											
Treasury bills	78,990	21,491	5,639	138	105	219	272	2,173	4,528	383	44,042
Treasury notes	85,414	37,107	19,975	1,208	149	856	1,613	1,042	2,106	140	21,217
Treasury bonds	70,625	13,027	19,063	1,626	3,042	1,781	2,156	2,021	2,890	3,395	21,624
Total	235,029	71,625	44,678	2,971	3,296	2,856	4,041	5,236	9,524	3,918	86,883
By maturity distribution:											
Call classes (due or first becoming callable):											
Within 1 year	116,549	35,675	15,487	411	275	736	841	3,688	5,597	489	53,350
1 to 5 years	67,763	17,541	22,222	1,256	234	1,269	1,869	1,323	2,247	297	19,504
5 to 10 years	29,015	12,862	6,150	427	382	359	605	203	684	439	6,904
10 to 15 years	4,504	1,462	341	126	513	94	160	12	302	252	1,240
15 to 20 years	6,711	1,917	219	173	458	268	340	6	184	1,204	1,944
20 years and over	10,487	2,167	258	578	1,434	129	226	4	512	1,236	3,941
Total	235,029	71,625	44,678	2,971	3,296	2,856	4,041	5,236	9,524	3,918	86,883
Maturity classes (final maturity):											
Within 1 year	109,550	35,073	12,339	351	158	536	686	3,317	5,396	450	51,244
1 to 5 years	74,762	18,144	25,370	1,316	351	1,470	2,024	1,694	2,447	336	21,610
5 to 10 years	26,247	12,239	5,989	373	84	303	531	202	492	118	5,917
10 to 15 years	6,058	1,654	428	151	778	120	211	13	419	435	1,849
15 to 20 years	2,305	626	124	56	231	60	127	*	126	203	752
20 years and over	16,107	3,891	427	725	1,694	367	462	10	643	2,376	5,511
Total	235,029	71,625	44,678	2,971	3,296	2,856	4,041	5,236	9,524	3,918	86,883

Figure A.24: October 1969 US Treasury Survey of Ownership, Securities by Type and Maturity. This figure shows which investors held securities of over 20 years to final maturity (second row from bottom), all of which were flower bonds. Estates were contained in "all other investors."

A.5 Auxiliary Data

A.5.1 Short Rate Data

In 3.4.2, we highlight that we discipline our discount function estimation to perfectly fit short-term (short rate) data. Here, we list the short rate data used in both our corporate and Treasury discount function estimation.

Treasurys

1. January 1860-March 1917: We do not fit any short rate data, since there is no well-defined government short rate during this time.

2. April 1917-December 1919: We construct a series of short rates from offering rates on Certificates of Indebtedness, documented in Garbade, 2012 p. 112-113. These securities were issued for cash management purposes around issuance of the long-term Liberty bonds. Certificates issued under the authority of the First Liberty Loan were fully tax exempt, while securities issued under the Second Liberty Loan and later were subject to partial taxation. We fix the duration of these securities at 60 days for April-June 1917, 90 days for July 1917-May 1918, and 120 days for June 1918-Dec 1919.

3. January 1920-November 1925: We use the NBER macrohistory series for yields on short-term government certificates and notes (FRED code: M1329AUSM193NNBR). Certificates issued during this time were issued under the authority of the Second Liberty Loan, and were thus subject to partial taxation. We set their duration at 120 days.

4. December 1925-Present: We use the CRSP Risk-Free file to fit 3-month certificates/Treasury Bills since December 1925, and 1-month Treasury Bills since January 1934. We drop all other securities with shorter maturity than the 3-month Treasury Bill from the Risk-Free series, and drop all coupon bonds maturing before the longest duration Treasury Bill.

5. March 1942-March 1951: During the yield curve control period, in addition to fitting 1-month and 3-month Treasury Bills, we fit 6-month and 1-year certificates. These rates were fixed by the Treasury and Federal Reserve which created a step-wise term structure at the very short end.

Corporate

1. January 1860-March 1971: We use the NBER macrohistory series for New York commercial paper rates (FRED code: M13002US35620M156NNBR) and set the duration to 3 months.

2. April 1971-December 1996: We use 3-month and 1-month prime commercial paper rates (FRED codes: H15H0RIFSPPCM03NB and H15H0RIFSPPCM01NB) which are associated with non-financial corporations. These markets became stressed during and around the 1974 recession, so we replace their values with projected values from a regression of 3-month/1-month prime commercial paper rates on 3-month/1-month high grade financial paper (FRED codes: H15H0RIFSPPFM03NB and H15H0RIFSPPFM01NB).

3. January 1997-present: We use 90-day and 30-day AA nonfinancial commercial paper rates (FRED codes: DCPN3M and DCPN30).

A.5.2 Index-based Aaa Corporate-Treasury Spread

The most widely used measure of historical government funding advantage on long-maturity bonds was initially proposed by Krishnamurthy and Vissing-Jorgensen (2012) and computed as the difference between two yield indices:

- Moody's Seasoned Aaa-rated long-maturity corporate bond index (FRED code: Aaa)– constructed from a sample of industrial and utility bonds (industrial only after 2002) with more than 20 years to maturity, and

- The Federal Reserve Bulletin's long-term US government bond yield index (FRED code: LTGOVTBD), constructed as the average yield on *all* outstanding government bonds neither due nor callable in less than 10 years.[70]

[70]More precisely, the Treasury bonds included are due or callable after 12 years for 1926–1941, 15 years for 1941–1951, 12 years for 1952, and 10 years for 1953–1999. The series was discontinued in 2000, after which point papers in the literature use the "market yield on US Treasury Securities at 20-year constant maturity" (FRED code: GS20).

B Loss Function and Estimation

In this Appendix, we specify the details of the yield curve estimation summarized in Section 3.4.2. For both Treasurys and highest-grade corporate bonds, we seek to find the yield curve which prices one dollar of income at time t in the future from a bond without options and, in the case of Treasurys, without any federal income tax exemptions. We therefore need an estimation method that makes the appropriate "corrections," allowing us to learn about the prices of tax-exempt, option-free bonds from the prices of tax-distorted, option-contaminated ones. We represent these corrections as wedges $z_i^{(j)}$ that multiply the cash flow $cf_i^{(j)}$ of bond i on future cash flow date j, for all i and j on a given date. Given our interest in estimating the funding advantage of the government, on every estimation date, we estimate the Treasury yield curve *then* the corporate yield curve. We do this by fitting the corporate discount function as deviation from the Treasury discount function, allowing us to estimate the spread directly.

In Subsections B.1, B.2, and B.3 we detail the loss function our yield curve estimator minimizes and the wedge specifications used in the estimation. In Appendix C, we show that the inclusion of these wedges in the estimation yields price errors independent of specific classes of tax exemptions and option-related features, and that the wedges themselves are individually necessary to improve the quality of fit relative to popular benchmark estimates.

B.1 Loss Function Construction and Estimation

Environment: Consider an estimation date normalized to be time zero. We consider pricing bonds of a single class, either Treasury or high grade corporate bonds. Let $\mathcal{I}$ be the set of bonds observed at time zero, with the number of bonds $|\mathcal{I}| = N$. We stack the corresponding bond prices $p_1, \ldots, p_N$ in vector P. We summarize the promised cash flows of all bonds in an $N \times M$ matrix C, where M is the maximum maturity over all bonds $i = 1, \ldots, N$ and $C_{i,j} = cf_i^{(j)}$. We seek to estimate the entire discount function $\mathsf{q} : [0, \infty] \to \mathbb{R}$ with initial condition $q^{(0)} = 1$.[71] We stack evaluations of q at $1, \ldots, M$ in vector $q = [q^{(1)}, \ldots, q^{(M)}]$ which apply to the M total promised cash flow dates over the N bonds. Each bond i has a multiplicative wedge $z_i^{(j)}$ on cash flow $cf_i^{(j)}$ at future date j. For each i, we stack cash flows $cf_i^{(j)}$ across j in row vector cf_i which drops the (j) superscript. We summarize the wedges in the $N \times M$ matrix valued function $Z = Z(q, \theta)$, depending on the vectorized discount function, q, and wedge parameters θ. We define the fitted price function $\widehat{P}(q, C, Z)$ which maps bonds with cash flows in the rows of C to their fitted prices. The fitted prices of the N

[71] The convention in this section is to denote any functional f in sans-serif font, $f^{(j)}$ an evaluation of f at j, and $f = [f^{(1)}, \ldots, f^{(M)}]$ a vector of evaluations over $j = 1, \ldots, M$.

bonds is the vector $\widehat{P} = (C \odot Z)q$, where $\odot$ denotes elementwise multiplication so element $i \leq N$ of $\hat{P}$ is:

$$\widehat{P}_i := \sum_{j=1}^{M} c f_i^{(j)} z_i^{(j)} q^{(j)}$$

The vector of price errors is the difference between the observed and fitted price vectors:

$$\epsilon := P - \widehat{P}(q, C, Z).$$

Kernel Ridge Estimation: To perform our discount function estimation, we extend the Kernel Ridge methods of Filipović et al. (2022) (henceforth FPY) to account for multiplicative wedges $Z(q, \theta)$ that depend on q and a parameter vector θ. The spirit of the Kernel Ridge estimation is to fit a flexible non-parametric discount function q from the large space of twice weakly-differentiable functions, while prioritizing smoothness. Whereas FPY fit only the discount function, we jointly estimate the discount function q and a bond-specific wedge matrix Z to simultaneously capture the underlying term structure as well as the various bond price distortions across our historical sample.

Feasible Discount Functions: Towards constructing our Kernel Ridge loss function, we decompose the discount function q into a linear combination of a baseline anchor function a and a non parametric deviation function h, with:

$$q^{(j)} = a^{(j)} + h^{(j)}, \ \forall j \geq 0$$

subject to $a^{(0)} = 1$ and $h^{(0)} = 0$.[72] The vectorized discount function q therefore satisfies $q = a + h$ for vectorized functions a and h. In principle, both a and h are choice variables of the minimization problem, but each is subject to a different constraint set. The form of the anchor a is chosen exogenously from a narrow class of functions, $\mathcal{A}$, and is potentially parametrized by vector $\gamma \in \Gamma$. a can be interpreted as a reference discount curve, encoding coarse information regarding the yield curve.[73] Naturally, the specification of the anchor's constraint set $\mathcal{A}$ is application specific. We discuss our choices in Subsection B.2.

In contrast, the deviation function h is chosen from a much larger functional space, embodying non-parametric deviations from the anchor along the term structure whenever the data warrant them. To avoid overfitting, we restrict this choice set by penalizing the curvature of h, discouraging sharp oscillations and kinks in the fitted discount curve. To this

[72] The technical condition $h^{(0)} = 0$ is necessary such that the Hilbert space of deviation functions we estimate over is well-defined.

[73] We use the convention that $\Gamma = \emptyset$ when the anchor a is completely exogenously specified.

end, following FPY, we define a smoothness norm $||\cdot||_{\alpha,\delta}$ as a weighted average of the first and second derivatives of a twice weakly differentiable function:

$$||\mathsf{h}||_{\alpha,\delta} = \left(\int_0^\infty (\delta h'(x)^2 + (1-\delta)h''(x)^2)e^{\alpha x}dx\right)^{1/2} \tag{B.1}$$

for maturity weight parameter $\alpha \geq 0$ and shape parameter $\delta \in [0,1]$. The inner product associated with the norm (B.1) defines the set $\mathcal{H}_{\alpha,\delta}$ of twice weakly differentiable functions that satisfy finite norm (B.1) and $h^{(0)} = 0$.

Additionally, we restrict the space of feasible discount functions by requiring that q satisfies two desirable properties: (i) it perfectly fits the price of short dated bonds chosen to encode the observable short rate to the fitted yield curve and (ii) the price of an infinite maturity zero-coupon bond is zero, $q^{(\infty)} = 0$. In case of the Treasurys, property (i) allows for greater curvature at the short-end of the yield curve by overriding the Kernel Ridge curvature penalty at very short maturities. This is advantageous because it is well known that Treasury Bill yields can exhibit a shared variation away from other short-maturity Treasurys (e.g. Duffee, 1996), thus requiring very sharp curvature to fit the Treasury's short term financing costs. In case of the Aaa$^+$ corporate bonds, property (i) disciplines the short-maturity yield curve estimates when there are no short maturity bonds in our Aaa bond universe. This ensures that our short maturity yield curves are disciplined by the high-grade commercial paper rate rather than extrapolation from very long-maturity Aaa bonds. Property (ii) ensures theoretical consistency of the fitted discount function by imposing the transversality condition.

To operationalize properties (i)-(ii), we choose the set $\mathcal{I}_0$ of short bonds and infinity maturity zero-coupon bond to fit perfectly, with $|\mathcal{I}_0| = N_{\mathcal{I}_0}$. We stack the corresponding $N_{\mathcal{I}_0}$ prices in vector $P_{\mathcal{I}_0}$. Let $\mathcal{M}_{\mathcal{I}_0}$ be the set of cash flow dates associated with bonds in $\mathcal{I}_0$, with $|\mathcal{M}_{\mathcal{I}_0}| = M_{\mathcal{I}_0}$ the corresponding number of cash flow dates. We define $C_{\mathcal{I}_0}$ the bonds' corresponding cash-flow matrix, and $Z_{\mathcal{I}_0}$ the bonds' corresponding wedge matrix each with $M_{\mathcal{I}_0}$ columns/cash-flow dates. The constraint function $\mathcal{G}$ is defined as the price errors associated with the bonds in $\mathcal{I}_0$:

$$\mathcal{G}(\mathsf{h},\gamma,\theta) := P_{\mathcal{I}_0} - \widehat{P}\Big(a_{\mathcal{I}_0}(\gamma) + h_{\mathcal{I}_0},\ C_{\mathcal{I}_0},\ Z_{\mathcal{I}_0}(a_{\mathcal{I}_0}(\gamma) + h_{\mathcal{I}_0},\theta)\Big)$$

where $a_{\mathcal{I}_0}(\gamma)$ and $h_{\mathcal{I}_0}$ denote the vectorized functions evaluated at cash-flow dates in $\mathcal{M}_{\mathcal{I}_0}$. Given the anchor parameters γ and wedge parameters θ, requiring that short-dated and infinite-maturity bond prices are fitted perfectly ($\mathcal{G} = 0$) constrains the values of the non-parametric deviation function h at the $M_{\mathcal{I}_0}$ cash-flow dates. This constraint therefore has

the effect of restricting the Hilbert space $\mathcal{H}_{\alpha,\delta}$ to functions h which perfectly price bonds in $\mathcal{I}_0$.

Loss Function and Estimation: Given an observed price vector P, cash-flow matrix C, and pre-specified weights $\{\rho_i\}_{i=1}^N$[74], we define our loss function $\mathcal{L}$ as:

$$\mathcal{L}\big(\mathsf{h},\gamma,\theta\big) := \underbrace{\sum_{i=1}^{N}\rho_i\Big(P_i - \hat{P}_i\big(a(\gamma)+h,\ C,\ Z(a(\gamma)+h,\ \theta)\big)\Big)^2}_{\text{weighted mean squared price errors}} + \underbrace{\lambda||\mathsf{h}||^2_{\alpha,\delta}}_{\text{smoothness penalty}}$$

the sum of the weighted mean squared price errors and a smoothness penalty on the non-parametric deviation function h. The penalty term has three tuning parameters (λ,α,δ): α and δ define the norm (B.1) that restricts the space of feasible deviations $\mathcal{H}_{\alpha,\delta}$ to sufficiently smooth functions, while λ governs the overall strength of the penalty.

We define the Kernel Ridge estimator as the minimizer of the following bilevel minimization problem:

$$\min_{\gamma\in\Gamma,\ \theta\in\Theta}\left\{\min_{\mathsf{h}\in\mathcal{H}_{\alpha,\delta}}\mathcal{L}(\mathsf{h},\gamma,\theta),\quad \text{s.t. } \mathcal{G}(\mathsf{h},\gamma,\theta)=0\right\} \tag{B.2}$$

Minimizing (B.2) without additional simplifications is challenging due to the bilevel structure and the fact that the inner minimization is over an *entire functional* h. However, as in FPY, the representer theorem for reproducing kernel Hilbert spaces applies to the inner minimization of (B.2), so given (γ,θ) the minimizing functional h can be expressed as a weighted sum of kernel basis functions:

$$\mathsf{h} = \sum_{j=1}^{M} k(\cdot,j)\beta^{(j)} + \sum_{x\in\mathcal{M}_{\mathcal{I}_0}} k(\cdot,x)\beta^{(x)}_{\mathcal{I}_0}$$

for $M\times 1$ vector β, $M_{\mathcal{I}_0}\times 1$ vector $\beta_{\mathcal{I}_0}$, and upon defining the kernel function $k:[0,\infty)\times[0,\infty)\to\mathbb{R}$ given by:

$$\begin{aligned} k(x,y) = -\frac{\alpha}{\delta\ell_2^2}\big(1-e^{-\ell_2 x}-e^{-\ell_2 y}\big) + \frac{1}{\alpha}(1-e^{-\alpha(x\wedge y)}) \\ + \frac{1}{\delta\sqrt{D}}\left(\frac{\ell_1^2}{\ell_2^2}e^{-\ell_2(x+y)} - e^{-\ell_1(x\wedge y)-\ell_2(x\vee y)}\right)\end{aligned}$$

[74]We set weights $\rho_i = \frac{1}{N}\frac{1}{DP_i^2}$ proportional to the inverse of duration times price squared, as is standard in the literature.

where we define

$$D = \alpha^2 + \frac{4\delta}{1-\delta}, \qquad \ell_1 = \frac{\alpha - \sqrt{D}}{2}, \quad \ell_2 = \frac{\alpha + \sqrt{D}}{2}$$
$$x \wedge y = \min\{x,y\}, \quad x \vee y = \max\{x,y\}$$

where we obtain additional limiting cases shown in FPY as $\alpha \to 0$, $\delta \to 0$, and/or $\delta \to 1$. The case $\alpha = 0, \delta = 0$ is not specified. Importantly, this representation allows us to turn the (potentially infinite dimensional) minimization over the entire functional h into minimization over finite dimensional coefficient vectors β and $\beta_{\mathcal{I}_0}$ because the number of observed cash flow dates M in our sample is always finite.

We use the above representation to restate the inner minimization problem (B.2) in terms of $\beta, \beta_{\mathcal{I}_0}$. We define the Gram matrix $\mathbf{K}$ with $\mathbf{K}_{xy} := k(x,y)$ in block form:

$$\mathbf{K} = \begin{bmatrix} \mathbf{K}_{\mathcal{I},\mathcal{I}^T} & \mathbf{K}_{\mathcal{I},\mathcal{I}_0^T} \\ \mathbf{K}_{\mathcal{I}_0,\mathcal{I}^T} & \mathbf{K}_{\mathcal{I}_0,\mathcal{I}_0^T} \end{bmatrix}$$

blocked according to the cash flow dates of bonds in $\mathcal{I}$ and $\mathcal{I}_0$, with $\mathbf{K}_{\mathcal{I},\mathcal{I}_0^T} = (\mathbf{K}_{\mathcal{I}_0,\mathcal{I}^T})^T$. The inner minimization of (B.2) now takes the form:

$$\min_{\beta,\beta_{\mathcal{I}_0}} \Big\{ \mathcal{L}\Big(\beta, \beta_{\mathcal{I}_0}, \gamma, \theta;\, \mathbf{K}\Big), \quad \text{s.t.} \quad \mathcal{G}(\beta, \beta_{\mathcal{I}_0}, \gamma, \theta; \mathbf{K}) = 0 \Big\} \tag{B.3}$$

$$\text{(Objective)} \qquad \mathcal{L}(\beta, \beta_{\mathcal{I}_0}, \gamma, \theta; \mathbf{K}) = \sum_{i=1}^{N} \rho_i (P_i - \hat{P}_i)^2 + \lambda \begin{bmatrix} \beta^T & \beta_{\mathcal{I}_0}^T \end{bmatrix} \mathbf{K} \begin{bmatrix} \beta \\ \beta_{\mathcal{I}_0} \end{bmatrix}$$

$$\text{(Constraint)} \qquad \mathcal{G}(\beta, \beta_{\mathcal{I}_0}, \gamma, \theta; \mathbf{K}) = P_{\mathcal{I}_0} - \widehat{P}(q_{\mathcal{I}_0}, C_{\mathcal{I}_0}, Z(q,\theta))$$

$$\text{(Fitted Prices)} \qquad \widehat{P} = \widehat{P}(q, C, Z(q,\theta)) = (C \odot Z(q,\theta))q$$

$$\text{(Discount Function)} \qquad q = q(\beta, \beta_{\mathcal{I}_0}, \gamma) = a(\gamma) + \mathbf{K}_{\mathcal{I},\mathcal{I}^T}\beta + \mathbf{K}_{\mathcal{I},\mathcal{I}_0^T}\beta_{\mathcal{I}_0},$$
$$q_{\mathcal{I}_0} = q_{\mathcal{I}_0}(\beta, \beta_{\mathcal{I}_0}, \gamma) = a_{\mathcal{I}_0}(\gamma) + \mathbf{K}_{\mathcal{I}_0,\mathcal{I}^T}\beta + \mathbf{K}_{\mathcal{I}_0,\mathcal{I}_0^T}\beta_{\mathcal{I}_0}$$

In order to obtain unconstrained minimization over β and (a, θ), we plug the constraint $\mathcal{G} = 0$ into the objective function $\mathcal{L}$ in (B.3). We do this by expressing $\beta_{\mathcal{I}_0}$ in terms of β and $\mathcal{I}$ information. We assume that the function $\widehat{P}(q_{\mathcal{I}_0}, C_{\mathcal{I}_0}, \theta)$ is invertible, such that $\mathcal{G} = 0$

implies:[75]

$$P_{\mathcal{I}_0} = \widehat{P}(q_{\mathcal{I}_0}, C_{\mathcal{I}_0}, \theta)$$
$$\widehat{P}^{-1}(P_{\mathcal{I}_0}, C_{\mathcal{I}_0}, \theta) = q_{\mathcal{I}_0} = a_{\mathcal{I}_0}(\gamma) + \mathbf{K}_{\mathcal{I}_0,\mathcal{I}^T}\beta + \mathbf{K}_{\mathcal{I}_0,\mathcal{I}_0^T}\beta_{\mathcal{I}_0}$$

which may be rearranged to obtain:

$$\beta_{\mathcal{I}_0} = \mathbf{K}^{-1}_{\mathcal{I}_0,\mathcal{I}_0^T}\Big(\widehat{P}^{-1}(P_{\mathcal{I}_0}, C_{\mathcal{I}_0}, \theta) - a_{\mathcal{I}_0}(\gamma) - \mathbf{K}_{\mathcal{I}_0,\mathcal{I}^T}\beta\Big) \tag{B.4}$$

Consequently, substitution of (B.4) into $\mathcal{L}$ in (B.3) reduces the inner minimization problem to an unconstrained minimization problem in β. Furthermore, since the inner minimization problem has an unconstrained representation, we can express the full minimization problem of (B.2) without a bilevel structure as:

$$\min_{\beta,\ \gamma\in\Gamma,\ \theta\in\Theta} \quad \sum_{i=1}^{N} \rho_i (P_i - \hat{P}_i)^2 + \lambda\beta^T\bar{\mathbf{K}}\beta \tag{B.5}$$
$$\text{where } \bar{\mathbf{K}} := \mathbf{K}_{\mathcal{I},\mathcal{I}^T} - \mathbf{K}_{\mathcal{I},\mathcal{I}_0^T}\mathbf{K}^{-1}_{\mathcal{I}_0,\mathcal{I}_0^T}\mathbf{K}_{\mathcal{I}_0,\mathcal{I}^T}$$

is the updated Gram matrix from having substituted in the constraint $\mathcal{G} = 0$. Similarly, the implied anchor function $\bar{\mathsf{a}}$ becomes:

$$\bar{a}^{(j)}(\gamma) = a^{(j)}(\gamma) + \mathbf{K}_{j,\mathcal{I}_0^T}\mathbf{K}^{-1}_{\mathcal{I}_0,\mathcal{I}_0^T}\Big(\widehat{P}^{-1}(P_{\mathcal{I}_0}, C_{\mathcal{I}_0}, \theta) - a_{\mathcal{I}_0}(\gamma)\Big), \quad \forall j \geq 0$$
$$\bar{a}(\gamma) = a(\gamma) + \mathbf{K}_{\mathcal{I},\mathcal{I}_0^T}\mathbf{K}^{-1}_{\mathcal{I}_0,\mathcal{I}_0^T}\Big(\widehat{P}^{-1}(P_{\mathcal{I}_0}, C_{\mathcal{I}_0}, \theta) - a_{\mathcal{I}_0}(\gamma)\Big)$$

Additionally, the corresponding vectorized discount function takes the form:

$$q = \bar{a}(\gamma) + \bar{\mathbf{K}}\beta$$

with the entire discount function q given by:

$$q^{(j)} = \bar{a}^{(j)}(\gamma) + \left(\mathbf{K}_{j,\mathcal{I}^T} - \mathbf{K}_{j,\mathcal{I}_0^T}\mathbf{K}^{-1}_{\mathcal{I}_0,\mathcal{I}_0^T}\mathbf{K}_{\mathcal{I}_0,\mathcal{I}^T}\right)\beta$$

We estimate β, anchor parameters γ, and wedge parameters θ jointly on a given date by

[75] $\widehat{P}$ is always invertible if the number of prices equals the number of non-zero cash flow dates, which trivially holds for a vector of zero-coupon bond prices.

minimizing (B.5). The first order conditions associated with (B.5) are:

$$
\begin{aligned}
(\beta) : 0 &= \bar{\mathbf{K}}\beta - \bar{\mathbf{K}}\left(\nabla_q(\hat{P})\right)^T \Lambda^{-1}\epsilon \\
(\gamma) : 0 &= (\nabla_\gamma(\bar{a}))^T \left(\nabla_q(\hat{P})\right)^T \Lambda^{-1}\epsilon \\
(\theta) : 0 &= \left(\nabla_\theta(\hat{P})\right)^T \Lambda^{-1}\epsilon
\end{aligned}
\tag{B.6}
$$

where ϵ are the price errors $P - \widehat{P}$, $\Lambda = \text{diag}(\lambda/\rho)$, and $\nabla_q(\hat{P}), \nabla_\theta(\hat{P})$ are the Jacobian matrices of the fitted bond prices $\widehat{P}$ with respect to q and θ, respectively. $\nabla_\gamma(\bar{a})$ is the Jacobian matrix of the implied anchor function $\bar{a}$. To obtain point estimates of β and θ, we find the pair $(\widehat{\beta}, \widehat{\theta})$ which solve the system (B.6).[76]

(B.6) gives "moment conditions" for β and θ which we use to form GMM standard errors. We parameterize the covariance matrix of errors: $Var(\epsilon) = s^2\,\text{diag}(DP^2) = \frac{s^2}{\lambda N}\Lambda$, $s^2 = \frac{1}{N}\sum_{i=1}^{N}\rho_i\epsilon_i^2$ such that errors are noise scaled by the duration times price of the bond and are uncorrelated in the cross section.[77]

B.2 Specifying the Anchor Curve

In B.1, we showed the formulation of the Kernel Ridge minimization problem over $\mathcal{L}$ with general anchor curve a and wedges $z_i^{(j)}$ for all i and j which we use to perform discount function estimation on a given date. In this Subsection, we describe how we specify the set of feasible anchor functions $\mathcal{A}$ and associated parameters $\gamma \in \Gamma$. Subsection B.3 describes how we parameterize wedges Z.

At a high level, when we fit the Treasury and corporate discount functions, we seek to obtain non-parametric deviations relative to a tightly parameterized function in the space in which we want to remain conservative. For the Treasury term structure, we consider non-parametric deviations from a flat yield curve which captures the approximate level across the term structure. This specification corresponds to an exponentially decaying discount function that satisfies the properties $a^{(0)} = 1$ and $a^{(\infty)} = 0$. These are properties that any discount function must satisfy, so not imposing them on the anchor forces the Kernel Ridge estimator to penalize curvature of the fitted discount function needed to achieve these necessary properties.[78] On the other hand, when constructing the corporate anchor curve,

[76] In the special case when there are no wedges (i.e. $Z = 1$ and θ is null), (B.6) is a linear system in β. Solving that system gives the same closed-form point estimate $\widehat{\beta}$ as in FPY.

[77] A first order Taylor expansion of each bond price i by its yield shows that s^2 is approximately the mean-squared error of the fitted bond yields.

[78] FPY chooses an anchor of $a^{(j)} = 1$ for all $j \geq 0$ (zero-yield anchor). This necessarily imposes that curvature of the discount function as $j \to \infty$ is penalized when the estimator attempts to deviate from the anchor to try to satisfy the transversality condition $q^{(\infty)} = 0$, which they also do not impose in their

we want a specification which captures the stylized observation from the yield-to-maturity plots that the corporate term structure is approximately a level shift above the Treasury term structure. Given our object of interest is the spread between these curves, we parameterize the corporate anchor function such that its shape is a level shift in the yield spread over Treasurys. The Kernel Ridge estimation then allows non-parametric deviations that change the shape of the corporate term structure only when warranted by the data. This allows us to remain conservative on the term structure of the funding advantage, ensuring that we do not impose unwarranted shape on our fitted spread due to interpolation across the corporate term structure where the data is sparse.

Treasurys: Formally, for our baseline Treasury anchor curve, we parameterize a by the exponential decay function:

$$a^{(j)} = \exp(-rj)$$

Beginning in 1926, we estimate r jointly with the other parameters to minimize pricing errors. Before 1926, we do not estimate r and instead specify a using the Dynamic Nelson-Siegel parameters of Payne et al. (2025)—i.e., the set $\mathcal{A}$ is a singleton. We do this because prior to World War I, there is often very few bonds on a given date, and so we lose yield curve shape when we perform period-by-period estimation. Payne et al. (2025) attempts to solve this problem by smoothing the yield curve shape across time, with a time varying smoothing parameter that is wider when the data is sparse. Adopting the Dynamic Neslson-Siegel as an anchor ensures that i) when there is sufficient data, the Kernel Ridge estimator fits a non-parametric deviation which weakly improves on the fit of the tightly parameterized Dynamic Nelson-Siegel function and ii) when there are only a few bonds on a given date which are well fitted by the Dynamic Nelson-Siegel curve, the Kernel Ridge chooses minimal non-parametric deviation and returns the Dynamic Nelson-Siegel curve as the discount function estimate. We choose 1926 as the date to change the anchor to try and minimize the disruption to the series. In principle, any date following the US entry into World War I in 1917 contains enough bond data to jointly estimate r and the other parameters. However, since 1917 coincides with large structural shifts in debt maturity management, the quantity of outstanding debt, and our funding advantage estimates, we choose the relatively calmer period of the mid-1920s to switch to our baseline Treasury anchor curve. Robustness exercises find that the exact switch date has negligible impact on our fitted Treasury discount function.

implementation.

Corporate Bonds: For corporate bonds, we construct the anchor curve by applying a level shift in yields relative to the Treasury curve. Specifically, defining the corporate anchor curve $\tilde{\mathsf{a}}$ we construct $\tilde{\mathsf{a}}$ according to:

$$\tilde{a}^{(j)} = q^{(j)} \exp(-\tilde{r} j)$$

where $q^{(j)}$ is the fitted Treasury discount function. This formulation of the corporate anchor curve allows us to keep the shape of the Treasury yield curve but allows a potential level shift in yields. This means that the Kernel Ridge estimator only chooses a different shape from the Treasury yield curve when warranted by the available data, and not from interpolation or extrapolation when there is sparse data at certain maturities.

B.3 Specifying the Wedges

We now describe the construction of our bond price wedge functions. All objects in this appendix are defined for a single sample (Treasury or corporate) on a single estimation date, normalized to time zero so that time subscripts are suppressed. The corporate analog of each object carries a tilde.

Assumption 3. The price of bond i of class k (tax exemption, callable, etc.), with coupon c_i, maturity m_i, observable characteristics x_i and promised cash flows $cf_i^{(j)}$, is

$$p_i = \sum_{j=1}^{m_i} cf_i^{(j)} z_i^{(j)} q^{(j)} + \varepsilon_i, \qquad z_i^{(j)} = z_k^{(j)}(\theta, \mathsf{q}; x_i) \tag{B.7}$$

$$cf_i^{(j)} = \begin{cases} c_i, & j < m_i \\ c_i + 100, & j = m_i \end{cases}$$

where $q^{(j)}$ is the $j-$maturity zero-coupon discount function common to all bonds in the sample, $z_i^{(j)}$ is the bond-price wedge of bond i with parameters θ, and ε_i is a mean-zero measurement error that we suppress in the derivations below.

Plain vanilla bonds have $z_i^{(j)} = 1$. The assumption says that a common discount function prices the promised payments of all bonds in a sample once cash flows are adjusted by wedges that depend on bond i only through its class and characteristics x_i. What identifies the decomposition between q and z is that q is the component common to every bond in the sample, while z captures the valuation of cash flows specific to a bond's class or clientele. Because each z is an explicit function of q and a few parameters, rather than an unrestricted residual, it can be identified from a single date's cross section.

Our primary goal is to construct a parametric model for $z_i^{(j)}$ that is sufficiently parsimonious to be identified in the data, sufficiently flexible to resolve the fit problems documented in Section 3, and ultimately able to successfully extract a common zero-coupon yield curve from the heterogeneous bonds in our sample. A secondary goal is to map our wedge functions into semi-structural models so that we can interpret our estimated parameters and, when appropriate, compare them to untargeted data series. We showcase these interpretations in Section 4.2 and Appendix C.

For each type of bond feature (i.e. tax exemptions, capital gains, call options, flower bonds, and the exchange privilege), we proceed in the following way. We first present the wedge specification and additional identifying assumptions as needed. We then discuss the high-level intuition behind the wedge, followed by a derivation of the wedge from a general environment. Ultimately, we obtain a Treasury wedge parameter vector θ consisting of 10 parameters and a corporate wedge parameter consisting of 3 parameters according to:[79]

$$\begin{aligned}
(\text{Treasury}) : \theta &= \Big(\underbrace{\theta_{fte}, \theta_{pte}, \eta}_{\substack{\text{Tax} \\ \text{Adjustments}}} , \quad \underbrace{\kappa, \sigma}_{\substack{\text{Option-Implied} \\ \text{Interest Rate Volatility}}} , \quad \underbrace{\bar{\alpha}, \bar{\beta}, \bar{\eta}}_{\substack{\text{Flower Bond} \\ \text{Early Exercise}}} , \quad \underbrace{\zeta_1, \zeta_2}_{\substack{\text{Exchange Privilege} \\ \text{Payout Adjustments}}} \Big) \\
(\text{Corporate}) : \tilde{\theta} &= \Big(\underbrace{\tilde{\eta}}_{\substack{\text{Capital Gains} \\ \text{Advantage}}} , \quad \underbrace{\tilde{\kappa}, \tilde{\sigma}}_{\substack{\text{Option-Implied} \\ \text{Interest Rate Volatility}}} \Big)
\end{aligned} \tag{B.8}$$

While we suppress time subscripts in (B.8), each parameter, when activated, is fit only to the cross sectional data on a given estimation date within the Treasury or corporate sample. When both Treasury and corporate samples share similar provisions (i.e. call options), we fit θ separately for the two datasets. Without loss of generality, we proceed in the following sections by showing wedges with parameters contained in θ since each element of $\tilde{\theta}$ contains an analogous element in θ.

B.3.1 Tax Exemptions

Federal income tax on Treasury interest between 1917 and 1941 had two layers, a normal tax and progressive surtaxes. Taxable bonds paid both, partially tax-exempt (*pte*) bonds paid only the surtaxes, and fully tax-exempt (*fte*) bonds paid neither. The Treasury issued no fully taxable securities until 1941, after which they traded alongside *fte* and *pte* bonds. To recover the common term structure information shared across this tax-heterogeneous sample, we must remove the effects of these exemptions. Here, we consider the wedges $z_i^{(j)}$ for bonds

[79] Throughout this section, we discuss parameters belonging to general parameter vector θ which may correspond to either the Treasury or corporate dataset. In practice, we consider Treasury parameter vector θ and corporate parameter vector $\tilde{\theta}$.

with tax exemptions.

Specification 1 (Tax Exemption Wedge)**.** Let $y^{(j)} = -\frac{1}{j}\log q^{(j)}$ be the zero-coupon yield curve applicable to fully taxable bonds. The yield curve applicable to class-k bonds is a scaled version of it, $y_k^{(j)} = \theta_k\, y^{(j)}$ for all j and $k \in \{fte,\ pte\}$, with scaling parameters $\theta_{fte}, \theta_{pte}$. This gives the wedge

$$z_{i,k}^{(j)} = (q^{(j)})^{\theta_k - 1}, \quad k \in \{fte,\ pte\}. \tag{B.9}$$

Discussion: The yield curve scaling is the behavior we see in the data: *fte* and *pte* yields look like scaled-down versions of the taxable yield curve at every maturity, as showcased in the left and middle panels of Figure 6. The scaling parameters carry a structural interpretation. If agents believe that tax rates follow a random walk independent of their own discount factor, and that all sources of return (income, capital gains/losses) are taxed continuously, then

$$\theta_k = \frac{1-\tau^{inc}}{1-\tau^{k}}, \qquad k \in \{fte,\ pte\}, \tag{B.10}$$

where τ^{inc} is the marginal federal income tax rate on taxable bonds, $\tau^{fte} = 0$, and τ^{pte} is the marginal (surtax) rate still paid on *pte* interest. Under the structural interpretation, the wedges are expressible in terms of tax rates:

$$z_{i,\text{fte}}^{(j)} = (q^{(j)})^{-\tau^{inc}}, \qquad z_{i,\text{pte}}^{(j)} = (q^{(j)})^{-\frac{\tau^{inc}-\tau^{pte}}{1-\tau^{pte}}}.$$

The random walk assumption allows us to identify the tax wedge across the term structure with time zero tax rates.[80] Figure 11 shows that the tax rates implied by our estimates of θ_{fte} and θ_{pte} closely track contemporaneous corporate and top personal income tax rates from IRS data over the period $1917 - 1953$.

Derivation of Tax Exemption Wedges: To derive the tax exemption wedge (B.9) and the mapping (B.10), we consider the following environment in which both taxable and tax-exempt bonds are traded. Let $q_t^{(j)}$ be a taxable zero-coupon bond of maturity j at time t and $\breve{q}_t^{(j)}$ be a tax-exempt bond of maturity j at time t, with $q_t^{(0)} = \breve{q}_t^{(0)} = 1$ for all t. We assume (1) over period Δ, the profit on the taxable bond from t to $t+\Delta$ is taxed at tax rate $\tau_{t+\Delta}$ and (2) that τ_t follows a random walk (i.e. $\tau_t = \mathbb{E}_t[\tau_{t+s}]$ for all $s \geq 0$) with changes in

[80] A Dickey-Fuller test on corporate income tax rates over 1917-1961, the time period with heterogeneous income tax treatment across Treasury bonds, fails to reject the null hypothesis of a unit root in the data.

the tax rate orthogonal to the SDF. The general Euler equations for the two bonds are:

$$\begin{aligned} q_t^{(j)} &= \mathbb{E}_t[\xi_{t,t+\Delta}(q_{t+\Delta}^{(j-\Delta)} - \tau_{t+\Delta}(q_{t+\Delta}^{(j-\Delta)} - q_t^{(j)})] \\ \breve{q}_t^{(j)} &= \mathbb{E}_t[\xi_{t,t+\Delta}\breve{q}_{t+\Delta}^{(j-\Delta)}] \end{aligned} \tag{B.11}$$

where $\xi_{t,t+\Delta}$ is the SDF from t to $t+\Delta$. Upon rearrangement of (B.11), we see that returns on the tax-exempt bond must match the after-tax returns on the taxable bond:[81]

$$\mathbb{E}_t\left[\xi_{t,t+\Delta}\underbrace{\left(\frac{q_{t+\Delta}^{(j-\Delta)}}{q_t^{(j)}} - \tau_t\left(\frac{q_{t+\Delta}^{(j-\Delta)}}{q_t^{(j)}} - 1\right)\right)}_{\text{after tax return on taxable bond}}\right] = \mathbb{E}_t\left[\xi_{t,t+\Delta}\underbrace{\frac{\breve{q}_{t+\Delta}^{(j-\Delta)}}{\breve{q}_t^{(j)}}}_{\text{return on tax exempt bond}}\right]$$

Since after-tax returns on the taxable bonds must equate with returns on the tax exempt bond for any $j-$maturity bond over period Δ, we impose no arbitrage across tax treatments within the date's cross section of Treasurys. This implies that at time 0, we must have:

$$(1-\tau)\left(\frac{q^{(j)}}{q^{(j+\Delta)}} - 1\right) = \frac{\breve{q}^{(j)}}{\breve{q}^{(j+\Delta)}} - 1 \tag{B.12}$$

where $q^{(j)} = q_0^{(j)}, \breve{q}^{(j)} = \breve{q}_0^{(j)}, \tau = \tau_0$ are time 0 objects. (B.12) states that the after-tax forward rate for taxable bonds over j to $j+\Delta$ equals the forward rate for tax-exempt bonds over j to $j+\Delta$. Since taxes are a random walk and tax changes are independent of the SDF, we can substitute time 0 tax rates in (B.12) for expected future tax rates at $j+\Delta$.

We now consider the limit where $\Delta \to 0$ (continuous taxation). Sending $\Delta \to 0$, (B.12) becomes a differential equation:

$$\begin{aligned} (1-\tau)\frac{\dot{q}^{(j)}}{q^{(j)}} &= \frac{\dot{\breve{q}}^{(j)}}{\breve{q}^{(j)}} \\ \text{s.t.} \quad q^{(0)} &= \breve{q}^{(0)} = 1 \end{aligned}$$

which may be integrated to obtain

$$\breve{q}^{(j)} = (q^{(j)})^{(1-\tau)}$$

Therefore, at future date j, if bond i is tax-exempt, we have the pricing wedge:

$$z_{\text{exemption}}^{(j)} = (q^{(j)})^{-\tau} \tag{B.13}$$

[81] The following equation applies $\tau_t = \mathbb{E}_t[\tau_{t+\Delta}]$.

since $z^{(j)}_{\text{exemption}} q^{(j)} = \breve{q}^{(j)}$. For a fully exempt bond, $\tau = \tau^{inc}$ in (B.13), so that $\breve{q}^{(j)} = (q^{(j)})^{1-\tau^{inc}}$, which is the scaling $y^{(j)}_{fte} = \theta_{fte}\, y^{(j)}$ with $\theta_{fte} = 1 - \tau^{inc}$. For a partially tax exempt bond, $\tau = \frac{\tau^{inc} - \tau^{pte}}{1-\tau^{pte}} = 1 - \theta_{pte}$. This gives $y^{(j)}_{pte} = \theta_{pte} y^{(j)}$, as needed.

A partially exempt bond still pays the surtax rate τ^{pte}, so the left-hand side of (B.12) becomes $(1-\tau^{inc})(\cdot)$ and the right-hand side $(1-\tau^{pte})(\cdot)$. The same steps give $\breve{q}^{(j)} = (q^{(j)})^{\theta_{pte}}$ with $\theta_{pte} = \frac{1-\tau^{inc}}{1-\tau^{pte}}$. This is (B.10), and $z^{(j)}_{i,k} = \breve{q}^{(j)}/q^{(j)}$ is (B.9).

Full Tax Rate Adjustments pre-1941: After 1941, taxable Treasurys trade alongside *fte* and *pte* bonds, so θ_{fte} identifies the marginal investor's income tax rate directly, $\tau^{inc} = 1 - \theta_{fte}$, and $\theta_{fte}/\theta_{pte} = 1 - \tau^{pte}$ identifies their surtax rate. Before 1941 no fully taxable Treasurys existed, so τ^{inc} cannot be inferred from bond prices alone, which prohibits us from directly estimating a Treasury curve subject to the same federal income tax provisions as corporate bonds. The *after-tax* curve can still be estimated: the *fte* and *pte* bonds identify the *fte* discount function $\breve{q}^{(j)} = (q^{(j)})^{\theta_{fte}}$ and the surtax rate τ^{pte}, with the estimator remaining agnostic to τ^{inc}. What is missing is τ^{inc}, which we need to scale the *fte* curve into a counterfactual fully taxable curve, $q^{(j)} = (\breve{q}^{(j)})^{1/(1-\tau^{inc})}$. Our estimated τ^{pte} is informative because it is the surtax rate of the marginal investor, and the tax code assigned individuals and corporations (including banks) very different surtax schedules. Corporate surtax rates were negligible throughout this period, whereas surtax rates for wealthy individuals were not, so we bound the latter with the marginal rates faced by the top 0.1% and 0.01% of earners (Piketty and Saez, 2003), for each year since 1913. If the estimated τ^{pte} falls inside this individual range, we take individuals to be the marginal investor that year. If it falls below the range, close to the near-zero corporate surtax rate, we take corporations. Given the inferred marginal investor, τ^{inc} is either a) our estimated surtax rate plus the matching statutory normal tax rate if the marginal investor is an individual or b) the statutory normal income tax rate plus surtax rate (0) if the marginal investor is a corporation. The resulting path of τ^{inc} delivers the fully taxable Treasury yield curve for 1917 to 1941. See the bottom panel of Figure 11 for the estimated τ^{pte} path and the identification of the marginal investor. Figure C.41 shows our inferred tax rates pre-1941.

B.3.2 Capital Gains

Handling capital gains tax distortions is more complicated than ordinary income tax exemptions because a) the capital gains rate τ^{cg} is generally lower than the income tax rate τ^{inc} and b) the payment of capital gains tax can be deferred until the bond is either resold, called or matured. We therefore desire a wedge which captures the two features investors save on

tax payments when they pay tax at the capital gains rate, and that they can defer payment of capital gains. Premium bonds carry no such advantage, since the premium is amortized against ordinary income, so the wedge must be inactive for them.

Specification 2 (Capital Gains Wedge)**.** We form our capital gains wedge from the following generalized bond pricing formula:

$$p_i = \sum_{j=1}^{m_i} c_i q^{(j)} + (100 + \eta(100 - p_i)^+)q^{(m_i)} = \sum_{j=1}^{m_i} cf_i^{(j)} q^{(j)} + \eta(100 - p_i)^+ q^{(m_i)} \tag{B.14}$$

where $(100 - p_i)^+ = \max\{100 - p_i, 0\}$ and we define our capital gains advantage parameter $\eta \geq 0$. (B.14) adds an additional discounted payout term to the standard bond pricing formula (B.7) which scales in the discount of the bond $(100 - p_i)^+$ as well as the discount function corresponding to the bond's maturity, $q^{(m_i)}$.[82]

To obtain the wedge $z_i^{(j)}$, we solve (B.14) for p_i:

$$z_{i,\text{cg}}^{(j)} = \begin{cases} 1, & \sum_{k=1}^{m_i} cf_i^{(k)} q^{(k)} \geq 100 \\ \frac{1+\eta(100/cf_i^{(j)})\mathbb{1}\{j=m_i\}}{1+\eta q^{(m_i)}}, & \sum_{k=1}^{m_i} cf_i^{(k)} q^{(k)} < 100 \end{cases} \qquad \forall j$$

Since (B.14) implies that $p_i < 100$ if and only if $\sum_k cf_i^{(k)} q^{(k)} < 100$, the two cases are the same as conditioning on the bond trading at a discount.

Discussion: If agents believe that the relative capital gains tax advantage η_t follows a random walk independent of their own discount factor, so that $\eta = \mathbb{E}_0[\eta_t]$, and hold the bond to maturity, then η maps onto the income and capital gains tax rates:

$$\eta = \frac{\tau^{inc} - \tau^{cg}}{1 - \tau^{inc}} \tag{B.15}$$

Upon relaxation of the held-to-maturity assumption, (B.15) is only approximately true, as we show in the derivation of the wedge formula. Taking this as an approximation, η can therefore absorb distortions related to the average holding period of bonds within the sample. Therefore, we remain agnostic with regards to the holding period of bonds in the sample when we fit (B.14).

Derivation of Capital Gains Wedge: To derive the pricing formula (B.14) associated with the wedges $z_{i,\text{cg}}^{(j)}$, we first describe the setting in which bond i earns a capital gains advantage. Let $p_{i,t}$ be the price of bond i at time $t \geq 0$ with coupon c_i and maturity m_i. Without loss

[82] Treasury Bills carry no capital gains advantage (Section 2.3) and receive no capital gains wedge.

of generality, assume that there is a single intermediate holding period length H between t and $t+H$ after which the bond i is held to maturity. At period $t+H$, the investor pays capital gains tax on the price change $(p_{i,t+H}-p_{i,t})^+$ at rate τ^{cg}_{t+H}. Their after tax savings from paying tax at rate $\tau^{cg}_{t+H} \leq \tau^{inc}_{t+H}$ are $(\tau^{inc}_{t+H}-\tau^{cg}_{t+H})(p_{i,t+H}-p_{i,t})^+$. We can convert these savings into a one-time before-tax payout at $t+H$ with before-tax savings:

$$\eta_{t+H}(p_{i,t+H}-p_{i,t})^+, \quad \eta_{t+H}=\frac{\tau^{inc}_{t+H}-\tau^{cg}_{t+H}}{1-\tau^{inc}_{t+H}}$$

upon defining the term η_{t+H}.

With this setup, the price of i at t, $p_{i,t}$, satisfies the Euler equation, where $\xi_{t,t+H}$ is the SDF from t to $t+H$ and, as in the Discussion, η_t is a random walk independent of ξ, so that η replaces η_t:

$$p_{i,t}=\sum_{j>t}^{t+H} c_i q_t^{(j)} + \mathbb{E}_t[\xi_{t,t+H}p_{i,t+H}] + \eta\mathbb{E}_t[\xi_{t,t+H}(p_{i,t+H}-p_{i,t})^+] \tag{B.16}$$

$$p_{i,t+H}=\sum_{j>t+H}^{m_i} cf_i^{(j)} q_{t+H}^{(j)} + \eta(100-p_{i,t+H})^+ q_{t+H}^{(m_i)} \tag{B.17}$$

An immediate consequence of (B.16) is that when $H=m_i$ such that the bond is priced to be held to maturity and $t=0$, we get back (B.14) with the exact relationship:

$$\eta=\eta_0=\frac{\tau^{inc}-\tau^{cg}}{1-\tau^{inc}}$$

While this relationship does not hold exactly when considering a general holding period, we show in the following steps that it holds approximately. To make progress, we first define the *forward price* of bond i at $t+H$ by $\widehat{p}_{i,t}$:

$$\widehat{p}_{i,t+H}=\frac{\mathbb{E}_t[\xi_{t,t+H}p_{i,t+H}]}{\mathbb{E}_t[\xi_{t,t+H}]}$$

For purposes of exposition, we assume a monotonic price path of $p_{i,t} \leq \widehat{p}_{i,t+H} \leq 100$.[83] After

[83] If $\widehat{p}_{i,t+H} > 100$, we recover $q_t^{(t+H)}\widehat{p}_{i,t+H} = \sum_{j>t+H}^{m_i} cf_i^{(j)} q_t^{(j)}$ which is independent of capital gains advantage parameter η. Approximating $\widehat{p}_{i,t+H}$ around par value gives $q_t^{(t+H)}\widehat{p}_{i,t+H} \approx \frac{\sum_{j>t+H}^{m_i} cf_i^{(j)} q_t^{(j)} + (1+\eta)100 q_t^{(m_i)}}{1+\eta}$, which is the approximation made later in the monotonic price path case in (B.20), as needed.

taking time $t-$expectations of (B.17), we have:

$$q_t^{(t+H)}\widehat{p}_{i,t+H} = \frac{\sum_{j>t+H}^{m_i} cf_i^{(j)} q_t^{(j)} + (1+\eta)100 q_t^{(m_i)}}{1+\eta\frac{q_t^{m_i}}{q_t^{t+H}}} \tag{B.18}$$

We next make two approximations. The first approximation is to approximate $p_{i,t}$ with a smaller value, and the second approximation approximates $\widehat{p}_{i,t+H}$ with a larger value. Substituting the approximate value of $\widehat{p}_{i,t+H}$ into the approximate pricing equation we obtain for $p_{i,t}$, we obtain the approximate pricing equation:

$$p_{i,t} \simeq \sum_{j>t+H}^{m_i} cf_i^{(j)} q_t^{(j)} + \eta(100 - p_{i,t})^+ q_t^{(m_i)} + (\text{error})$$

where the sign on the error term is indeterminate since our approximations carry different signs. Specifically, we first approximate (B.16) using Jensen's inequality:

$$\begin{aligned} p_{i,t} &= \sum_{j>t}^{t+H} c_i q_t^{(j)} + q_t^{(t+H)}\widehat{p}_{i,t+H} + \eta\mathbb{E}_t[\xi_{t,t+H}(p_{i,t+H} - p_{i,t})^+] \\ &\approx \sum_{j>t}^{t+H} c_i q_t^{(j)} + q_t^{(t+H)}\widehat{p}_{i,t+H} + \eta q_t^{(t+H)}(\widehat{p}_{i,t+H} - p_{i,t})^+ \\ &= \sum_{j>t}^{t+H} c_i q_t^{(j)} + q_t^{(t+H)}\widehat{p}_{i,t+H} + \eta q_t^{(t+H)}(\widehat{p}_{i,t+H} - p_{i,t}) \end{aligned} \tag{B.19}$$

where the final line drops the $(\cdot)^+$ in the final term because we assumed that $\widehat{p}_{i,t+H} \geq p_{i,t}$. The second line applies Jensen's inequality on the final term, with $\mathbb{E}_t[\xi_{t,t+H}(p_{i,t+H} - p_{i,t})^+] \geq q_t^{(t+H)}(\widehat{p}_{i,t+H} - p_{i,t})^+$ since $(\cdot)^+$ is convex. This approximation essentially shuts down the desire of a time t investor to purchase bond i for potential capital gains advantage if bond i is "at-the-money" compared to its projected time $t+H$ price, $\widehat{p}_{i,t+H}$. Given that the capital gains advantage plays its largest role when prices are well below par with large projected future price increases, this approximation likely has a small effect on deep discount bonds purchased for their capital gains advantage.

The second approximation we make is to approximate the term in the denominator of (B.18), $q_t^{m_i}/q_t^{t+H}$ with 1. As $t+H \to m_i$ such that the final holding period from $t+H$ to m_i is small, we mechanically obtain $q_t^{m_i}/q_t^{t+H} \to 1$ exactly. The approximate value for $\widehat{p}_{i,t+H}$ is

obtained from:

$$q_{t+H}^{(m_i)}\widehat{p}_{i,t+H} = \frac{\sum_{j>t+H}^{m_i} cf_i^{(j)} q_t^{(j)} + (1+\eta)100 q_t^{(m_i)}}{1+\eta\frac{q_t^{m_i}}{q_t^{t+H}}} \approx \frac{\sum_{j>t+H}^{m_i} cf_i^{(j)} q_t^{(j)} + (1+\eta)100 q_t^{(m_i)}}{1+\eta} \tag{B.20}$$

Upon substitution of (B.20) into (B.19), we obtain the approximate formula for $p_{i,t}$:

$$p_{i,t} \simeq \sum_{j>t}^{m_i} cf_i^{(j)} q_t^{(j)} + \eta(100 - p_{i,t})^+$$

which looks like the familiar held-to-maturity formula. This formula holds approximately for η, but importantly, the sign of the approximation error is ambiguous because we made two approximations taking on opposite signs. This is an important observation because it means our capital gains pricing formula does not necessarily exhibit bias in one direction if one does not fully embrace a held-to-maturity assumption.

We may apply our approximation procedure iteratively for arbitrary holding periods $0 < H_1 < \cdots < t$ such that we obtain the time zero approximate pricing equation:

$$p_i \simeq \sum_{j=1}^{m_i} cf_i^{(j)} q^{(j)} + \eta(100 - p_i)^+ q^{(m_i)}$$

which is (B.14), as needed.

B.3.3 Callable Bonds

The most prevalent feature across bonds in both our Treasury and corporate samples is callability. We therefore need a wedge $z_{i,\text{call}}^{(j)}$ which reflects the fact that callable bonds are derivatives of non-callable bonds, spanned by the discount function q we seek to estimate, as well as additional parameters relating to the forecast of future interest rates, which determine the probability of early call.

Specification 3 (Callable Bond Wedge)**.** Let p_i be the time zero price of a callable bond i, with promised cash flows $cf_i^{(j)}$ and call price schedule $cp_i^{(j)}$ for $j = 1, \ldots, m_i$, where $cp_i^{(m_i)} = 100$ is the principal.[84] We assume that issuers (i.e. the US Treasury, Aaa corporations) make optimal calls.[85] That is, given that bond i is callable at j and has not yet been called, the

[84]The call price schedule for i is data, and is known at the bond's issuance.

[85]For discussion of optimal call policy, as well as other non-optimal call rules, see Bliss and Ronn (1998).

issuer calls the bond at $j < m_i$ if the call price does not exceed the price of the bond if not called (continuation price) $p_{i,j}$:

$$i \text{ is called at } j \text{ if } cp_i^{(j)} \leq p_{i,j}$$

Let $NC_i^{(j)} := \left\{cp_i^{(s)} > p_{i,s}, \forall s < j\right\}$ be the event that i is not called before j. The terminal condition is that a bond not called before maturity is redeemed at its principal, $\mathbb{P}^{Q^{m_i}}\left(i \text{ called at } m_i \mid NC_i^{(m_i)}\right) = 1$. With this environment, we obtain our callable bond price wedge from the present value identity:

$$p_i = \sum_{j=1}^{m_i} q^{(j)}\, \mathbb{P}^{Q^j}\left(NC_i^{(j)}\right)\left[c_i + \mathbb{P}^{Q^j}\left(i \text{ called at } j \mid NC_i^{(j)}\right) cp_i^{(j)}\right] \tag{B.21}$$

with call probabilities:

$$\begin{aligned} \mathbb{P}^{Q^j}\left(NC_i^{(j)}\right) &= \mathbb{P}^{Q^j}\left(cp_i^{(1)} > p_{i,1}, \ldots, cp_i^{(j-1)} > p_{i,j-1}\right) \\ \mathbb{P}^{Q^j}\left(i \text{ called at } j \mid NC_i^{(j)}\right) &= \mathbb{P}^{Q^j}\left(cp_i^{(j)} \leq p_{i,j} \,\middle|\, cp_i^{(1)} > p_{i,1}, \ldots, cp_i^{(j-1)} > p_{i,j-1}\right), \quad j < m_i \end{aligned} \tag{B.22}$$

and $\mathbb{P}^{Q^{m_i}}(i \text{ called at } m_i \mid NC_i^{(m_i)}) = 1$. We write $\mathbb{P}^{Q^j}(i \text{ called at } j \text{ and } NC_i^{(j)})$ for the product of the two. Each probability is taken under the $j-$forward measure Q^j defined by the Radon-Nikodym derivative:

$$\frac{dQ^j}{dP} = \frac{\xi_j}{\mathbb{E}[\xi_j]}$$

where ξ_j is the SDF from 0 to j and P is the physical measure. This is the relevant risk-neutral measure for maturity j since it treats the $j-$maturity risk free bond as the numeraire.[86] We compute the probabilities (B.22) using a Hull-White short rate process (Hull and White, 1990) with mean-reversion κ and volatility σ, calibrated so that it reproduces the discount function q.

Rearranging (B.21) allows us to express the price p_i at time 0 in the form of:[87]

$$p_i = \sum_{j=1}^{m_i} cf_i^{(j)} z_{i,\text{call}}^{(j)} q^{(j)}$$

[86] This property allows us to remain agnostic about the degree to which term premium or the like are reflected in the probability that the bond is called at date j.

[87] The price of a callable bond is made up of the price on a non-callable bond with the promised cash flows of bond i through maturity less a call option. Accordingly, we can obtain the call option value of any option bond i by taking the complement of (B.23): call option$_i = \sum_{j=1}^{m_i} cf_i^{(j)}(1 - z_{i,\text{call}}^{(j)})q^{(j)}$.

where $z^{(j)}_{i,\text{call}}$ is defined by:

$$z^{(j)}_{i,\text{call}} = \mathbb{P}^{Q^j}\left(NC_i^{(j)}\right)\frac{c_i}{cf_i^{(j)}} + \mathbb{P}^{Q^j}\left(i \text{ called at } j \text{ and } NC_i^{(j)}\right)\frac{cp_i^{(j)}}{cf_i^{(j)}}, \qquad j = 1, \ldots, m_i \tag{B.23}$$

This is the callable bond wedge of Section 3.4.1, for all j. At $j = m_i$ the terminal condition and $cp_i^{(m_i)} = 100$ give $z^{(m_i)}_{i,\text{call}} = \mathbb{P}^{Q^{m_i}}(NC_i^{(m_i)})$, the probability that i is not called early. We derive the present value formula (B.21) in the derivation below.

Discussion: The wedge $z^{(j)}_{i,\text{call}}$ is a duration adjustment to bond i through the probability of early call. A deep in-the-money bond that is called with probability 1 on its first cash flow date has $z^{(1)}_{i,\text{call}} = 1 + cp_i^{(1)}/c_i$ and $z^{(j)}_{i,\text{call}} = 0, j \geq 2$, so that it is effectively a one-period bond. A bond that is deep out of the money on every cash flow date is never called, $z^{(j)}_{i,\text{call}} = 1, \forall j$, and is priced like a non-callable bond.

We compute the call probabilities with a Hull-White short rate model whose mean function is chosen to reproduce q (specified below). It delivers call probabilities in terms of well-studied multivariate normal CDFs. Since we fit κ and σ period-by-period, we obtain a time series of implied interest rate volatility, akin to Black-Scholes. An innovation is that we obtain this implied volatility while jointly fitting the non-callable term structure from the cross section of callable and non-callable bonds.

Completing the parameterization of $z^{(j)}_{i,\text{call}}$ boils down to parameterizing the call probabilities in (B.22), and we obtain them from an interest rate process. A reduced-form parameterization of the call probabilities, as a function of observable bond characteristics, would (a) restrict the flexibility of the call probability functional form, whereas the interest rate process lets the probabilities respond jointly to each bond's call schedule, coupon and maturity and to the shape of q, and (b) not impose consistency within the sample. On (b), if we priced non-callable bonds with an interest rate process consistent with reduced-form call probabilities, we would not necessarily recover q. That would violate our assumption that a common discount function prices all bonds in the sample. Non-callable bond pricing under our Hull-White process is governed by the function q by construction, thereby satisfying Assumption 3. Tying the call probabilities to q also allows counterfactuals and lets us fit periods with only callable bonds.

Finally, evaluating (B.21) given *any* short rate process requires $\mathcal{O}(2^{m_i})$ integrals to recursively compute continuation prices $p_{i,1}, \ldots, p_{i,m_i-1}$ under m_i probability measures, which is computationally infeasible for bonds with many call dates.[88] We therefore replace each con-

[88] For example, evaluation of (B.21) for a semi-annually compounded callable corporate bond maturing in 30 years with no call protection requires approximately 10^{18} nested integrals.

tinuation price $p_{i,j}$ with $p^*_{i,j}$, the time$-j$ price of the remaining promised cash flows of bond i with the additional option to call the bond at time $j+1$. While approximating continuation prices is not new (e.g. Longstaff and Schwartz (2001)), ours keeps the call probabilities informed by the yield curve information in the jointly estimated discount function q. We derive the resulting call probabilities in the sections below.

Derivation of the General Callable Bond Wedge: We first derive the general present value form from a standard Euler equation. Throughout the derivation, $NC_i^{(j)} := \left\{cp_i^{(s)} > p_{i,s}, \forall s < j\right\}$ denotes the event that bond i is not called before date j, i.e. that at every earlier call date the call price exceeded the continuation price. Its indicator is the product $D(NC_i^{(j)}) = \prod_{s=1}^{j-1} D(cp_i^{(s)} > p_{i,s})$. The general Euler equation for callable bonds, where $\xi_{t,t+1}$ is the SDF from t to $t+1$, is given for $t+1 < m_i$ by:

$$\begin{aligned} p_{i,t} &= \mathbb{E}_t[\xi_{t,t+1}(c_i + \min\{cp_i^{(t+1)}, p_{i,t+1}\})] \qquad \text{(B.24)} \\ &= \underbrace{q_t^{(t+1)} c_i}_{(i)} + \underbrace{\mathbb{E}_t\left[\xi_{t,t+1} D\left(cp_i^{(t+1)} \le p_{i,t+1}\right)\right] cp_i^{(t+1)}}_{(ii)} + \underbrace{\mathbb{E}_t\left[\xi_{t,t+1} D\left(cp_i^{(t+1)} > p_{i,t+1}\right) p_{i,t+1}\right]}_{(iii)} \end{aligned}$$

where $D(\cdot)$ is the indicator function. The three terms in (B.24) are (i) the discounted one period coupon next period which is guaranteed to pay out at $t+1$ given i has not been called by t (ii) the discounted event that the bond is called at $t+1$ times its call price and (iii) the discounted event that the bond is not called at $t+1$ times its continuation price. Equation (B.24) is subject to the terminal condition that a bond not called before m_i is redeemed at its principal $cp_i^{(m_i)} = 100$ with probability one, $\mathbb{P}(i \text{ called at } m_i \mid NC_i^{(m_i)}) = 1$:

$$p_{i,m_i-1} = q_{m_i-1}^{(m_i)}(c_i + 100) \qquad \text{(B.25)}$$

which states that the continuation value on the bond's final call date is equal to a zero-coupon bond paying coupon plus principal on the bond's maturity date m_i. Iterating (B.24) from 0 to m_i gives the present discounted value formula under the physical measure $\mathbb{P}$:[89] [90]

$$p_i = \sum_{j=1}^{m_i} \left(\mathbb{E}\left[\xi_j D(NC_i^{(j)})\right] c_i + \mathbb{E}\left[\xi_j D(NC_i^{(j)}, cp_i^{(j)} \le p_{i,j})\right] cp_i^{(j)} \right) \qquad \text{(B.26)}$$

[89] Without loss of generality, equation (B.26) assumes that bond i is callable on all remaining cash flow dates. (B.26) is easily adjusted if i contains call protection through date $J_i < m_i$.

[90] Callable bonds often have notice periods which force an issuer to give an explicit notice that they will call the bond a few months in advance of a coupon date. For callable Treasury bonds, this was usually four months notice (TNOTICE = 4 in CRSP). Therefore, even without call protection, if bond i's first coupon date occurs within the notice period, bond i is not callable on this first coupon date. We adjust (B.26) accordingly to account for this, when needed.

where the call indicator is one at $j = m_i$ by the terminal condition. Upon changing into the $j-$forward measure on each cash flow date, using $\mathbb{E}[\xi_j] = q^{(j)}$, expectations of indicators become probabilities:

$$p_i = \sum_{j=1}^{m_i} \left(\mathbb{P}^{Q^j}\left(NC_i^{(j)}\right) c_i + \mathbb{P}^{Q^j}\left(NC_i^{(j)},\, cp_i^{(j)} \leq p_{i,j}\right) cp_i^{(j)} \right) q^{(j)}$$

Writing the joint probability in the second term as $\mathbb{P}^{Q^j}(NC_i^{(j)})\,\mathbb{P}^{Q^j}(cp_i^{(j)} \leq p_{i,j} \mid NC_i^{(j)})$ and collecting terms gives the general callable bond present value identity (B.21). It is an intuitive asset pricing formula which tells us that callable bonds are probability weighted discounted payouts: the coupon with the forward-measure probability that the bond is still active at j, and the call price with the probability that it is then called. We emphasize that the derivation of this present value formula is done under minimal assumptions or restrictions.

Approximating the Call Option Continuation Value: Here, we provide additional details regarding our continuation value approximation. This approximation is entirely separate from the choice of short rate model needed to complete the parameterization. To make progress on a tractable callable bond pricing formula, we make note of what the true event to call the bond $cp_i^{(j)} \leq p_{i,j}$ captures: there is some refinancing strategy available to the issuer at j that dominates waiting any longer than j to call the bond. When there are many call dates, there are many refinancing strategies to consider. While the continuation price $p_{i,j}$ encodes information on when the bond should be called and with which new securities the bond should be refinanced with, the key determinant of $p_{i,j}$ is whether it is worth waiting until a future date to call the bond, not the construction of the refinancing strategy itself. Put differently, what matters for the timing of the call date is whether waiting until that particular future call date is contained within the dominant strategy to refinance the bond. To this end, we approximate the decision to call the bond at j with the event that the time-j value of the cash flows over all candidate stopping times considered by the issuer exceeds the call price $cp_i^{(j)}$. Formally, let $\mathcal{T}_i^{(j)}$ be the set of stopping times considered by the issuer of bond i at time j. We let $p_{i,j}^*$ be the time j price of a bond with the same promised cash flows and call prices as the original bond i from $j+1$ to $\max\{\mathcal{T}_i^{(j)}\}$ with the right to call at all intermediate times $t \in \mathcal{T}_i^{(j)} \setminus \max\{\mathcal{T}_i^{(j)}\}$. The price of this security is given by the fixed

point:

$$
\begin{aligned}
p^*_{i,j} = & \sum_{k>j}^{\max\{\mathcal{T}_i^{(j)}\}} \mathbb{P}_j^{Q^k}(NC_{i,j}^{(k)}) c_i q_j^{(k)} \\
& + \sum_{k\in\{\mathcal{T}_i^{(j)}\setminus\{\max\{\mathcal{T}_i^{(j)}\}\}\}} \mathbb{P}_j^{Q^k}(i \text{ called at } k \text{ and } NC_{i,j}^{(k)}) cp_i^{(k)} q_j^{(k)} \\
& + \mathbb{P}_j^{Q^t}(NC_{i,j}^{(t)}) cp_i^{(t)} q_j^{(t)} \Big|_{t=\max\{\mathcal{T}_i^{(j)}\}}
\end{aligned}
\tag{B.27}
$$

where $NC_{i,j}^{(k)} := \left\{cp_i^{(s)} > p^*_{i,s},\, j < s < k\right\}$ is the forward-looking version of $NC_i^{(k)}$, the event that i is not called between j and k (with $p^*_{i,s}$ replacing the continuation price), and the event to call at time k is given by:

$$cp_i^{(k)} \leq p^*_{i,k}$$

Equation (B.27) approximates the true continuation price $p_{i,j}$ by filtering potential future call dates to those in set $\mathcal{T}_i^{(j)}$; its final term is the redemption at $t = \max\{\mathcal{T}_i^{(j)}\}$, which is the principal $cp_i^{(m_i)} = 100$ when $t = m_i$. Equation (B.27) nests two important cases. When j is the only call date, (B.27) trivially returns the terminal condition (B.25) and the approximate formula (B.27) exactly matches the original formula (B.21). On the other hand, when all future call/coupon dates after j are included in $\mathcal{T}_i^{(j)}$, then (B.27) returns the exact continuation value $p_{i,j}$.

Having constructed $p^*_{i,j}$ for each j, we approximate the event when bond i is called at j in the original present value formula by:

$$D(cp_i^{(j)} \leq p_{i,j}) \approx D\left(cp_i^{(j)} \leq p^*_{i,j}\right)$$

Let $NC_i^{*(j)} := \left\{cp_i^{(s)} > p^*_{i,s}, \forall s < j\right\}$ be the approximate event that i is not called before j (the case $NC_{i,0}^{(j)}$ of the previous definition). Upon substitution into (B.21) we obtain an approximate present value formula for the callable bond price p_i:

$$p_i \simeq \sum_{j=1}^{m_i} q^{(j)}\, \mathbb{P}^{Q^j}\left(NC_i^{*(j)}\right)\left[c_i + \mathbb{P}^{Q^j}\left(cp_i^{(j)} \leq p^*_{i,j} \mid NC_i^{*(j)}\right) cp_i^{(j)}\right] \tag{B.28}$$

with the conditional call probability equal to one at $j = m_i$. Equation (B.28) greatly reduces the complexity of estimating callable bond prices by reducing the problem of estimating nested continuation prices many times over to the problem of how long the issuer should wait to call if they do not call the bond the bond at j. Under assumptions about the distribution of future realized bond prices, the approximate probabilities in (B.28) can be

tractably computed given a choice set for stopping times $\mathcal{T}_i^{(j)}$. We form $\mathcal{T}_i^{(j)}$ at date j by:

$$\mathcal{T}_i^{(j)} = \{j+1, m_i\}$$

such that issuers essentially compare calling the bond at j with call price $cp_i^{(j)}$ to the price of the remaining cash flows with the additional option to call the bond at $j+1$ with call price $cp_i^{(j+1)}$.[91] Most importantly, allowing only two elements in $\mathcal{T}_i^{(j)}$ dramatically reduces the number of required integrals needed to compute p_i to complexity $\mathcal{O}(m_i^3)$, allowing for computational feasibility.

Parameterizing the Call Probabilities: To compute the call probabilities in (B.28), we assume that at time 0 the short rate follows, under the risk-neutral measure Q, the Hull-White process

$$dr_s = \kappa(\mu^{(s)} - r_s)ds + \sigma dW_s^Q \tag{B.29}$$

over future times $s \geq 0$ for mean-reversion κ, interest rate volatility σ, and W_s^Q is standard Brownian motion under Q. $\mu^{(s)}$ is a deterministic function of time governing the path of long-run short rates under Q, which we back out of the bond pricing equation $q^{(j)} = \mathbb{E}^Q\left[\exp\left(-\int_0^j r_s ds\right)\right]$ so that the process reproduces q for all $j \geq 0$. The resulting distribution of future bond prices under Q is then moved to the Q^j forward measure to obtain the probabilities in (B.28).

Bond pricing using (B.29) is obtained exploiting log-normality, where:

$$\begin{aligned} x^{(t,T)} &= \mathbb{E}_t^Q[\nu^{(t,T)}] + \frac{1}{2}Var_t^Q(\nu^{(t,T)}) \\ \nu^{(t,T)} &= -\int_t^T r_s ds \end{aligned} \tag{B.30}$$

where $x^{(t,T)} = \log(q_t^{(T)})$ is the log bond price at t maturing at T.[92] Since the integrated path of short rates $v_{t,T}$ is normal, the future log bond price is normal. By manipulation of (B.30) and using the law of iterated expectations back to time 0, we obtain an expression for the mean function of future log bond prices under Q:

$$\mathbb{E}_0^Q[x^{(t,T)}] = \underbrace{x^{(0,T)} - x^{(0,t)}}_{\text{(-) forward rate } \log\left(q^{(T)}/q^{(t)}\right)} \underbrace{-Cov_0(\nu^{(0,t)}, x^{(t,T)}) - \frac{1}{2}Var_0(x^{(t,T)})}_{\text{covariance adjustment}}$$

[91] For corporate bonds with annually decaying call schedules, we let $\mathcal{T}_i^{(j)} = \{j+2, m_i\}$ if $cp_i^{(j)} = cp_i^{(j+1)}$, such that the issuer is always considering calling the bond on the next date on which the call price falls.

[92] We can set $t = 0$ and differentiate (B.30) with respect to T to back out the function μ from κ, σ and q.

which is a linear combination of the forward rate from t to T obtained from the time zero discount curve and a covariance adjustment term which is an expression of κ and σ. We also obtain a covariance function for future log-bond prices by evaluating:

$$\begin{aligned} Cov_0^Q(x^{(t_1,T_1)}, x^{(t_2,T_2)}) &= Cov_0^Q\left(\mathbb{E}_{t_1}^Q[\nu^{(t_1,T_1)}], \mathbb{E}_{t_2}^Q[\nu^{(t_2,T_2)}]\right) \\ &= B(T_1 - t_1)B(T_2 - t_2)Cov_0^Q(r_{t_1}, r_{t_2}) \end{aligned}$$

for $B(t) = \frac{(1-\exp(-\kappa t))}{\kappa}$ which links bond price volatility to interest rate volatility.

In order to obtain call probabilities under the relevant forward measures, we need to change measures to obtain the distribution of future prices $x^{(t,T)}$ under Q^j. For any starting time t, the Radon-Nikodym derivative for the change of measure is given by:

$$\frac{dQ^j}{dQ} = \frac{\exp\left(-\int_t^T r_s ds\right)}{\mathbb{E}_t^Q\left[\exp\left(-\int_t^T r_s ds\right)\right]}$$

which allows us to obtain the following mean and covariance functions for $x^{(t,T)}$ under Q^j in terms of $Q-$moments:

$$\begin{aligned} \mathbb{E}_0^{Q^j}[x^{(t,T)}] &= \mathbb{E}_0^Q[x^{(t,T)}] \quad \underbrace{+Cov_0^Q(\nu^{(0,j)}, x^{(t,T)})}_{\text{mean shift induced by measure change}} \\ Cov_0^{Q^j}(x^{(t_1,T_1)}, x^{(t_2,T_2)}) &= Cov_0^Q(x^{(t_1,T_1)}, x^{(t_2,T_2)}) \end{aligned}$$

where we require $j \geq t$ (prices at t are j measurable). The explicit distribution of future log bond prices $x_{t,T} = \log(q_t^{(T)})$ under Q^j is:

$$x_{t,T} \overset{Q^j}{\sim} \mathcal{N}\left(\log\left(\frac{q^{(T)}}{q^{(t)}}\right) + \left(B(j-t)B(T-t) - \frac{1}{2}B(T-t)^2\right)v_0^{(t)}, B(T-t)^2 v_0^{(t)}\right) \tag{B.31}$$

where we define:

$$v_{t_0}^{(t)} = Var_{t_0}(r_t) = \frac{\sigma^2}{2\kappa}(1 - \exp(-2\kappa(t - t_0)))$$

We now have a distribution for zero-coupon bond prices under the forward measure Q^j which we use to compute the call probabilities in (B.28). To compute the probabilities, we first express $p_{i,j}^*$ in terms of standard normal $Y_i^{(j)} \sim \mathcal{N}(0,1)$ by standardizing (B.31). Under

measure Q^T with $T \geq j$, we can plug $Y^{(j)}$ into (B.27) and apply $\mathcal{T}_i^{(j)} = \{j+1, m_i\}$ to obtain:

$$p_{i,j}^* \overset{Q^T}{=} c_i \frac{q^{(j+1)}}{q^{(j)}} G(Y_i^{(j)}, j+1; T, j, 0) + cp_i^{(j+1)} \frac{q^{(j+1)}}{q^{(j)}} G(Y_i^{(j)}, j+1; T, j, 0) \Phi(-W_i^{(j)}) + \sum_{k=2}^{m_i} cf_i^{(k)} \frac{q^{(j+k)}}{q^{(j)}} G(Y_i^{(j)}, j+k; T, j, 0) \Phi\left(W_i^{(j)} - B((j+k)-(j+1))\sqrt{v_j^{(j+1)}}\right) \quad \text{(B.32)}$$

where we define $G(Y, s; T, t)$ by:

$$G(Y, s; T, t, t_0) = \exp\left(\left(B(T-t)B(s-t) - \frac{1}{2}B(s-t)^2\right)v_{t_0}^{(t)} + B(s-t)\sqrt{v_{t_0}^{(t)}}Y\right) \quad \text{(B.33)}$$

and $W_i^{(j)} = W_i^{(j)}\left(Y_i^{(j)}\right)$ is an implicit function of $Y_i^{(j)}$ from the equation:

$$cp_i^{(j+1)} = \sum_{k=j+2}^{m_i} cf_i^{(k)} \frac{q^{(k)}}{q^{(j+1)}} \frac{G(Y_i^{(j)}, j+k; j, j, 0)}{G(Y_i^{(j)}, j+1; j, j, 0)} G(W_i^{(j)}, j+k; j+1, j+1, j)$$

Since we are interested in the approximate call event, $cp_i^{(j)} \leq p_{i,j}^*$, we must identify the $Y_i^{(j)}$ for which $cp_i^{(j)} = p_{i,j}^*$ in (B.32). Doing so, without loss of generality, we obtain the approximate probability that bond i is called at j and not before by computing cutoffs $y_i^{(1)}, \ldots, y_i^{(j)}$:

$$\begin{aligned}\mathbb{P}^{Q^j}(i \text{ called at } j \text{ and } NC_i^{(j)}) &\simeq \mathbb{P}^{Q^j}(cp_i^{(1)} > p_{i,1}^*, \ldots, cp_i^{(j-1)} > p_{i,j-1}^*, cp_i^{(j)} \leq p_{i,j}^*) \quad \text{(B.34)} \\ &= \mathbf{\Phi}_j\left(y_i^{(1)} - B(j-1)\sqrt{v_0^{(1)}}, \ldots, y_i^{(j-1)} - B(j-(j-1))\sqrt{v_0^{(j-1)}}, -y_i^{(j)}; R\right)\end{aligned}$$

where $\Phi_j(\cdot; \Sigma)$ is the multivariate normal CDF of dimension j with zero mean and covariance matrix Σ. The probability $\mathbb{P}^{Q^j}(NC_i^{*(j)})$ is the same expression of dimension $j-1$, dropping the last argument (and $\mathbb{P}^{Q^1}(NC_i^{(1)}) = 1$), and the conditional call probability in (B.22) is their ratio. The matrix R in (B.34) is the covariance matrix of the cutoffs $y_i^{(1)}, \ldots, y_i^{(j)}$, which is equal to the correlation matrix of future interest rates/log bond prices with:

$$R_{jk} = Corr_0(r_j, r_k) = \exp(-\kappa(k-j))\sqrt{\frac{1-\exp(-2\kappa j)}{1-\exp(-2\kappa k)}}$$

assuming $k \geq j$. The cutoff $y_i^{(j)}$ is obtained from the solution pair $(y_i^{(j)}, w_i^{(j)})$ of the non-linear

system:

$$
\begin{aligned}
cp_i^{(j)} &= c_i \frac{q^{(j+1)}}{q^{(j)}} G(y_i^{(j)}, j+1; j, j, 0) + cp_i^{(j+1)} \frac{q^{(j+1)}}{q^{(j)}} G(y_i^{(j)}, j+1; j, j, 0) \Phi(-w_i^{(j)}) \\
&\quad + \sum_{k=2}^{m_i} cf_i^{(k)} \frac{q^{(j+k)}}{q^{(j)}} G(y_i^{(j)}, j+k; j, j, 0) \Phi\left(w_i^{(j)} - B((j+k)-(j+1)) \sqrt{v_j^{(j+1)}} \right) \\
cp_i^{(j+1)} &= \sum_{k=2}^{m_i} cf_i^{(k)} \frac{q^{(j+k)}}{q^{(j+1)}} \frac{G(y_i^{(j)}, j+k; j, j, 0)}{G(y_i^{(j)}, j+1; j, j, 0)} G(w_i^{(j)}, j+k; j+1, j+1, j)
\end{aligned}
$$

which allows us to form the call option $z_{i,\text{call}}^{(j)}$ in terms of q, κ and σ, as needed.

B.3.4 Flower Bonds

Flower bonds were low-coupon, long-term bonds issued by the Treasury between 1918 and 1971 which allowed the bondholder to use the par value of the bond, not the market value, to pay their estate taxes upon death.[93] Additionally, before the Tax Reform Act of 1976, the redemption value of the bond (the difference between par and market value) was exempt from capital gains taxes, so flower bonds enjoyed additional capital gains advantage beyond straight bonds purchased at a discount. Finally, because only holders with large nominal estate tax liabilities benefit from the option, aggregate demand for it is satiable. With a no-shorting constraint, this demand is allocated only to the flower bonds trading furthest below par. As documented by Figure A.22, in the 1960s only the deepest-discount longest-maturity flower bonds were actually purchased for their flower option value. In the 1970s demand spread across many more bonds, as inflation eroded the fixed nominal estate tax exemption and pushed more low-coupon bonds to deep discounts.

These features motivate the need to model four components of the flower bond wedge: (i) the reference death probability $e^{(j)} = \mathbb{P}(\text{death at } j)$, assumed to be independent of the pricing kernel, which each bond's clientele tilts into its own eligibility probability $e_i^{(j)}$, (ii) the probability that the bond is redeemed on death date j, $\mathbb{P}^{Q^j}(p_{i,j} \leq 100)$, governed by the same interest rate volatility parameters κ, σ as callable bonds, (iii) the additional capital gains advantage of flower bonds over straight bonds defined by $\bar{\eta}$ with $\bar{\eta} \geq \eta$ and (iv) whether the option value was actively priced, given by the tilting function $A_i : [0, m_i] \to [0, 1]$ with $A_i^{(j)} = 0$ corresponding to no option value being priced at death date j and $A_i^{(j)} = 1$

[93] The pricing of flower bonds seemed to first exhibit price distortions following a wave of new flower bond issuance in the mid-to-late 1950s. Accordingly, we only fit the flower bond adjustment beginning in 1955 when the Treasury started issuing flower bonds again (bonds with IFLWR=2 in CRSP). We do not fit callable flower bonds, since the full term structure is always spanned by either non-callable flower bonds or callable non-flower bonds.

corresponding to full option value being priced at death date j. Features (i) and (ii) liken the flower bond to a put option where the exercise date is determined by future death probabilities, (iii) requires an adjustment of our capital gains formula upon early redemption to handle the additional capital gains advantage, and (iv) determines the relative premium investors were willing to pay for the most attractive flower bonds. We see this feature in the price/yield space throughout the 1960s and 70s as investors bid up the price of the most attractive bond, the 3% bond of 1995, relative to other flower bonds of similar maturity, as shown in Figure 5.

Specification 4 (Flower Bond Wedge)**.** We assume that the eligibility events are independent of the discount factor, such that eligibility probabilities can be separated from the discount function. Let $e_i^{(j)} = e^{(j)} A_i^{(j)}$ denote the eligibility probability of bond i at horizon j, i.e. $e_i^{(j)} = \mathbb{P}(i \text{ first eligible at } j)$, where $e^{(j)} = \mathbb{P}(\text{death at } j)$ is a reference death probability and $A_i : [0, m_i] \to [0, 1]$ is a tilting function to capture bond specific clientele effects. A bond is redeemed at $j < m_i$ if its holder is eligible at j and $p_{i,j} \leq 100$. We specify that the price p_i of flower bond i is given by:

$$
\begin{aligned}
p_i &= \sum_{j=1}^{m_i} \left(e^{(j)} \left(A_i^{(j)} \bar{p}_i^{(j)} + (1 - A_i^{(j)}) p_i^{st} \right) \right) + \left(1 - \sum_{j=1}^{m_i} e^{(j)} \right) p_i^{st} \\
\bar{p}_i^{(j)} &= \sum_{k=1}^{j} c_i q^{(k)} + \left(100 + \bar{\eta} (100 - p_i)^+ \right) \mathbb{P}^{Q^j}(p_{i,j} \leq 100) q^{(j)} \\
&\quad + \sum_{k>j}^{m_i} c f_i^{(k)} \mathbb{P}^{Q^k}(p_{i,j} > 100) q^{(k)} + \eta (100 - p_i)^+ \mathbb{P}^{Q^{m_i}}(p_{i,j} > 100) q^{(m_i)} \\
p_i^{st} &= \sum_{k=1}^{m_i} c_i q^{(k)} + (100 + \eta (100 - p_i)^+) q^{(m_i)}
\end{aligned}
\tag{B.35}
$$

where $\bar{p}_i^{(j)}$ is the price of a claim on i's cash flows conditional on death at j and p_i^{st} is the hypothetical value of the bond if it contained no option value. The construction of $\bar{p}_i^{(j)}$ follows from the fact that given death date j, the bond holder has a put option expiring at j to redeem the bond for its par value, thus the put option is in-the-money at j if the price of the remaining cash flows is below par value. We provide details on the probability of redemption in the subsequent section. The three equations in (B.35) can be rearranged to isolate p_i. Upon isolating p_i, we obtain $p_i = \sum_{j=1}^{m_i} c f_i^{(j)} z_{i,\text{flower}}^{(j)} q^{(j)}$ with $z_{i,\text{flower}}^{(j)}$ given in (B.42) below. Upon setting $\bar{\eta} = 0, \eta = 0$ and rearrangement of (B.35), we obtain the same $z_{i,\text{flower}}^{(j)}$ as in (3.6), as needed.

We provide a derivation of (B.35) from a general Euler equation below.

Discussion: We borrow from the mortality literature and parameterize the probability of

death in (B.35) with parameters $\bar{\alpha}, \bar{\beta}$ according to a Gompertz hazard rate function $\bar{f}$:

$$\bar{f}^{(j)} = \bar{\alpha} \exp(\bar{\beta} j)$$

This specification is popular in the mortality literature because it captures the empirical observation that mortality rates grow exponentially as people age.[94] The reference death probabilities are the probability of being alive at $j-1$ less the probability of being alive at j under this hazard:

$$e^{(j)} = \exp\left(-\frac{\bar{\alpha}}{\bar{\beta}}\left(\exp(\bar{\beta}(j-1)) - 1\right)\right) - \exp\left(-\frac{\bar{\alpha}}{\bar{\beta}}\left(\exp(\bar{\beta} j) - 1\right)\right)$$

so that $\bar{f}$ is a hazard rate while $e^{(j)}$ is a probability mass function.

We specify $A_i^{(j)}$ in (B.35) as a logistic function with elasticity ψ which activates option value around date j when the price of the claim $\bar{p}_i^{(j)}$ per unit investment in bond i exceeds that of the reference bond ref, the deepest-discount flower bond, which is the 3% bond of 1995 over most of the sample shown in Figure 5:[95]

$$A_i^{(j)} = \frac{1}{1 + \exp(-\psi(j - J_i))}, \qquad J_i = \inf\left\{ j : \frac{\bar{p}_i^{(j)}}{p_i} \geq \frac{\bar{p}_{ref}^{(j)}}{p_{ref}} \right\}$$

We set ref to be the 3% bond of 1995 from 1955 onward, which encompasses the entire period when embedded option value in flower bonds were priced. We choose this specification of $A_i^{(j)}$ to capture the empirical observation that the reference bond consistently demands an additional yield premium relative to other flower bonds, as shown in Figure 5. Intuitively, this is because the 3% bond of 1995 contained the lowest coupon of the longest maturity flower bonds, which caused this bond to trade at the deepest discount of all flower bonds over the 1960s and 70s. Since the flower option value was increasing in the discount of the bond, $(100 - p_i)$, and aggregate demand for the option was satiable, investors seeking exposure to this option provision, who expect to become eligible soonest and so have the highest willingness to pay, purchased the reference bond first since it exhibited the largest discount of all flower bonds.[96] Option value in other flower bonds $i \neq ref$ for a given death date j therefore came down to whether it was relatively more attractive to hold i than the reference

[94] This function may be written as $\lambda \exp(\beta(t + \text{age}_0))$, for structural parameters λ, β. However, for any fixed date we may absorb the age component in a constant $\bar{\alpha} = \lambda \exp(\beta \text{age}_0)$.

[95] For the reference bond, this specification leads to $A_{ref}^{(j)} = 1$ for all $j \geq 0$, which implies investors are pricing option value of this bond for all potential death dates during the bond's lifetime.

[96] In Figure A.22, we see that investors were indeed redeeming the 3% bond of 1995 in the early and mid-1960s well before other flower bonds began to be redeemed.

bond. Bond i became more attractive when the price of a claim on its cash flows given death at j per unit investment, $\bar{p}_i^{(j)}/p_i$, exceeded that of the reference bond, $\bar{p}_{ref}^{(j)}/p_{ref}$, and J_i is the first horizon at which this holds. Flower bonds whose discounts are small relative to the reference bond's have a large J_i, so only the reference bond is priced for its flower option, as in the 1960s. As other discounts close in on the reference bond's, J_i falls and multiple bonds are priced for the option, as in the 1970s. Our specification of $A_i^{(j)}$ softens the rigid condition that option value is priced in i at all death dates $j \geq J_i$, i.e. $A_i^{(j)} \approx D\Big(j \geq J_i\Big)$. We do not estimate ψ; we take it high, which approximates this step function. The specification also softens the even more rigid portfolio choice condition with frictionless markets and flexible supply where for each death date j, there is a *single* flower bond for which option value is priced given death at j, and the bond which contains this option value is the bond $\bar{i}^{(j)}$ defined by:

$$\bar{i}^{(j)} = \arg\max_i \left(\frac{\bar{p}_i^{(j)}}{p_i} \right)$$

Derivation of Flower Bond Wedge: Consider a flower bond i at time t with price $p_{i,t}$, coupon c_i, and maturity m_i. The event that the bond is redeemed at $t+1$ given it is not redeemed at t is given by the intersection $100 \geq p_{i,t+1} \ \cap \ i$ eligible at $t+1$, a moneyness condition and an eligibility condition. The eligibility condition absorbs all of the other factors and necessary conditions for redemption (i.e. agent is estate, holder dies at $t+1$, supply effects, etc.). We assume that this event is independent of the discount factor. This setup gives a general Euler equation for this flower bond that is analogous to the callable bond Euler equation:

$$\begin{aligned} p_{i,t} = \underbrace{q_t^{(t+1)} c_i}_{(i)} + \underbrace{\mathbb{E}_t\Big[\xi_{t,t+1} D\Big(100 \geq p_{i,t+1} \ \cap \ i \text{ eligible at } t+1\Big) 100\Big]}_{(ii)} \\ + \underbrace{\mathbb{E}_t\Big[\xi_{t,t+1} D\Big(100 < p_{i,t+1} \ \cup \ i \text{ not eligible at } t+1\Big) p_{i,t+1}\Big]}_{(iii)} \end{aligned} \tag{B.36}$$

where $D(\cdot)$ is the indicator function. The three terms in (B.36) are (i) the discounted one period coupon next period which is guaranteed to pay out whether the bond is redeemed or not at $t+1$, (ii) the discounted event that the bond is redeemed next period times redemption payout (par value = 100) and (iii) the discounted event that the bond is not redeemed at $t+1$ times its continuation price. The terminal condition is again:

$$p_{i,m_i-1} = q_{m_i-1}^{(m_i)}(c_i + 100)$$

Iterating (B.36) back to time 0 gives the present discounted value formula under the physical measure:

$$p_i = \sum_{j=1}^{m_i} \underbrace{\mathbb{E}\left[\xi_j \left(\prod_{s=1}^{j-1} D\Big(100 < p_{i,s} \ \cup \ i \text{ not eligible at } s\Big)\right)\right] cf_i^{(j)}}_{PV \text{ of cash flow, bond not redeemed before } j} \tag{B.37}$$

$$+ \sum_{j=1}^{m_i-1} \underbrace{\mathbb{E}\left[\xi_j \left(\prod_{s=1}^{j-1} D\Big(100 < p_{i,s} \ \cup \ i \text{ not eligible at } s\Big)\right) D\Big(100 \geq p_{i,j} \ \cap \ i \text{ eligible at } j\Big)\right] 100}_{PV \text{ of early redemption at } j}$$

To make progress on simplifying (B.37), we need to separate the moneyness and eligibility dummies. To do this, we expand the union and make an approximation. First, we note that:

$$\begin{aligned}
&D\Big(100 < p_{i,j} \ \cup \ i \text{ not eligible at } j\Big) \\
&= D\Big(100 < p_{i,j}\Big) + D\Big(i \text{ not eligible at } j\Big) - D\Big(100 < p_{i,j}\Big) D\Big(i \text{ not eligible at } j\Big) \\
&= D\Big(100 < p_{i,j}\Big) D\Big(i \text{ eligible at } j\Big) + D\Big(i \text{ not eligible at } j\Big)
\end{aligned}$$

utilizing the identity $D\Big(i \text{ eligible at } j\Big) = 1 - D\Big(i \text{ not eligible at } j\Big)$. We then take an approximation of the products inside (B.37) around the dominant terms:

$$\begin{aligned}
&\prod_{s=1}^{j-1} D\Big(100 < p_{i,s} \ \cup \ i \text{ not eligible at } s\Big) \qquad (B.38) \\
=&\prod_{s=1}^{j-1} \Big(D\Big(100 < p_{i,s}\Big) D\Big(i \text{ eligible at } s\Big) + D\Big(i \text{ not eligible at } s\Big) \Big) \\
\simeq&\prod_{s=1}^{j-1} D\Big(i \text{ not eligible at } s\Big) \\
&+ \sum_{s=1}^{j-1} D\Big(100 < p_{i,s}\Big) D\Big(i \text{ eligible at } s\Big) \left(\prod_{k=1}^{s-1} D\Big(i \text{ not eligible at } k\Big) \right) \\
=&D\Big(i \text{ first eligible after } j-1\Big) + \sum_{s=1}^{j-1} D\Big(100 < p_{i,s}\Big) D\Big(i \text{ first eligible at } s\Big)
\end{aligned}$$

where we define first eligibility dummies

$$\begin{aligned}
D(i \text{ first eligible after } j-1) &= \prod_{s=1}^{j-1} D(i \text{ not eligible at } s) \\
D(i \text{ first eligible at } j) &= D(i \text{ eligible at } j) D(i \text{ first eligible after } j-1).
\end{aligned}$$

This approximation, done by collecting dominant terms and truncating products in the second term to the horizon of first eligibility, allows us to obtain a simplified expression for (B.37) in terms of probabilities of first redemption. Substituting (B.38) into (B.37), we obtain the approximate present value formula:[97]

$$p_i = \sum_{j=1}^{m_i} \mathbb{E}\left[\xi_j \left(D\Big(i \text{ first eligible after } j-1\Big) + \sum_{s=1}^{j-1} D\Big(100 < p_{i,s}\Big) D\Big(i \text{ first eligible at } s\Big)\right)\right] cf_i^{(j)} \tag{B.39}$$
$$+ \sum_{j=1}^{m_i-1} \mathbb{E}\left[\xi_j D\Big(i \text{ first eligible at } j\Big) D\Big(100 \geq p_{i,j}\Big)\right] 100$$

in terms of first redemption probabilities.[98] Rewriting (B.39) in terms of forward probabilities as in the call option formula, and using $\mathbb{P}(i \text{ first eligible at } s) = e_i^{(s)}$, we obtain the present value formula:

$$p_i = \sum_{j=1}^{m_i} \left(\mathbb{P}\Big(i \text{ first eligible after } j-1\Big) + \sum_{s=1}^{j-1} \mathbb{P}^{Q^j}(p_{i,s} > 100)\, e_i^{(s)} \right) cf_i^{(j)} q^{(j)} \tag{B.40}$$
$$+ \sum_{j=1}^{m_i-1} \mathbb{P}^{Q^j}(p_{i,j} \leq 100)\, e_i^{(j)}\, 100\, q^{(j)}$$

where $\mathbb{P}(i \text{ first eligible after } j-1) = 1 - \sum_{s=1}^{j-1} e_i^{(s)}$. Let $NR_i^{(j)} := \{i \text{ not eligible at } s \cup p_{i,s} > 100,\ \forall s < j\}$ be the event that i is not redeemed before j, as in Section 3.4.1. Its indicator is the product in (B.37), which (B.38) replaces by the term in brackets above, so that $\mathbb{P}^{Q^j}(NR_i^{(j)}) = 1 - \sum_{s=1}^{j-1} e_i^{(s)}\, \mathbb{P}^{Q^j}(p_{i,s} \leq 100)$. Let $d_i^{(j)} = e_i^{(j)}\, \mathbb{P}^{Q^j}(p_{i,j} \leq 100)$ be the probability that i is redeemed at j. Equation (B.40) gives a general flower bond pricing formula in terms of moneyness probabilities and eligibility probabilities, and is the price implied by the specification (B.35) when $\bar{\eta} = \eta = 0$. This formula does not add any extra adjustments for the fact that capital gains adjustments occur on flower bonds, and with greater advantage over plain vanilla bonds selling at a discount. In order to address this advantage, we introduce flower bond capital gains advantage $\bar{\eta}$, satisfying $\bar{\eta} \geq \eta$, and insert it into (B.40) analogously

[97] Note that when interest rates are high and $100 \geq p_{i,j}$ almost surely for all $j < m_i$, equations (B.37) and (B.38) match exactly.

[98] In the final term we assumed that the cross term $\left(\sum_{s=1}^{j-1} D\big(100 < p_{i,s}\big) D\big(i \text{ first eligible at } s\big)\right) D(100 \geq p_{i,j})$ is negligible.

to our capital gains bond pricing formula (B.14):

$$p_i = \sum_{j=1}^{m_i} \mathbb{P}^{Q^j}(NR_i^{(j)})\, cf_i^{(j)} q^{(j)} + \sum_{j=1}^{m_i-1} d_i^{(j)}\, 100\, q^{(j)} \tag{B.41}$$
$$+ (100 - p_i)^+ \Big[\bar{\eta} \sum_{j=1}^{m_i} d_i^{(j)} q^{(j)} + \eta \left(\mathbb{P}^{Q^{m_i}}(NR_i^{(m_i)}) - e_i^{(m_i)} \right) q^{(m_i)} \Big]$$

where the last line is the flower bond capital gains adjustment and $d_i^{(m_i)} = e_i^{(m_i)}$. This adjustment inserts an additional probability-weighted discounted capital gains advantage payout. There are higher payouts at intermediate dates $j < m_i$ if the bond is redeemed early, because $\bar{\eta} \geq \eta$, and the ordinary advantage η is paid at maturity only if the holder is not eligible at m_i and the bond was not redeemed earlier. Collecting terms in (B.35) gives the same expression, so the specification and the Euler equation derivation coincide. Finally, from (B.41) we obtain a formula for $z_{i,\text{flower}}^{(j)}$. Define E by:

$$E = \bar{\eta} \sum_{j=1}^{m_i} d_i^{(j)} q^{(j)} + \eta \left(\mathbb{P}^{Q^{m_i}}(NR_i^{(m_i)}) - e_i^{(m_i)} \right) q^{(m_i)}$$

We obtain the following cases:

$$z_{i,\text{flower}}^{(j)} = \frac{\mathbb{P}^{Q^j}(NR_i^{(j)}) + \left(100/cf_i^{(j)}\right)(1+\bar{\eta})\, d_i^{(j)}}{1+E}, \quad (j < m_i) \tag{B.42}$$
$$z_{i,\text{flower}}^{(m_i)} = \frac{\mathbb{P}^{Q^{m_i}}(NR_i^{(m_i)}) + \left(100/cf_i^{(m_i)}\right)\left(\bar{\eta}\, d_i^{(m_i)} + \eta \left(\mathbb{P}^{Q^{m_i}}(NR_i^{(m_i)}) - e_i^{(m_i)} \right)\right)}{1+E}, \quad (j = m_i)$$

if $p_i < 100$, and

$$z_{i,\text{flower}}^{(j)} = \mathbb{P}^{Q^j}(NR_i^{(j)}) + \left(100/cf_i^{(j)}\right) d_i^{(j)}, \quad (j < m_i)$$
$$z_{i,\text{flower}}^{(m_i)} = \mathbb{P}^{Q^{m_i}}(NR_i^{(m_i)}), \quad (j = m_i)$$

if $p_i \geq 100$, where $p_i < 100$ if and only if the right-hand side of (B.40) is below 100. The second case is the wedge (3.6) of Section 3.4.1, in which $\bar{\eta}$ and η are set to zero and $c_i/cf_i^{(j)} = 1$ for $j < m_i$. At $j = m_i$ the principal is part of $cf_i^{(m_i)}$ and is paid unless the bond was redeemed earlier, so no separate redemption term appears. Setting $A_i^{(j)} = 0$ for all j gives $\mathbb{P}^{Q^j}(NR_i^{(j)}) = 1$, $d_i^{(j)} = 0$, and returns the capital gains wedge implied by (B.14).

Parameterizing Moneyness Probabilities: We next consider how to parameterize the moneyness probabilities $\mathbb{P}^{Q^j}\left(p_{i,j} \leq 100\right)$. We do this by approximating the moneyness event

$p_{i,j} \leq 100$ dependent on future prices with the event that the price of the remaining cash flows is below par:[99]

$$100 \geq \sum_{l>j}^{m_i} cf_i^{(l)} q_j^{(l)}$$

and using the same interest rate parameters κ, σ from our call option formula.[100] From this formulation, we obtain the moneyness probability:

$$\mathbb{P}^{Q^j}(p_{i,j} \leq 100) \simeq \mathbb{P}^{Q^j}\left(\sum_{l>j}^{m_i} cf_i^{(l)} q_j^{(l)} \leq 100\right) = \Phi(y_i^{(j)})$$

where $y_i^{(j)}$ solves the equation:

$$100 = \sum_{l>j}^{m_i} cf_i^{(l)} \frac{q^{(l)}}{q^{(j)}} G(y_i^{(j)}, l; j, j, 0)$$

with G defined according to (B.33). Similarly, for $k \geq j$, we obtain:

$$\mathbb{P}^{Q^k}(p_{i,j} > 100) \simeq \Phi\left(-y_i^{(j)} + B(k-j)\sqrt{v_0^{(j)}}\right)$$

where the additional term in the argument accounts for the probability taken under forward measure Q^k with $k \geq j$ (B and $v_0^{(j)}$ are as in Appendix B.3.3).

B.3.5 Exchange Privilege

The Exchange Privilege was an important option affecting all coupon-bearing Treasury securities throughout much of the 1930s and early 1940s.[101] During this time, when the yields of coupon-bearing Treasury securities are plotted against their time to maturity, it appears as if all yields peel off sharply toward $-\infty$ as their time-to-maturity approaches 0, as seen in Figure 3. The Exchange Privilege essentially paid a hidden option value to the holder of coupon-bearing Treasury security i to subscribe to the latest issue of Treasury bonds. To this end, specifying the exchange privilege wedge adjustment comes down to specifying $z_{i,\text{ex priv}}^{(m_i)}$ at maturity m_i.

While Cecchetti (1988) provides a framework to estimate $z_{i,\text{ex priv}}^{(m_i)}$, his implementation

[99] As a practical matter, these are essentially the same events for deep in-the-money flower bonds that exhibited price distortions.

[100] In principle (κ, σ) are identified from callables and flower bonds jointly, but after the early 1960s flower bonds traded at such deep discounts that $p_{i,j} \leq 100$ held almost surely for all j irrespective of (κ, σ).

[101] Treasury Bills did not contain this exchange option.

relies on computing the *realized* values of the bond specific exchange privilege option when the bond has three months to maturity (at time $m_i - \frac{1}{4}$ in the future) then discounting the realized option value back to time zero using perfect foresight. Instead of imposing perfect foresight of realized exchange options, we utilize a parametric specification for $z^{(m_i)}_{i,\text{ex priv}}$ to capture the time-varying value of the exchange privilege for all coupon-bearing Treasury bonds. We choose a specification that captures the features seen in the yield-to-maturity plots: the effect of the exchange option on bond i's yield dies off quickly and the presence of the exchange privilege shifts yields uniformly downwards towards $-\infty$ as $m_i \to 0$ regardless of coupon level or price. Together, these observations suggest parameterizing the level and persistence of the exchange option adjustment.

Specification 5 (Exchange Privilege Wedge)**.** We specify the exchange option wedge with parameters ζ_1 and ζ_2 given by:

$$z^{(m_i)}_{i,\text{ex priv}} = \exp\Big(\underbrace{\zeta_1}_{\text{level adjustment}} \underbrace{\exp(-\zeta_2 m_i)}_{\text{persistence}} \Big), \qquad \zeta_1 \geq 0, \zeta_2 \geq 0$$

Discussion: Since we impose that our fitted discount function q perfectly fits the three-month Treasury Bill rate, we impose the same cross-sectional no-arbitrage condition from the Cecchetti (1988) procedure without imposing perfect foresight across time. Our fitted option values are only marginally different from Cecchetti (1988).

B.3.6 Interactions Between Wedges:

Many bonds in our sample exhibit more than one distortion. For example, we observe many callable, partially tax exempt Treasury bonds during the exchange privilege period of the 1930s, as well as deep discount callable Aaa bonds during the 1970s when capital gains received favorable tax treatment. We construct an itemized list detailing how we interact each of of various wedges below.

1. Without loss of generality, for Treasury bonds with full income tax exemptions and a call option, we scale the mean and standard deviation terms associated with log bond price $x^{t,T}$ by $(1 - \tau^{inc})$. Specifically, we scale $\log\left(\frac{q^{(T)}}{q^{(t)}}\right)$ by $(1 - \tau^{inc})$ and $v_0^{(t)}$ by $(1 - \tau^{inc})^2$ in (B.31). Under the Q^T forward measure (i.e. $j = T$ in (B.31)), this adjustment to the callable bond formula delivers consistent non-callable bond pricing to our income tax adjustment formula. Using these terms to create a schedule of $z^{(j)}_{i,\text{call, fully exempt}}$ for fully tax exempt callable bonds, we obtain the price by: $p_i =$

$\sum_{j=1}^{m_i} cf_i^{(j)} z^{(j)}_{i,\text{call, fully exempt}} (q^{(j)})^{1-\tau^{inc}}$. If i is partially tax exempt we obtain an analogous formula with scaling factor $\frac{1-\tau^{inc}}{1-\tau^{pte}}$.

2. For Treasury bonds with a call option during the exchange privilege period (i.e. 1932-1942), we specify that the payout if i is called at j, $c_i + 100$, is multiplied by the exchange privilege wedge at time j: $z^{(j)}_{i,\text{ex priv}}$. The Treasury's decision to call the bond early is not affected. In practice, this interaction occurs for bonds which are also tax exempt, requiring us to consider the interaction of the three wedges. However, whereas the tax adjustment is a discount rate adjustment, the exchange privilege wedge is a payout adjustment, and thus there is no additional nonlinear interaction between the tax adjustment and the exchange privilege. We show the reconciliation of these interactions in Figure C.36.

3. For both datasets, we face the need to price callable bonds trading at a discount exposed to the capital gains adjustment. To handle this interaction, we specify that if bond i is called at j, the investor receives additional payout of $\eta(cp_i^{(j)} - p_i)^+$ at j. The capital gains advantage does not have any influence on the issuer's decision to call the bond since call prices are either at or above par value when there is no capital gains advantage term affecting the traded price. We show how our estimator handles this interaction for Aaa bonds in Figure C.37.

C Additional Results on Fit and Robustness

In this section, we present results from the time series of our estimates, the quality of the fit along different dimensions, and the structural interpretations of the fit. In both our Treasury and corporate samples, price error sorts along any dimension of interest indicate that we can successfully price key distortions in our estimation samples. These results deliver confidence that we can extract a common discount function within each sample, as desired.

C.1 Statistical Fit

We assess the quality of our yield curve fit by examining price errors across different dimensions of interest. In Figure C.25, we sort price errors of both of our samples by coupon rate and observe that the price errors do not exhibit any apparent bias and that the dispersion remains relatively constant. These results are particularly important since bonds with relatively low coupon rates historically enjoyed a capital gains advantage and thus a significant price distortion relative to higher coupon bonds.



Figure C.25: Weighted Price Errors Across Coupons

Notes: The top panel shows weighted price errors since 1909 by coupon rate for the Treasury sample, and the bottom panel shows weighted price errors since 1909 by coupon rate for the Aaa sample. Boxes are interquartile ranges within the specified bin.

In the top panels of C.26, we report price errors by different option features in both

samples. We observe that regardless of option feature, the price errors are centered around zero and do not suggest any bias. In the bottom panels, we plot price errors against a general notion of "moneyness" for option bonds within each sample. For these plots, we define moneyness of an option bond i at time t as the risk-neutral probability weighted time to maturity/call/redemption of the bond normalized to $[0, 1]$ by the length of bond i's exercise window. Specifically, if $m_{i,t}$ is the maturity of bond i at t and $T^c_{i,t}$ is the first call/redemption date of bond i after time t, then we define $moneyness_i$ by:

$$moneyness_{i,t} = \frac{m_{i,t} - \sum_{j \geq T^c_{i,t}}^{m_{i,t}} j\mathbb{P}^Q(\text{called/redeemed/matured at } j)}{m_{i,t} - T^c_{i,t}} \tag{C.1}$$

where each risk neutral probability of early exercise is computed according to the call probability in Appendix B.3.3 or flower bond redemption probability in Appendix B.3.4 where appropriate. We choose this definition of moneyness because it is comparable across bonds of different coupon, maturity, and call protection.[102] (C.1) captures the normalized duration adjustment to bond i at time t induced by the bond's probability of early exercise, with $moneyness_{i,t} = 1$ corresponding to bond i being called on its next call date with probability 1, and $moneyness_{i,t} = 0$ corresponding to option bond i trading like a non-option bond at t. A bond with $moneyness_{i,t} = .5$ may be interpreted as trading "at-the-money." In the bottom panels of Figure C.26 we again see no apparent pattern in the price errors across the moneyness spectrum and for both samples. These plots suggest that our estimator can handle pricing in-the-money, at-the-money, and out-of-the-money bonds well, and that there is no predictable variation in the price errors for different values of moneyness.

C.2 Fitting the Short Rate Exactly

As discussed in Section B.1, we ensure that our fitted yield curve matches the observed short rate exactly. We do this for maximal flexibility at the short end of the yield curve because short-term securities can exhibit extreme curvature away from the term structure traced out by longer-term securities. We highlight this feature in Figure C.27 for the Treasury sample in March 1947, during the middle of the yield curve control period when the Federal Reserve took part in fixing rates along the yield curve.[103] Specifically, it abided by a 2.5% cap on long bonds, a 2% cap on 7-9 year notes, a 7/8% cap on 1 year certificates, and a 3/8% peg

[102] A coarser, observable measure of moneyness is a comparisons of the bond's price to its call price, or its coupon to the prevailing par-yield. However, this does not account for call protection.

[103] For superb treatment of the policy deliberations and actions, as well as their impact on the term structure of interest rates, see Chaurushiya and Kuttner, 2003.

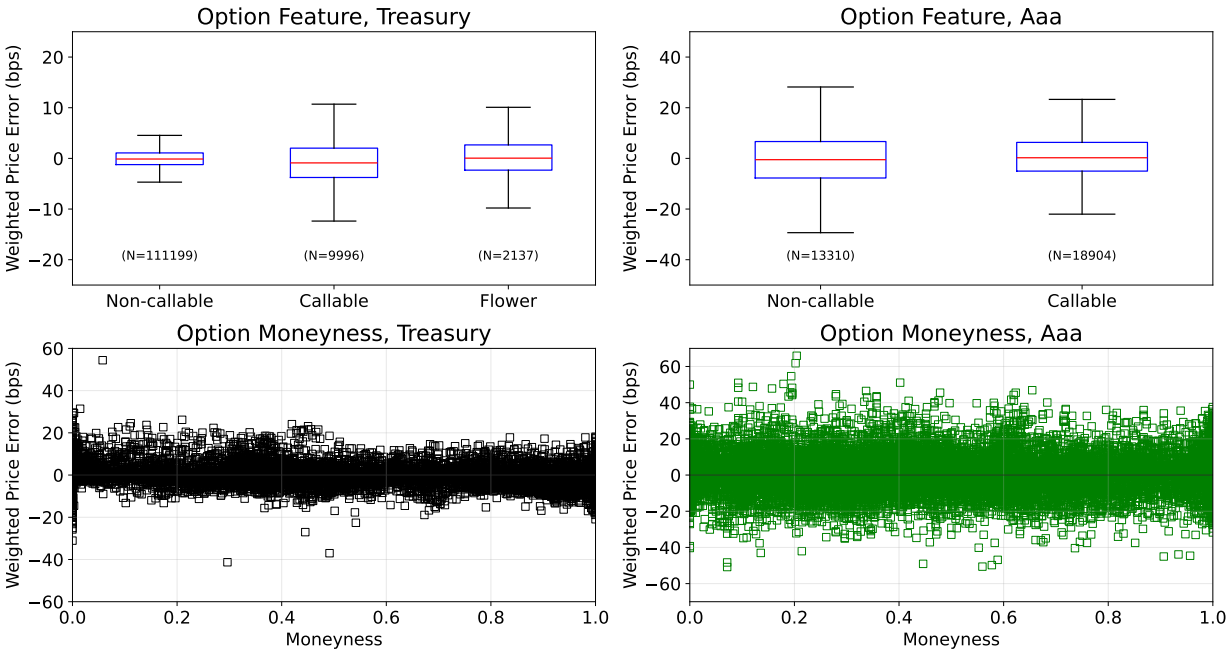


Figure C.26: Weighted Price Errors Across Option Features

Notes: The top panels show weighted price errors by option feature for the Treasury and Aaa samples, respectively, since 1909. The bottom panels show weighted price errors against our measure of option moneyness implied by fitted option parameters for all option bonds in the respective samples since 1909.

on short term Treasury Bills. By 1947, postwar inflationary pressure placed great strain on the Fed to defend the Bill peg, as well as the rates on short term certificates. Consequently, this sent yields on short term securities toward their respective caps, as seen by the blue dots (Treasury Bills) and black dots (certificates) in the bottom left panel of C.27, creating a step-wise term structure at the short end of the yield curve.

The implication of this policy for yield curve estimation is that fitting a smooth yield curve through short maturities is misspecified because the step-wise yield curve shape was a deliberate policy choice. Evidently, we want a yield curve estimator that can nest this step-wise behavior at the short end while providing a smooth fit for longer maturities. By perfectly fitting the Bill rate and the certificate rate with the Kernel Ridge estimator, we achieve this desire, as seen in the middle column of Figure C.27. The fit is clearly excellent at both long maturities and short maturities.

We compare our estimates to an estimate of a Nelson-Siegel yield curve using the implementation of Gürkaynak et al., 2007 in the right column of C.27. We perform this estimation conditional on our fitted tax and option adjustments. The consequences of using this method are twofold. The first is that despite policy fixing the first year of the yield curve, the Nelson-

Siegel curve does not have the flexibility to fit this shape, which leads to large price errors at short maturities as seen in the bottom right panel of C.27. The second consequence is that because the estimator attempts to fit the short maturity data, it sacrifices fit at long maturities because it allocates its flexibility to the short end where there is the greatest curvature. The result is that the Nelson-Siegel functional form biases the estimated yield curve because it is not flexible to pick up the shape of the yield curve at all maturities.

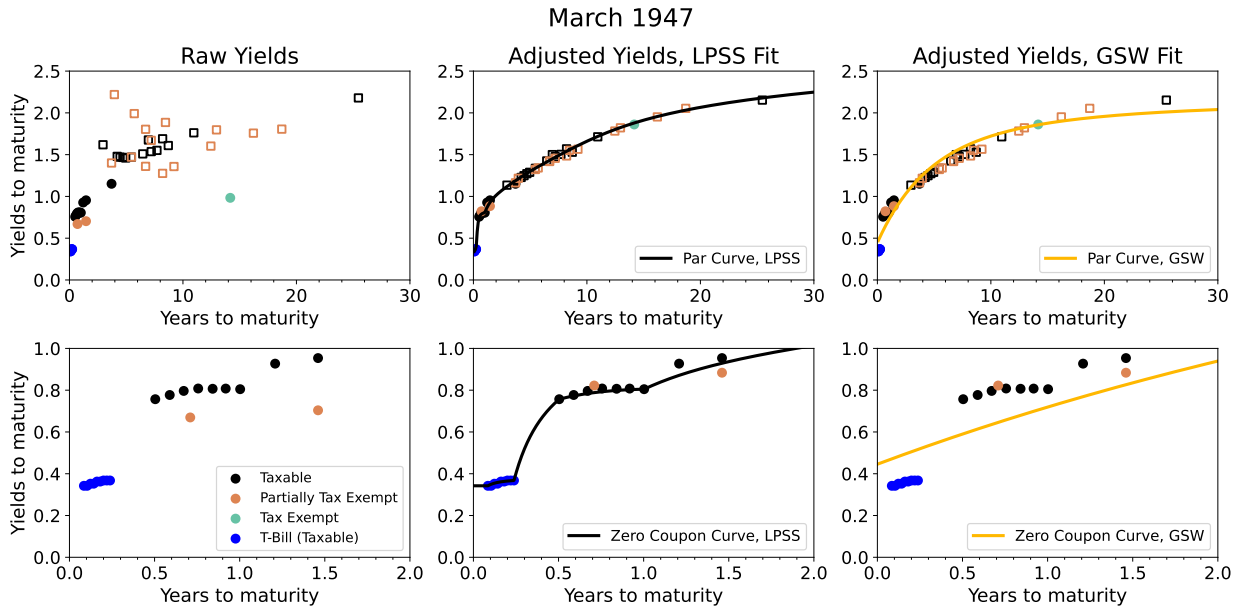


Figure C.27: Yield Curve Model Selection, March 1947

Notes: First column shows raw yields-to-maturity for Treasury securities in March 1947 during the yield curve control period. Second column shows our tax/option adjusted yields along with our fitted yield curves. Third column shows an estimate of a Nelson-Siegel yield curve using the implementation of Gürkaynak et al., 2007 *conditional* on our tax/option adjustments. Dots are non-callable securities, squares are callable.

C.3 Yield Curve Information in Option Bonds

Option bonds make up the majority of the data in both our corporate and Treasury samples, particularly at long maturities. Without them, the longest available maturities would be too short to estimate long-term yields. Our yield curve estimation must therefore separate and extract discount function information from embedded option features reliably. This section shows that it does, which lets us extend the maturity spectrum for both Treasurys and corporate bonds.[104] We validate the separation by applying our procedure to subsamples

[104]We are grateful to Greg Duffee for his insightful suggestions, which inspired this section.

of option bonds sorted by their characteristics, in months where option features are heterogeneous along the term structure. The fitted yield curves differ little from our baseline estimates and across the sorting dimensions.

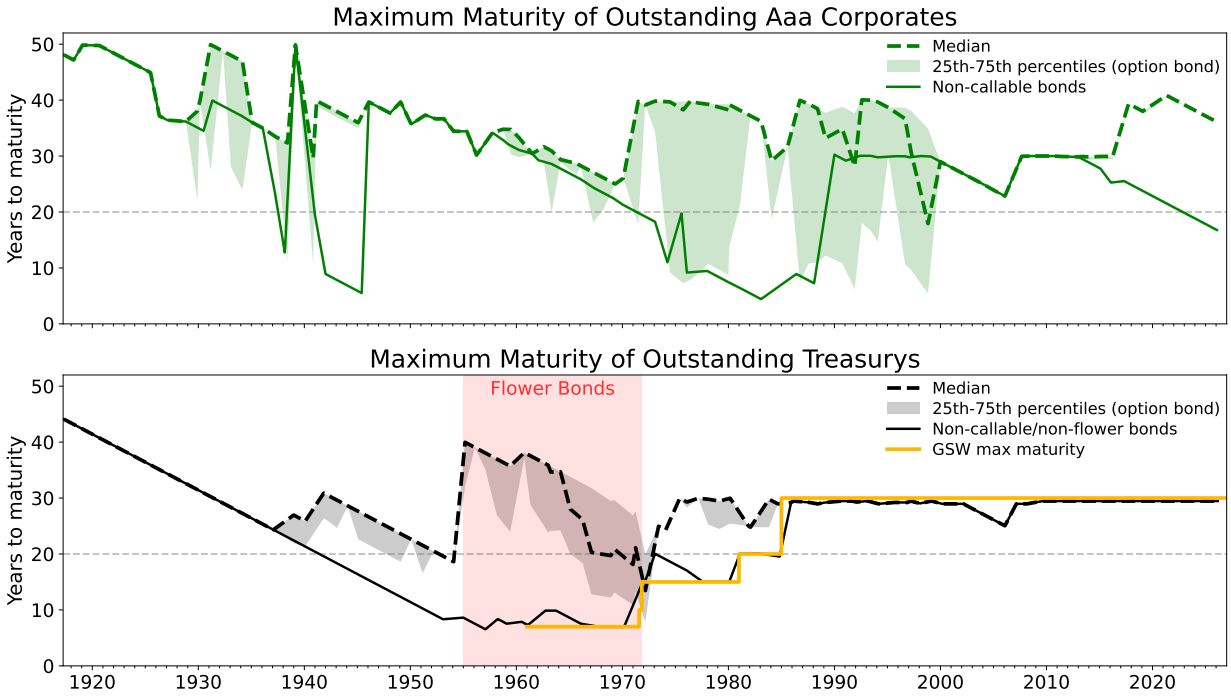


Figure C.28: Effective Maximum Maturities of Aaa$^+$ and Treasury Bonds, Modern Sample

Notes. The figure shows the median option-adjusted time to maturity of the maximum maturity Aaa$^+$ bond (dashed green line, top panel) and Treasury bond (dashed black line, bottom panel). When the maximum maturity bond is an option bond, we show the interquartile range of its option-adjusted maturity dates, computed using our adjustment functions under measure $\mathbb{Q}$. Solid lines plot the maximum maturity non-option bond within each sample.

Figure C.28 shows the time series of maximum option-adjusted "effective" maturities in both datasets from 1917.[105] When the maximum maturity bond is an option bond, we show the interquartile range of its projected call or redemption dates, computed under measure $\mathbb{Q}$ with the probabilities in our adjustment functions. Beyond the 75th percentile of these dates, our adjustment functions assign $z_{i,t}^{(j)} \approx 0$, because the bond is likely to be called or redeemed earlier. High moneyness bonds therefore carry little information about long-term yields, even at long maturities. The solid lines show the maximum maturity that would remain without option bonds. The effective maximum maturity marks the horizon

[105]Before 1917, maximum maturities for Treasurys and highest-grade corporates were long and free of option features. See Figure 1.

beyond which an estimated yield curve is extrapolated.[106] In Figure C.28, effective maximum maturities of Treasury and Aaa$^+$ bonds are generally well above 20 years, despite decades in which option bonds dominate each sample. Callable bonds in the Aaa$^+$ sample during the 1970s and 1980s behave like true long-term bonds, and flower bonds in the Treasury sample behave like bonds with 20 or more years to maturity when they are the only long-maturity Treasurys outstanding over 1955–71.[107] Effective maturities of Aaa$^+$ and Treasury bonds are therefore long enough to encode long-term yield information without implicit extrapolation. The following sections examine how well we infer this information from option bonds.

C.3.1 Callable Bonds

In this section, we show that our Aaa$^+$ yield curve estimates are robust to callability, as well as the "moneyness" of the embedded call options. These tests serve as important verification because the majority of our corporate bond observations are callable, rendering the corporate discount function estimation procedure largely an options pricing problem.

In Figure C.29, we show that our estimation procedure can extract discount function estimation from callable bonds by comparing our baseline fit to estimates fitted to a subsets of only non-callable and callable bonds in a representative month, March 1928. We choose this month because there are non-callable and callable bonds across the full term structure. Moreover, the data suggests that there is heterogeneity in the moneyness of the callable bonds because many callable bond yields trade well above those of non-callables, as seen in the top panels of Figure C.29. The left panels highlight non-callable bonds which we use to estimate a non-callable yield curve shown in the bottom left panel. The right panels highlight callable bonds which we use to estimate a callable yield curve shown in the bottom right panel, in addition to computing option-adjusted yields. We show the associated anchor yield curve (the Treasury yield curve plus a level shift) for the non-callable and callable yield curves in the bottom left and right panels, respectively. We observe the following:

- The yield curves fitted to subsamples of non-callable/callable bonds are largely similar to our baseline estimate and are therefore similar to each other. Differences along most of the term structure are on the order of 3 basis points.

- The yield curve fitted to callable bonds is able to find the same humped shape as the

[106] The yield curve literature typically does not report extrapolated yields, because they are not inferred from the data. The FEDS working paper version of GSW explicitly advises against constructing them, "Although there is nothing to stop a researcher from constructing yields at any horizons from [our] parameter estimates, we strongly advise against doing so, since there is no reason to expect the extrapolation to yield reasonable results."

[107] We explore long-term Treasury yield inference from flower bonds further in Appendix C.3.2.

yield curve fitted to non-callable bonds despite this shape not being present in the anchor yield curve.

The second observation highlights the importance of embedding the yield curve in the Hull-White interest rate forecast used to determine probability of early call. This allows the estimator to extract discount function information encoded in theoretically consistent call probabilities, not just from standard zero-coupon bond pricing. Figure C.29 suggests that imposing this pricing consistency in our estimation is not a constraint on the fit; rather, it allows for identification of the discount function when we face only callable bonds in the sample, as is often the case.

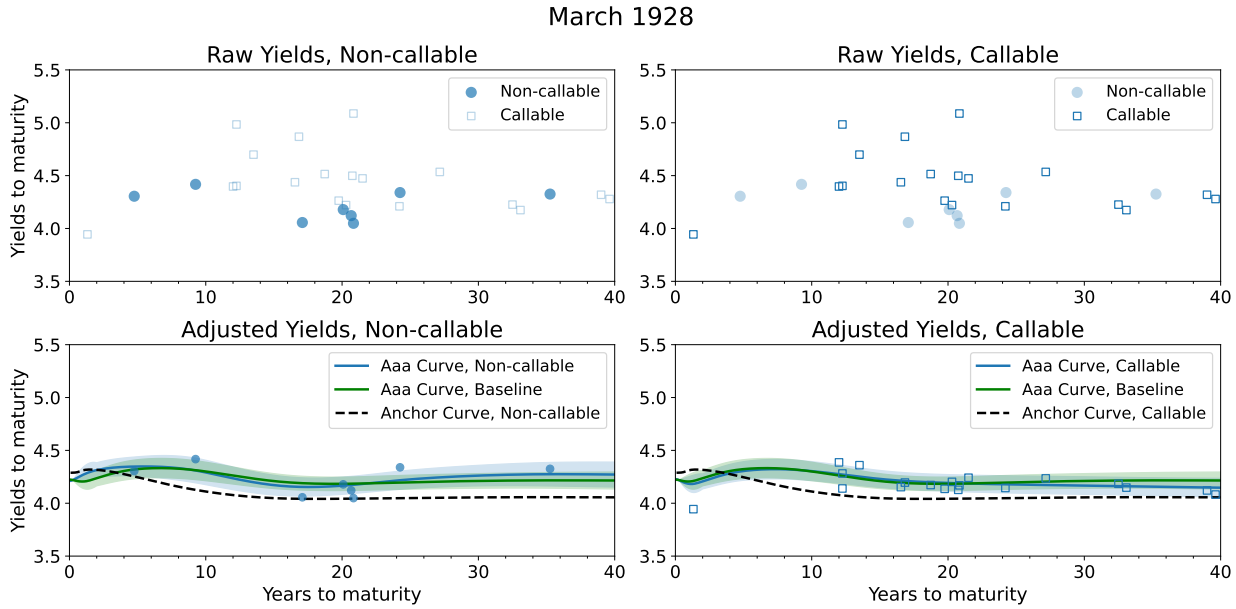


Figure C.29: Discount Function Information in Callable Bonds

The figure compares our baseline estimate of the March 1928 Aaa yield curve (in green) to estimates fitted to only non-callable bonds (left panels) and callable bonds (right panels). Separate anchors associated with each estimation are plotted in black.

We perform an additional exercise in Figure C.30 showing that our estimation procedure can extract a common discount function when fitted to only low, medium, or high moneyness callable bonds in February 1991, a month with substantial variation in the embedded moneyness of the observed bonds. Since we often observe bond data with a common degree of moneyness depending on the interest rate environment, it is useful to verify that our procedure produces similar yield curves when fitted to bond subsets sorted by a proxy for their moneyness. In this exercise, we let the observed price be a proxy for the moneyness, with par value = 100 a stand-in for "at-the-money." We perform three estimations against

our baseline, fitted to sorts of a) low moneyness (price <85), b) medium moneyness (price between 85 and 98), and c) high moneyness (price >98). We show these fitted curves in the bottom panels of Figure C.30 and observe that they are largely indistinguishable from our baseline estimate, and are therefore indistinguishable from each other. The difference between the low moneyness and baseline 20 year estimates is one basis point, the difference between the medium moneyness and baseline 20 year estimate is seven basis points. The difference between the high moneyness and baseline estimates is six basis points at long maturities, though we cap the maximum inferred maturity at 15 years, corresponding to the maximum option-adjusted maturity of the high moneyness fit. This is because high moneyness bonds have the highest chance of being called early, and thus they receive the largest duration adjustment from our methodology. Figure C.28 makes clear that we are never only restricted to high moneyness bonds in our sample, so we can always infer yield information at 20+ year maturities, as desired.

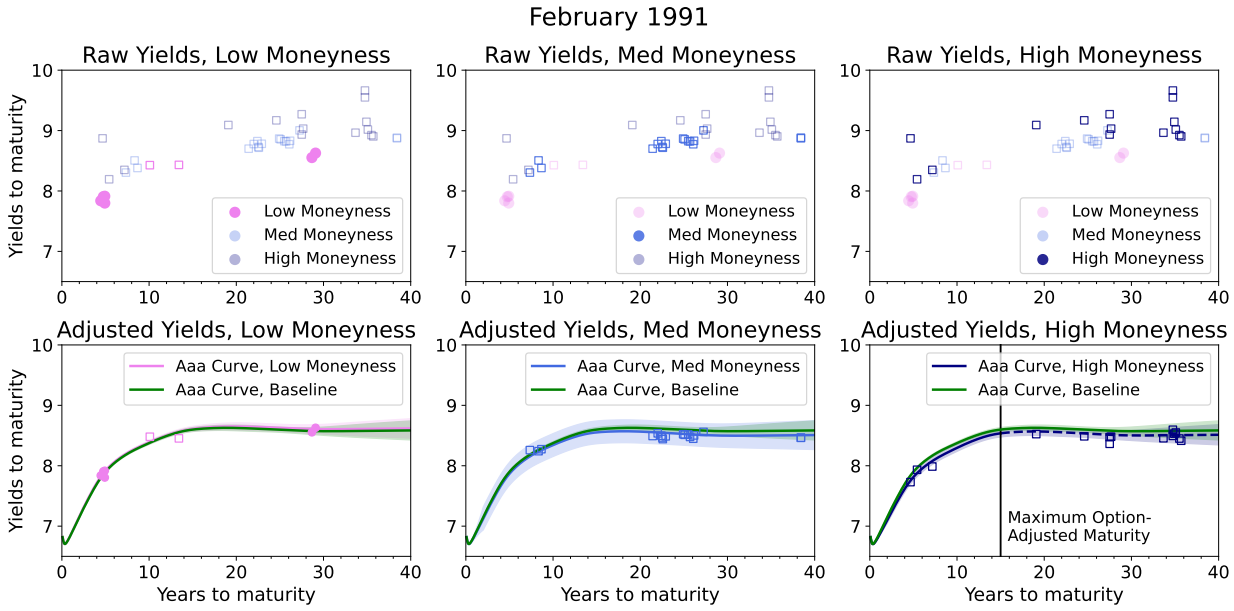


Figure C.30: Discount Function Information in In/Out-of-the-Money Callable Bonds

The figure compares our baseline estimate of the February 1991 Aaa yield curve (in green) to estimates fitted to only deep out-of-the-money (low moneyness) callable/non-callable bonds (pink, left panels), moderately out-of-the-money (medium moneyness) callable bonds (medium blue, middle panels), and at/in-the-money (high moneyness) callable bonds (dark blue, right panels). Low moneyness bonds have traded price < 85, medium moneyness bonds have traded price between 85 and 98, and high moneyness bonds have traded price > 98.

These are two illustrative months. To check that the property holds more generally, we repeat the moneyness exercise on every estimation date since 1920 with enough bonds in

each cell of a two-by-two sort.[108] We split bonds by price into low moneyness ($<$ \$90) and high moneyness ($\geq$ \$90), and by remaining maturity into under and over 10 years, and we require at least two bonds in each of the four cells. The coarser price split, relative to the three-way split above, lets more dates meet this requirement. On each date that satisfies this requirement, we estimate the Aaa$^+$ yield curve separately on the low-moneyness bonds and on the high-moneyness bonds and compare each curve with our baseline estimate. The bottom panel of Figure C.31 shows the share of dates in each decade that satisfy the condition. These dates cluster in the 1970s–90s, a period of high interest rates and noisy bond yields, which makes the test demanding.

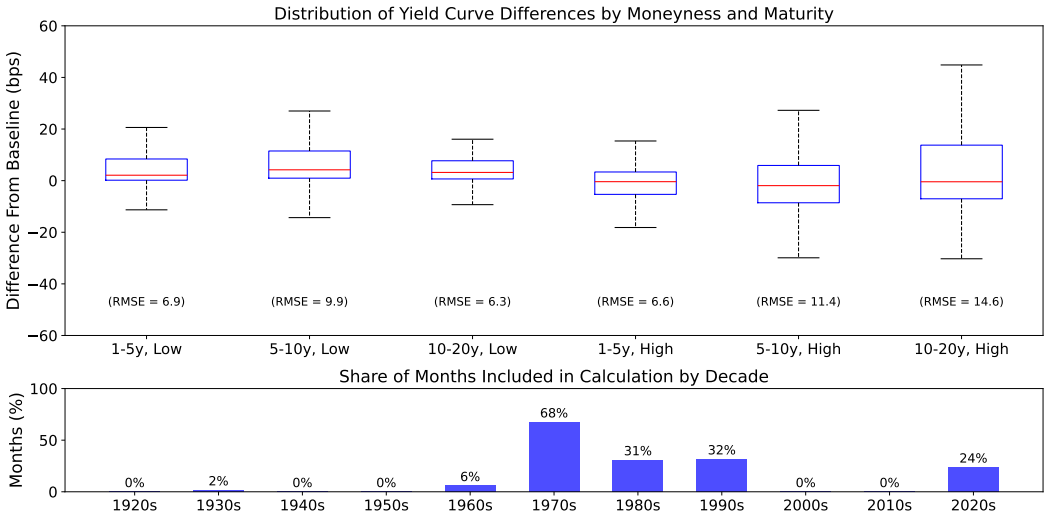


Figure C.31: Distribution of Yield Curve Differences by Moneyness and Maturity

Notes: Each box/whisker shows the distribution over estimated dates the difference between the estimated yields along the low/high moneyness sort and our baseline estimates, binned for short maturities (average of 1-5 year maturities), medium maturities (average of 5-10 year maturities), and long maturities (average of 10-20 year maturities). The bottom panel shows the share of dates each decade which we can compute estimates sorted along moneyness.

The top panel of Figure C.31 shows the distributions of yield differences at short (1–5 years), medium (5–10 years), and long (10–20 years) maturities for the low-moneyness subsample (left three box plots) and the high-moneyness subsample (right three box plots). Root mean square differences across dates appear in parentheses under each box plot. Yields estimated on either subsample are close to the baseline. The average difference is 3.2 basis

[108]This exercise contains the callable vs non-callable comparison. Low-moneyness bonds have options far from exercise, so the call adjustment barely affects their prices and they play the role of the noncallable sample. High-moneyness bonds have options close to exercise, so the adjustment drives their prices and they play the role of the callable sample.

points for low-moneyness bonds and -0.9 basis points for high-moneyness bonds. The interquartile ranges are 8.6 and 14.6 basis points, and the root mean square differences are 7 and 11 basis points, the same order of magnitude as our in-sample pricing errors. Each subsample contains fewer bonds than our full sample, so these differences include sampling noise from smaller sample size and overstate any error from the call adjustment. These differences are also small relative to the average long term Aaa^+ yields, which averaged around 10% in the 1970s and 1980s.

As expected, for high-moneyness bonds, the differences grow with maturity. A long-maturity bond with high moneyness is likely to be called early and behaves like a short-maturity bond. It carries little information about long maturities, and our adjustment functions reflect this by assigning it a high call probability that truncates its effective cash flows. The long-maturity estimates from the high-moneyness sort are therefore relatively imprecise. In the full sample, however, long-term yield information comes from bonds with long effective maturities, which this subsample excludes. Figure C.28 above shows that the full sample generally contains bonds with effective maturities well above 20 years, so in practice we do not need to rely on high-moneyness bonds for long-term yields.

C.3.2 Flower Bonds

In this section we study how the inclusion of flower bonds may affect our baseline yield curve estimate. Ideally, our procedure enables us to extract information about non-callable long-term "plain vanilla" bonds embedded within flower bonds, thereby allowing to construct a comprehensive estimate of the term structure of Treasury yields during the 1960s and early 1970s—a period in which only flower bonds are observable at the long end. The reliability of our ability to extract long-maturity yield curve information from our estimation procedure depends upon the effective duration of the flower bonds. That is, when the value of early redemption was low, the flower bonds traded like regular long-term bonds with bountiful long-term yield information. Conversely, when the early redemption value was highest, flower bonds traded like short maturity bonds, and nested very little long-maturity yield information. In this case, all flower bonds have $z_{i,t}^{(j)} \approx 0$ for large j, and thus yield curve estimation with effective long-term cash flows of zero mechanically results in extrapolation. We show that throughout the 1960s and early 70s, our method produces "long enough" effective durations in the flower bonds to identify long-term yield information from the data.

Figure C.32 shows our adjusted time-to-maturity for the longest maturity flower bond, the 3.5% bond of 1998. We show the quartiles of the effective maturity such that, without loss of generality, the median adjusted years-to-maturity on a given date corresponds to the

maturity at which half of the remaining principal is projected to be redeemed by, according to the bond's fitted eligibility distribution. Evidently, during the 1960s and 70s, effective maturity declined faster than its promised time to maturity, with the black line suggesting that the bond traded like an 8 year bond by 1978 (20 years before its promised maturity). This meant that by 1978, very little long-term yield information was left embedded in this bond. However, as seen in the figure, this is of little concern after the Treasury began issuing non-callable/non-flower long-term bonds in late 1971. In the preceding period, during which all long-term bonds were flower bonds, the 3.5% bond of 1998 consistently traded like a 20+ years to maturity bond after option adjustment, its effective maturity bottoming out (around 13 years) in mid-1971 after Congress fixed the supply of remaining flower bonds. The question remains whether this effective maturity was sufficient to infer long-term yield information from the data.

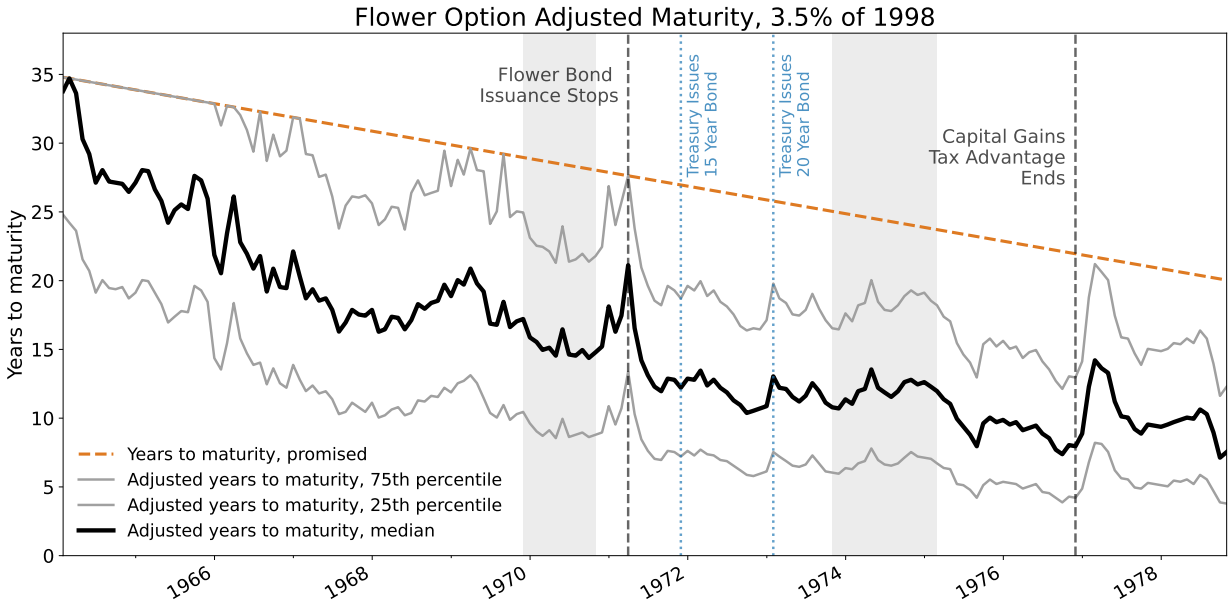


Figure C.32: Option Adjusted Maturity in the Longest Maturity Flower Bond

Figure shows the effective maturity quartiles of the longest maturity flower bond upon option adjustment.

To ensure that our estimator captures genuine yield curve information rather than extrapolation, we conduct a targeted "sense check" upon the Treasury's issuance of a 20 year plain vanilla bond. Specifically, we re-estimate the Treasury yield curve for January 1973 using only non-flower bonds at maturities under 7 years (even though the Treasury had just issued long-term non-flower bonds), supplemented by longer-term flower bonds. In addition to having plain vanilla long bonds, we choose this month because our baseline estimate of

the effective maturity of the 3.5% bond of 1998 was approximately 13 years, suggesting that there was some long-term yield information left in the bonds. In the left panel of Figure C.33, we compare our sense check estimates to our baseline specification for the same date, which does incorporate the newly issued plain-vanilla long-term bonds, and assess the estimator's ability to fit these longer-term non-flower bond yields that were excluded from the restricted sample. We observe the following:

1. The adjusted yields on the four longest flower bonds are directly in line with the two long dated plain vanilla bonds which are dropped from the sense check exercise.

2. The yield differential between our baseline and the sense check are minimal, 6 basis points at 15 years to maturity and 10 basis points at 20 years to maturity.

3. Even with the 13 year effective maturity for the longest dated flower bond, the flower bonds pick up the same curvature at the long end in our baseline estimate, and the baseline is within one standard deviation of the sense check.

This exercise serves as an out-of-sample confirmation that even as the effective maturity of the flower bonds shrank to as little as 13 years, there was sufficient long-term yield information left in the bonds that is picked up by our estimator. This provides greater assurance that we can identify long-term bond yields from the flower bonds when they comprise all 7+ year Treasury securities from 1962-1971, as desired.

To highlight when we lose long-term yield information in the flower bonds, we perform the same sense check for January 1976. In this month, our baseline estimates of effective maturities on flower bonds do not exceed 7.5 years. We show the result of this sense check in the right panel of Figure C.33. In this panel, the sense check appears to perform horribly after 7 years to maturity, with the fitted yield curve some 70 basis points above the baseline at long maturities. However, this result is driven our z_{flower} adjustment assigning effective maturities of less than 8 years to the remaining flower bonds (i.e. $z^{(j)}_{flower} \approx 0$ for $j > 8$ years to maturity for all i). This mechanically fixes the sense check exercise to be an extrapolation, and in the face of a nearly 3 percent yield curve slope out to 7 years, the estimator chooses a shape which continues this same steep slope. Fortunately, this is of little relevance for our baseline estimate, which does incorporate non-flower long-term bonds because it can use the information in the observed plain vanilla bonds to find the flower adjustment consistent with the observed term structure. We only lose the ability to fit observed long-term bond yields from the flower bonds when they behave like short-term bonds, but we conclude that i) when they behaved like long-term bonds we find evidence that we can infer the correct long-term yield information from them and ii) when they behaved like short-term bonds, there were

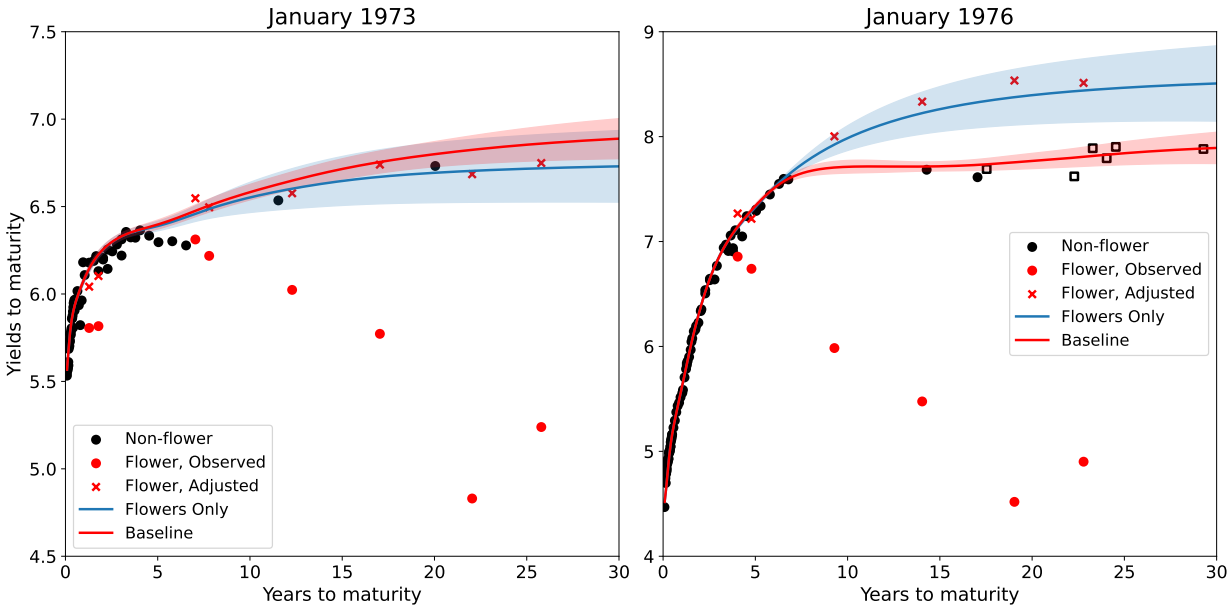


Figure C.33: Fitting Flower Bonds

The figure compares our baseline estimate of the Treasury yield curve (in red) to a "sense check" exercise (in blue) re-estimated to exclude non-flower bonds maturing beyond 7 years. Dots are observed yields-to-maturity. We compute the flower bond adjusted yields (x's) by removing the fitted option value in each sense check exercise from the observed bond price. Yield curves shown are par yields. We drop bonds which previously incurred large capital gains.

non-flower long-term bonds that allowed us to pin down the flower option adjustment and to fit long-term yields.

C.4 Duration adjustment in option bonds

As a case study, Figure C.34 shows how our estimator adjusts bond yields for call options in August 1989. The top left panel shows raw yields-to-maturity of Aaa bonds, all of which are callable and exhibit little apparent structure across the maturity spectrum. The top right panel shows the yields-to-maturity on the same bonds after we remove the fitted call option price from the observed price. Clearly, we gain back the desired structure of the yields-to-maturity after removing the call options and the shape of the fitted yield curve is roughly the same as the Treasury yield curve after 5 years-to-maturity.[109] In the bottom figure, we showcase how the option adjustment $z_{i,\text{call}}$ adjusts the implied cash flow stream of a particular bond, the Southern Bell Tel & Tel (SB&T) 8% bond of 2014, which is highlighted

[109] We include the commercial paper rate on these plots to highlight how the corporate short rate disciplines the short end of the corporate yield curve.

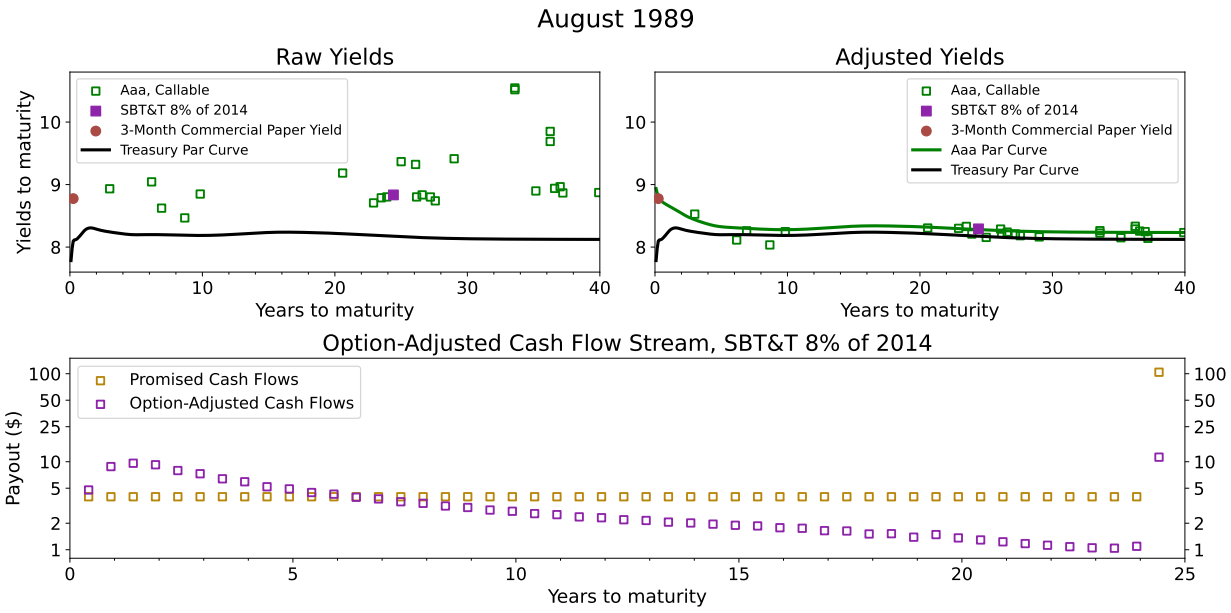


Figure C.34: Call Option-Adjusted Cash Flow Stream, August 1989

Notes: The top left panel shows raw yields-to-maturity of all Aaa bonds in our sample in August 1989, all of which are callable. The top right panel shows option-adjusted yields-to-maturity, as well as the fitted corporate par curve. The bottom panel highlights the "option-adjusted cash flows" of the Southern Bell Tel & Tel (SBT&T) 8% bond of 2014, highlighted in lavender in the top panels. For each future cash flow date j, gold squares correspond to its promised cash flow at j, $cf^{(j)}_{SBT\&T}$, and lavender squares correspond to the fitted $cf^{(j)}_{SBT\&T} z^{(j)}_{SBT\&T,\text{call}}$, its option-adjusted cash flow at j.

in the top figures in lavender. At each cash flow date j, we compare the promised cash flow stream $cf^{(j)}_{SBT\%T}$, which is a semiannual coupon of \$4 with final maturity payment \$104, to the option-adjusted cash flow stream $cf^{(j)}_{SBT\%T} z^{(j)}_{SBT\&T}$. Not only do we see from the top panels that the option-adjusted bond price is well fitted by the Aaa yield curve, the bottom panel shows that the method to achieve this good fit is an economically meaningful duration adjustment reflecting the probability of early call. Our option adjustment assigns the highest probability of the bond being called 1-5 years in the future, leading to a peak in the option-adjusted cash flow stream over this window.[110] This is easily reconciled by comparing the fitted yield curve to the bond's coupon: current market interest rates are above the bond's coupon, making it just out-of-the-money, but the downward sloping yield curve forecasts a decline in future interest rates making the call option most valuable in 1-5 years time. After this peak, the adjusted cash flow stream steadily declines towards the maturity date, reflecting positive probability that the bond is called at each intermediate cash flow date and

[110]Indeed, the bond was actually called just over 4 years later on September 27, 1993, which is consistent with the highest probability call dates fitted by our option adjustment.

therefore lower probability that the bond continues to pay out on future dates. However, the option-adjustment still places a relatively high adjusted payout on the maturity date due to a significant probability that interest rates stay high over the bond's lifetime and that the bond is never called. We obtain this result because the fitted parameter $\tilde{\kappa}$, the persistence parameter for the short-term corporate rate process in (B.29), is low, which leads to high autocorrelation in the forecast of future corporate interest rates.

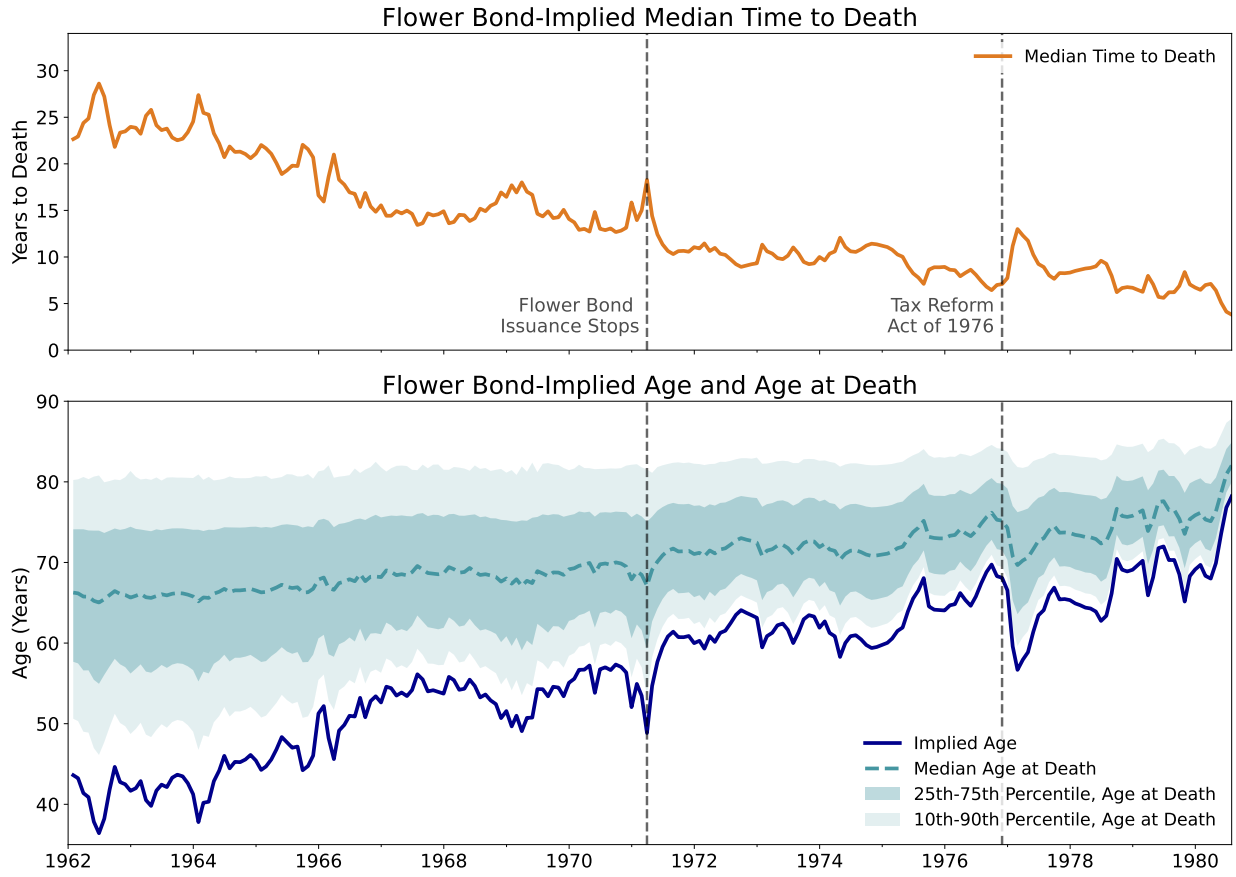


Figure C.35: Flower Bond Reference Clientele Implied Age

Notes: The top panel shows the flower bond reference clientele median time to death, a measure of the effective duration on flower bonds. The bottom panel shows the implied age (dark blue) of the flower bond reference clientele: those buying the 3% bond of 1995. The teal dashed line depicts the median age at death of the reference clientele buyer. Percentile bands are shown in faded shades of teal. The average male life expectancy, according to the Social Security Administration, was around 70 years.

Next, we show that the flower bond reference death probabilities impact the bonds' effective durations. The top panel of (C.35) shows the reference clientele's median time to death, computed using parameters $\bar{\alpha}_t, \bar{\beta}_t$. As mentioned in B.3, $\bar{\alpha}_t, \bar{\beta}_t$ form a Gompertz distribution for future death dates of the reference clientele. Under this interpretation, the

probability that the clientele is alive at horizon j is given by the function:

$$\mathbb{P}_t(ref \text{ alive at } t+j) = 1 - \sum_{s=1}^{j} e_t^{(s)} = \exp\left(-\frac{\bar{\alpha}_t}{\bar{\beta}_t}\left(\exp(\bar{\beta}_t j) - 1\right)\right)$$

from which it follows that the median time to death is given by the function $\log\Big(1 + (\bar{\beta}_t/\bar{\alpha}_t)\log(2)\Big)/\bar{\beta}_t$. Given the relation of $\mathbb{P}_t(ref \text{ alive at } t+j)$ to the eligibility probabilities $e_t^{(j)}$, the median time to death is interpretable as a duration adjustment, with high times to death corresponding to longer durations and vice versa. As expected, we see that this adjusted duration declined over time as because rising interest rates made the value of early redemption more valuable. Importantly, when flower bonds were the only long-term Treasury bonds between 1962-71, we see that this effective duration was between 15 and 25 years, indicating that flower bonds still traded like relatively long-term bonds during this period though not as long as the roughly 30 years left until their final maturity. [111] We also use our estimated $\bar{\alpha}_t, \bar{\beta}_t$ to back out an implied marginal age of the reference clientele. We do this by fitting a structural Gompertz model to US male mortality data over 1950-1980.[112] A structural Gompertz mortality model with parameters λ, β says that the probability that an individual $\tilde{i}$ is alive at time $t+j$, given they are of age $a_{\tilde{i},t}$ at time t is given by:

$$\mathbb{P}_t(\tilde{i} \text{ alive at } t+j | a_{\tilde{i},t}) = \exp\left(-\frac{\lambda}{\beta}\exp(\beta a_{i,t})\left(\exp(\beta j) - 1\right)\right)$$

We therefore take the implied age of the reference clientele at t, age_t, to be the age which minimizes the L^2 norm between the bond-implied and structural probabilities of staying alive over horizon j:

$$age_t = \arg\min_a \int_0^\infty \left[\exp\left(-\frac{\bar{\alpha}_t}{\bar{\beta}_t}\left(\exp(\bar{\beta}_t j) - 1\right)\right) - \exp\left(-\frac{\lambda}{\beta}\exp(\beta a)\left(\exp(\beta j) - 1\right)\right)\right]^2 dj$$

We plot the implied age age_t in dark blue in the bottom panel of Figure C.35. The ages are economically reasonable, and could plausibly map to ages of estate owners looking to buy flower bonds for their anticipated redemption value. The steady increase in the implied age over the 1960s and 70s suggests that the reference clientele *behaved like* an ever older estate owner whose death drew nearer and nearer. We compliment this line by showing the distributions of reference clientele age at death in teal, computed by adding the time to death Gompertz distribution under each $\bar{\alpha}_t, \bar{\beta}_t$ to the implied age age_t. Over the 1960s and 70s, the

[111] We explore the implications of this duration adjustment for inference in C.3.

[112] We obtain US mortality data from the Human Mortality Database.

projected median age at death stayed relatively constant around 70 years old, consistent with life expectancy at the time as computed by the Social Security Administration. Altogether, this suggests that buyers of flower bonds i) behaved like estate owners who expected to die around their life expectancy and ii) behaved more like elderly estate owners anticipating sooner redemption as inflation and interest rates rose over the 1960s and 70s.

C.5 Interactions between adjustment functions

We next show our ability to fit wedge interactions in both of our datasets. We first show this for the Treasury yield curve in October 1938 in C.36 when there are bonds containing at most three different distortions: tax-exemptions, call options, and exchange privilege.[113] The left panel of C.36 shows the raw yields-to-maturity, which are clearly disconnected at short maturities from the (tax-exempt) Treasury Bill rate. Moreover, partially tax exempt securities seem to exhibit uniformly higher yields than tax-exempt securities, and a number of the callable bonds appear distorted, reflecting a high likelihood that many of them will be called early. In the right panel, we show our adjusted yields which correct for interactions between wedges. The adjusted yields create a homogeneous term structure in the undistorted tax-exempt yield space.[114] The short end of the yield curve is reconnected with the Treasury Bill yield, yields on fully tax-exempt securities line up with yields on adjusted yields of partially tax-exempt securities, and adjusted yields on partially tax-exempt callable bonds line up with adjusted yields on partially tax-exempt non-callable bonds.

We also show how our estimator handles the interaction of callability and capital gains tax advantage in the Aaa$^+$ dataset, which is the prominent interaction in the 1970s as interest rates rose and callable bond prices fell below par. In the top left panel of C.37, we show the raw yields-to-maturity of Aaa$^+$ bonds in August 1979 with the Treasury par yield curve and commercial paper yield drawn as additional reference points. We color the yields by the price of the bond, which gives a sense of both (i) the capital gains tax advantage embedded in the bond through a larger discount and (ii) the approximate moneyness of the call option as measured by distance from par value \$100. Disregarding the commercial paper yield, the raw yields-to-maturity seem to exhibit an upward-sloping term structure, the exact opposite shape of the Treasury term structure. However, the price heat map suggests that deep discount bonds are considerably more favorable than higher-price bonds, which drives the term structure shape traced out by raw yields at 5-20 year maturities. Moreover, the longest maturity callable bonds with prices close to or exceeding par appear to trade at elevated

[113] Each callable bond in C.36 was partially tax exempt. Since they were coupon bearing bonds, they were also subject to the exchange privilege.

[114] We plot a par yield curve using the tax-exempt discount function $\breve{q}_t^{(j)} = (q_t^{(j)})^{1-\tau^{inc}}$.

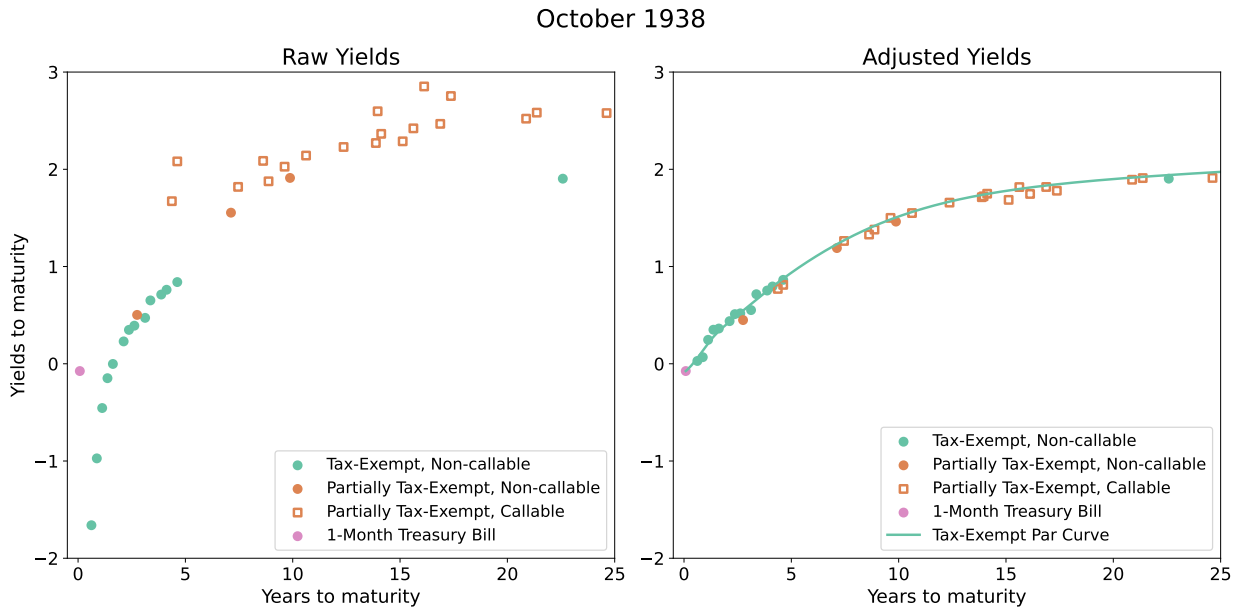


Figure C.36: Treasury Wedge Interactions, October 1938

Notes: The figure shows raw and adjusted yields-to-maturity for Treasury securities in October 1938. Three price distortions are present in this sample: tax exemptions, call options, and the exchange privilege. We show yields relative to unadulterated tax-exempt securities. Green dots are subject to the exchange privilege. Orange dots are subject to the exchange privilege and partial taxation. Orange squares are subject to the exchange privilege, partial taxation, and a call option.

yields, suggesting that those bonds have positive probability that they will be called soon.

The right panel shows our capital gains and call option-adjusted yields after fitting the Aaa$^+$ discount curve which is disciplined by the commercial paper yield and the shape of the Treasury yield curve. Evidently, yields on deep discount bonds (including deep out-of-the-money callable bonds) move up in line with yields on higher-price bonds after adjusting for capital gains advantage. Additionally, the yields on the highest price (at-the-money and in-the-money) long-maturity callable bonds are brought down in line with out-of-the-money bond yields upon adjusting for their valuable call options.[115] The resulting fitted yield curve is one that regains the shape of the Treasury yield curve and exactly reflects the price of short term highest grade corporate borrowing costs. We report the price errors associated with the fitted yield curve in the bottom figure of C.37 again colored by price. It is clear that the price errors do not exhibit any clear predictabiltity by the bond's traded price through the capital gains tax advantage or option moneyness, as desired.

[115]Out-of-the-money callable bonds are treated for both their embedded call option and capital gains advantage, though the adjustment for capital gains advantage dominates if the bond's coupon is well below the prevailing coupon on new issues.

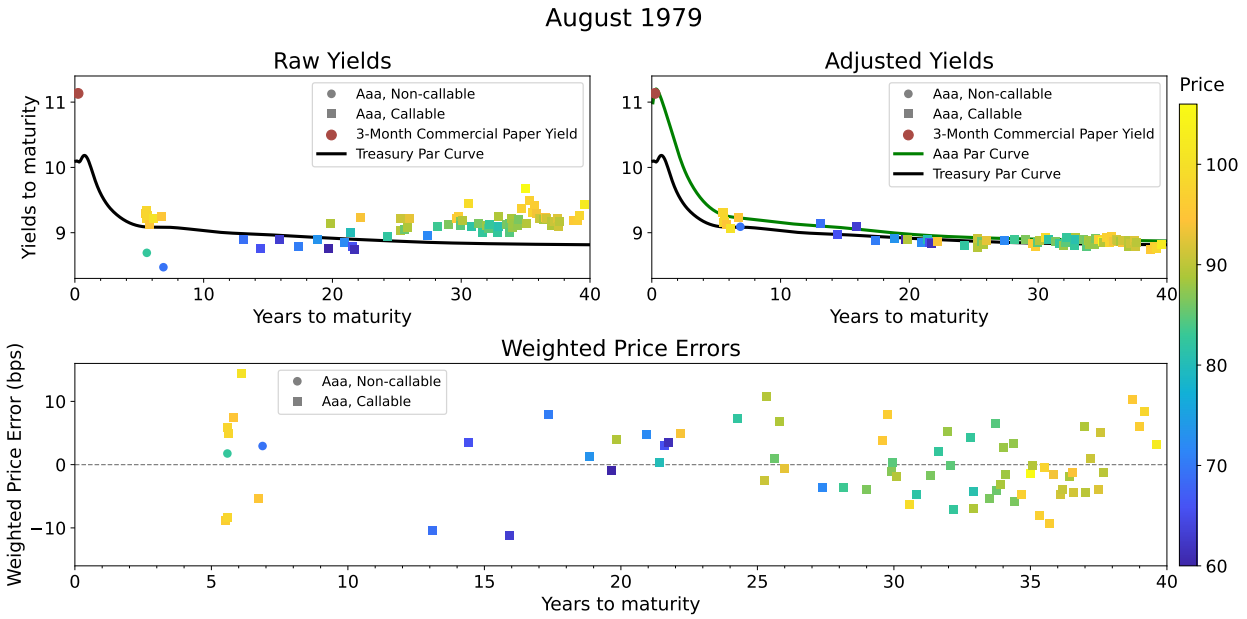


Figure C.37: Aaa Wedge Interactions, August 1979

Notes: The figure shows raw and adjusted yields-to-maturity for Aaa securities in August 1979. Yields are colored by a heatmap according to the bond's traded price, with cool colors corresponding to low prices and bright colors corresponding to high prices. Capital gains advantage and call options are present in the sample. Dots (non-callable) are subject to capital gains advantage, and squares (callable) are subject to capital gains advantage and a call option.

C.6 Comparison to Gilchrist and Zakrajšek (2012)

To account for embedded options, we introduce adjustment functions derived from first principles. Despite being parsimonious, they take complicated functional forms, and one may ask whether such forms are necessary to remove option effects and recover undistorted Aaa$^+$ and Treasury yields. We show here that the simpler and commonly used option adjustment of Gilchrist and Zakrajšek (2012) (GZ-12) can imply negative option values, which asset pricing theory rules out, while our adjustments work to turn noisy distorted data into one homogeneous yield curve. This does not mean that the GZ-12 regression fails for its intended purpose, an index of credit spreads across a large universe of bonds. It does mean, however, that it cannot serve as a bond-level call adjustment for building a yield curve.

A common technique in the literature adjusts yields and yield spreads for call option values with the panel regression approach of GZ-12. GZ-12 purge the bond-level corporate-to-Treasury yield spread $S_{it}[k]$, for bond k issued by firm i at time t, of expected default, the embedded call option, and other controls. In particular, they regress $\log(S_{it}[k])$ on their distance-to-default measure DFT_{it}, a dummy for callability D_i^c, aggregate, bond-level, and

firm-level characteristics $\mathbf{X}_{it}[k]$, and interactions of the call dummy with DFT_{it} and $\mathbf{X}_{it}[k]$:[116]

$$\log(S_{it}[k]) = \beta_1 DFT_{it} + \gamma_1' \mathbf{X}_{it}[k] + \phi D_i^c + \beta_2 D_i^c \times DFT_{it} + \gamma_2 (D_i^c \times \mathbf{X}_{it}[k]) + \epsilon_{it}[k] \quad \text{(C.2)}$$

Provided that bond k is callable, and assuming normal errors with variance $\hat{\sigma}^2$, the contribution of callability to the GZ spread, $\hat{S}_{it}^{call}[k]$, is given by

$$\hat{S}_{it}^{call}[k] = \exp\left(\hat{\phi} + (\hat{\beta}_1 + \hat{\beta}_2) DFT_{it} + (\hat{\gamma}_1 + \hat{\gamma}_2)' \mathbf{X}_{it}[k] + \frac{\hat{\sigma}^2}{2}\right) - \exp\left(\hat{\beta}_1 DFT_{it} + \hat{\gamma}_1' \mathbf{X}_{it}[k] + \frac{\hat{\sigma}^2}{2}\right)$$

where $\{\hat{\beta}_1, \hat{\beta}_2, \hat{\gamma}_1, \hat{\gamma}_2, \hat{\phi}\}$ are fitted panel regression coefficients. We interpret $\hat{S}_{it}^{call}[k]$ as the panel-regression-implied call option value (in terms of yields) of bond k at date t, so the call-adjusted yield to maturity is $y_{it}[k] - \hat{S}_{it}^{call}[k]$, where $y_{it}[k]$ is the raw yield to maturity. Because embedded call options have non-negative value, theory requires $\hat{S}_{it}^{call}[k] \geq 0$.

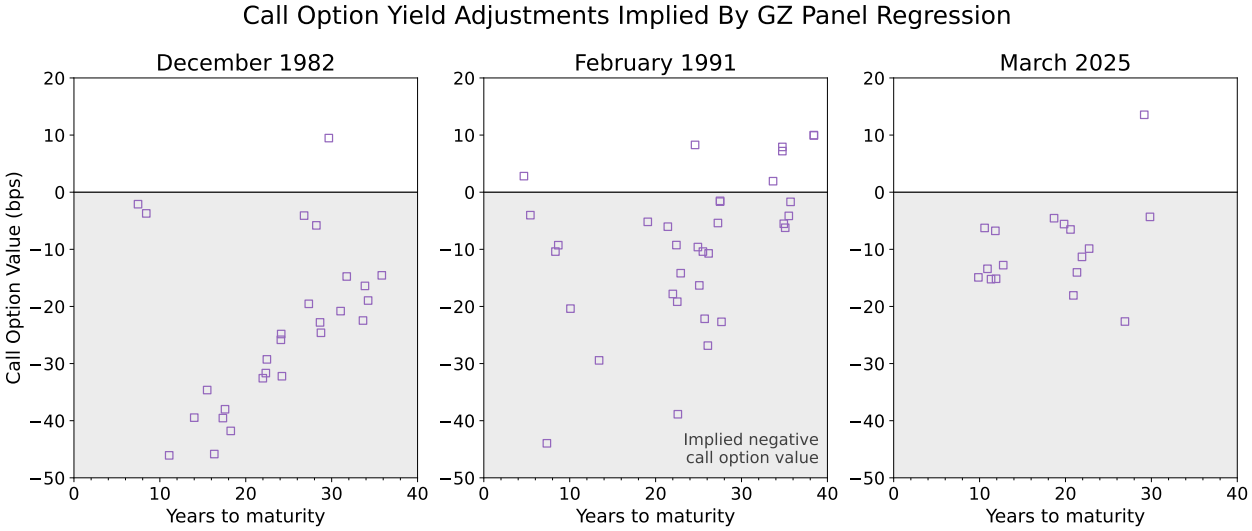


Figure C.38: Call Option Values Implied by Gilchrist and Zakrajšek, 2012

We estimate the coefficients in (C.2) using the GZ-12 bond universe and characteristics $\mathbf{X}_{it}[k]$ over January 1973 to September 2026. Figure C.38 shows the estimated call option values $\hat{S}_{it}^{call}[k]$ for the subset of bonds that appear in our Aaa$^+$ sample in December 1982, February 1991, and March 2025. In all three months, most $\hat{S}_{it}^{call}[k]$ are negative and violate the restriction that call option values be weakly positive.[117] The panel regression (C.2) does

[116] Unlike our call adjustment function, a call dummy cannot distinguish bond-specific call protection or call price schedules. GZ-12 also include industry and rating fixed effects.

[117] This does not necessarily mean that the panel regression adjustment does not work well to adjust

not impose this restriction, because it is a heavily reduced-form yield adjustment for callable bonds, and nothing ensures that its coefficients produce weakly positive option values. GZ-style panel regressions can therefore do more harm than good to adjust observed yield spreads for embedded call options.

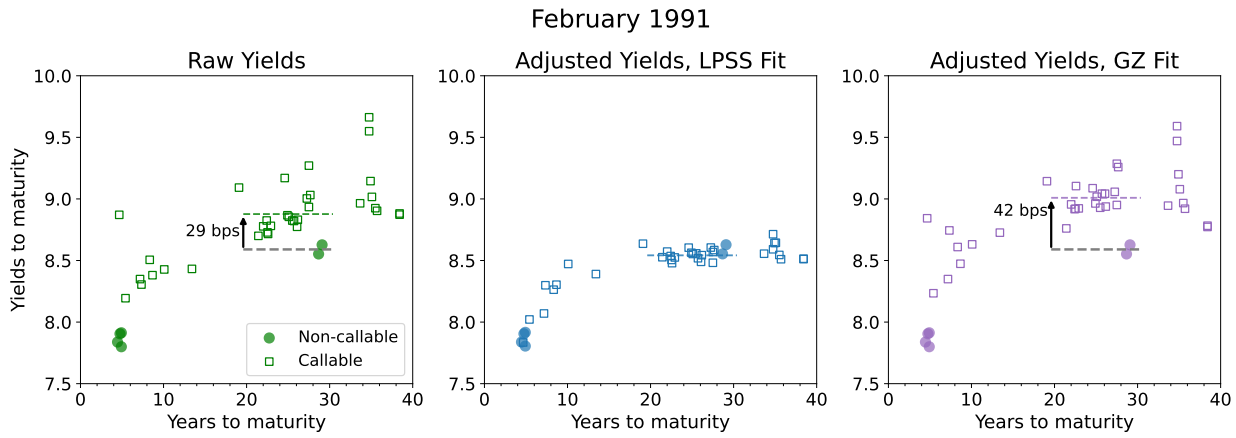


Figure C.39: Call Option Adjustment Comparison to Benchmark

Notes: The left panel shows raw Aaa$^+$ yields-to-maturity in February 1991. The middle panel shows our call option-adjusted yields using our callable bond adjustment function. The right panel shows the panel regression call option adjustment from Gilchrist and Zakrajšek, 2012. Dots are non-callable bonds and squares are callable bonds.

Figure C.39 plots the call option adjustments for February 1991. The left panel shows raw yields to maturity, which are widely dispersed. The right panel shows yields adjusted with the GZ-12 option adjustment. Besides implying negative call option values, this adjustment does little to reduce the dispersion of yields in the left panel, so any yield curve fitted through the adjusted yields inherits it. The middle panel shows our call option-adjusted yields. They deliver both (i) non-negative option values, since adjusted yields fall relative to the left panel, and (ii) homogeneous adjusted yields across callability, allowing us to infer a smooth yield curve across the term structure. Our adjustments, though complicated in form, outperform simpler panel regression methods on both theoretical and empirical grounds.

C.7 Additional Results

Figure C.40 plots our funding advantage at several maturities over 1860–2026. During the Civil War, the funding advantage falls from about 1.5–2% to zero by 1862 and turns negative

for callability across the entire GZ bond universe, nor that their default/call option adjusted forecasting variable (the Excess Bond Premium) is mismeasured. However, it does show that a reduced form call option adjustment gives little reason to expect theoretically consistent or empirically compelling results.

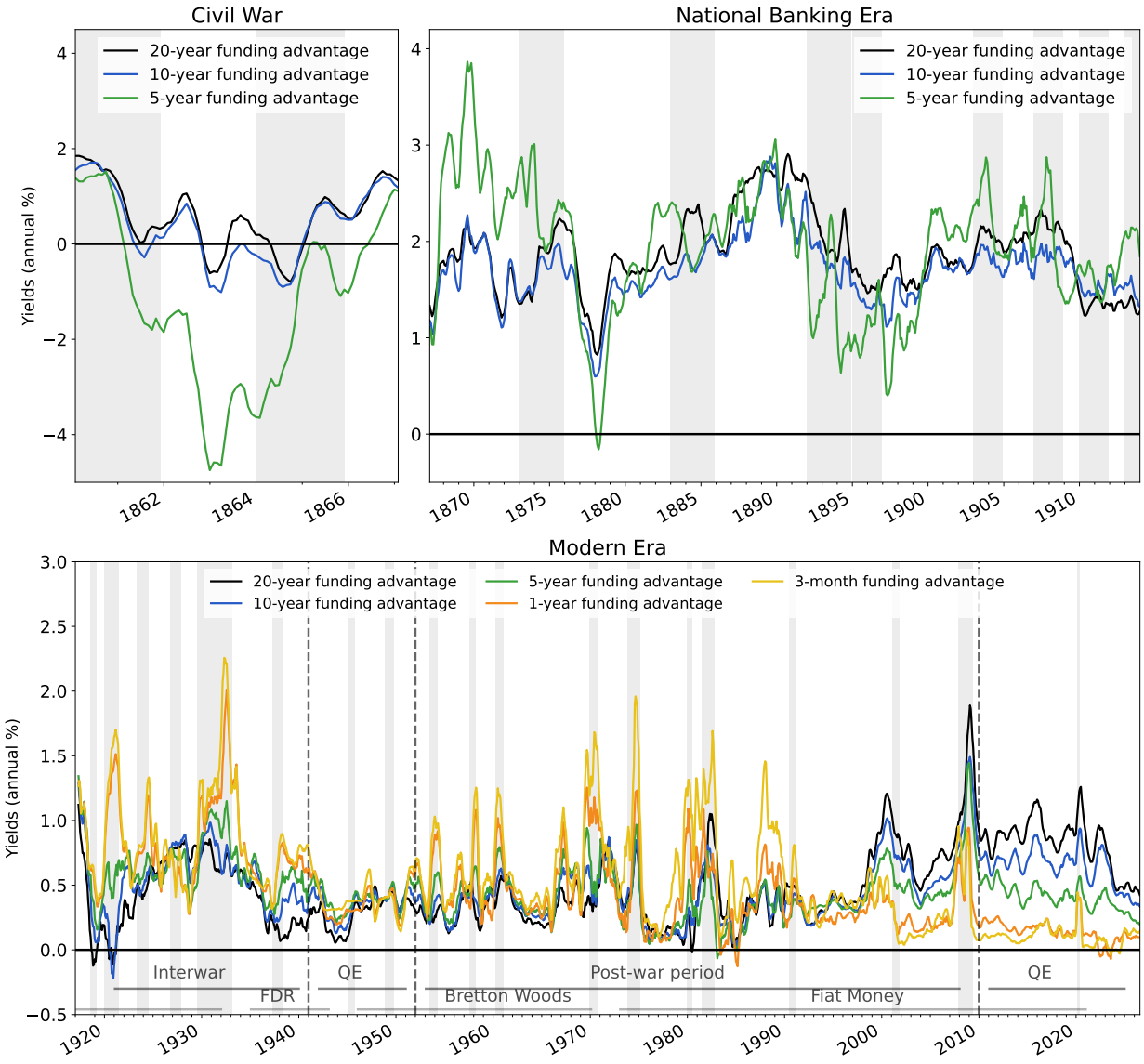


Figure C.40: Funding Advantage Across Maturities, 1860–2026

Notes. Our estimates of the funding advantage, the Aaa$^+$ corporate minus Treasury zero-coupon yield, in annual percent. The panels cover the Civil War (top left), the National Banking Era (top right), and the Modern Era (bottom). The top panels show 5-year (green), 10-year (blue), and 20-year (black) maturities. The bottom panel adds 1-year (orange) and 3-month (yellow) maturities. Light gray bands are NBER recessions. In the bottom panel, dashed vertical lines denote financial regulatory eras and the labels at the bottom mark monetary and policy regimes.

in 1863–65, most sharply at 5 years, where it reaches about −5%. It recovers to 1–1.5% by 1867. Throughout the National Banking Era the three maturities move together between roughly 1% and 3%, with the 5-year the most volatile. Since 1917, the funding advantage averages about 0.5% with brief spikes, largest at short maturities, in 1920–21, 1933, and the

1970s and early 1980s. After the GFC the term structure fans out, as the 20-year funding advantage stays between 0.5% and 1.2% while the 1-year and 3-month fall to near zero.

Figure C.41 shows the tax rates we infer for Treasurys over 1917–41, before fully taxable bonds were issued. Individuals are the marginal buyers until about 1927, banks until about 1931, and individuals again afterward. The inferred income tax rate peaks near 39% in 1918, stays near 30% in the early 1920s, and falls to about 15% by 1927. While banks are marginal, it equals the maximum corporate income tax rate, between 11% and 13.5%. It then jumps to about 20% in 1932 and settles at 25–30% from 1935 to 1941. These rates enter our tax adjustment functions for the tax-exempt Treasurys of this period.

Figure C.42 shows how callable bonds distort the commonly-used index-based spread in October 2001, the smaller episode noted in Subsection 4.1. Treasury yields rise from about 2% at the short end to 5% at 20 years and are flat beyond. Only five Aaa corporate bonds mature beyond 20 years. The two non-callable bonds yield about 6%, while the three callable bonds, which are well in the money, yield between 7.1% and 7.4%. Moody's Aaa index, which averages all five, stands 176 basis points above the 20-year Treasury yield, but the two non-callable bonds alone imply a spread of 100 basis points. The call options account for the remaining 76 basis points.

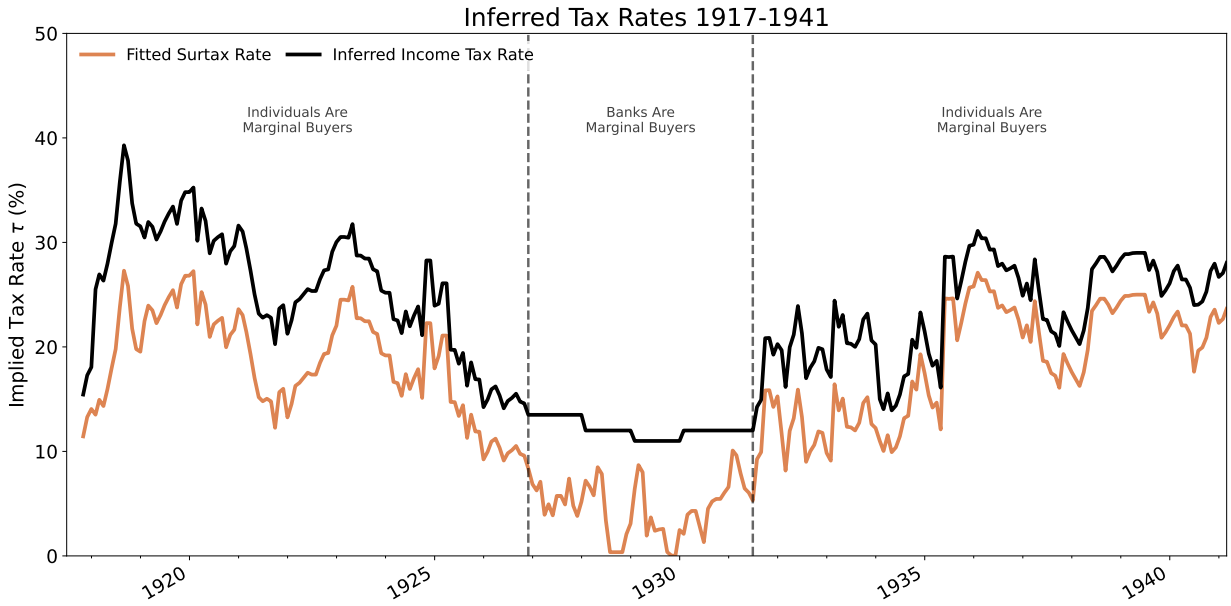


Figure C.41: Pre-1941 Tax Rates on Treasurys

Notes: Figure shows our inferred income tax rate (black) and estimated surtax rate (orange) on Treasurys before fully taxable bonds were issued. When individuals were the marginal buyer, we compute the black line by adding the normal income tax rate from the IRS to the orange line. When banks were the marginal buyer, we take the black line to be the maximum IRS corporate income tax rate.

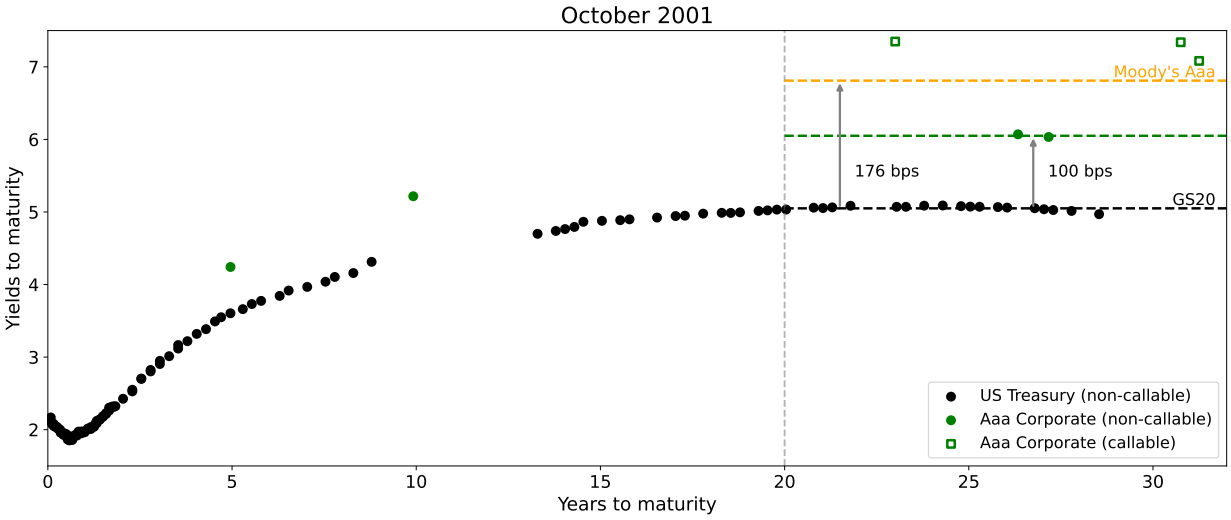


Figure C.42: Callable Bonds in the Moody's Aaa Index, October 2001.

Notes: The figure plots yields-to-maturity (y-axis) against years-to-maturity (x-axis) for Treasurys and Aaa corporate bonds in October 2001. Black markers are regular taxable Treasurys and green markers are Aaa corporate bonds. Dots denote non-callable bonds and squares denote callable bonds. The dashed yellow line is the Moody's Seasoned Aaa yield index, and the dashed black line is GS20 on October 31, 2001.

D Government Budget Constraint Arithmetic

We use the government's discount function to set up the government's intertemporal budget constraint. We can express the period-t market value of the government debt portfolio in the following (equivalent) forms:

$$\sum_{i\in\mathcal{N}_t} p_{i,t}B_{i,t} = \sum_{i\in\mathcal{N}_t}\sum_{j=1}^{\infty} q_t^{(j)} cf_{i,t}^{(j)} B_{i,t} = \sum_{j=1}^{\infty} q_t^{(j)} \sum_{i\in\mathcal{N}_t} cf_{i,t}^{(j)} B_{i,t} =: \sum_{j=1}^{\infty} q_t^{(j)} b_t^{(j)},$$

where the last expression defines $b_t^{(j)} := \sum_{i\in\mathcal{N}_t} cf_{i,t}^{(j)} B_{i,t}$ as the number of $t+j$ dollars that the government has at time t promised to deliver. We call the sequence $\mathsf{b}_t := \{b_t^{(j)}\}_{j\geq 1}$ the *zero-coupon equivalent* government debt portfolio.

The government consists of a fiscal authority (Treasury) and a monetary authority (Fed). Each period t, the Treasury enters with a stock of promised payments b_{t-1}, spends g_t, raises taxes τ_t and finances the resulting deficit/surplus by "restructuring" its debt portfolio in the form of new issues of zero-coupon bonds b_t. Simultaneously at period t, the Fed creates/contracts new money $\Delta M_t = M_t - M_{t-1}$ and pays interest on the outstanding money supply (if applicable) at the predetermined rate ι_{t-1}.[118] It follows that the period t government budget constraint (Treasury + Fed) can be written as

$$b_{t-1}^{(1)} + g_t - \tau_t + \iota_{t-1}M_{t-1} = \sum_{j=1}^{\infty} q_t^{(j)} \left(b_t^{(j)} - b_{t-1}^{(j+1)} \right) + \Delta M_t$$

that is, period-t interest payments, $b_{t-1}^{(1)}$ and $\iota_{t-1}M_{t-1}$, and primary deficit, $(g_t - \tau_t)$, must be funded by refinancing the government debt portfolio at market prices $\{q_t^{(j)}\}_{j\geq 1}$ or by printing money ΔM_t.

We add and subtract corporate bond prices and rearrange terms to obtain:

$$\sum_{j=1}^{\infty} q_t^{(j-1)} b_{t-1}^{(j)} = \tau_t - g_t + \sum_{j=1}^{\infty} (q_t^{(j)} - \tilde{q}_t^{(j)}) b_t^{(j)} + \sum_{j=1}^{\infty} \tilde{q}_t^{(j)} b_t^{(j)} + \Delta\mathcal{M}_t \tag{D.1}$$

where we define:

$$\Delta\mathcal{M}_t = \Delta M_t - \iota_{t-1}M_{t-1}$$

new money created by the monetary authority less interest paid on outstanding money.

Assumption 4. Assume $\xi_{t,t+1}$

To get the lifetime budget constraint, we need to iterate Equation (D.1) forward. This

[118] ι_t plays the role of the (net) interest rate on reserves.

requires forecasting future prices, which we do by imposing the corporate bond Euler equation:

$$\tilde{q}_t^{(j)} = \mathbb{E}_t\left[\xi_{t,t+1}\tilde{q}_{t+1}^{(j-1)}\right], \quad \forall j$$

in the government budget constraint. This gives:

$$\begin{aligned}\sum_{j=1}^{\infty} q_t^{(j-1)} b_{t-1}^{(j)} &= \tau_t - g_t + \Delta\mathcal{M}_t + \sum_{j=1}^{\infty}(q_t^{(j)} - \tilde{q}_t^{(j)})b_t^{(j)} + \sum_{j=1}^{\infty}\mathbb{E}_t\left[\xi_{t,t+1}\tilde{q}_{t+1}^{(j-1)}\right] b_t^{(j)} \\ &= \tau_t - g_t + \Delta\mathcal{M}_t + \sum_{j=1}^{\infty}(q_t^{(j)} - \tilde{q}_t^{(j)})b_t^{(j)} + \mathbb{E}_t\left[\xi_{t,t+1}\sum_{j=1}^{\infty}(\tilde{q}_{t+1}^{(j-1)} - q_{t+1}^{(j-1)})b_t^{(j)}\right] \\ &\quad + \mathbb{E}_t\left[\xi_{t,t+1}\sum_{j=1}^{\infty} q_{t+1}^{(j-1)} b_t^{(j)}\right].\end{aligned}$$

By adding and subtracting corporate bond prices, this can be written as

$$\begin{aligned}\sum_{j=1}^{\infty} q_t^{(j-1)} b_{t-1}^{(j)} &= \tau_t - g_t + \Delta\mathcal{M}_t + \sum_{j=1}^{\infty}(q_t^{(j)} - \tilde{q}_t^{(j)})b_t^{(j)} \\ &\quad + \mathbb{E}_t\left[\xi_{t,t+1}\sum_{j=1}^{\infty}(\tilde{q}_{t+1}^{(j)} - q_{t+1}^{(j)})b_t^{(j+1)}\right] + \mathbb{E}_t\left[\xi_{t,t+1}\sum_{j=1}^{\infty} q_{t+1}^{(j-1)} b_t^{(j)}\right].\end{aligned}$$

Iterating the final term forward and imposing the transversality condition gives:

$$\begin{aligned}\sum_{j=1}^{\infty} q_t^{(j-1)} b_{t-1}^{(j)} &= \mathbb{E}_t\left[\sum_{s=0}^{\infty}\xi_{t,t+s}(\tau_{t+s} - g_{t+s})\right] + \sum_{j=1}^{\infty}(q_t^{(j)} - \tilde{q}_t^{(j)})b_t^{(j)} \\ &\quad + \mathbb{E}_t\left[\sum_{s=0}^{\infty}\xi_{t,t+s}\Delta\mathcal{M}_{t+s}\right] + \mathbb{E}_t\left[\sum_{s=1}^{\infty}\xi_{t,t+s}\sum_{j=1}^{\infty}\left(q_{t+s}^{(j)} - \tilde{q}_{t+s}^{(j)}\right)\left(b_{t+s}^{(j)} - b_{t-1+s}^{(j+1)}\right)\right]\end{aligned}$$

which after rearranging terms gives:

$$\begin{aligned}\sum_{j=1}^{\infty} q_t^{(j-1)} b_{t-1}^{(j)} &= \underbrace{\mathbb{E}_t\left[\sum_{s=0}^{\infty}\xi_{t,t+s}\Big(\tau_{t+s} - g_{t+s}\Big)\right]}_{(i)} + \underbrace{\sum_{j=1}^{\infty}\Big(q_t^{(j)} - \tilde{q}_t^{(j)}\Big)b_{t-1}^{(j+1)}}_{(ii)} + \underbrace{\mathbb{E}_t\left[\sum_{s=0}^{\infty}\xi_{t,t+s}\Delta\mathcal{M}_t\right]}_{(iii)} \\ &\quad + \underbrace{\mathbb{E}_t\left[\sum_{s=0}^{\infty}\xi_{t,t+s}\underbrace{\left\{\sum_{j=1}^{\infty}\Big(q_{t+s}^{(j)} - \tilde{q}_{t+s}^{(j)}\Big)\Big(b_{t+s}^{(j)} - b_{t-1+s}^{(j+1)}\Big)\right\}}_{\text{non-pecuniary revenue in period } t+s}\right]}_{(iv)}. \qquad \text{(D.2)}\end{aligned}$$

where $\Delta\mathcal{M}_t$ is the new money created by the monetary authority net of interest paid on reserves. The market value of outstanding debt, interest included, is thus the sum of three terms: (i) the present value of future primary surpluses; (ii) the premium on the *stock* of existing long-term debt, which vanishes if the government issues only one-period debt; (iii) the present value of seigniorage from future government money creation; and (iv) the present value of the "non-pecuniary revenue" earned by issuing *new* debt more cheaply than the private sector could. Assuming the government keeps issuing new long-term debt, terms (ii) and (iv) are both positive whenever the funding advantage is positive ($\mathsf{q}_t > \widetilde{\mathsf{q}}_t$), so part of the value of government debt is backed not by future surpluses but by the implicit tax generated by the government's funding advantage on new debt.

D.1 Flow Revenue

We also decompose the flow revenue from government funding advantage. To do this, we rearrange the intertemporal government budget constraint to identify the capital gains on existing debt and the premia on new issuance. The intertemporal budget constraint at time t can be written as:

$$\begin{aligned}
\sum_{j=1}^{\infty} q_t^{(j-1)} b_{t-1}^{(j)} &= \tau_t - g_t + \Delta\mathcal{M}_t + \sum_{j=1}^{\infty} q_t^{(j)} b_t^{(j)} \\
\Rightarrow g_t - \tau_t - \Delta\mathcal{M}_t &= -\sum_{j=1}^{\infty} q_t^{(j-1)} b_{t-1}^{(j)} + \sum_{j=1}^{\infty} q_t^{(j)} b_t^{(j)} \\
&= -\sum_{j=1}^{\infty} (q_t^{(j-1)} - q_{t-1}^{(j-1)}) b_{t-1}^{(j)} + \sum_{j=1}^{\infty} (q_t^{(j)} b_t^{(j)} - q_{t-1}^{(j-1)} b_{t-1}^{(j)})
\end{aligned}$$

where the first term is the capital gains/loss from the revaluation of debt between $t-1$ and t and the second term is the market value of new issuance. We can write this in spreads to get:

$$\begin{aligned}
g_t - \tau_t - \Delta\mathcal{M}_t = & -\sum_{j=1}^{\infty} (\tilde{q}_t^{(j-1)} - \tilde{q}_{t-1}^{(j-1)}) b_{t-1}^{(j)} - \sum_{j=1}^{\infty} ((q_t^{(j)} - \tilde{q}_t^{(j)}) - (q_{t-1}^{(j)} - \tilde{q}_{t-1}^{(j)})) b_{t-1}^{(j+1)} \\
& + \sum_{j=1}^{\infty} (\tilde{q}_t^{(j)} b_t^{(j)} - \tilde{q}_{t-1}^{(j-1)} b_{t-1}^{(j)}) + \sum_{j=1}^{\infty} ((q_t^{(j)} - \tilde{q}_t^{(j)}) b_t^{(j)} - (q_{t-1}^{(j)} - \tilde{q}_{t-1}^{(j)}) b_{t-1}^{(j+1)})
\end{aligned}$$

where the second and fourth terms are the contributions from premium changes and non-pecuniary benefit on new issuance.

We show the decomposition of the normalized flow revenue to the government (the black line in the bottom panel of Figure 7) in Figure D.43. The teal and orange bars decompose

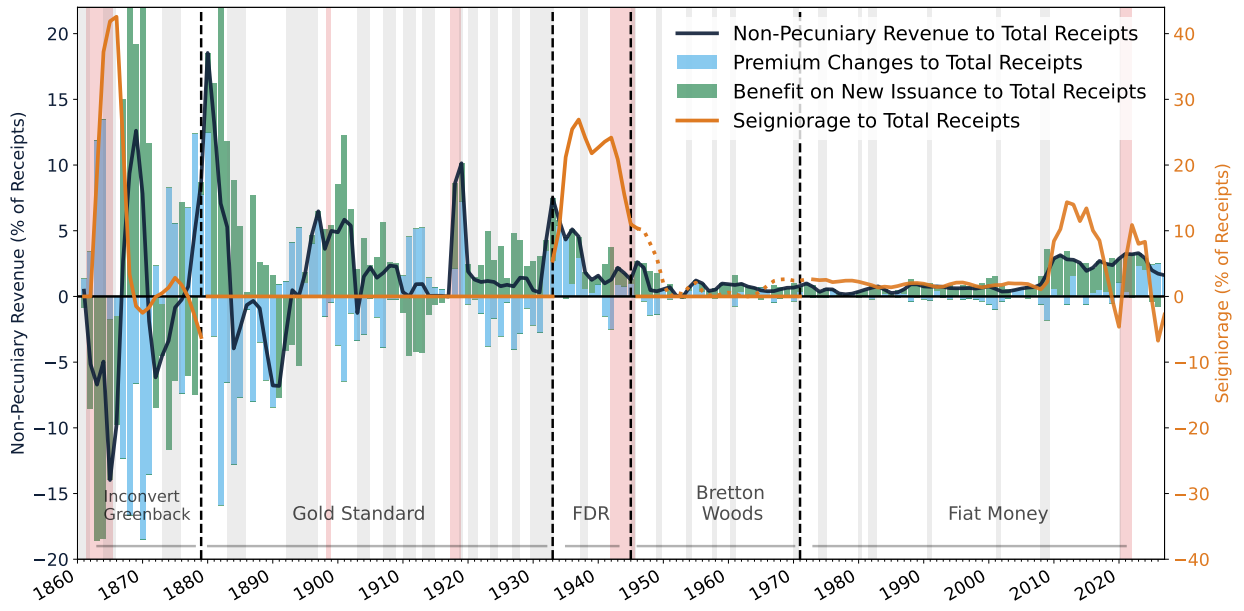


Figure D.43: Non-Pecuniary Revenue as percentage of Total Federal Revenue.

Notes: Components of the non-pecuniary revenue is expressed relative to total annual federal revenue: receipts plus seigniorage plus non-pecuniary revenue. "Premium changes" are defined as $-\sum_{j=1}^{\infty}((q_t^{(j)}-\tilde{q}_t^{(j)})-(q_{t-1}^{(j)}-\tilde{q}_{t-1}^{(j)}))b_{t-1}^{(j+1)}$. "Benefit on new-issuance" is defined as $\sum_{j=1}^{\infty}((q_t^{(j)}-\tilde{q}_t^{(j)})b_t^{(j)}-(q_{t-1}^{(j)}-\tilde{q}_{t-1}^{(j)})b_{t-1}^{(j+1)})$. The two terms add up to the period-t non-pecuniary revenue, i.e. $\sum_{j=1}^{\infty}(q_t^{(j)}-\tilde{q}_t^{(j)})(b_t^{(j)}-b_{t-1}^{(j+1)})$ from term (iv) in Equation (D.2). The light gray intervals depict NBER recessions. The light red bands, from left to right, depict the Civil War, the Spanish-American War, WW1, WW2, and COVID.

the period-t non-pecuniary revenue into the contribution from premium changes (the light blue bar) and the spread on new issuance (the green bar). The figure shows that it was costly for the government to establish its funding advantage at the end of the Civil War: the government had to absorb capital losses when Treasurys started to trade at a premium in the late 1860s. However, this was a short-term cost that reaped significant long-term gains. During periods of crisis in the twentieth century (World Wars, New Deal, GFC, COVID), the government used its funding advantage as a significant form of revenue. This suggests that in these periods, the government preferred financing expansionary measures by having the private sector hold its debt as opposed to raising taxes.

E Default Risk

Throughout the paper we interpret the Aaa-Treasury spread as a funding advantage of the government that reflects the liquidity, convenience, and/or regulatory benefits of holding US Treasurys relative to Aaa corporate bonds. An alternative explanation of the spread is that Aaa corporate bonds may yield more than Treasurys because they carry default risk. In this section we argue that realized default risk in our sample is very small and that market prices of default risk cannot account for more than a modest fraction of the spread unless agents have very high risk aversion (a "default-risk-premium puzzle") or believe in a latent risk state which has never realized (a "rare-disaster puzzle"). This section proceeds in five steps. First, we document realized default rates and realized credit losses in our sample. We then ask what price the market could plausibly attach to this risk from four angles. The market prices of default protection, the theoretical capacity of risk premia, firm-level measures of default risk and the worst realized stress episode in our sample.

E.1 Realized Default Rates

In this section we calculate realized default rates on every bond that was ever included in our Aaa^+ sample. We find that for bonds rated Aaa in 1920 or later, the cumulative twenty-year default rate is 0.65 percent. Comparing this to Moody's published rate of 1.3 percent for its broader Aaa issuer universe over the same period, this brackets default risk between 0.65 and 1.3 percent at the twenty-year horizon.

The sample underlying this calculation is our estimation panel, the same set of bonds from which the paper's Aaa^+ corporate yield curve is estimated. The panel only contains senior bonds of non-financial issuers, excluding sinking fund and convertible bonds, that carry an Aaa rating from every agency rating them at the time we observe their prices. It is thus a strict subset of Moody's Aaa issuer universe, which explains why its realized default rate sits below Moody's. A bond leaves our panel once downgraded and may default many years later. We accordingly track every bond from the month it first enters the panel until its maturity or call, through any downgrades along the way and record whether it ever defaults. We restrict to bonds whose full lives fall inside the sample period so that no outcome is unobserved.

Defaults are identified by matching our bond level observations to the Moody's Default and Recovery Database (DRD), which records default events both on the issuer and the instrument level as well as the type of credit event (e.g. missed principal payment, missed interest payment, bankruptcy, etc.) and the instrument's market price thirty days after default. If we cannot reliably match a bond to the DRD we rely on the issuer match and

treat any issuer level default event as a default of the bond in our sample, irrespective of whether the bond was actually affected.[119] To define defaults we follow Moody's definition: a missed or deferred interest or principal payment, a payment moratorium, receivership, bankruptcy, or a distressed exchange. We do not count maturity extensions and other indenture modifications or preferred dividend omissions. A default triggered by a missed interest payment places the entire bond in default. We therefore measure the loss on the whole position as one minus the bond's market price thirty days after the event (following Moody's loss-given-default measure) not as the value of the missed coupons alone. The average recovery rate for bonds in our sample for which the DRD reports a recovery value is 40.1% over the 1909-2024 period. For defaulted bonds with no recorded recovery price, we assume a recovery rate of 40%.[120]

We calculate default rates for bond cohorts formed on rating status at the 5-year horizon, 20-year horizon, and over the entire lifetime of bonds. For the lifetime calculation we form, for each start year s, the expanding "cohort" $\mathcal{C}(s)$ of all bonds that we observe with an Aaa rating at any date in year s or later. This includes seasoned bonds that entered the panel before s and still carried the rating in year s. Bond i's entry date e_i is its first Aaa observation on or after the start of year s, and we assume the bond is held from e_i until it matures, is called, or defaults, through any rating downgrades along the way. We include only bonds whose stated maturity precedes the end of our sample, so that every bond's outcome is observed. Writing T_i for the date of bond i's exit at maturity or call, and τ_i for the date of the first default event recorded on the bond after e_i, with $\tau_i = \infty$ if the bond is repaid without defaulting, the *lifetime* default rate of a cohort $\mathcal{C}(s)$ is the pooled share

$$\mathrm{DR}_{\mathrm{life}}(s) = \frac{1}{|\mathcal{C}(s)|} \sum_{i \in \mathcal{C}(s)} \mathbf{1}_{\{\tau_i \le T_i\}}. \tag{E.1}$$

The denominator is fixed at cohort formation. A bond that is repaid at maturity or call counts as a non-default rather than leaving the sample, because its life is complete and its outcome known.

The five-year and twenty-year rates follow the methodology behind Moody's published cumulative *issuer-level* default rates, described in Hamilton and Cantor (2006), which we

[119] This convention can only overstate default rates in our sample. Our reported default rates are therefore upper bounds.

[120] Using the thirty-day price after default as the realized recovery likely overstates ultimate losses because prices often increase after thirty days. This is especially true when the default event is only a missed coupon payment rather than a missed principal or bankruptcy. For instance in September 1938 the Baltimore & Ohio Railroad deferred interest payments under a voluntary readjustment plan rather than enter receivership resumed but ultimately resumed payments and cleared its deferred coupons in full.

apply to our *bond-level* data. For every formation year s from 1909 through 2005 we form the annual sample $\mathcal{S}(s)$ of all bonds rated Aaa in year s (1909 is the first year Moody's ratings exist and 2005 is the last year whose twenty-year window closes within our sample). A bond re-enters every annual sample it survives with the Aaa rating. Time after qualification is divided into annual intervals $h = 1, \ldots, n$ where n is the horizon of five or twenty years. Let $x_s(h)$ denote the number of defaults among the bonds of $\mathcal{S}(s)$ in interval h, and $n_s(h)$ the number of its bonds at risk at the start of the interval. The "marginal" default rate in interval h, computed across all annual samples formed in year s_0 or later, and the cumulative n-year rate that annualizes it, are as in Hamilton and Cantor (2006), equations (2.4) and (2.5):

$$\bar{d}(h) = \frac{\sum_{s \geq s_0} x_s(h)}{\sum_{s \geq s_0} n_s(h)}, \qquad \mathrm{DR}_n(s_0) = 1 - \prod_{h=1}^{n} \Big(1 - \bar{d}(h)\Big). \tag{E.2}$$

What remains to specify is the at-risk denominator $n_s(h)$, and here we follow Hamilton and Cantor's two methods in turn. Under the *unadjusted* method (their equation (3.1)), the at-risk population shrinks only through default,

$$n_s(h) = n_s(0) - \sum_{k<h} x_s(k),$$

so a bond repaid at maturity or call remains in the denominator to the end of the window as an observed non-default. The resulting rate is thus intuitively the share of cohort bonds observed to default within n years, the multi-cohort analog of equation E.1.

In contrast, the *withdrawal-adjusted* method recognizes that there are three possible end-of-period outcomes: default, survival and rating withdrawal. Moody's records a withdrawal (WR) when an issuer's rated debt is extinguished, so repaid bonds leave its data through withdrawals. In our panel the withdrawals $w_s(h)$ are therefore both actual rating withdrawals and retirements at maturity or call. The adjusted denominator removes them from the population at risk, counting withdrawals within the current interval for half, as in their equation (3.2):

$$n_s(h) = n_s(0) - \sum_{k<h} x_s(k) - \sum_{k<h} w_s(k) - \tfrac{1}{2}\, w_s(h).$$

Hamilton and Cantor (2006) favor the adjusted rate for two reasons. First, defaults that occur after a rating is withdrawn are unobserved, so unadjusted rates may understate true default risk. Second, the unadjusted rate confounds credit risk with the timing of retirements. A bond that matures early cannot default in the remaining years of the window, yet stays in the denominator as a survivor, so cohorts of shorter-lived bonds show mechanically lower unadjusted rates at any given credit quality. Moody's publishes its rates on the adjusted

basis, so we report adjusted rates for comparability, with the unadjusted rates in parentheses.

We calculate loss rates using an analogous methodology, weighting each default by the loss it caused rather than counting it as one. For the lifetime rate, the loss analog of equation (E.1) replaces the indicator with $\mathbf{1}_{\{\tau_i \le T_i\}}(1 - R_i)$, where R_i is bond i's recovery. For the n-year rates, the "marginal" loss rate $\bar{\ell}(h)$ replaces the default count $x_s(h)$ in equation (E.2) with the sum of losses $1 - R_i$ over the interval-h defaulters, and the cumulative loss rate accumulates it under the survival factors,

$$\mathrm{LR}_n(s_0) = \sum_{h=1}^{n} \bar{\ell}(h) \prod_{k<h} \Big(1 - \bar{d}(k)\Big),$$

which reduces to $\mathrm{DR}_n(s_0)$ when recoveries are zero. The recovery R_i is the bond's market price thirty days after default as a fraction of par. When no price is recorded we set R_i to 0.40, the average of the recorded recovery prices over 1920-2025. Bonds whose issuer defaulted but which are verified to have paid in full contribute no loss.

Table E.7 reports the 5-year, 20-year and lifetime default and loss rates for five cohort start dates. Five-year cumulative default rates are exactly zero throughout. For bonds entering the panel since 1909, which is the first year Moody's ratings exist, the twenty-year withdrawal adjusted cumulative default rate is 0.73% and the loss rate is 0.36% of par. Starting instead in 1925, once Moody's coverage was well established, the rate falls to 0.40% (0.27% without the withdrawal adjustment). For every cohort of bonds formed after World War II realized default rates at any horizon are exactly zero. Of the 367 sample bonds rated Aaa in 1945 or later, each followed from the day it entered to the day it matured or was called, including through any subsequent downgrades, not one defaulted, nor did any issuer default while one of these bonds was outstanding. The entire measured rate is thus driven by railroad bonds that were rated Aaa in the 1910s and 1920s but subsequently failed during the Great Depression or many years later, long after being downgraded below investment grade.[121]

These realized rates sit well below Moody's own published cumulative default rate of 1.3% for Aaa rated *issuers* over a twenty-year horizon between 1920-2024. One reason for this is that our unit of observation is the bond, not the issuer and bond seniority matters in failure states. Moody's records an issuer as defaulted when it misses a payment on *any* of its rated debt, however junior the obligation, even when holders of its senior bonds are paid in full. Indeed, several senior bonds in our sample paid in full while their issuer defaulted on subordinated debt.[122] Another reason is that our sample is a strict subset of the Moody's

[121]The last one being the Penn Central bankruptcy of 1970.

[122]The average Moody's *issuer-level rating* at the time of default was B. This rating reflects the average

Table E.7: Default and loss rates of the Aaa$^+$ sample

		Default rate				Loss rate			
Cohort Rated Aaa	N	5-year	20-year		lifetime	5-year	20-year		lifetime
in 1909 or later	528	0.00	0.73	(0.51)	1.14	0.00	0.36	(0.25)	0.55
in 1920 or later	513	0.00	0.65	(0.44)	0.78	0.00	0.31	(0.21)	0.35
in 1925 or later	492	0.00	0.40	(0.27)	0.81	0.00	0.21	(0.14)	0.37
in 1930 or later	439	0.00	0.05	(0.04)	0.23	0.00	0.03	(0.02)	0.14
in 1945 or later	367	0.00	0.00	(0.00)	0.00	0.00	0.00	(0.00)	0.00
Moody's Aaa issuers 1920-2024	issuer cohorts	0.13	1.3						

Notes: The 5-year and 20-year columns follow the cumulative default rate methodology used by Moody's, described in Hamilton and Cantor (2006) with the at-risk denominator adjusted for rating withdrawals, which in our panel are retirements at maturity or call. Figures in parentheses are the unadjusted variant, which keeps repaid bonds in the denominator as observed non-defaults. The lifetime columns are the pooled shares of equation (E.1), following every qualifying bond to maturity, call, or default. Loss rates value each default at one minus its recovery, the bond's market price thirty days after default (we assume the sample average of 40 percent where no price is recorded). All rates use the issuer-attributed upper-bound convention. The Moody's row shows the published issuer-weighted average cumulative 20-year rates, 1920-2024.

Aaa universe. To include a bond, we require a unanimous Aaa rating on the specific bond across all rating agencies we observe, seniority, no sinking fund and no financial issuers. Even among Aaa-rated bonds our sample thus includes only the safest issues, minimizing default risk. Indeed, every default in our sample is a railroad bond rated Aaa pre-1930 and defaulting during the Great Depression or many years later after it has long been downgraded (the average rating at default is Baa).

There are several possible interpretations of the declining default rates in our sample over time. First, the quality and accuracy of ratings may have improved over time as the market evolved and Moody's methodology became more sophisticated. The Great Depression likely also caused a reevaluation of credit risk. Second, the risk itself may have changed, because monetary and financial policy became more protective of the corporate bond market after the Depression, so that pre-WWII default realizations overstate post-war risk. Third, since our default rates are driven by railroads failing in the 1930s or later, they may reflect a sectoral change that eventually brought down even top rated issuers. We do not take a stand among these interpretations. The expected loss they must price is bounded by the same numbers in either case. Even taking the full 1909 history at face value, realized credit losses amount

rating of the issuer's senior unsecured debt, not the rating of any particular bond.

to at most 0.80 percent of par over the bonds' entire lifetime, which is a few basis points per year, and the following sections show that no plausible premium on losses of this size can explain the observed spread.

E.2 The Price of Default Protection

This section asks how much of the Aaa$^+$-Treasury spread can be accounted for by the price at which the market actually sells protection against default risk. For the modern period we can observe this price via credit default swaps (CDS). A CDS premium is the annual rate at which the market sells protection against a credit event on the issuer's senior unsecured debt. We follow Longstaff et al. (2005) (henceforth LMN) to subtract CDS premia from Aaa-Treasury spread to split the spread into a default and a non-default component. LMN conclude that over their March 2001 to October 2002 sample period the default component accounts for the majority of corporate spreads in every rating class, including their high grade bucket. However, since few Aaa issuers had traded CDS in 2001 and 2002, their high grade bucket pools Aaa and Aa firms and of its seven companies only one carries a Aaa rating, with most of the companies being financials. Their conclusion for "Aaa/Aa" credits therefore mainly describes financial Aa issuers.

We first replicate LMN's estimates on their own sample to verify our implementation. We then apply the identical construction to non-financial Aaa issuers over their original March 2001 to October 2002 window, and finally over January 2007 to December 2009 (GFC), when default fears and the price of default protection were at their highest. Few non-financial Aaa issuers had traded CDS in either window, so both samples are necessarily small and the results should be interpreted with caution.

The sample selection and construction follows LMN. CDS premia are Markit end-of-day composite par spreads for five-year protection on senior unsecured debt in U.S. dollars, averaged within the month. The five-year tenor is the most liquid point of the CDS curve. Bond prices come from our bond panel. The unit of observation is a firm-month, which enters the sample when the firm has a five-year CDS quote and two eligible bonds whose maturities bracket the five-year horizon. Eligible bonds are senior unsecured, fixed-coupon, dollar-denominated issues with at least $10 million outstanding and less than ten years to maturity, excluding bonds with embedded options other than make-whole calls and those with equity-linked or asset-backed features. Because no bond typically matures exactly five years out, the five-year horizon is bracketed by bonds on both sides of the horizon and the five-year bond spread is obtained by linearly interpolating between them. This avoids extrapolation and reduces the influence of idiosyncratic pricing errors in any single issue.

We follow LMN and summarize the price of default protection in two ways. The first is a model-free ratio that directly compares the firm's CDS premium with its five-year bond spread. For each eligible bond i of firm j in month t, let y_{ijt} denote its yield and let $y_t^f(c_{ij}, \tau_{ijt})$ denote the yield of a riskless bond with the identical coupon rate and maturity, priced off the benchmark riskless curve. The bond's spread is $S_{ijt} = y_{ijt} - y_t^f(c_{ij}, \tau_{ijt})$. Comparing bonds with identical promised cash flows keeps coupon and term-structure effects out of the spread. To obtain a five-year spread, we follow LMN and regress the bracketing bonds' spreads on their remaining maturities within each firm-month, $S_{ijt} = a_{jt} + b_{jt}\,\tau_{ijt} + \varepsilon_{ijt}$, and take the fitted value at the five-year horizon, $S_{jt}(5) = \hat{a}_{jt} + 5\,\hat{b}_{jt}$. The model-free ratio for firm j is

$$R_j = \frac{\overline{\mathrm{CDS}}_j}{\overline{S}_j(5)}, \tag{E.3}$$

the ratio of the firm's mean five-year CDS premium to its mean five-year spread across its firm-months. If the entire spread compensated for default, R_j would be close to one and the shortfall of R_j from one measures the non-default residual. We compute spreads against two riskless benchmarks, the constant-maturity Treasury curve that LMN use and the Treasury zero-coupon curve constructed in this paper, so that the results are comparable with the original paper with the rest of our analysis.

The raw ratio is informative but biased, because its numerator and denominator are not spreads on the same instrument. Duffie (1999) shows that a CDS premium equals the spread of a par floating-rate note issued by the same credit. Our denominator $S_{jt}(5)$, however, is the yield spread of a *fixed-coupon* bond, and Duffie and Liu (2001) show that the two spreads differ, because a fixed-coupon bond's yield spread also depends on its coupon size and the shape of the yield curve. For investment-grade issuers the fixed-coupon spread is the larger of the two, so the raw ratio understates the default share. LMN attempt to remove the bias with a parametric model that prices the observed fixed-coupon bonds off the default intensity implied by the CDS, so that the default component and the total spread refer to the same instrument. The model works as follows. Under the risk-neutral measure the firm defaults at the first jump of a Poisson process with intensity λ_t, and bondholders then recover a fraction $1 - w$ of par. Corporate bond cash flows are discounted at $r_t + \lambda_t + \gamma_t$, where r_t is the riskless rate and γ_t is a non-default spread that captures whatever extra return investors require for holding the corporate bond beyond compensation for default, most naturally for its illiquidity. The three processes are assumed mutually independent. Independence implies that riskless discounting enters only through the observable discount function $D(T) = E[\exp(-\int_0^T r_t\,dt)]$, which we read directly off the benchmark curve, so the dynamics of r_t never need to be specified. The default intensity follows the square-root

process

$$d\lambda{=}(\alpha-\beta\lambda)\,dt{+}\sigma\sqrt{\lambda}\,dZ_\lambda, \qquad\qquad d\gamma{=}\eta\,dZ_\gamma, \tag{E.4}$$

which permits mean reversion and keeps λ_t nonnegative, while the driftless specification for γ_t allows the non-default spread to take either sign. Under these assumptions the price of a corporate bond with coupon rate c and maturity T, written with continuous coupons for expositional clarity, is

$$\begin{aligned} CB(c,w,T) = {} & E\left[c\int_0^T \exp\left(-\int_0^t (r_s+\lambda_s+\gamma_s)ds\right)dt\right] \\ & + E\left[\exp\left(-\int_0^T (r_t+\lambda_t+\gamma_t)dt\right)\right] \\ & + E\left[(1-w)\int_0^T \lambda_t \exp\left(-\int_0^t (r_s+\lambda_s+\gamma_s)ds\right)dt\right], \end{aligned} \tag{E.5}$$

whose three terms are the present values of the promised coupons, of the principal, and of the recovery payment in default. The CDS is priced from its two legs. The premium leg, $s\,E[\int_0^T \exp(-\int_0^t (r_s+\lambda_s)\,ds)\,dt]$, is the value of the premium annuity paid until default or maturity, and the protection leg, $E[w\int_0^T \lambda_t \exp(-\int_0^t (r_s+\lambda_s)\,ds)\,dt]$, is the value of the loss payment at default. Setting the two equal and solving for the premium gives the par CDS spread

$$s = \frac{E\left[w\int_0^T \lambda_t \exp\left(-\int_0^t (r_s+\lambda_s)ds\right)dt\right]}{E\left[\int_0^T \exp\left(-\int_0^t (r_s+\lambda_s)ds\right)dt\right]}. \tag{E.6}$$

The identifying assumption of the model is that the non-default spread γ_t appears in equation (E.5) but not in equation (E.6). The corporate bond's cash flows are discounted at $r_t{+}\lambda_t{+}\gamma_t$, while both legs of the CDS are discounted at $r_t+\lambda_t$ only. This asymmetry is what allows the two processes to be separated. The CDS premium depends on λ_t alone, so it pins down the default component, and whatever part of bond yields the default component cannot explain is absorbed by γ_t. LMN justify the asymmetry by the contractual nature of a CDS. A bond is a security in fixed supply that can become scarce or special. A CDS is a contract in zero net supply, so new protection can be written at any time, positions are exited by offset rather than sale, and selling protection is as easy as buying it. The frictions that generate a non-default component in bond prices thus largely do not apply to CDS. Because the dynamics in (E.4) are affine, equations (E.5) and (E.6) have closed-form solutions, derived in LMN's Appendix A. The continuous-payment formulas are for readability only. In estimation we match the actual semiannual coupon and quarterly premium timing, as LMN do.

Estimation of λ_t and γ_t proceeds firm by firm. Given candidate parameters $(\alpha,\beta,\sigma,\eta)$, in

each month of the firm's sample we invert E.6 for the value of λ_t that reprices the observed five-year CDS premium exactly, and we then choose γ_t so that the bond prices implied by E.5 fit the observed yields of the bracketing bonds as closely as possible. The parameters are chosen to minimize the root-mean-squared yield error across all of the firm's months, with the recovery parameter fixed at $w = 0.50$ as in LMN and with σ and η required to be consistent with the sample volatilities of the fitted λ_t and γ_t series. The results are insensitive to w, which enters the bond and the CDS nearly symmetrically. By construction λ_t carries exactly the default risk that the CDS market prices, while γ_t absorbs the part of bond yields that the CDS cannot explain. The model fraction we report is

$$F_j = \frac{\bar{\lambda}_j}{\bar{\lambda}_j + \bar{\gamma}_j}, \tag{E.7}$$

where bars denote time-series means of the fitted processes for firm j. This is LMN's measure of the default share of the spread. Separating γ_t from bond-specific pricing noise requires a cross-section of at least two bonds in each month. With a single bond, γ_t mechanically absorbs that bond's entire pricing error and E.7 is not identified. Two firms in Panel A of Table E.8 have only one eligible bond. We relax LMN's two-bond rule to retain them for the model-free ratio, measuring $S_{jt}(5)$ from the single bond, but we do not report F_j for them.

Panel A of Table E.8 shows that in LMN's original 2001-2002 window the price of default protection accounts for between 0.21 and 0.40 of the spread at the firm level, with averages of 0.27 against the constant maturity Treasury benchmark and 0.30 against our zero-coupon curve. The corresponding figure in LMN is 0.49 for their pooled Aaa/Aa bucket, or 0.51 for their model-based measure. Where the model fraction is identified it confirms the model-free reading, ranging from 0.21 to 0.44, and the choice of riskless benchmark barely matters. For genuinely non-financial Aaa credits, the market thus priced roughly a quarter to a third of the spread as default over a distressed sample window from which LMN conclude that default dominates.

Panel B repeats the exercise for 2007-2009. The model-free ratio averages 0.53 under either benchmark, and the model fractions, estimated from few months, range from 0.55 to 0.99. Even during the deepest crisis in our sample, at the moment when the price of default protection was itself most inflated, roughly half of the Aaa spread was not priced as default.

Three features of the exercise make these estimates of the non-default residual conservative. First, the CDS premium is not a pure price of default losses. It also compensates the protection seller for the costs of supplying protection, including dealer balance-sheet and intermediation costs, so subtracting the full premium if anything attributes too much of the spread to default.

Table E.8: Default and non-default components of Aaa spreads: LMN-style decomposition

			Model-free ratio (CDS/Spread)		Model fraction
Firm	Months	5-year CDS (bp)	CMT	Ours	(CMT / ours)
Panel A: Aaa non-financial, March 2001 to October 2002					
Pfizer	5	20.4	0.23	0.24	0.21 / 0.22
Merck	7	25.9	0.34	0.40	0.37 / 0.44
J&J †	6	31.0	0.31	0.34	n.i.
ExxonMobil †	4	13.8	0.21	0.22	n.i.
Average			**0.27**	**0.30**	
Panel B: Aaa non-financial, 2007 to 2009 (GFC)					
J&J	29	33.8	0.41	0.42	0.65 / 0.55
Microsoft	7	32.7	0.65	0.65	0.66 / 0.99
Average			**0.53**	**0.53**	

This table decomposes Aaa corporate spreads into default and non-default components following Longstaff et al. (2005) (LMN). *Months* is the number of monthly observations in which the firm has a five-year CDS quote and eligible bonds bracketing the five-year horizon. *CDS (bp)* is the firm's mean five-year CDS premium over those months. The *model-free ratio* is R_j in equation (E.3), the mean CDS premium divided by the mean five-year yield spread over the riskless benchmark, and measures the share of the spread that the market's price of default protection can account for. "CMT" benchmarks spreads against constant-maturity Treasury yields (FRED H.15), the riskless curve used by LMN. "Ours" uses the paper's Treasury zero-coupon curves. The *model fraction* is F_j in equation (E.7), the LMN structural estimate of the default share of the spread. † indicates firms with a single eligible bond, a relaxation of LMN's two-bond rule. For these firms the model fraction is not identified (n.i.).

Second, the premium bundles expected default losses with the compensation investors demand for bearing default risk. Our five-year realized default rates from Section E are zero in every cohort and even Moody's issuer-level rate of 0.13 percent implies, at a 40 percent recovery, an expected loss of about 2 bp per year of insuring an Aaa bond over five years. However, the five-year CDS premia in Table E.8, the annual rates for a five year default insurance, range from 14 to 34 bp, an order of magnitude larger. This pattern is also reported in Berndt et al. (2018) who compare CDS premia with expected default losses estimated firm by firm from historical default frequencies and find that the market charges about twice the actuarial loss for the average investment-grade credit, with the multiple rising steeply in higher credit quality. Expected losses are thus a small fraction of the CDS

premium. To rationalize the premium investors would thus have to be extremely risk averse and attach a high price to bearing a tiny credit risk or they would have to fear a latent risk much larger than anything realized in the data. The next subsection formalizes both possibilities, by deriving a bound on the volatility of the pricing kernel, by asking how volatile the pricing kernel or equivalently how risk averse a representative investor would have to be to generate a compensation of this size.

Third, the collateral posted against CDS positions makes the causality between the premium and the funding advantage unclear. Protection sellers post collateral, typically cash or Treasury securities. This practice was standard under bilateral credit-support annexes in both of our sample windows and that the post-crisis reforms, through the Dodd-Frank clearing mandate and the margin requirements for uncleared swaps, have since made mandatory. The marginal cost of selling protection therefore includes the opportunity cost of holding Treasurys, which is precisely the funding advantage this paper measures. Causality can then run in either direction. One direction is that high default risk may raise the premium, which supports a large default share of the spread. Alternatively, a rise in the funding advantage, in a flight to safety episode for example, may raise the cost of posting collateral and push up CDS premia with no change in default expectations. Prices alone cannot distinguish these two interpretations. For this reason in Section E.3 we next ask how large of a default risk premium investor preferences can plausibly support.

E.3 How large a default risk premium can preferences support?

The previous subsection showed that the market price of Aaa default protection cannot account for most of the Aaa-Treasury spread. One possible interpretation is that CDS markets may misprice default risk and that under *some* specification of preferences a default risk premium could instead rationalize the funding advantage $\chi^{(j)}$ itself. In this subsection we bound that possibility. Following the logic of Hansen and Jagannathan (1991) our argument is general and does not rely on any particular model of preferences or beliefs. Under any pricing kernel, the default risk premium is given by the covariance between the pricing kernel and the default event. By the Cauchy-Schwarz inequality, this covariance cannot exceed the product of their standard deviations. Since the volatility of the default event is pinned down by observable default probabilities, the observed spread implies a lower bound on the volatility of the pricing kernel.

Environment. There is a representative investor with equilibrium stochastic discount factor $\xi_{0,t}$ who can invest at date zero in N zero-coupon Aaa bonds of maturity t. Default risk

has two primitive sources: a systematic default event $Z_t \in \{0,1\}$, in which all Aaa bonds default together and idiosyncratic default events $\tilde{D}_t^i \in \{0,1\}$, independent across bonds, independent of Z_t, and independent of $\xi_{0,t}$. Bond i defaults by time t, indicated by D_t^i, if either event occurs:

$$D_t^i = \underbrace{Z_t}_{\text{systematic}} + \underbrace{(1 - Z_t)\,\tilde{D}_t^i}_{\text{idiosyncratic}}.$$

We parameterize the two primitives by the resulting default probability and pairwise default correlation,

$$\mathbb{E}_0^P[D_t^i] = p \;\; \forall i, \qquad Corr_0^P(D_t^i, D_t^j) = \phi \;\; \text{for } i \neq j,$$

which correspond to $\mathbb{E}_0^P[Z_t] = \phi p/(1 - p(1-\phi))$ and $\mathbb{E}_0^P[\tilde{D}_t^i] = p(1-\phi)$. In default, bondholders recover a fraction δ of principal. Because the bonds are zero-coupon and priced to maturity, interim downgrades affect payoffs only through eventual default by t. The measured cumulative default probabilities we use below include this downgrade-then-default channel, so no separate downgrade state is needed.

Portfolio choice and equilibrium prices. The investor allocates wealth across the N bonds to maximize expected utility. Under the representative agent's SDF, the equilibrium asset pricing for each bond is:

$$\tilde{q}(t) = \mathbb{E}_0^P\Big[\xi_{0,t}\Big(1 - (1-\delta)\,D_t^i\Big)\Big], \tag{E.8}$$

and by symmetry the optimum is the equal-weighted portfolio, whose price equals that of any individual bond. What the investor is exposed to in equilibrium is therefore the aggregate default rate $D_t^{pf} = Z_t + (1 - Z_t)\frac{1}{N}\sum_i \tilde{D}_t^i$. It has the same mean p as a single bond but a smaller standard deviation,

$$\sigma_0^P\Big(D_t^{pf}\Big) = \sqrt{p(1-p)}\,\sqrt{\phi + \frac{1-\phi}{N}} \xrightarrow{N\to\infty} \sqrt{p(1-p)\,\phi}. \tag{E.9}$$

because the idiosyncratic events average out. With a single bond or perfectly correlated defaults ($N = 1$ or $\phi = 1$), the portfolio inherits the full Bernoulli volatility of an individual default indicator. With mostly idiosyncratic default risk and many bonds, the portfolio payoff approaches risklessness even though every bond can default. Only the systematic component survives diversification and only what survives diversification can earn a pre-

mium.

The default risk premium in equilibrium. Let Q^t denote the t-forward measure, the risk-neutral measure associated with the t-maturity numeraire, defined by $dQ^t/dP = \xi_{0,t}/\mathbb{E}_0^P[\xi_{0,t}]$. Combining this change of measure with the pricing equation (E.8) gives the risk-neutral default probability and expresses the default risk premium as a covariance:

$$\mathbb{E}_0^{Q^t}[D_t^{pf}] = \frac{1}{1-\delta}\left(1 - \frac{\tilde{q}(t)}{\mathbb{E}_0^P[\xi_{0,t}]}\right), \qquad \mathbb{E}_0^{Q^t}[D_t^{pf}] - p = Cov_0^P\left(\frac{\xi_{0,t}}{\mathbb{E}_0^P[\xi_{0,t}]},\, D_t^{pf}\right). \tag{E.10}$$

Applying the Cauchy-Schwarz inequality to the covariance, using (E.9), and assuming only that the premium is nonnegative, delivers a Hansen-Jagannathan bound for bond default risk:[123]

$$\frac{\mathbb{E}_0^{Q^t}[D_t^{pf}] - p}{\sqrt{p(1-p)}\sqrt{\phi + \frac{1-\phi}{N}}} \leq \frac{\sigma_0^P(\xi_{0,t})}{\mathbb{E}_0^P[\xi_{0,t}]}. \tag{E.11}$$

No structure has been imposed on the kernel beyond its existence. Equation (E.11) must hold for any preferences and any beliefs consistent with the observed prices.

Confronting the bound with the funding advantage. Suppose, as the hypothesis to be tested, that the entire funding advantage compensates for default risk. Then the wedge in the Euler equations (**??**) is one, $\Omega_{0,t} = 1$, so the Treasury discount function prices the hypothetical default-free bond, $q(t) = \mathbb{E}_0^P[\xi_{0,t}]$. Substituting this and the definition of the funding advantage (3.1), $\tilde{q}(t)/q(t) = e^{-t\chi^{(t)}}$, into the first equation of (E.10) pins down the risk-neutral default probability the Aaa-Treasury spread requires:

$$\mathbb{E}_0^Q[D_t^{pf}] = \frac{1 - e^{-t\chi^{(t)}}}{1-\delta} \tag{E.12}$$

Inserting (E.12) into (E.11), taking the limit $N \to \infty$, and annualizing the kernel volatility by $\sqrt{t}$ scaling, $\sigma_0^P(\xi_{0,t})/\mathbb{E}_0^P[\xi_{0,t}] = \sqrt{t}\,\sigma_0^P(\xi_{0,1})/\mathbb{E}_0^P[\xi_{0,1}]$, yields the object Figure E.44 plots. For each hypothetical physical default probability p, the minimum annualized kernel volatility

[123] Indeed, if we define $R_t - R_t^f = \frac{1-(1-\delta)D_t^{pf}}{\tilde{q}(t)} - \frac{1}{\mathbb{E}_0[\xi_{0,t}]}$, the return on a zero-cost long-short portfolio which goes long in the corporate bond portfolio and borrows at the risk free rate, then the LHS of (E.11) is the Sharpe ratio of this portfolio $\frac{\mathbb{E}_0[R_t - R_t^f]}{\sigma(R_t - R_t^f)}$ as in the classic Hansen-Jagannathan bound.

consistent with a spread of $\chi^{(t)}$,

$$\frac{\sigma_0^P(\xi_{0,1})}{\mathbb{E}_0^P[\xi_{0,1}]} \geq s(p;\chi) := \frac{1}{\sqrt{t}} \frac{\frac{1-e^{-t\chi}}{1-\delta} - p}{\sqrt{p(1-p)}\sqrt{\phi}}. \tag{E.13}$$

To translate volatilities into preferences, we back out risk aversion from the one-year kernel under power utility, $\sigma_0^P(\xi_{0,1})/\mathbb{E}_0^P[\xi_{0,1}] = \sqrt{e^{\gamma^2\sigma_c^2} - 1}$, with lognormal consumption growth of annual volatility $\sigma_c = 2\%$, as in Cochrane and Hansen (1992) and Cochrane (2005).

The calibrated inputs to (E.13) are chosen in the hypothesis's favor, i.e. to give default every chance to explain the spread. We set the recovery rate to $\delta = 40\%$. We set the default correlation to $\phi = 1/4$, at the ceiling the Basel framework imposes on the correlation parameter for corporate exposures.[124] Finally, we take the limit $N \to \infty$, as appropriate for the diversified marginal investor (the diversification factor in (E.9) converges quickly in N).

Figure E.44 reports the frontier at the twenty-year horizon in the top panel and at the five-year horizon in the bottom panel. The horizontal axis is the physical cumulative default probability p that an investor might assume for the Aaa portfolio over the horizon, plotted on a log scale. The left vertical axis is the minimum annualized volatility of the pricing kernel, $\sigma_0^P(\xi_{0,1})/\mathbb{E}_0^P[\xi_{0,1}]$, that the bound (E.13) requires at each p for default risk to explain the period-average spread. Combinations above the curve (shaded) can rationalize the spread, combinations below cannot. The right vertical axis restates the same magnitudes as the CRRA risk aversion γ that would generate them under power utility.

To rationalize the funding advantage as a default risk premium, an investor must either be extremely risk averse or believe in a latent risk that has never been realized (i.e. worse than the Great Depression). At the twenty-year horizon, evaluating the frontiers at either measured default rate delivers extreme requirements. The full-sample spread requires an annualized kernel volatility above the Hansen-Jagannathan bound implied by the equity risk premium itself and the GFC crisis period spreads require several times that bound. In units of risk aversion, the requirements run from roughly one and a half to three times the level at which the equity premium is already deemed a puzzle, about $\gamma = 20$ under our calibration, the value implied by the postwar US equity Sharpe ratio of roughly 0.4 (Hansen

[124] The Basel capital framework caps the supervisory correlation parameter for corporate exposures at 0.24 (Basel Framework, CRE31). The parameter comes from a one-factor Merton model where each firm's asset value loads on a common factor, the parameter is the share of asset-value variance explained by that factor, and a firm defaults when its asset value falls below a threshold calibrated to its default probability. This structure maps asset correlations into default correlations. The correlation between two default events, the ϕ in our bound, is determined by the probability that both firms' asset values cross their thresholds together. For Aaa issuers the thresholds sit far in the tail, so joint defaults are rare even when asset values move together. $\phi = 1/4$ therefore is a conservative assumption about the undiversifiable share of Aaa default risk.

and Jagannathan, 1991). At the five-year horizon the requirements are even larger. At Moody's measured rate the required volatility increases and at our sample's realized rate of exactly zero the frontier diverges, since no kernel can price a premium on an event that has never occurred.

The bound also carries back to the CDS evidence. The same logic that limits a default interpretation of the spread applies to the premia themselves. The exercise above asks whether default risk can explain the Aaa-Treasury spread. A weaker requirement is whether it can explain the price of default protection itself. Figure 12 therefore applies the bound (E.11) directly to the CDS premium. The horizontal axis converts the risk-neutral default expectation into the annualized premium of the corresponding CDS contract, so that observed quotes can be read against the frontier.

A premium $CDS^{(t)}$ buys protection against the same loss $1-\delta$ that the bondholder suffers, so it implies the risk-neutral default probability

$$\mathbb{E}_0^{Q^t}[D_t^{pf}] = \frac{1 - e^{-t\,CDS^{(t)}}}{1-\delta},$$

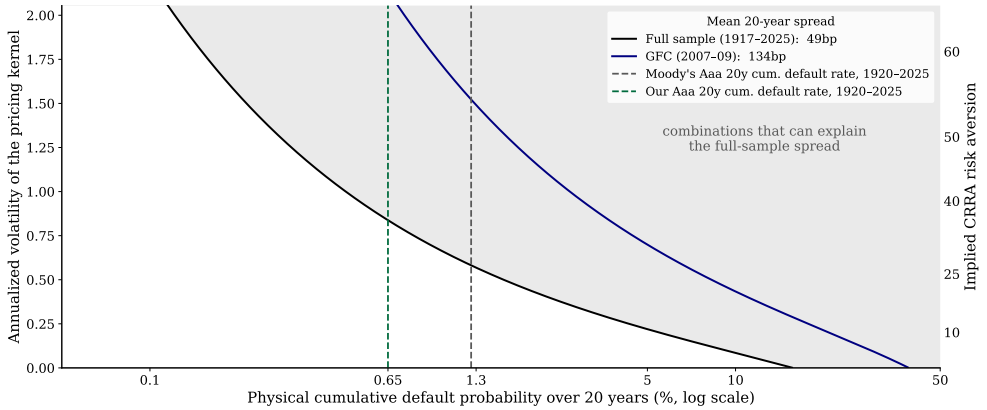
the analogue of (E.12) with the premium in place of the funding advantage. The minimum annualized kernel volatility consistent with a premium of $CDS^{(t)}$ is therefore $s\big(p; CDS^{(t)}\big)$, the frontier (E.13) evaluated at $\chi = CDS^{(t)}$. Figure 12 plots it at the five-year horizon with p fixed at Moody's measured Aaa rate and the premium on the horizontal axis, so that observed quotes can be read directly against the frontier.

Even this weaker requirement fails. At 14 to 34 bp against an expected loss of about 2 bp, Aaa CDS premia would embed a price of default risk that is difficult to reconcile with plausible risk aversion and kernel volatilities. This suggests that a substantial part of these premia reflects the costs of supplying protection rather than default risk alone (**?**). To the extent this is the case, subtracting the full premium in the previous subsection would have overstated the default component.

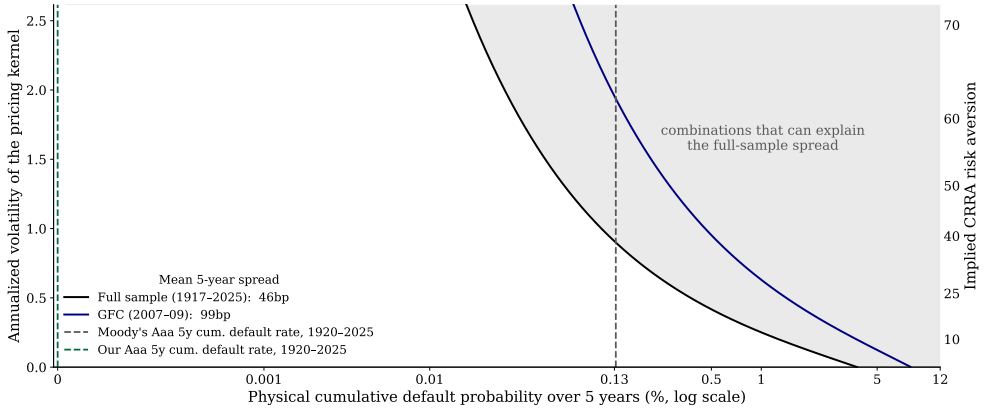
E.4 Distance to Default and the Excess Bond Premium

This section asks how much of the Aaa-Treasury spread can be attributed to default risk using the approach of Gilchrist and Zakrajšek (2012). We regress bond-level spreads on an observable, equity-based measure of the issuer's default risk, call the fitted part the default-risk component, and call the residual the excess bond premium. We run this decomposition on our own Aaa sample back to 1970.

For each bond and month we compute the spread as the bond's yield to maturity from its



(a) Twenty-year horizon



(b) Five-year horizon

Figure E.44: "Hansen-Jagannathan" bound for bond default risk.

Notes: Each solid curve traces the minimum annualized volatility of the pricing kernel required to rationalize the period-average Aaa-Treasury spread as compensation for default risk, computed from equation (E.13) at the spread reported in the legend. The horizontal axis is the physical cumulative default probability p of the Aaa portfolio over the horizon in log scale. The left vertical axis is the minimum annualized volatility of the pricing kernel required to rationalize the period-average Aaa-Treasury spread as compensation for default risk. The right vertical axis restates the same magnitudes as the CRRA risk aversion implied by power utility with two percent annual consumption growth volatility. The calibration sets the recovery rate to $\delta = 40\%$, the default correlation to $\phi = 1/4$, a diversified portfolio ($N \to \infty$) and the correlation between the pricing kernel and the portfolio default rate to its maximum of one. The shaded region collects the belief combinations (p, s) that can rationalize the full-sample spread while in the white region the bound rules the default-only hypothesis out. The dashed vertical lines mark the cumulative Aaa default rate over 1920-2025 as measured by Moody's (gray) and as realized by the bonds in our own Aaa sample (green). Those are 1.3 and 0.65 percent respectively at the twenty-year horizon and 0.13 and exactly zero percent at the five-year horizon, where the bound diverges (no finite kernel volatility can generate a default premium on an event with no measured probability).

observed price minus the yield to maturity of the same promised cash flows discounted on the Treasury curve implied by our estimates (the Aaa zero curve minus the estimated funding advantage), so that the spread object is exactly the paper's funding advantage measured at the level of individual bonds.

The default risk measure is the distance to default taken from the Merton model for a monthly panel of firms included in our sample. In the model, a firm defaults at horizon T if the value of its assets V_{it}, which has volatility σ_{it} and expected return μ_{it}, falls below the face value of its debt D_{it}. The distance to default is the number of standard deviations of asset value movement separating the firm from that point,

$$DD_{it} = \frac{\ln(V_{it}/D_{it}) + (\mu_{it} - \frac{1}{2}\sigma_{it}^2)\,T}{\sigma_{it}\sqrt{T}}$$

with T equal to one year. Asset value and asset volatility are recovered from the market value and volatility of equity by treating equity as a call option on the firm's assets. Each bond is assigned the distance to default of the firm that stands behind its credit. For subsidiaries and financing vehicles this is the listed parent.[125] Bonds in the estimation sample that cannot be mapped to a parent company are dropped.

Following Gilchrist and Zakrajšek (2012) we estimate

$$\ln S_{it} = \alpha + \beta\, DD_{it} + \gamma_1 \ln T_{it} + \gamma_2\, c_i + \varepsilon_{it},$$

where S_{it} is the bond-month spread in bp, DD_{it} the issuer's distance to default, T_{it} remaining maturity, and c_i the coupon. We estimate equation (E.4) separately on the two halves of the sample corresponding to our underlying price sources: 1970-1997 and 1998-2025.

From the fitted spread

$$\hat{S}_{it} = \hat{\kappa}\exp\Big(\hat{\alpha} + \hat{\beta}\, DD_{it} + \hat{\gamma}_1 \ln T_{it} + \hat{\gamma}_2\, c_i\Big), \qquad \hat{\kappa} = \tfrac{1}{N}\textstyle\sum_{i,t}\exp(\hat{\varepsilon}_{it}),$$

where $\hat{\kappa}$ corrects the retransformation from logs, we construct two objects. The first is the excess bond premium, the monthly average gap between observed and fitted spreads,

$$EBP_t = \frac{1}{N_t}\sum_{i\in t}\Big(S_{it} - \hat{S}_{it}\Big).$$

Because residuals average to zero within each sample, EBP_t captures the variation of the

[125] This matters in particular for the Bell System. All operating companies are mapped to AT&T through 1983 and to their regional holding company (Ameritech, Bell Atlantic, BellSouth) from 1984. Mobil is a separate firm until its 1999 merger with Exxon.

non-default component over time, not its level. The second object answers the level question. We set the distance to default to a reference value DD^*, the 95th percentile of each sample (16.5 and 26.5), at which the model-implied default probability is zero, and define the default-attributable spread as the difference

$$D_t = \frac{1}{N_t} \sum_{i \in t} \Big(\hat{S}_{it} - \hat{S}_{it} \big|_{DD_{it}=DD^*} \Big). \tag{E.14}$$

D_t measures how many bp of spread exist only because issuers' measured default risk sits at its actual level rather than at a level where model implied default risk is zero.

Figure E.45 presents the results. The top panel plots the observed spread in black, the fitted spread $\hat{S}_{it}$ in blue, and the excess bond premium EBP_t in red, all as monthly averages across bonds. The blue line is the spread that default risk and bond characteristics can explain and the red line is the part of the observed spread left over. The bottom panel addresses the level of the spread rather than its movements. It plots the part of the spread attributable to default risk D_t in red against the observed spread in gray. D_t remains far below the observed spread throughout and the vertical gap between the two lines is the part of the spread that measured default risk cannot account for. Table E.9 reports these numbers as decade averages. The default-attributable spread ranges from 8 bp in the 2010s to 27 bp in the 2000s, against observed spreads of 50 bp to 97 bp, respectively. The share of the spread attributable to default risk therefore lies between 0.14 and 0.28 in every decade since 1970.

Table E.9: Decade averages of observed and default-attributable spreads

	Observed spread	Default-attributable	Share
1970s	50bp	10bp	0.21
1980s	75bp	21bp	0.28
1990s	67bp	14bp	0.20
2000s	97bp	27bp	0.28
2010s	58bp	8bp	0.14
2020s	52bp	12bp	0.23

Notes: Monthly cross-bond averages of the observed spread S_{it} and the default-attributable spread D_t of equation (E.14), averaged within each decade. The share is the ratio of the two.

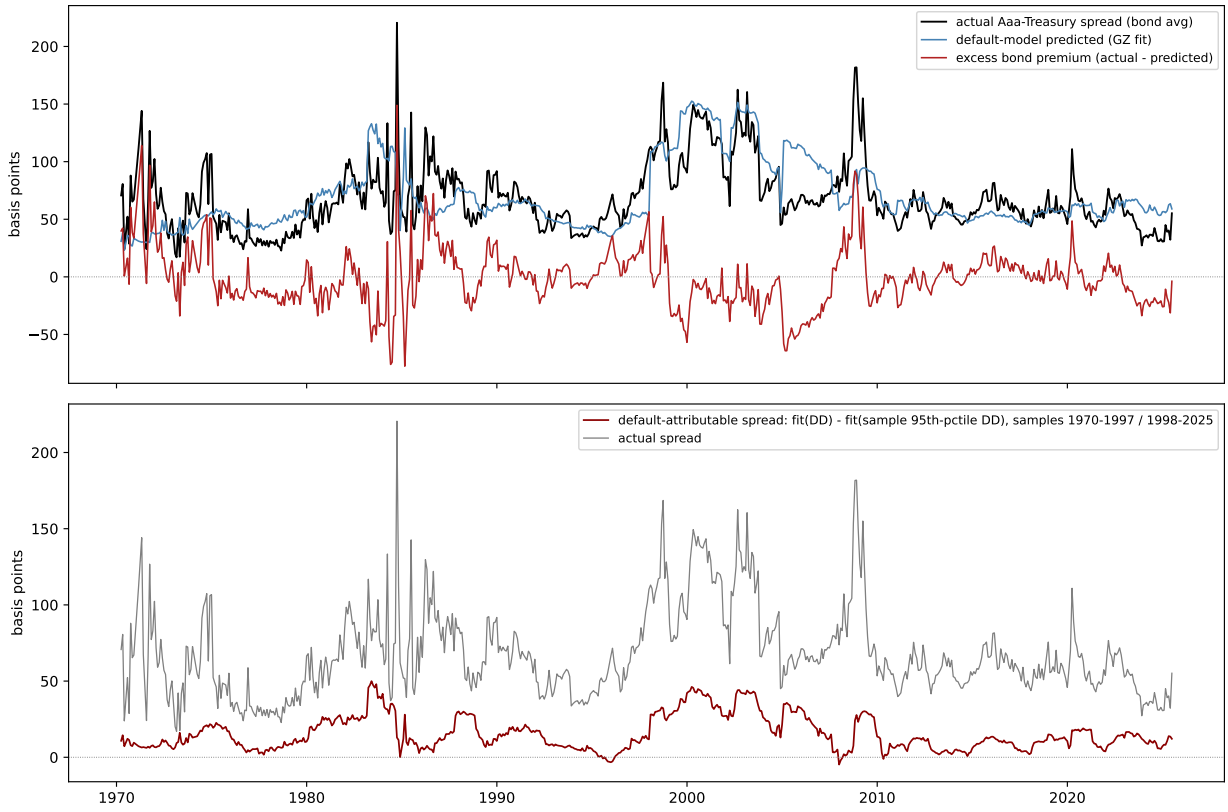


Figure E.45: Default risk and the Aaa-Treasury spread, 1970-2025

Notes: Monthly averages across the bonds of the Aaa estimation sample mapped to issuers with a Merton distance to default. Top panel: observed Aaa-Treasury spread (black), the fitted spread from equation (E.4) estimated separately on 1970-1997 and 1998-2025 (blue) and the excess bond premium, defined as observed minus fitted (red). Bottom panel: the default-attributable spread (dark red), defined as the fitted spread at the issuer's actual distance to default minus the fitted spread at the sample's 95th-percentile distance to default, with the observed spread (gray) repeated for scale. Spreads are computed against the Treasury curve implied by our estimates. Issuers without traded equity and General Electric (whose consolidated Merton measure reflects GE Capital) are excluded.

E.5 The Great Depression as a Stress Test

If default risk explains the Aaa-Treasury spread anywhere, it should be during the Great Depression, the worst corporate credit episode in our sample period. Thus, we conclude by measuring realized defaults of our Aaa sample over this episode specifically. Similar to Section E.1 we form two cohorts from every sample bond rated Aaa in 1925 and every sample bond rated Aaa in 1930 and follow each bond to the earlier of its maturity, call, default, or year-end 1940, covering the entire Depression. As in Section E.1, default events are taken from Moody's DRD and we manually cross-check and validate this information against the original Moody's Manuals who start to report bond-level default events in 1931 to make sure

we cover every default event in our sample. Figure E.46 shows an example entry from the Moody's DRD versus the original Moody's Manual entry. The default definition, the loss measurement, and the recovery conventions all carry over unchanged from Section E.1.

Table E.10 reports the results alongside the same calculation for Moody's own issuer-level Aaa universe over identical windows. Even through the Great Depression our Aaa sample's cumulative default rate was 2-3%, roughly half that of Moody's own issuer-Aaa universe, and its recovery-adjusted credit loss amounted to 1.2-1.6% of par, or about 10 basis points per annum spread over the worst fifteen credit years of the century. Set against the Aaa-Treasury spread that averages roughly 100 basis points over the Depression years realized credit losses are an order of magnitude too small to account for the spread, even under assumptions that treat almost any credit event as a full default.

Table E.10: Cumulative default rates and credit losses through the Great Depression (%)

Cohort (window)	N	Defaults	Loss rate (default $\times$ (1$-$recovery))
Sample bonds Aaa 1925 (1925-40)	105 bonds	2.9	1.6
Sample bonds Aaa 1930 (1930-40)	51 bonds	2.0	1.2
Moody's issuer-Aaa 1925 (1925-40)	96 issuers	5.2	3.7
Moody's issuer-Aaa 1930 (1930-40)	84 issuers	3.6	2.3

Table E.11 lists the events behind the realized default rates.

Table E.11: Default events underlying the sample cohorts' default rates

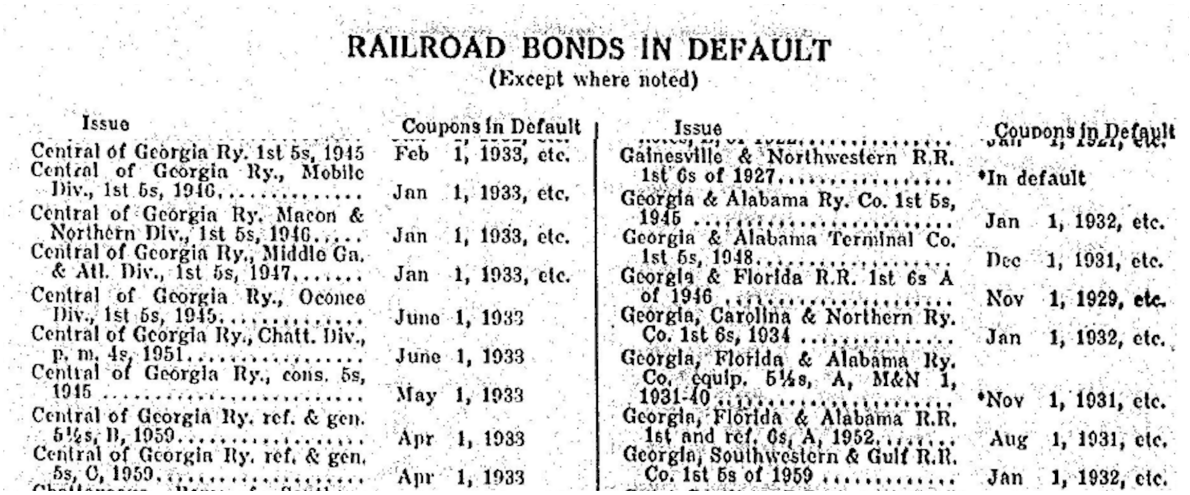
Bond	Event	Date
Aaa cohort 1925 (3 of 105 bonds)		
Florida East Coast first 4.5s of 1959	Receivership	Sep. 1931
Central of Georgia consolidated 5s of 1945	Missed interest	Feb. 1933
Chicago & North Western secured 6.5s of 1936	Missed interest	May 1935
Aaa cohort 1930 (1 of 51 bonds)		
Baltimore & Ohio first 4s of 1948	Interest deferred[a]	Sep. 1938

[a]Deferred interest under B&O's voluntary readjustment plan of September 1938; no bankruptcy followed, payments resumed, and the deferred coupons were ultimately payed in full.

Panel A: the record in Moody's Default & Recovery Database

Issuer	Instrument	Seniority	Coupon	Event type	Issuer default date	Bond default date	Issuer Rating	Price 30 days after default
Central of Georgia Railway	Consolidated mortgage gold ("cons"), due 1945	Senior secured	5%	Missed interest	Feb. 1, 1933	May 1, 1933	Caa	51.5

Panel B: the original record, Moody's Railroad Manual 1933



RAILROAD BONDS IN DEFAULT
(Except where noted)

Issue	Coupons in Default
Central of Georgia Ry. 1st 5s, 1945	Feb 1, 1933, etc.
Central of Georgia Ry., Mobile Div., 1st 5s, 1946	Jan 1, 1933, etc.
Central of Georgia Ry. Macon & Northern Div., 1st 5s, 1946	Jan 1, 1933, etc.
Central of Georgia Ry., Middle Ga. & Atl. Div., 1st 5s, 1947	Jan 1, 1933, etc.
Central of Georgia Ry., Oconee Div., 1st 5s, 1945	June 1, 1933
Central of Georgia Ry., Chatt. Div., p. m. 4s, 1951	June 1, 1933
Central of Georgia Ry., cons. 5s, 1945	May 1, 1933
Central of Georgia Ry. ref. & gen. 5½s, B, 1959	Apr 1, 1933
Central of Georgia Ry. ref. & gen. 5s, C, 1959	Apr 1, 1933

Issue	Coupons in Default
Gainesville & Northwestern R.R. 1st 6s of 1927	*In default
Georgia & Alabama Ry. Co. 1st 5s, 1945	Jan 1, 1932, etc.
Georgia & Alabama Terminal Co. 1st 5s, 1948	Dec 1, 1931, etc.
Georgia & Florida R.R. 1st 6s A of 1946	Nov 1, 1929, etc.
Georgia, Carolina & Northern Ry. Co. 1st 6s, 1934	Jan 1, 1932, etc.
Georgia, Florida & Alabama Ry. Co. equip. 5½s, A, M&N 1, 1931-40	*Nov 1, 1931, etc.
Georgia, Florida & Alabama R.R. 1st and ref. 6s, A, 1952	Aug 1, 1931, etc.
Georgia, Southwestern & Gulf R.R. Co. 1st 5s of 1959	Jan 1, 1932, etc.

Figure E.46: An example of a default record in the DRD and its original source. Panel A shows the DRD's machine readable record for the Central of Georgia consolidated 5s of 1945, a bond that defaulted on interest in our 1925 Aaa cohort. Panel B shows the "Railroad Bonds in Default" listing from the original Moody's Manual. The consolidated 5s appear with coupons in default from May 1, 1933, the instrument-level date in Panel A, while the earliest coupon default in the railway's structure (the first 5s of 1945, February 1, 1933) corresponds to the DRD's issuer-level event date. The listing dates each instrument's default separately, division by division, exactly as the DRD's instrument-level records do. Source: Moody's Manual of Investments, Railroad Securities, 1933.

F Details on Literature Comparison in Section 6

In this Appendix, we provide additional details on our re-evaluation of the macro-finance literature in Section 6.1. For convenience, we use the abbreviations in Table F.12 below throughout.

Abbreviation	Paper
KVJ	Krishnamurthy and Vissing-Jorgensen (2012)
N	Nagel (2016)
KL	Krishnamurthy and Li (2023)
CKP	Choi et al. (2026)

Table F.12: Abbreviations for Papers

F.1 A Structural Asset Pricing Model

To make our analysis concrete and self-contained, we construct an illustrative model that nests the equilibrium equations in key papers in the macro-finance literature, namely those in Table F.12.

Setting: The economy is in discrete time $t = 0, 1, 2, \ldots$ with infinite horizon. There is one consumption good produced by a Lucas tree that produces stochastic output Y_t each period.

The economy consists of a representative household, a firm, a bank, and a government. The household is ultimately the residual claimant to all output from the Lucas tree. However, the representative firm controls the Lucas tree and the agents issue a collection of different assets that are directly or indirectly claims to the Lucas tree. The government issues M_t^s money. We use money as the numeraire and let P_t denote the number of units of money required to buy 1 good (i.e. the price level in the economy). The government and firms issue nominal zero-coupon bonds with maturities $j \in [1, \ldots, \mathcal{J}]$ that pay 1 unit of money at maturity. We let $(b_t^{s,(j)}, q_t^{(j)})$ denote the supply and price of government-issued bonds with maturity j and let $(\tilde{b}_t^{s,(j)}, \tilde{q}_t^{(j)})$ denote the supply and price of corporate bonds with maturity j. The banking sector issues one-period deposits D_t^s with interest rate i_t^d. We assume that all bonds and deposits are default risk free. The firms and banks issue equity supplies $k_t^{s,f}$ and $k_t^{s,b}$. We denote the aggregate claim to all equity by $k_t^s := k_t^{s,f} + k_t^{s,b}$, its price by q_t^k, and its aggregate dividend by Π_t. Throughout, we use the superscript "s" to explicitly distinguish supply of assets from household demand. The government levies lump-sum taxes τ_t on the household to satisfy its period budget constraint:

$$b_{t-1}^{s,(1)} = \tau_t + \left(M_t^s - M_{t-1}^s\right) + \sum_{j \leq \mathcal{J}} q_t^{(j)} \left(b_t^{s,(j)} - b_{t-1}^{s,(j+1)}\right), \qquad b_{t-1}^{s,(\mathcal{J}+1)} := 0. \tag{F.1}$$

F.1.1 Household Optimization and Treasury Demand

To nest the literature, we assume households derive a non-pecuniary benefit from holding different classes of fixed income assets. We use this to derive asset demand functions.

Household preferences: A representative household ranks allocations according to:

$$\mathbb{E}_0 \left[\sum_{t=0}^{\infty} \beta^t \left(u(C_t) + \alpha \nu_t(\mathcal{A}_t) \right) \right]$$

where

$$\mathcal{A}_t = f_t\left(\frac{M_t}{P_t}, \frac{D_t}{P_t}, \left\{\frac{q_t^{(j)} b_t^{(j)}}{P_t}\right\}_{j\leq\mathcal{J}}\right), \qquad f_t(\cdot) := f(\cdot;\lambda_t), \qquad \nu_t(\cdot) := \nu(\cdot;\zeta_t),$$

Here, $\mathcal{A}_t$ is an asset composite constructed by the function f_t from money, deposits, government bonds, and the stochastic weight λ_t, $\nu_t(\cdot)$ is the non-pecuniary benefit of the asset composite $\mathcal{A}_t$ to households, and ζ_t is a stochastic demand shifter. We exclude corporate bonds from $\mathcal{A}_t$ because none of the specifications that we replicate explicitly assigns non-pecuniary services to corporate bonds.[126] For convenience, define $\mathcal{Q}_t := \sum_{j\leq\mathcal{J}} q_t^{(j)} b_t^{(j)}$ as the market value of the household's government bond holdings. Papers in the literature differ in how they specify both $\mathcal{A}_t$ and $\nu_t(\cdot)$. We summarize the differences across the key papers in Table F.13 below.

Paper	Asset composite	Non-pecuniary benefit
KVJ	$\mathcal{A}_t = \frac{1}{P_t}\mathcal{Q}_t + \kappa\theta_t^P$	$\nu_t'(\mathcal{A}_t) = \frac{u'(Y_t)}{\alpha}\left[\beta_0 + \beta_1 \log\left(\frac{\mathcal{A}_t}{Y_t}\right) + \log(\zeta_t)\right]$
N	$\mathcal{A}_t = \frac{1}{P_t}\left((1-\lambda_t)D_t^\rho + \lambda_t\left(q_t^{(1)}b_t^{(1)}\right)^\rho\right)^{\frac{1}{\rho}}$	$\nu_t(\mathcal{A}_t) = \log(\mathcal{A}_t)$
KL	$\mathcal{A}_t = \frac{1}{P_t}\left((1-\lambda_t)D_t^\rho + \lambda_t\mathcal{Q}_t^\rho\right)^{\frac{1}{\rho}}$	$\nu_t(\mathcal{A}_t) = \log(\mathcal{A}_t)$
CKP	$\mathcal{A}_t = \frac{1}{P_t}\mathcal{Q}_t$	$\nu_t(\mathcal{A}_t) = \frac{u'(Y_t)}{\alpha} Y_t(\beta + \gamma v_t)\frac{\mathcal{A}_t}{Y_t}\left[\log\left(\frac{\mathcal{A}_t}{Y_t}\right) - \zeta_t\right]$

Table F.13: Non-pecuniary benefit functions across different papers. KVJ includes a variable θ_t^P in the composite $\mathcal{A}_t$ that they describe as capturing all privately issued assets that generate non-pecuniary benefits but do not provide the exact specification of θ_t^P. KVJ put the non-pecuniary benefit inside the consumption utility function as $u(C_t + \nu_{t,KVJ}(\mathcal{A}_t))$ with $\nu'_{t,KVJ} = \beta_0 + \beta_1 \log(\mathcal{A}_t/Y_t) + \log(\zeta_t)$. We scale their non-pecuniary benefit by the equilibrium value of $u'(Y_t)/\alpha$ so that we get the analogous asset-pricing first order condition when evaluated at equilibrium consumption. CKP put the non-pecuniary benefit in the next-period household budget constraint rather than in utility. We scale their benefit by the equilibrium value of $u'(Y_t)/\alpha$ so that our additively separable formulation gives the same one-period funding advantage when evaluated at equilibrium consumption. In KL, $\mathcal{Q}_t$ is the market value of Treasurys held by households (their net Treasurys, which exclude holdings of banks and the Federal Reserve), not total Treasurys outstanding.

Household optimization problem: Let a_t denote the household's nominal wealth at the beginning of period t. For convenience, define the one-period-ahead payoffs from government

[126] Corporate bonds could in principle enter $\mathcal{A}_t$ and provide non-pecuniary services. KVJ allow their private convenience-asset aggregate θ_t^P to contain privately issued assets without specifying precisely which assets it contains. However, the corporate bond used to derive their estimation equation (equation (12) in their paper) is a corporate bond that provides no convenience services.

and firm bonds purchased with maturity j as

$$n_{t+1}^{(j)} := \begin{cases} 1, & j = 1, \\ q_{t+1}^{(j-1)}, & j > 1 \end{cases}, \qquad \tilde{n}_{t+1}^{(j)} := \begin{cases} 1, & j = 1, \\ \tilde{q}_{t+1}^{(j-1)}, & j > 1. \end{cases}$$

Let $V_t(a_t)$ denote the value function for a household with wealth a_t at time t. We use the t subscript to capture the dependence on the aggregate state space rather than specifying an aggregate state space explicitly. The household Bellman equation is:

$$\begin{aligned} V_t(a_t) = \max_{C_t, M_t, D_t, \{b_t^{(j)}, \tilde{b}_t^{(j)}\}_{j \leq \mathcal{J}}, k_t} & \{u(C_t) + \alpha \nu_t(\mathcal{A}_t) + \beta \mathbb{E}_t \left[V_{t+1}(a_{t+1})\right]\} \\ s.t. \quad & P_t C_t + M_t + D_t + \sum_{j \leq \mathcal{J}} q_t^{(j)} b_t^{(j)} + \sum_{j \leq \mathcal{J}} \tilde{q}_t^{(j)} \tilde{b}_t^{(j)} + q_t^k k_t \leq a_t - \tau_t, \\ & a_{t+1} = M_t + \left(1 + i_t^d\right) D_t + \sum_{j \leq \mathcal{J}} n_{t+1}^{(j)} b_t^{(j)} + \sum_{j \leq \mathcal{J}} \tilde{n}_{t+1}^{(j)} \tilde{b}_t^{(j)} + \left(\Pi_{t+1} + q_{t+1}^k\right) k_t. \end{aligned} \tag{F.2}$$

First order conditions: Let μ_t^c denote the multiplier on the household budget constraint and define

$$f_{M,t} := \frac{\partial f_t}{\partial (M_t / P_t)}, \qquad f_{D,t} := \frac{\partial f_t}{\partial (D_t / P_t)}, \qquad f_{b,t}^{(j)} := \frac{\partial f_t}{\partial (q_t^{(j)} b_t^{(j)} / P_t)}.$$

The household FOCs are:

$$\begin{aligned} [C_t]: \quad & 0 = u'(C_t) - P_t \mu_t^c, \\ [M_t]: \quad & 0 = \frac{\alpha}{P_t} \nu_t'(\mathcal{A}_t) f_{M,t} + \beta \mathbb{E}_t \left[\partial_a V_{t+1}(a_{t+1})\right] - \mu_t^c, \\ [D_t]: \quad & 0 = \frac{\alpha}{P_t} \nu_t'(\mathcal{A}_t) f_{D,t} + \beta \mathbb{E}_t \left[\left(1 + i_t^d\right) \partial_a V_{t+1}(a_{t+1})\right] - \mu_t^c, \\ [b_t^{(j)}]: \quad & 0 = \frac{\alpha q_t^{(j)}}{P_t} \nu_t'(\mathcal{A}_t) f_{b,t}^{(j)} + \beta \mathbb{E}_t \left[n_{t+1}^{(j)} \partial_a V_{t+1}(a_{t+1})\right] - q_t^{(j)} \mu_t^c, \\ [\tilde{b}_t^{(j)}]: \quad & 0 = \beta \mathbb{E}_t \left[\tilde{n}_{t+1}^{(j)} \partial_a V_{t+1}(a_{t+1})\right] - \tilde{q}_t^{(j)} \mu_t^c, \\ [k_t]: \quad & 0 = \beta \mathbb{E}_t \left[\left(\Pi_{t+1} + q_{t+1}^k\right) \partial_a V_{t+1}(a_{t+1})\right] - q_t^k \mu_t^c. \end{aligned}$$

The envelope theorem gives

$$\partial_a V_t(a_t) = \mu_t^c = \frac{u'(C_t)}{P_t}.$$

Define the nominal stochastic discount factor:

$$\xi_{t,t+1} := \beta \frac{u'(C_{t+1})}{u'(C_t)} \frac{P_t}{P_{t+1}}.$$

Substituting the envelope condition and the definition of $\xi_{t,t+1}$ into the household first order conditions gives

$$1 - \mathbb{E}_t \left[\xi_{t,t+1} \frac{\tilde{n}_{t+1}^{(j)}}{\tilde{q}_t^{(j)}} \right] = 0, \tag{F.3}$$

$$1 - \mathbb{E}_t \left[\xi_{t,t+1} \frac{n_{t+1}^{(j)}}{q_t^{(j)}} \right] = \frac{\alpha}{u'(C_t)} \nu_t'(\mathcal{A}_t) f_{b,t}^{(j)}, \tag{F.4}$$

$$1 - \mathbb{E}_t \left[\xi_{t,t+1} \right] = \frac{\alpha}{u'(C_t)} \nu_t'(\mathcal{A}_t) f_{M,t}, \tag{F.5}$$

$$1 - \mathbb{E}_t \left[\xi_{t,t+1} \left(1 + i_t^d \right) \right] = \frac{\alpha}{u'(C_t)} \nu_t'(\mathcal{A}_t) f_{D,t}, \tag{F.6}$$

$$1 = \mathbb{E}_t \left[\xi_{t,t+1} \frac{\Pi_{t+1} + q_{t+1}^k}{q_t^k} \right].$$

Thus, the non-pecuniary benefit of an asset creates a wedge between its pecuniary return and the return implied by the household stochastic discount factor.

Rewriting as a price wedge: Rearranging equations (F.4) and (F.3) gives:

$$\tilde{q}_t^{(j)} = \mathbb{E}_t \left[\xi_{t,t+1} \tilde{n}_{t+1}^{(j)} \right], \tag{F.7}$$

$$q_t^{(j)} = \frac{\mathbb{E}_t \left[\xi_{t,t+1} n_{t+1}^{(j)} \right]}{1 - \frac{\alpha}{u'(C_t)} \nu_t'(\mathcal{A}_t) f_{b,t}^{(j)}} \approx \mathbb{E}_t \left[\xi_{t,t+1} \exp \left(\frac{\alpha}{u'(C_t)} \nu_t'(\mathcal{A}_t) f_{b,t}^{(j)} \right) n_{t+1}^{(j)} \right] \tag{F.8}$$

so, using the notation from the main text, the Treasury wedge is:

$$\Omega_{t,t+1}^{(j)} = \exp \left(\frac{\alpha}{u'(C_t)} \nu_t'(\mathcal{A}_t) f_{b,t}^{(j)} \right). \tag{F.9}$$

Rearranging equations (F.5) and (F.6) gives the corresponding price equations for money

and deposits:

$$1 = \frac{\mathbb{E}_t\left[\xi_{t,t+1}\right]}{1 - \frac{\alpha}{u'(C_t)}\nu_t'(\mathcal{A}_t)f_{M,t}} \approx \mathbb{E}_t\left[\xi_{t,t+1}\exp\left(\frac{\alpha}{u'(C_t)}\nu_t'(\mathcal{A}_t)f_{M,t}\right)\right],$$
$$1 = \frac{\mathbb{E}_t\left[\xi_{t,t+1}\left(1+i_t^d\right)\right]}{1 - \frac{\alpha}{u'(C_t)}\nu_t'(\mathcal{A}_t)f_{D,t}} \approx \mathbb{E}_t\left[\xi_{t,t+1}\exp\left(\frac{\alpha}{u'(C_t)}\nu_t'(\mathcal{A}_t)f_{D,t}\right)\left(1+i_t^d\right)\right],$$

while equity, which does not enter the asset-services aggregator, satisfies the standard pricing equation

$$q_t^k = \mathbb{E}_t\left[\xi_{t,t+1}\left(\Pi_{t+1} + q_{t+1}^k\right)\right].$$

Rewriting in yield space: Define the one-period nominal reference rate i_t by

$$\frac{1}{1+i_t} := \mathbb{E}_t\left[\xi_{t,t+1}\right].$$

Define the period-by-period compounding interest rates on government and corporate bonds, $i_t^{(j)}$ and $\tilde{i}_t^{(j)}$, by:

$$1 + i_t^{(j)} = \left(\frac{1}{q_t^{(j)}}\right)^{1/j}, \qquad 1 + \tilde{i}_t^{(j)} = \left(\frac{1}{\tilde{q}_t^{(j)}}\right)^{1/j},$$

the yields analogously to the main text:

$$y_t^{(j)} = -\frac{1}{j}\log\left(q_t^{(j)}\right) \approx i_t^{(j)}, \qquad \tilde{y}_t^{(j)} = -\frac{1}{j}\log\left(\tilde{q}_t^{(j)}\right) \approx \tilde{i}_t^{(j)},$$

and the pricing-kernel-weighted one-period holding interest rates on government and corporate bonds, $\hat{i}_t^{(j)}$ and $\tilde{\hat{i}}_t^{(j)}$, by:

$$1 + \hat{i}_t^{(j)} := \frac{\mathbb{E}_t\left[\xi_{t,t+1}\frac{n_{t+1}^{(j)}}{q_t^{(j)}}\right]}{\mathbb{E}_t\left[\xi_{t,t+1}\right]}, \qquad 1 + \tilde{\hat{i}}_t^{(j)} := \frac{\mathbb{E}_t\left[\xi_{t,t+1}\frac{\tilde{n}_{t+1}^{(j)}}{\tilde{q}_t^{(j)}}\right]}{\mathbb{E}_t\left[\xi_{t,t+1}\right]}.$$

For the special case of one-period debt we have $n_{t+1}^{(1)} = \tilde{n}_{t+1}^{(1)} = 1$ and so the pricing-kernel-weighted returns become the unweighted returns:

$$1 + \hat{i}_t^{(1)} := \frac{1}{q_t^{(1)}} \qquad 1 + \tilde{\hat{i}}_t^{(1)} := \frac{1}{\tilde{q}_t^{(1)}}$$

Using the definition of i_t and the pricing-kernel-weighted holding interest rates in equations

(F.5)–(F.3) gives

$$\frac{u'(C_t)}{1+i_t} i_t = \alpha \nu_t'(\mathcal{A}_t) f_{M,t},$$

$$\frac{u'(C_t)}{1+i_t} \left(i_t - i_t^d\right) = \alpha \nu_t'(\mathcal{A}_t) f_{D,t}, \tag{F.10}$$

$$\frac{u'(C_t)}{1+i_t} \left(i_t - \hat{i}_t^{(j)}\right) = \alpha \nu_t'(\mathcal{A}_t) f_{b,t}^{(j)}, \tag{F.11}$$

$$\frac{u'(C_t)}{1+i_t} \left(i_t - \tilde{\hat{i}}_t^{(j)}\right) = 0. \tag{F.12}$$

Subtracting equation (F.11) from equation (F.12) gives the pricing-kernel-weighted bond return difference:

$$\tilde{\hat{i}}_t^{(j)} - \hat{i}_t^{(j)} = \frac{1+i_t}{u'(C_t)} \alpha \nu_t'(\mathcal{A}_t) f_{b,t}^{(j)}$$

Treasury funding advantage: For the special case of $j = 1$, we get that the Treasury funding advantage on 1-period bonds is:

$$\chi_t^{(1)} = \tilde{y}_t^{(1)} - y_t^{(1)} = \log\left(\frac{1+\tilde{\hat{i}}_t^{(1)}}{1+i_t^{(1)}}\right) \approx \frac{\alpha \nu_t'(\mathcal{A}_t)}{u'(C_t)} f_{b,t}^{(1)}. \tag{F.13}$$

For the general case of $j > 1$, the expression for funding advantage is more complicated. Substituting equation (F.9) into equations (F.8) and (F.7), and using that $\Omega_{t,t+1}^{(j)}$ is known at time t, gives:

$$\begin{aligned} q_t^{(j)} &\approx \Omega_{t,t+1}^{(j)} \mathbb{E}_t\left[\xi_{t,t+1} q_{t+1}^{(j-1)}\right] \\ &= \Omega_{t,t+1}^{(j)} \mathbb{E}_t\left[\xi_{t,t+1}\right] \mathbb{E}_t\left[q_{t+1}^{(j-1)}\right] \left(1 + \frac{\mathrm{Cov}_t\left[\xi_{t,t+1}, q_{t+1}^{(j-1)}\right]}{\mathbb{E}_t\left[\xi_{t,t+1}\right] \mathbb{E}_t\left[q_{t+1}^{(j-1)}\right]}\right) \end{aligned} \tag{F.14}$$

$$\begin{aligned} \tilde{q}_t^{(j)} &\approx \mathbb{E}_t\left[\xi_{t,t+1} \tilde{q}_{t+1}^{(j-1)}\right] \\ &= \mathbb{E}_t\left[\xi_{t,t+1}\right] \mathbb{E}_t\left[\tilde{q}_{t+1}^{(j-1)}\right] \left(1 + \frac{\mathrm{Cov}_t\left[\xi_{t,t+1}, \tilde{q}_{t+1}^{(j-1)}\right]}{\mathbb{E}_t\left[\xi_{t,t+1}\right] \mathbb{E}_t\left[\tilde{q}_{t+1}^{(j-1)}\right]}\right). \end{aligned} \tag{F.15}$$

So, the funding advantage is:

$$\begin{aligned}
\chi_t^{(j)} &= \frac{1}{j}\log\left(q_t^{(j)}\right) - \frac{1}{j}\log\left(\tilde{q}_t^{(j)}\right) \\
&= \frac{1}{j}\log\left(\Omega_{t,t+1}^{(j)}\right) + \frac{1}{j}\log\left(\mathbb{E}_t\left[q_{t+1}^{(j-1)}\right]\right) - \frac{1}{j}\log\left(\mathbb{E}_t\left[\tilde{q}_{t+1}^{(j-1)}\right]\right) \\
&\quad + \frac{1}{j}\log\left(1 + \frac{\mathrm{Cov}_t\left[\xi_{t,t+1}, q_{t+1}^{(j-1)}\right]}{\mathbb{E}_t\left[\xi_{t,t+1}\right]\mathbb{E}_t\left[q_{t+1}^{(j-1)}\right]}\right) - \frac{1}{j}\log\left(1 + \frac{\mathrm{Cov}_t\left[\xi_{t,t+1}, \tilde{q}_{t+1}^{(j-1)}\right]}{\mathbb{E}_t\left[\xi_{t,t+1}\right]\mathbb{E}_t\left[\tilde{q}_{t+1}^{(j-1)}\right]}\right). \\
&\approx \frac{\alpha\nu_t'(\mathcal{A}_t)}{ju'(C_t)} f_{b,t}^{(j)} + \frac{1}{j}\left(\log\left(\mathbb{E}_t\left[q_{t+1}^{(j-1)}\right]\right) - \log\left(\mathbb{E}_t\left[\tilde{q}_{t+1}^{(j-1)}\right]\right)\right) \\
&\quad + \frac{1}{j}\left(\frac{\mathrm{Cov}_t\left[\xi_{t,t+1}, q_{t+1}^{(j-1)}\right]}{\mathbb{E}_t\left[\xi_{t,t+1}\right]\mathbb{E}_t\left[q_{t+1}^{(j-1)}\right]} - \frac{\mathrm{Cov}_t\left[\xi_{t,t+1}, \tilde{q}_{t+1}^{(j-1)}\right]}{\mathbb{E}_t\left[\xi_{t,t+1}\right]\mathbb{E}_t\left[\tilde{q}_{t+1}^{(j-1)}\right]}\right).
\end{aligned}$$

where we have used that $\log(1+x) \approx x$ in the last line.

F.1.2 Equilibrium

To close the model, we need to specify asset supply rules and market clearing. We do this for a general supply process in Definition 1 below. The different papers in Table F.12 differ in how they specify asset supply rules and impose equilibrium (or not impose equilibrium). We discuss this in the next Subsection when we replicate the empirical exercises in the different papers.

Definition 1. An equilibrium is a collection of price processes $(P_t, i_t^d, (q_t^{(j)}, \tilde{q}_t^{(j)}, q_t^k)_{j\geq 1})_{t\geq 0}$, household allocation processes $(C_t, M_t, D_t, (b_t^{(j)}, \tilde{b}_t^{(j)})_{j\geq 1}, k_t)_{t\geq 0}$, and supply processes $(Y_t, M_t^s, D_t^s, (b_t^{s,(j)}, \tilde{b}_t^{s,(j)})_{j\geq 1}, k_t^{s,f}, k_t^{s,b}, \tau_t)_{t\geq 0}$ such that:

1. Taking the price processes, the household solves Problem (F.2).
2. The firms supply corporate bonds and equity according to the process: $(\tilde{b}_t^{s,(j)}, k_t^{s,f})_{t\geq 0}$.
3. The banks supply deposits and equity according to the process: $(D_t^s, k_t^{s,b})_{t\geq 0}$.
4. The government supplies money and bonds according to the process $(M_t^s, b_t^{s,(j)})_{t\geq 0}$ and its budget constraint (F.1) is satisfied.
5. Money, deposit, bond, equity, and goods markets clear:

$$M_t = M_t^s, \quad D_t = D_t^s, \quad b_t^{(j)} = b_t^{s,(j)}, \quad \tilde{b}_t^{(j)} = \tilde{b}_t^{s,(j)}, \quad k_t = k_t^{s,f} + k_t^{s,b}, \quad C_t = Y_t.$$

F.2 Re-evaluation of the Empirical Analysis in the Literature

We now study how the literature has taken models of non-pecuniary Treasury demand to the data. For each paper in Table F.12 we use their functional specification from Table F.13 to derive their main estimation equation for the corporate-Treasury spread. We then replicate their empirical strategies both with their original data and with our new estimates for government funding advantage. This allows us to highlight which results are confirmed and which results are overturned by our new yield curves.

F.2.1 KVJ

Equilibrium spread equations: The key funding advantage equations in KVJ are the short-maturity equation (12) in their Prediction 1 and the long-maturity equation (17) for their Prediction 2. To derive these equations, KVJ consider a government bond that provides one unit of non-pecuniary benefit and a corporate bond that provides no non-pecuniary benefit (i.e. corporate bonds are not in θ_t^P). Using the definitions of $\mathcal{A}_t$ and $\nu_t'(\mathcal{A}_t)$ in Table F.13, this implies that $f_{b,t}^{(j)} = 1$ for all $j \leq J$. KVJ impose equilibrium in the bond market so that $\mathcal{Q}_t = \mathcal{Q}_t^s$ without imposing a government bond issuance policy rule. Under these restrictions, the one-period spread demand equation (F.13) becomes the equilibrium relationship:

$$\chi_t^{(1)} \approx \frac{\alpha \nu_t'(\mathcal{A}_t)}{u'(C_t)} f_{b,t}^{(1)} = \beta_0 + \beta_1 \log\left(\frac{\mathcal{Q}_t^s / P_t + \kappa \theta_t^P}{Y_t}\right) + \log(\zeta_t), \tag{F.16}$$

where the final line in equation (F.16) is KVJ's equation (12) under the condition of no default ($\tilde{L}_t = 0$ for all t in KVJ-12's notation). For long-maturity bonds, iterating the bond-pricing recursion in equations (F.14) and (F.15) expresses the spread as the expected average of future marginal convenience benefits plus the corresponding expected-payoff and pricing-kernel covariance terms, as in their equation (17).

Empirical test: KVJ interpret $\beta_1 < 0$ and equation (F.16) as predicting that an increase in the ratio of the market value of government debt to GDP will be negatively related to the equilibrium Aaa corporate-Treasury spread if demand shocks and the supply of convenience-generating private assets are held fixed. They test this prediction by regressing different corporate-Treasury spreads on the market value of government debt-to-GDP and controls. The regressions are summarized in Table 1 of their paper. For the spread, they consider a long-maturity spread (the index-based spread defined and discussed in Section 3.3) and a short-maturity spread (the commercial paper rate minus the Treasury bill rate).[127] They

[127] They also consider spreads for lower rated corporate bonds but they are not relevant for our paper.

control for the volatility of the US stock market, which they interpret as a control for corporate default risk, and control for the slope of the Treasury yield curve, which they interpret as a control for the business and/or credit cycle. KVJ do not construct an instrument for aggregate government debt-to-GDP.[128] So, their OLS coefficient can only be interpreted as identifying a Treasury demand function under the assumption that, conditional on the controls, Treasury supply is uncorrelated with omitted shocks to non-pecuniary demand, default risk, and risk premia. Under that interpretation, the coefficient measures the effect of Treasury supply inclusive of the induced response of privately supplied convenience assets, rather than the primitive slope of the non-pecuniary-benefit function.

In Table F.14 we reproduce the KVJ regressions in panels A and C of their Table 1 using their data (the top panel) and replicate the analysis using our new yield curve estimates (the bottom panel). To maintain comparability with KVJ, we estimate the coefficients by OLS and compute standard errors using a stationary AR(1) covariance structure estimated from the OLS residuals.[129] For the regressions using our new yield curve estimates, we construct the non-spread controls as follows. The volatility control, the analog to KVJ's own CRSP-based stock-return volatility, is built from Global Financial Data's "Top 100" U.S. total-return stock index, available monthly without gaps from 1792 onward. We compute monthly log returns on this index and take the annualized standard deviation of the trailing twelve months of log returns, with the window ending each June 30 to match our fiscal-year annual convention. The debt-to-GDP ratio uses the market value of outstanding Treasury debt relative to GDP, sampled at its native June 30 annual frequency. Finally, the Treasury slope control is constructed directly from our own zero-coupon Treasury yield curve as the spread between the 10-year and 3-month yields or the spread between the 10-year and 1-year yields. This is again sampled annually as of June 30.

Replicating the KVJ analysis with our data significantly changes the results. For the long-maturity spread regressions, the strongly significant negative coefficient on Log(debt/GDP) using the KVJ data becomes insignificant using our data. In addition, R^2 falls substantially from 0.573 with KVJ's data to 0.166 with our yield curve estimates. For the short-maturity spread regressions, our data reveals a similar relationship to debt-to-GDP as in KVJ when we control using the 10-year to 3-month spread. However, our yield curve estimates allow us to try a wide range of yield curve slopes. In particular, we find that when we control for the 10-year to 1-year spread in column (8') the slope is highly significant and debt-to-GDP is significant only at the 10% level. This shows that the estimated coefficient on debt/GDP

[128] Although in Table 2 they do use total Debt-to-GDP as an instrument for the supply of Treasurys with long or short maturities.

[129] These are not feasible generalized least squares estimates. The AR(1) specification changes the estimated covariance matrix but not the coefficient estimates.

	A. Aaa-Treasury spread				C. CP-TBills spread				
	1919-2008	1926-2008	1941-2008	1926-2008	1920-2008	1926-2008	1926-2008	1926-2008	1926-2008
	(1)	(3)	(3*)	(3′)	(6)	(8)	(8′)		
KVJ data									
Dependent spread	Aaa-Tsy	Aaa-Tsy	Aaa-Tsy		CP-TBills	CP-TBills			
Log(debt/GDP)	-0.746***	-0.800***	-0.779***		-0.730***	-0.554***			
	(0.171)	(0.156)	(0.144)		(0.165)	(0.156)			
Volatility		1.313*	4.272***			1.902**			
		(0.681)	(1.043)			(0.835)			
Slope (10yr-3mo)		0.081*	0.084**			-0.083			
		(0.043)	(0.041)			(0.060)			
Intercept	0.109	0.077	0.098		0.095	0.227			
	(0.177)	(0.157)	(0.134)		(0.168)	(0.153)			
Observations	90	83	68		89	83			
Residual AR(1) coef.	0.604	0.552	0.375		0.175	0.016			
R^2	0.449	0.573	0.637		0.226	0.258			
Our data									
Dependent spread	20yr par	20yr par	20yr par	20yr par	3mo	3mo	3mo	6mo	1yr
Log(debt/GDP)	-0.203	-0.153	-0.081	-0.135	-0.484***	-0.349**	-0.251*	-0.275**	-0.130
	(0.135)	(0.133)	(0.144)	(0.137)	(0.167)	(0.145)	(0.141)	(0.129)	(0.114)
Volatility		0.758*	0.634	0.808*		2.169***	2.568***	2.556***	2.488***
		(0.443)	(0.682)	(0.444)		(0.621)	(0.606)	(0.549)	(0.486)
Slope (10yr-3mo)		-5.611*	-5.237			-7.142		-6.621	-6.203*
		(3.042)	(3.212)			(4.710)		(4.158)	(3.683)
Slope (10yr-1yr)				-6.701*			-15.606***		
				(3.744)			(5.421)		
Intercept	0.211	0.274*	0.331*	0.295*	0.145	0.285	0.398**	0.262*	0.388***
	(0.167)	(0.161)	(0.170)	(0.166)	(0.205)	(0.173)	(0.169)	(0.155)	(0.137)
Observations	90	83	68	83	89	83	83	83	83
Residual AR(1) coef.	0.647	0.591	0.557	0.603	0.289	0.118	0.090	0.139	0.140
R^2	0.088	0.174	0.095	0.173	0.141	0.249	0.302	0.311	0.302

Table F.14: Table 1 of Krishnamurthy and Vissing-Jorgensen (2012), Panel A (Aaa-Treasury) columns (1) and (3), and Panel C (CP-Bills) columns (6) and (8), replicated on the paper's own data and on this project's own yield curves, plus columns (3′) and (8′) – "our data" only, same specification as (3)/(8) but with the slope control built 10yr-1yr instead of KVJ-12's own 10yr-3mo – and two further CP-Bills columns (6mo and 1yr zero-coupon spread) with no KVJ-12 counterpart, otherwise identical to column (8)'s own specification (controls, 1926-2008, 10yr-3mo slope). Columns (2) and (7), which require Moody's EDF data, and Panel B (Baa-Treasury), for which this project has no alternative estimate, are omitted. Our data uses the 20-year par-yield spread for the Aaa-Treasury columns and the 3-month zero-coupon spread for the CP-Bills columns, matching KVJ-12's own 3-month commercial paper maturity exactly. Standard errors, reported in parentheses, are computed using the stationary AR(1) covariance structure estimated from the OLS residuals, following KVJ-12. KVJ-12 instead report the corresponding t-statistics. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (based on $|t|$).

is sensitive to the particular yield-curve slope used as a control. This could be because the 10-year to 3-month spread and the 10 year to 1 year spread have very different effectiveness as proxies for business cycle risk.

F.2.2 N and KL

N and KL use very similar specifications of non-pecuniary demand but impose different supply rules and equilibrium characterizations, which leads them to adopt different empirical tests for their theories.

N equilibrium spread equation: Using N's specification of $\mathcal{A}_t$ from Table F.13, the marginal contributions of deposits and government bonds to the asset composite are

$$f_{D,t} = \mathcal{A}_t^{1-\rho}(1-\lambda_t)\left(\frac{D_t}{P_t}\right)^{\rho-1}, \qquad f_{b,t}^{(1)} = \mathcal{A}_t^{1-\rho}\lambda_t\left(\frac{q_t^{(1)}b_t^{(1)}}{P_t}\right)^{\rho-1}. \tag{F.17}$$

Because $\nu_t'(\mathcal{A}_t) = 1/\mathcal{A}_t$, substituting equation (F.17) into equations (F.10) and (F.11) gives

$$\frac{\alpha}{u'(C_t)}\mathcal{A}_t^{-\rho}(1-\lambda_t)\left(\frac{D_t}{P_t}\right)^{\rho-1} = \frac{i_t - i_t^d}{1+i_t}, \tag{F.18}$$

$$\frac{\alpha}{u'(C_t)}\mathcal{A}_t^{-\rho}\lambda_t\left(\frac{q_t^{(1)}b_t^{(1)}}{P_t}\right)^{\rho-1} = \frac{i_t - i_t^{(1)}}{1+i_t}, \tag{F.19}$$

which are are equations (5) and (6) in N respectively. Dividing equation (F.19) by equation (F.18) gives

$$i_t - i_t^{(1)} = \frac{\lambda_t}{1-\lambda_t}\left(\frac{q_t^{(1)}b_t^{(1)}}{D_t}\right)^{\rho-1}(i_t - i_t^d), \tag{F.20}$$

which is the liquidity-premium equation in N.

To close equilibrium, N makes additional assumptions on the behavior of monetary policy makers and the banking sector. On the banking side, N imposes that the deposit supply and rate are mechanically linked to government money supply and the undistorted interest rate:

$$\begin{aligned} D_t^s &= \phi M_t^s \\ i_t^d &= \delta i_t, \quad 0 \le \delta \le 1 - 1/\phi. \end{aligned}$$

On the government side, N imposes that the government sets money supply M_t^s to target an

interest rate i_t^* according to a policy rule:

$$i_t^* = \psi\left(P_t/P_t^*; \zeta_t^i\right)$$

where P_t^* represents a target path for the price level and ζ_t^i is an exogenous random shock to monetary policy. In equilibrium, the central bank adjusts M_t^s so that $i_t = i_t^*$. This policy rule means that equilibrium can be expressed for a target path $\{P_t, i_t\}$ given the supply of other assets, with M_t^s adjusting in the background to implement the interest-rate target. For the general case $0 < \rho < 1$, N log-linearizes the household FOCs and the banking and monetary-policy conditions. Define $Q_t^{s,(1)} := q_t^{(1)} b_t^{s,(1)}$. Bond and deposit market clearing imply $q_t^{(1)} b_t^{(1)} = Q_t^{s,(1)}$ and $D_t = D_t^s$ Define $(\widehat{M^s/P})_t := \log((M_t^s/P_t)/(\bar{M}^s/\bar{P}))$, $(\widehat{Q^{s,(1)}/P})_t := \log((Q_t^{s,(1)}/P_t)/(\bar{Q}^{s,(1)}/\bar{P}))$, $\widehat{Y}_t := \log(Y_t/\bar{Y})$, $\widehat{\lambda}_t := \lambda_t - \bar{\lambda}$, $\widehat{i}_t := \log((1+i_t)/(1+\bar{i}))$, and $\widehat{i}_t^{(1)} := \log((1+i_t^{(1)})/(1+\bar{i}^{(1)}))$. Their log-linearized equilibrium system is

$$(\widehat{M^s/P})_t = -\iota_i \widehat{i}_t - \iota_\lambda \widehat{\lambda}_t - \iota_Q (\widehat{Q^{s,(1)}/P})_t + \iota_Y \widehat{Y}_t, \tag{F.21}$$

$$\widehat{i}_t - \widehat{i}_t^{(1)} = \varphi_i \widehat{i}_t + \varphi_\lambda \widehat{\lambda}_t - \varphi_Q (1-\rho)(\widehat{Q^{s,(1)}/P})_t + \varphi_Y (1-\rho)\widehat{Y}_t, \tag{F.22}$$

where ι_i, ι_λ, ι_Q, ι_Y, φ_i, φ_λ, φ_Q, and φ_Y are positive constants defined by the steady state. Equations (F.21) and (F.22) correspond to equations (17) and (18) in N. Equation (F.21) gives the reserve supply required to implement the target path for the short rate, while equation (F.22) gives the resulting equilibrium T-bill liquidity premium.

When $\rho = 1$, deposits and T-bills are perfect substitutes and equations (F.20) and $i_t^d = \delta i_t$ give the exact equilibrium spread

$$i_t - i_t^{(1)} = \frac{\lambda_t}{1-\lambda_t}(1-\delta) i_t. \tag{F.23}$$

The reserve supply that implements the interest-rate target in this case is

$$M_t^s = \frac{\alpha(1+i_t)P_t}{(1-\delta) i_t \phi u'(Y_t)} - \frac{\lambda_t}{\phi(1-\lambda_t)} Q_t^{s,(1)}. \tag{F.24}$$

Equations (F.23) and (F.24) correspond to equations (13) and (14) in N. Conditional on i_t and λ_t, the central bank offsets a change in T-bill supply by changing reserve supply, so T-bill supply does not enter equation (F.23). At the other limit, as $\rho \to 0$, deposits and T-bills are Cobb–Douglas and the exact equilibrium spread is

$$i_t - i_t^{(1)} = \frac{\alpha \lambda_t (1+i_t) P_t}{u'(Y_t) Q_t^{s,(1)}}. \tag{F.25}$$

Equation (F.25) corresponds to equation (15) in N. In the log-linearized equation (F.22), the coefficients on T-bill supply and liquidity demand are proportional to $1-\rho$: they vanish in the perfect-substitutes case and attain their limiting values in the Cobb–Douglas case.

KL equilibrium spread equation: The KL specification of $\mathcal{A}_t$ and ν from Table F.13 uses the same CES structure for deposits and government bonds as N but includes all government debt rather than only short-term government debt. The marginal contributions of deposits and government bonds to the asset composite are

$$f_{D,t} = \mathcal{A}_t^{1-\rho}(1-\lambda_t)\left(\frac{D_t}{P_t}\right)^{\rho-1}, \qquad f_{b,t}^{(1)} = \mathcal{A}_t^{1-\rho}\lambda_t\left(\frac{Q_t}{P_t}\right)^{\rho-1}. \tag{F.26}$$

Because $\nu_t'(\mathcal{A}_t) = 1/\mathcal{A}_t$, substituting equation (F.26) into equations (F.10) and (F.11) gives

$$\frac{\alpha}{u'(C_t)}\mathcal{A}_t^{-\rho}(1-\lambda_t)\left(\frac{D_t}{P_t}\right)^{\rho-1} = \frac{i_t - i_t^d}{1+i_t}, \tag{F.27}$$

$$\frac{\alpha}{u'(C_t)}\mathcal{A}_t^{-\rho}\lambda_t\left(\frac{Q_t}{P_t}\right)^{\rho-1} = \frac{i_t - i_t^{(1)}}{1+i_t}. \tag{F.28}$$

Dividing equation (F.28) by equation (F.27) and imposing market clearing gives

$$i_t - i_t^{(1)} = \frac{\lambda_t}{1-\lambda_t}\left(\frac{Q_t^s}{D_t^s}\right)^{\rho-1}(i_t - i_t^d), \tag{F.29}$$

which is equation (11) in KL.

N and KL empirical specifications: N interprets equation (18) (our equation F.22) as predicting that, once the target short rate i_t^* is controlled for, Treasury quantities are only relevant for predicting spreads if ρ differs from one. N tests this prediction by regressing a short-maturity spread on different measures of debt-to-GDP, the federal funds rate, and the VIX (in their Table III). Specifically, the short-maturity spread is the three-month bankers' acceptance rate minus the three-month T-bill yield from January 1920 through April 1991, spliced with the three-month general-collateral repo rate minus the three-month T-bill yield from May 1991 through December 2011. N uses the monthly average of the daily effective federal funds rate from FRED (H.15 release) [`DFF`] from July 1954 onward, and the Federal Reserve Bank of New York discount rate (NBER Macro History Database, via FRED) before then.

KL augments the empirical analysis in N by constructing the additional quantity measures that enter equation (F.29). As noted in the definition of $\mathcal{Q}_t$ in Appendix F.1.1, KL

measure $\mathcal{Q}_t$ with net Treasurys held by households, not Treasurys outstanding. They construct their Net Treasury measure by starting with Treasury debt held outside the federal government and subtracting the holdings of banks and the Federal Reserve to approximate the Treasury holdings of the non-bank private sector. They convert this measure to market value and construct deposits as the sum of checking deposits, savings deposits, and small time deposits. This produces Net Treasurys/GDP and Net Treasurys/Deposits. Their baseline sample runs from 1934 through 2016, using annual observations before 1953 and quarterly observations thereafter. For the other series, KL uses the series from N extended to 2016. KL also run a collection of additional tests beyond the analysis in N in order to estimate ρ. In particular, they run an IV in which the instrument for Log(NetTsy/Deposits) is Total Debt/GDP.

Replication: To replicate the analysis in N and KL, we construct our own version of the relevant series. For the spread, we use our 3-month funding advantage. Since our funding advantage is a continuously-compounded zero-coupon yield sampled at the last trading day of each calendar month, for consistency we construct our own end-of-month Federal Funds Rate series, which we label as FFR(LPSS). We splice from three sources, each taking precedence over the others wherever their coverage overlaps. The first is the daily midpoint of high/low federal funds quotations reported in the *New York Herald-Tribune* (Anbil, Carlson, Hanes, and Wheelock, 2021), covering April 1928 to March 1938. The second is the daily midpoint of high/low quotations from the *Wall Street Journal*, covering June 1932 to June 1954 and preferred over the Herald-Tribune leg wherever both are available (June 1932 to March 1938). The third is the FRED daily effective federal funds rate (series `DFF`), covering July 1954 onward. These three legs are combined into a single daily series and resampled to the last available trading-day observation in each calendar month. FFR($LPSS$) is undefined prior to April 1928, before any of the three source series begins. We use the quantity series from N and KL.

The top panel of Table F.15 reconstructs Table III from N, which summarizes the main regression results (columns 1-6). It also reconstructs the additional regressions run by KL to include debt/deposit ratios (columns 7-8). This summarizes the main results in both papers. Columns (1-6) show N's argument that the coefficients on Log(TotalDebt/GDP) and Log(TBills/GDP) become non-significant once FFR is added to the regressions. This is the result that leads N to conclude that ρ must be close to 1. Columns (7-8) show KL's response that once the ratio of Net Treasurys to deposits is used as the quantity regressor, it becomes significant. The bottom panel of Table F.15 replicates the analysis using our data. These regressions show that the conclusions in N and KL are sensitive to the 3-month spread

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)
N and KL data								
Dependent variable	LiqPrem	LiqPrem	LiqPrem	LiqPrem	LiqPrem	LiqPrem	LiqPrem	LiqPrem
	1926-2011	1926-2011	1926-2011	1947-2011	1947-2011	1947-2011	1926-2011	1934-2011
FFR	10.778***		10.758***		10.999***	10.628***	10.431***	10.395***
	(0.766)		(0.824)		(0.843)	(0.845)	(0.745)	(0.702)
VIX	1.250***	0.396	1.247***	1.668***	1.660***	1.649***	1.097***	1.398***
	(0.178)	(0.255)	(0.186)	(0.550)	(0.350)	(0.350)	(0.192)	(0.281)
Log(TotalDebt/GDP)		-48.648***	-0.290			-7.258		
		(7.547)	(3.689)			(7.626)		
Log(TBills/GDP)				-90.966***	-3.956	-0.341		
				(16.842)	(7.643)	(8.229)		
Log(NetTsy/GDP)							-8.471***	
							(2.874)	
Log(NetTsy/Deposits)								-12.865***
								(2.685)
Intercept	-29.432***	-12.886*	-29.560***	-208.861***	-48.264**	-44.059**	-38.179***	-39.921***
	(5.323)	(7.114)	(5.403)	(43.131)	(20.365)	(20.076)	(6.290)	(7.305)
Observations	1032	1032	1032	780	780	780	1032	936
R^2	0.601	0.198	0.601	0.185	0.616	0.617	0.610	0.636
Our data								
Dependent variable	3mo spread	3mo spread	3mo spread	3mo spread	3mo spread	3mo spread	3mo spread	3mo spread
	1928-2011	1926-2011	1928-2011	1947-2011	1947-2011	1947-2011	1928-2011	1934-2011
FFR(LPSS)	4.262***		3.400***		5.463***	5.848***	2.950***	3.916***
	(0.684)		(0.838)		(0.765)	(0.886)	(0.676)	(0.599)
VIX	2.737***	2.022***	2.545***	1.112**	1.107**	1.119**	2.239***	1.370***
	(0.406)	(0.342)	(0.386)	(0.561)	(0.510)	(0.511)	(0.352)	(0.441)
Log(TotalDebt/GDP)		-30.200***	-14.184**			7.980		
		(4.935)	(6.593)			(9.575)		
Log(TBills/GDP)				-36.986***	7.726	3.676		
				(12.664)	(11.553)	(11.697)		
Log(NetTsy/GDP)							-27.335***	
							(4.704)	
Log(NetTsy/Deposits)								-21.195***
								(4.259)
Intercept	-13.523*	-9.305	-19.354**	-57.173*	26.005	21.290	-38.713***	-3.924
	(8.002)	(7.436)	(8.547)	(33.647)	(29.606)	(29.042)	(8.992)	(9.444)
Observations	912	1032	912	780	780	780	912	843
R^2	0.261	0.222	0.271	0.055	0.219	0.221	0.329	0.264

Table F.15: Table III of N, as replicated in Table 10 of KL, run here on KL's own data and on our 3-month corporate-Treasury spread. Dependent spread ("LiqPrem"/"3mo spread") is 100 times the raw spread (basis points), not logged. Columns (1)-(6) replicate N's own Table III specifications (Total Debt/GDP, then T-Bills/GDP); columns (7)-(8) are KL's own additions (NetTsy/GDP, NetTsy/Deposits). Sample starts 1926, not N's 1920, per KL's own VIX-data constraint. Signs in columns (4)-(6) match N's own text discussion of these results. HAC SEs (4 lags) in parentheses. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (based on $|t|$).

that is used. We find that, even after conditioning on the FFR, Log(TotalDebt/GDP) is still significant at the 5% level.

Table F.16 replicates Table 2 from KL, which uses both OLS and IV regressions to estimate ρ. The top panel reconstructs their results, which shows that they find that ρ well below one, in contrast to N. The bottom panel replicates their analysis using our series. To minimize the differences, we use their exact quantity series and instrument. We find that estimated ρ is higher (in fact closer to N's estimates) and the IV is no longer significant. To understand why, we split the sample by era. The coefficient is negative and significant before 1960, indistinguishable from zero from 1960–2006, and strongly positive and significant for both spreads during the 2007–2016 crisis and its aftermath. The full-sample IV estimate is therefore highly sensitive to how these offsetting eras happen to average out, which differs slightly between the two series given their different observation counts by era, rather than to which short rate enters the regression.

F.2.3 CKP

CKP uses a similar Treasury demand function to the other papers discussed in the Appendix. However, the authors impose more structure on Treasury supply by assuming that the Treasury acts as a monopolist.

Equilibrium spread equation: Using CKP's specification of $\mathcal{A}_t$ and $\nu_t'(\mathcal{A}_t)$ in Table F.13, the one-period spread demand equation (F.13) becomes:

$$\chi_t^{(1)} \approx \frac{\alpha \nu_t'(\mathcal{A}_t)}{u'(C_t)} f_{b,t}^{(1)} = (\beta + \gamma v_t) \left(\log \left(\frac{\mathcal{Q}_t / P_t}{Y_t} \right) - \zeta_t + 1 \right), \tag{F.30}$$

which is the same demand function as in KVJ except that the authors have introduced the stochastic demand "rotator" variable v_t and set $\theta_t^P = 0$ so only government debt carries a non-pecuniary benefit. CKP define $\zeta_t = 1 - (\alpha_0 + \delta X_t + \varepsilon_t)/(\beta + \gamma v_t)$, where α_0 denotes the regression intercept, X_t is a vector of controls, and ε_t is the residual. Therefore, equation (F.30) becomes their empirical inverse-demand equation (8):

$$\chi_t^{(1)} = \alpha_0 + \beta \log \left(\frac{\mathcal{Q}_t / P_t}{Y_t} \right) + \gamma v_t \log \left(\frac{\mathcal{Q}_t / P_t}{Y_t} \right) + \delta X_t + \varepsilon_t. \tag{F.31}$$

CKP close the model by allowing the US government to potentially act as the monopoly supplier of the asset providing non-pecuniary benefits to the household. To state their government problem in the notation of this Appendix, let $\chi_t^{(1)}(Q_t^s / P_t)$ denote the inverse

demand function for the spread in equation (F.31), and let $\mathcal{C}_t(Q_t^s/P_t)$ denote the increasing and convex resource cost of issuing debt. Let k_t^K denote the US capital stock, $r_{K,t}$ its rental rate, δ_K its depreciation rate, and w_t labor income. CKP define the US net asset position as $a_{t+1} := k_{t+1}^K - Q_{t+1}^s/P_{t+1}$. After this change of variables and imposing market clearing, the government's problem is

$$\begin{aligned} &\max_{\{C_t, a_{t+1}, Q_t^s/P_t\}_{t\geq 0}} \sum_{t=0}^{\infty} \beta^t u(C_t) \\ &\text{s.t.} \quad C_t + a_{t+1} = w_t + (1 - \delta_K + r_{K,t})a_t + S_t(Q_t^s/P_t)Q_t^s/P_t - \mathcal{C}_t(Q_t^s/P_t). \end{aligned} \tag{F.32}$$

The government takes w_t and $r_{K,t}$ as given but internalizes the effect of its issuance on $S_t(Q_t^s/P_t)$. Because debt is short term, the issuance decision in equation (F.32) is static. The FOC determining government debt supply is

$$\chi_t^{(1)}(Q_t^s/P_t) = \mathcal{C}_t'(Q_t^s/P_t) - (\chi_t^{(1)})'(Q_t^s/P_t)Q_t^s/P_t,$$

which in equilibrium gives the explicit government supply equation:

$$(\beta + \gamma v_t)\left(\log\left(\frac{Q_t^s/P_t}{Y_t}\right) - \zeta_t + 2\right) = \mathcal{C}_t'(Q_t^s/P_t). \tag{F.33}$$

Equations (F.31) and (F.33) jointly determine the equilibrium spread and government debt supply.

Empirical demand estimation: CKP start by estimating the demand function (F.31). Rather than focusing on the spread at a particular maturity, they construct a weighted average funding advantage measure which is a fixed-weight combination of two single-maturity proxy spreads. The long-maturity leg is the index-based spread defined and discussed in Section 3.3. That is, the spread between the Moody's Aaa corporate bond yield (FRED `Aaa`) and a long-term Treasury yield, spliced from the long-term U.S. government bond yield (FRED `LTGOVTBD`) for 1935–2000 and the 20-year constant-maturity Treasury yield (FRED `GS20`) for 2001–2019. The short-maturity leg is the spread between a 3-month commercial paper yield and a Treasury bill yield. The commercial paper leg splices 3-month AA-or-equivalent dealer-placed commercial paper (FRED `CP3M`) for 1971–1996 and 3-month AA nonfinancial commercial paper (FRED `CPN3M`) for 1997–2019, while the bill leg splices 3–6-month Treasury bills from NBER Macrohistory series `m13029b` for 1935–1958, 6-month Treasury bills (FRED `TB6MS`) for 1959–1970, and 3-month Treasury bills (FRED `TB3MS`) for 1971–2019. Both legs are converted to quarterly averages of the underlying monthly series, and the baseline spread

	OLS		IV	
	1934-2016	1934-2016	1934-2016	1934-2016
	(1)	(2)	(3)	(4)
KL data				
Dependent spread	log(LiqPrem)	log(LiqPrem)	log(LiqPrem)	log(LiqPrem)
Log(FFR)	0.541***	0.508***	0.461***	0.473***
	(0.078)	(0.076)	(0.109)	(0.091)
Log(VIX)	0.728***	0.566**	0.590*	0.464
	(0.269)	(0.270)	(0.328)	(0.289)
Log(NetTsy/Deposits)	-0.347***		-0.866***	
	(0.116)		(0.325)	
Log(NetTsy/KVJDeposits)		-0.517***		-0.723***
		(0.158)		(0.278)
Intercept	-4.210***	-3.979***	-4.109***	-3.860***
	(0.777)	(0.766)	(0.912)	(0.785)
Observations	300	300	300	300
R^2	0.431	0.443	0.364	0.436
ρ	0.653	0.483	0.134	0.277
	(0.116)	(0.158)	(0.325)	(0.278)
Our data				
Dependent spread	log(3mo spread)	log(3mo spread)	log(3mo spread)	log(3mo spread)
Log(FFR)	0.254***	0.293***	0.330***	0.338***
	(0.038)	(0.056)	(0.058)	(0.060)
Log(VIX)	0.322	0.264	0.321	0.380
	(0.246)	(0.260)	(0.292)	(0.265)
Log(NetTsy/Deposits)	-0.496***		0.119	
	(0.118)		(0.203)	
Log(NetTsy/KVJDeposits)		-0.161		0.165
		(0.183)		(0.198)
Intercept	-2.316***	-2.013***	-1.980**	-2.077***
	(0.703)	(0.776)	(0.872)	(0.797)
Observations	299	299	299	299
R^2	0.387	0.314	0.266	0.291
ρ	0.504	0.839	1.119	1.165
	(0.118)	(0.183)	(0.203)	(0.198)

Table F.16: Table 2 of KL, replicated on the paper's own data and on this project's own 3-month zero-coupon corporate-Treasury spread (KL's own maturity), otherwise identical to the paper's own specification. Columns (1)-(2) are OLS; columns (3)-(4) instrument the quantity ratio with log(Total Tsy/GDP) and Total Tsy/GDP (level). Net Tsy and the instrument (Total Tsy/GDP) are KL's own series, exactly as published. ρ is KL's own CES substitution parameter between money/deposits and Treasury debt (their Equation 20), and the corresponding elasticity of substitution is $1/(1-\rho)$. We compute ρ as 1+ the coefficient on whichever quantity variable is in that column. HAC standard errors are reported in parentheses below each coefficient, rather than the t-statistics KL-23's own Table 2 reports. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$ (based on $|t|$, coefficient rows only).

is

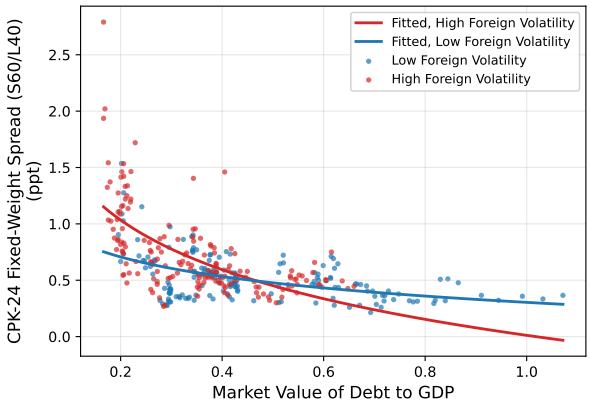
$$\text{spread_S60_L40} = 0.6 \times \text{spread_short} + 0.4 \times \text{spread_long},$$

with the 0.6/0.4 weights held fixed over the full 1935–2019 sample.

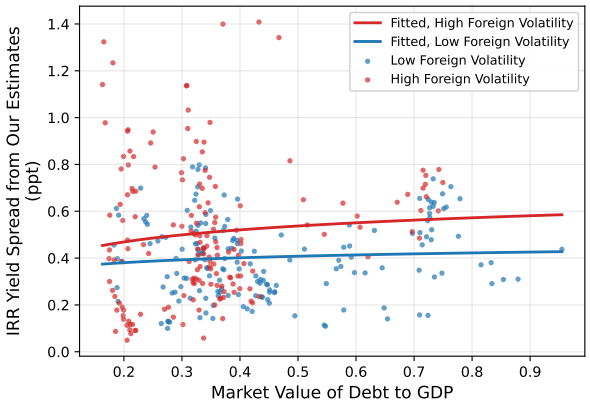
Our replication of CKP's Table 1 substitutes their weighted average spread with our internal rate of return (IRR) spread that takes into account the full term structure. Our spread is the gap between two IRRs computed on the government's full panel of promised par cash flows across every outstanding maturity. The first IRR is the single constant yield that, when used to discount those cash flows, reproduces the actual observed market value of Treasury debt outstanding. The second IRR is the constant yield that discounts the identical cash-flow structure as the first IRR but reproduces the market value of Treasury debt outstanding priced using the corporate zero-coupon bond prices. That is, the second IRR reproduces a hypothetical market value representing what the debt would be worth if priced with no funding advantage attached. This construction is preferred over an approach that averages corporate and Treasury yields directly, weighted by cash-flow size, because cash-flow weighting is valid only in price space not in the yield space. The debt-to-GDP denominator paired with this spread is likewise our own market-value-consistent series, built from the same market-value debt-quantity total underlying the spread itself, rather than CKP's own par-value debt-to-GDP measure, so that the numerator and denominator of the regressor reflect a single, internally consistent debt-quantity construction. All other elements of the specification are held fixed at CKP's own values: the quarterly sampling frequency, the 1935-onward estimation window, the foreign-volatility demand rotator (residualized against U.S. volatility and real GDP growth), the volatility and slope controls, and, for the instrumented column, the Ramey military-news-shock instrument.

Figure F.47 depicts the relationship between the Aaa Corporate-Treasury spread and debt issuance using their data and our data. Panel (a) reproduces the plot from Figure 2 in CKP. Panel (b) replicate the plot using our estimate of the IRR yield spread. Panels (c) and (d) show the same plot but using the weighted average spread at short maturities (less than 1 year) and long maturities (greater than 1 year). On all plots, the red dots depict periods with high UK equity return volatility while the blue dots depict periods with low UK equity return volatility. CKP interpret the changes to the shape in their scatter plot as suggestive evidence of a rotation in US debt demand. Contrary to CKP, we find little evidence that the foreign volatility acts as a rotator, except for very short-term maturities.

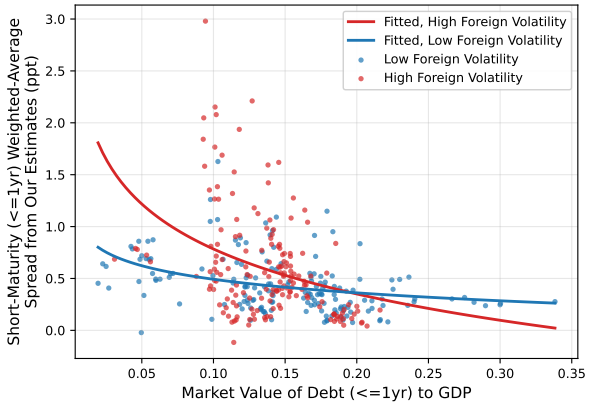
Table F.17 then replicates the main demand estimation regressions in CKP, which are in Table 1 in their paper. The top panel reproduces the CKP results while the bottom panel replicates them using our IRR spreads. Evidently, there are a number of major differences.



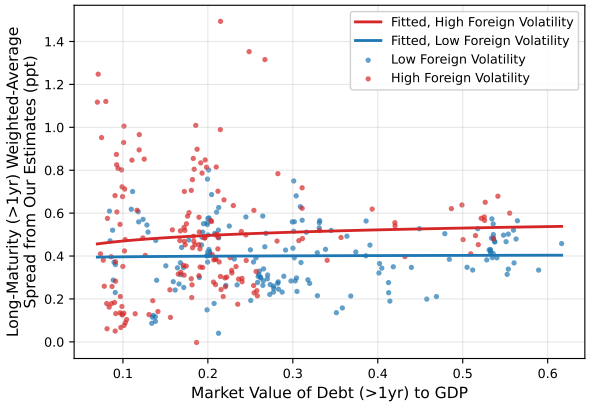
(a) CKP fixed weight ave. spread



(b) Our IRR yield spread



(c) Our short maturity spread ($\leq$1yr)



(d) Our long maturity (>1yr)

Figure F.47: Spread vs. market value of debt/GDP by foreign-volatility regime, maturity-bucketed. Our short maturity spread is a quantity weighted average spread for maturities less than or equal to one year. Our long maturity spread is a quantity weighted average spread for maturities greater than one year.

In general, our regressions find a positive OLS coefficient on Log(debt/GDP) and an IV coefficient that is at most marginally negative. More importantly, we find that the coefficient on the Foreign Volatility Dummy and Log(Debt/GDP) interaction term is insignificant. This is consistent with the differences we observed visually in Figure F.47.

Market power estimation: CKP additionally test whether the US government behaves as a monopolist, using a Rivers-Vuong (2002) model-selection test that compares how strongly each candidate model's implied marginal-cost shifter ω_t correlates with the demand rotator

v_t. Rearranging the policy rule for ω_t gives

$$\omega_t(\mathsf{M}) = \log(\chi_t^{(1)}) + \log\left(1 - \mathsf{M}\,\mathcal{E}_t^{-1}(\sigma_t)\right) - \upsilon \log\left(\frac{\mathcal{Q}_t}{P_t Y_t}\right),$$

where $\mathsf{M} \in \{0, 1\}$ is an indicator for monopoly behavior so that $\omega_t^{PT} := \omega_t(0)$ under price-taking and $\omega_t^{ME} := \omega_t(1)$ under monopoly. Since $\log(1 - 0) = 0$, the cost shifters for the models with $\mathsf{M} = 0$ and $\mathsf{M} = 1$ differ only through

$$\omega_t^{ME} - \omega_t^{PT} = \log\left(1 - \mathcal{E}_t^{-1}(\sigma_t)\right). \tag{F.34}$$

The wedge in equation (F.34) varies with t only through $\mathcal{E}_t^{-1}(\sigma_t)$'s dependence on the volatility regime $\sigma_t \in \{\sigma_L, \sigma_H\}$, and that regime-dependence is exactly the FVD $\times$ Log(debt/gdp) interaction coefficient γ estimated in Table F.17. If $\gamma = 0$, then $\mathcal{E}^{-1}(\sigma_H) = \mathcal{E}^{-1}(\sigma_L)$, the wedge is a single constant across the whole sample, and it is removed exactly once ω_t is demeaned against an instrument set that includes an intercept. In that case ω_t^{ME} and ω_t^{PT} are identical after demeaning, the Rivers-Vuong statistic is exactly zero for every assumed cost elasticity υ, and the test has no power to distinguish the two models. The test's entire discriminating power is inherited from whatever precision Table F.17 already has on γ.

On our IRR spread, the γ coefficient is not statistically distinguishable from zero in either the OLS or IV column ($\hat{\gamma} = 0.062$, $p = 0.30$ and $\hat{\gamma} = -0.165$, $p = 0.38$ respectively). The conduct test therefore has no power to discriminate price-taking from monopoly issuance on our yield curve estimates.

	With FVD interaction		No interaction	
	(1) OLS	(2) IV	(3) OLS	(4) IV
	1935:Q1-2019:Q4			
CKP data (S60/L40 spread)				
Log(debt/gdp)	-0.233***	-0.238***	-0.383***	-0.282***
	(0.043)	(0.057)	(0.033)	(0.055)
Foreign Volatility Dummy (FVD)	-0.314***	-0.142	0.047*	0.072**
	(0.068)	(0.128)	(0.026)	(0.029)
FVD x Log(debt/gdp)	-0.338***	-0.184		
	(0.062)	(0.118)		
US VIX	0.018***	0.018***	0.019***	0.021***
	(0.002)	(0.002)	(0.002)	(0.002)
Yield curve slope	-0.058***	-0.062***	-0.055***	-0.063***
	(0.012)	(0.012)	(0.012)	(0.013)
Constant	0.115**	0.110	-0.049	0.021
	(0.058)	(0.067)	(0.053)	(0.062)
Observations	339	339	339	339
R^2	0.532		0.494	
Demand elasticity, high foreign vol	1.092	1.479	1.629	2.213
Demand elasticity, low foreign vol	2.676	2.622	1.629	2.213
Markup, high foreign vol	0.916	0.676	0.614	0.452
Markup, low foreign vol	0.374	0.381	0.614	0.452
Our data (IRR spread)				
Log(debt/gdp)	0.088**	-0.038	0.109***	-0.131*
	(0.043)	(0.071)	(0.032)	(0.073)
Foreign Volatility Dummy (FVD)	0.125*	-0.165	0.038	-0.016
	(0.066)	(0.213)	(0.024)	(0.030)
FVD x Log(debt/gdp)	0.062	-0.165		
	(0.060)	(0.189)		
US VIX	0.016***	0.016***	0.019***	0.016***
	(0.002)	(0.002)	(0.002)	(0.002)
Yield curve slope	-0.076***	-0.048***	-0.075***	-0.048***
	(0.012)	(0.017)	(0.012)	(0.014)
Constant	0.326***	0.178**	0.314***	0.114
	(0.057)	(0.081)	(0.052)	(0.077)
Observations	336	336	336	336
R^2	0.247		0.237	
Demand elasticity, high foreign vol	-3.004	2.226	-4.139	3.442
Demand elasticity, low foreign vol	-5.132	11.765	-4.139	3.442

Table F.17: Table 1 of Choi et al. (2026), replicated on the paper's own data (CKP-24's fixed 0.6/0.4 short/long weighting of privately-held debt) and on our IRR spread, paired with the market-value-consistent debt/GDP built from the same debt-quantity panel. Columns (1)/(2) include the debt/GDP × Foreign Volatility Dummy (FVD) interaction, letting the demand elasticity differ by volatility regime; columns (3)/(4) impose a single pooled elasticity. Columns (2)/(4) instrument log(debt/gdp) (and its FVD interaction, in column (2)) with a Ramey military-news-shock lag window chosen by first-stage F-stat, plus the change in the dependency ratio. R^2 is only meaningful for the OLS columns and left blank for IV, matching CKP's own published Table 1. Standard errors are reported in parentheses below each coefficient. Significance: * $p < 0.1$, ** $p < 0.05$, *** $p < 0.01$.

G The Affine Term Structure Model From Section 6.2

This Appendix completes the description of our Affine Term Structure Model (ATSM). We start by constructing the expressions for bond prices and holding returns. We next discuss the interpretation of the yield curve factors. We then show the output from the model selection exercise. Finally, we explain the shock identification for our impulse response exercise.

G.1 Yield Curve Principal Components

The top row of Figure G.48 shows the loadings of the corporate and Treasury yield curves on the first three principal component factors at each maturity. The bottom row shows the funding advantage on the first three principal components of the funding advantage. From this we can make a number of observations:

(i) The corporate yield curve principle components have a very similar structure to the Treasury yield curve. In particular, the first principal component is effectively a "level" factor that has a similar loading across maturity, the second principal component is a "slope" factor that is monotonic in the maturity, and the third principal component is a "curvature" factor that is convex in the maturity structure.

(ii) The funding advantage has a similar but not identical factor structure to the yield curves. The first principal component appears to approximately capture the level at long maturities while the second and third principal components look like slope and curvature factors for the funding advantage.

G.2 Bond Prices and Holding Returns

We complete the characterization of bond prices and holding returns from the affine term structure model described in Subsection 6.2 of the main text.

G.2.1 Corporate Bond Pricing

Bond prices: Standard results from the literature (e.g. Piazzesi, 2010) show that corporate bond prices take the exponential affine form:

$$\log \tilde{q}_t^{(j)} = \tilde{A}^{(j)} + (\tilde{B}^{(j)})^T X_t. \tag{G.1}$$

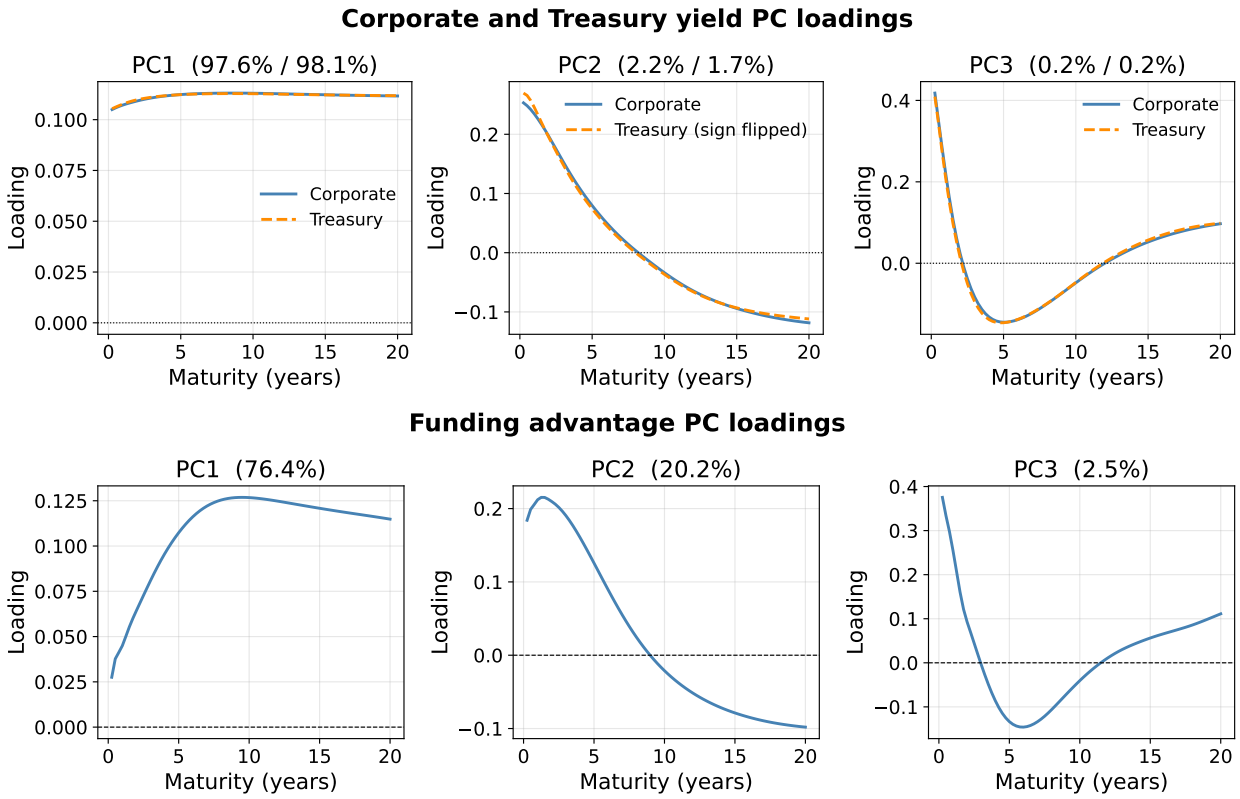


Figure G.48: Principal component loadings. The top row shows the loadings of the first three principal components of the corporate and Treasury yield curves on maturity (0 to 20 years). The sign of the second Treasury component is reversed, to improve the comparison. The bottom row shows the loadings of the first three principal components of the funding advantage, the corporate yield minus the Treasury yield at each maturity.

where $\tilde{A}^{(j)}$ and $\tilde{B}^{(j)}$ satisfy $\tilde{A}^{(0)} = \tilde{B}^{(0)} = 0$ for $j = 0$ and the following recursion for $j \geq 1$:

$$\begin{aligned} \tilde{A}^{(j)} &= -\delta_0 + \tilde{A}^{(j-1)} + (\tilde{B}^{(j-1)})^T(\mu_X - \Sigma\lambda_0) + 0.5(\tilde{B}^{(j-1)})^T\Sigma\tilde{B}^{(j-1)} \\ (\tilde{B}^{(j)})^T &= -\delta_1^T + (\tilde{B}^{(j-1)})^T(\Phi_X - \Sigma\lambda_1) \end{aligned}$$

Holding returns: Define the excess holding period return $\widetilde{rx}_{t+1}^{(j-1)}$ on a j maturity corporate bond by:

$$\frac{\tilde{q}_{t+1}^{(j-1)}}{\tilde{q}_t^{(j)}} =: \exp\left(\tilde{r}_t + \widetilde{rx}_{t+1}^{(j-1)}\right). \tag{G.2}$$

Substituting (G.1) into (G.2) implies that:

$$\widetilde{rx}_{t+1}^{(j-1)} = \alpha^{(j-1)} + (\beta^{(j-1)})^T V_{t+1} + (\gamma^{(j-1)})^T X_t + e_{t+1}^{(j-1)}, \tag{G.3}$$

where $e_{t+1}^{(j-1)}$ is a mean-zero pricing error that is independent of V_{t+1} and normally distributed with variance σ^2, and:

$$\begin{aligned}\alpha^{(j-1)} &= \tilde{A}^{(j-1)} + (\tilde{B}^{(j-1)})^T \mu_X - \tilde{A}^{(j)} + \tilde{A}^{(1)},\\ \beta^{(j-1)} &= \tilde{B}^{(j-1)},\\ (\gamma^{(j-1)})^T &= (\tilde{B}^{(j-1)})^T \Phi_X - (\tilde{B}^{(j)})^T + (\tilde{B}^{(1)})^T.\end{aligned}$$

Combining (G.3) with the corporate bond Euler equation (6.1) gives the direct connection between the excess holding return loading and the risk prices in the SDF:

$$\begin{aligned}\alpha^{(j-1)} &= (\beta^{(j-1)})^T \Sigma \lambda_0 - 0.5(\beta^{(j-1)})^T \Sigma \beta^{(j-1)} - 0.5\sigma^2\\ (\gamma^{(j-1)})^T &= (\beta^{(j-1)})^T \Sigma \lambda_1.\end{aligned}$$

G.2.2 Treasury Pricing Kernel

The pricing kernel for j maturity Treasurys, $j \geq 1$, is $\Xi_{t,t+1}^{(j)} := \xi_{t,t+1}\Omega_{t,t+1}^{(j)}$, which also has an exponential affine form:

$$\Xi_{t,t+1}^{(j)} = \exp\left(-r_t^{(j)} - \tfrac{1}{2}(\Lambda_t^{(j)})^T \Sigma \Lambda_t^{(j)} - (\Lambda_t^{(j)})^T V_{t+1}\right),$$

where:

$$\begin{aligned} r_t^{(j)} &= \Delta_0^{(j)} + (\Delta_1^{(j)})^T X_t, & \Delta_0^{(j)} &:= \delta_0 + \mu_0^{(j)}, & \Delta_1^{(j)} &:= \delta_1 + \mu_1^{(j)},\\ \Lambda_t^{(j)} &= \Lambda_0^{(j)} + \Lambda_1^{(j)} X_t, & \Lambda_0^{(j)} &:= \lambda_0 + \omega_0^{(j)}, & \Lambda_1^{(j)} &:= \lambda_1 + \omega_1^{(j)}.\end{aligned}$$

This highlights that, unlike for corporate bonds, we potentially allow the Treasury pricing kernel to differ across different maturities. The Treasury pricing kernel must satisfy the Treasury Euler equation:

$$q_t^{(j)} = \mathbb{E}_t\left[\xi_{t,t+1}\Omega_{t,t+1}^{(j)} q_{t+1}^{(j-1)}\right] = \mathbb{E}_t\left[\Xi_{t,t+1}^{(j)} q_{t+1}^{(j-1)}\right]. \tag{G.4}$$

Bond prices: Analogously to for corporate bonds, we can guess and verify that there exist $A^{(j)}$ and $B^{(j)}$ such that Treasury bond prices take the form:

$$\log q_t^{(j)} = A^{(j)} + (B^{(j)})^T X_t. \tag{G.5}$$

For $j = 0$, $q_t^{(0)} = 1$ so $A^{(0)} = B^{(0)} = 0$. For $j = 1$, the bond pricing equation implies that:

$$\begin{aligned} q_t^{(1)} &= \mathbb{E}_t \left[\Xi_{t,t+1}^{(1)} \right] \\ &= \mathbb{E}_t \left[\exp\left(-r_t^{(1)} - \tfrac{1}{2} (\Lambda_t^{(1)})^T \Sigma \Lambda_t^{(1)} - (\Lambda_t^{(1)})^T V_{t+1} \right) \right] \\ &= \exp(-r_t^{(1)}). \end{aligned}$$

so we have that $A^{(1)} = -\Delta_0^{(1)}$ and $B^{(1)} = -\Delta_1^{(1)}$. For $j \geq 2$, the bond pricing equation implies that:

$$\begin{aligned} q_t^{(j)} &= \mathbb{E}_t \left[\Xi_{t,t+1}^{(j)} q_{t+1}^{(j-1)} \right] \\ &= \mathbb{E}_t \Big[\exp\left(-r_t^{(j)} - \tfrac{1}{2} (\Lambda_t^{(j)})^T \Sigma \Lambda_t^{(j)} - (\Lambda_t^{(j)})^T V_{t+1} \right) \\ &\quad \times \exp\left(A^{(j-1)} + (B^{(j-1)})^T X_{t+1} \right) \Big] \\ &= \exp\left(-r_t^{(j)} + A^{(j-1)} + (B^{(j-1)})^T \mu_X + (B^{(j-1)})^T \Phi_X X_t \right) \\ &\quad \times \exp\left(-(B^{(j-1)})^T \Sigma \Lambda_t^{(j)} + \tfrac{1}{2} (B^{(j-1)})^T \Sigma B^{(j-1)} \right). \end{aligned}$$

Imposing the state dependence for $\Lambda_t^{(j)}$ we have the recursion:

$$\begin{aligned} A^{(j)} &= -\Delta_0^{(j)} + A^{(j-1)} + (B^{(j-1)})^T \mu_X - (B^{(j-1)})^T \Sigma \Lambda_0^{(j)} + \tfrac{1}{2} (B^{(j-1)})^T \Sigma B^{(j-1)} \\ (B^{(j)})^T &= -(\Delta_1^{(j)})^T + (B^{(j-1)})^T \left(\Phi_X - \Sigma \Lambda_1^{(j)} \right). \end{aligned}$$

Holding returns: Unlike for the corporate bonds, it is no longer helpful to consider the excess holding return because each maturity potentially has its own, unobserved, level $r_t^{(j)}$. Instead, we define the holding return by:

$$\frac{q_{t+1}^{(j-1)}}{q_t^{(j)}} =: \exp\left(h_{t+1}^{(j-1)} \right). \tag{G.6}$$

Substituting (G.5) into (G.6), and adding a mean-zero pricing error as for corporate bonds, the holding period return takes the form:

$$h_{t+1}^{(j-1)} = \alpha^{(j-1)} + (\beta^{(j-1)})^T V_{t+1} + (\gamma^{(j-1)})^T X_t + e_{t+1}^{(j-1)}, \tag{G.7}$$

where $e_{t+1}^{(j-1)}$ is a mean-zero pricing error that is independent of V_{t+1} and normally distributed with variance σ^2, and:

$$\begin{aligned}\alpha^{(j-1)} &= A^{(j-1)} + (B^{(j-1)})^T \mu_X - A^{(j)},\\ \beta^{(j-1)} &= B^{(j-1)},\\ (\gamma^{(j-1)})^T &= (B^{(j-1)})^T \Phi_X - (B^{(j)})^T.\end{aligned}$$

Combining (G.7) with the Euler equation (G.4) gives the direct connection between the holding return loadings and the maturity-specific Treasury pricing kernel:

$$\begin{aligned}1 &= \mathbb{E}_t\left[\Xi_{t,t+1}^{(j)} \exp\left(h_{t+1}^{(j-1)}\right)\right]\\ &= \mathbb{E}_t\Big[\exp\left(-r_t^{(j)} - \tfrac{1}{2}(\Lambda_t^{(j)})^T \Sigma \Lambda_t^{(j)} - (\Lambda_t^{(j)})^T V_{t+1}\right)\\ &\quad \times \exp\left(\alpha^{(j-1)} + (\beta^{(j-1)})^T V_{t+1} + (\gamma^{(j-1)})^T X_t + e_{t+1}^{(j-1)}\right)\Big]\\ &= \exp\left(-r_t^{(j)} + \alpha^{(j-1)} + (\gamma^{(j-1)})^T X_t - (\beta^{(j-1)})^T \Sigma \Lambda_t^{(j)} + \tfrac{1}{2}(\beta^{(j-1)})^T \Sigma \beta^{(j-1)} + \tfrac{1}{2}\sigma^2\right).\end{aligned}$$

It follows that:

$$\begin{aligned}\alpha^{(j-1)} &= \Delta_0^{(j)} + (\beta^{(j-1)})^T \Sigma \Lambda_0^{(j)} - 0.5(\beta^{(j-1)})^T \Sigma \beta^{(j-1)} - 0.5\sigma^2\\ (\gamma^{(j-1)})^T &= (\Delta_1^{(j)})^T + (\beta^{(j-1)})^T \Sigma \Lambda_1^{(j)}.\end{aligned}$$

Observe that now the loadings α and γ depend upon the maturity j through both the level and risk-price terms in the Treasury pricing kernel.

G.3 Estimation Approach

We estimate each model specification by maximum likelihood on the quarterly panels of corporate and Treasury yields ($T = 396$) and compare specifications by BIC. A specification is a triple of choices along the three dimensions in Table 4: the wedge level μ_t (M), the exposure of the wedge to the shocks ω_t (O), and the maturity-specific wedge (S).

We construct the likelihood following the approach from Joslin et al. (2011). We apply the canonical form of Joslin et al. (2011) to the corporate yield curve. Its free risk-neutral parameters are the eigenvalues of the risk-neutral transition matrix $\Phi_X - \Sigma\Lambda_1$ restricted to the yield-curve states, and the long-run mean of the short rate r_∞. The first three corporate principal components are priced exactly, which fixes the loadings of the corporate yields on the state X_t. The first three funding-advantage principal components are also priced exactly. The corporate yields not spanned by the three corporate principal components, and the

spread coordinates not spanned by the three funding-advantage principal components, have AR(1) pricing errors, with a separate persistence for the two blocks and maturity-specific variances. Conditional on the first date, the likelihood is the sum of three independent parts: the density of the corporate pricing errors, the density of the spread pricing errors, and the Gaussian transition density of the state X_t.

We estimate parameters by maximizing the likelihood. To keep the parameter search small, we follow the approach in Joslin et al. (2011) and solve in closed form for every parameter that enters the likelihood linearly given the others. These are $r_\infty = \delta_0 + \delta_1^T (I - \Phi_X + \Sigma\Lambda_1)^{-1}(\mu_X - \Sigma\Lambda_0)$ (the long run level of the short rate using Joslin et al. (2011)'s decomposition), the free entries of Λ_0 and Λ_1, the pricing-error variances and the constant μ_0 in the wedge level. The search is over the remaining parameters: the eigenvalue parameters, the Cholesky factor of Σ and the pricing-error persistences. We maximize this concentrated likelihood from several starting points, using Nelder–Mead followed by BFGS. Richer specifications start from the fitted values of the specifications they nest. We discard parameter values at which the map from the canonical factors to the observed state is nearly singular or the dynamics are explosive.

We estimate every model twice. The first estimation is unrestricted maximum likelihood estimation. The second estimation follows Duffee (2010) and maximizes the likelihood subject to a bound on the sample average of the maximum conditional Sharpe ratio. We bound the average for both the corporate and the Treasury kernel at $0.25\sqrt{3} \approx 0.433$ per quarter, which is Duffee's monthly bound of 0.25 scaled to the quarterly frequency (Duffee, 2010). We solve the constrained problem by a penalty method with multipliers on the two averages, started from the unconstrained solution, and compute the BIC at the constrained likelihood with the same k.

We compare specifications by $\text{BIC} = -2\ell + k \ln T$, where ℓ is the log-likelihood and k the number of free parameters. The comparison is valid across these specifications because they all use the same data vector. Figures G.49 and G.50 report the BIC of the M1 and M2 cells without the bound, and Figures G.51 and G.52 report it with the bound. In each figure, arrows show the change in BIC from adding a maturity-specific wedge (left to right) or a richer exposure (top to bottom), and the M2 figures also show each cell's BIC relative to the corresponding M1 cell. We omit S5, which frees every maturity leaving the exposure unidentified and is never selected. We also omit M0, in which every Treasury yield equals the corporate yield less a constant μ_0 and the spread carries no information. We estimate M0 on the corporate principal components alone, so its BIC of $-139{,}469.6$ is only indicative; it is about 300 points above the least flexible M1 cell and about 535 above the best cell.

The BIC favors a parsimonious wedge: a rank-one exposure (O2), a wedge level that

loads on the yield-curve states only (M1), and maturity-specific wedges at the shortest maturities (S3). Table 4 shades the selected triple, M1-O2-S3. Without the Sharpe-ratio bound (Figures G.49 and G.50), this is the best cell, with a BIC of −140,005.7; M1-O2-S4 is within five points of it. Adding a constant exposure (O1) to a model with none (O0) improves the BIC by roughly 75 to 90 points at every maturity specification, so exposure of the wedge to the risk factors matters. Richer exposures do not pay for their extra parameters. Moving from rank one (O2) to rank two (O3) worsens the BIC by 15 to 55 points at every maturity specification, and the full-rank exposures (O4 and O5) are worse still by 20 to 170 points at each step. Freeing the 3-month wedge (S1) is the largest maturity gain, at about 115 points for the models without exposure or with a constant exposure. Beyond that the gains shrink, and in the best model they vanish by the 2-year wedge (S4). Letting the wedge level also load on growth and the bond balances (M2) rarely helps. Under the Sharpe-ratio bound (Figures G.51 and G.52), M2 is worse than M1 in 14 of 16 comparable cells, and it wins only at O3-S3 and O4-S3, by 24 and 17 points, which still leaves it well behind the best model. The bound itself costs about 155 BIC points at the best cell and moves the preferred specification from S3 to S4 (M1-O2-S4, −139,850.6), but it leaves the qualitative ranking, a rank-one exposure and a yield-curve-only wedge level, unchanged.

G.4 Shock Identification

The reduced-form innovation $V_{t+1} = X_{t+1} - c - \Phi_X X_t$ has covariance Σ and partitions innovations associated with the growth rate, the three bond balances, and the yield curve principal components: $V = (V^g, V^{n\prime}, V^{s\prime})'$. Net of the mechanical effect of growth on these shares, unexpected issuance of the three bonds is $u = V^n + \ell V^g$, where $\ell \in \mathbb{R}^3$ collects the average effect of growth on the balance shares. Total issuance is $e_I = \mathbf{1}'u$, and unexpected issuance of bond i in excess of its average share $\bar{\sigma}_i$ of total issuance is $e_{li} = u_i - \bar{\sigma}_i e_I$. The composition innovations sum to zero, so we keep $e_l = (e_{l1}, e_{l2})'$ and $e_{l3} = -e_{l1} - e_{l2}$. Hence $V^n = \bar{\sigma} e_I + N e_l - \ell V^g$ with $N = [I_2; -\mathbf{1}']$, and stacking $v = (V^g, e_I, e_l', V^{s\prime})'$ gives $V = Jz$ with

$$J = \begin{pmatrix} 0 & 0 & 0 & I_6 \\ 1 & 0 & 0 & 0 \\ -\ell & \bar{\sigma} & N & 0 \end{pmatrix},$$

which is invertible. Let $\Omega_z = J^{-1}\Sigma J^{-\prime} = CC'$ with C the lower-triangular Cholesky factor. The structural shocks are $\varepsilon = C^{-1}v$, ordered $\varepsilon_g, \varepsilon_I, \varepsilon_{l1}, \varepsilon_{l2}$ and then six yield-curve shocks, and the impact matrix is $\mathcal{S} = JC$ with $\mathcal{S}\mathcal{S}' = \Sigma$. Because C is lower triangular, growth

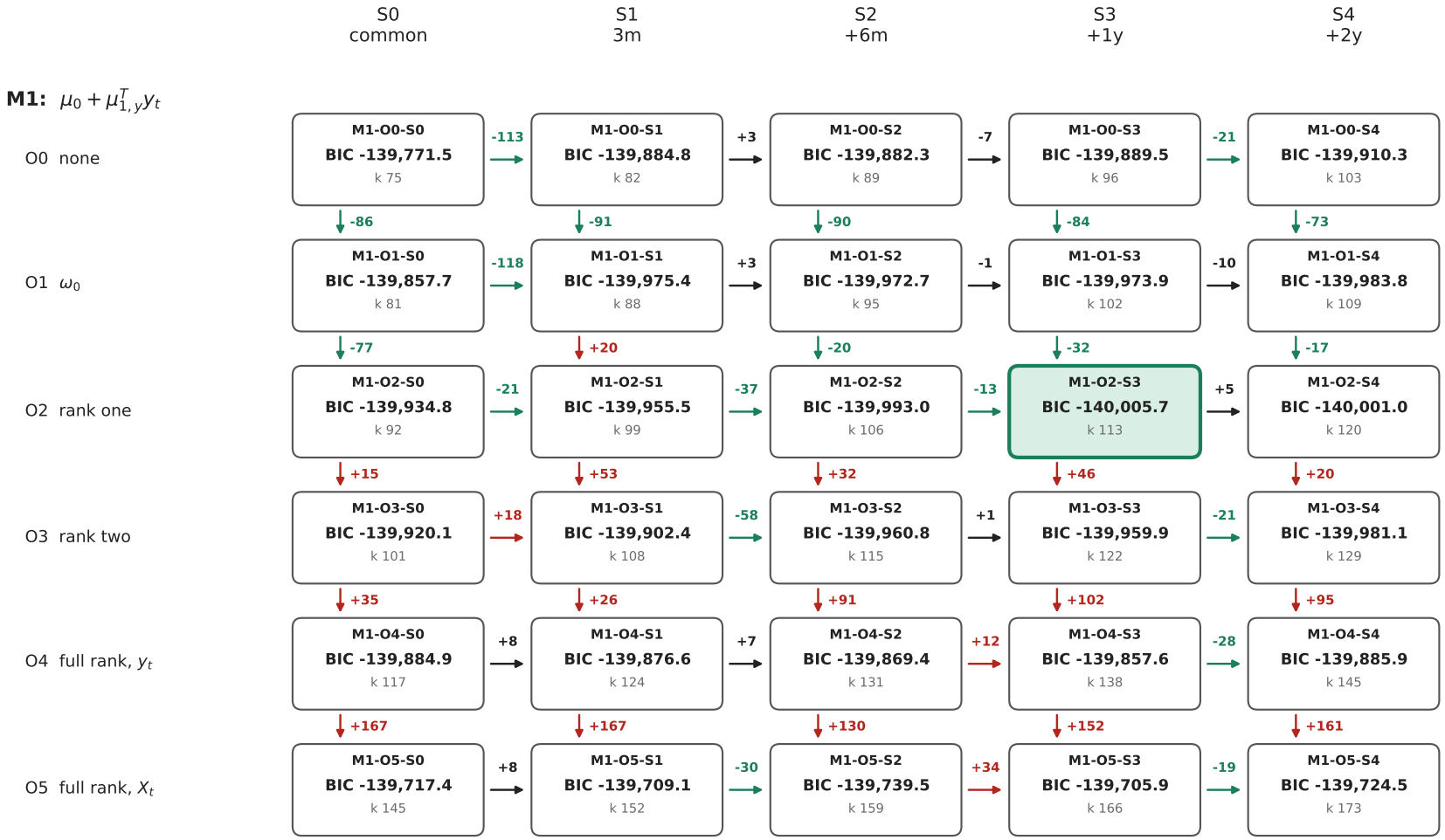


Figure G.49: Wedge model comparison by maximum-likelihood BIC: no drift loading on debt or growth states (specification M1 from Table 4). Each box is one specification, with the maturity-specific wedge S0 to S4 in the columns (the common wedge, then the Treasury loadings freed at the 3-month, 6-month, 1-year and 2-year maturities in turn) and the exposure O0 to O5 in the rows. A box reports the BIC and the number of free parameters k. Each arrow gives the change in BIC from adding a maturity-specific wedge (horizontal, to the next column) or a richer exposure (vertical, to the next row), that is, the BIC of the box at the head of the arrow minus the BIC of the box at its tail. Green arrows lower the BIC by 10 or more and red arrows raise it by 10. The shaded box has the lowest BIC of all M1 and M2 specifications.



Figure G.50: Wedge model comparison by maximum-likelihood BIC: with drift loading on debt and growth states (specification M2 from Table 4). Each box is one specification, with the maturity-specific wedge S0 to S4 in the columns (the common wedge, then the Treasury loadings freed at the 3-month, 6-month, 1-year and 2-year maturities in turn) and the exposure O0 to O5 in the rows. A box reports the BIC and the number of free parameters k. Each arrow gives the change in BIC from adding a maturity-specific wedge (horizontal, to the next column) or a richer exposure (vertical, to the next row), that is, the BIC of the box at the head of the arrow minus the BIC of the box at its tail. Green arrows lower the BIC by 10 or more and red arrows raise it by 10. The shaded box has the lowest BIC of all M1 and M2 specifications.

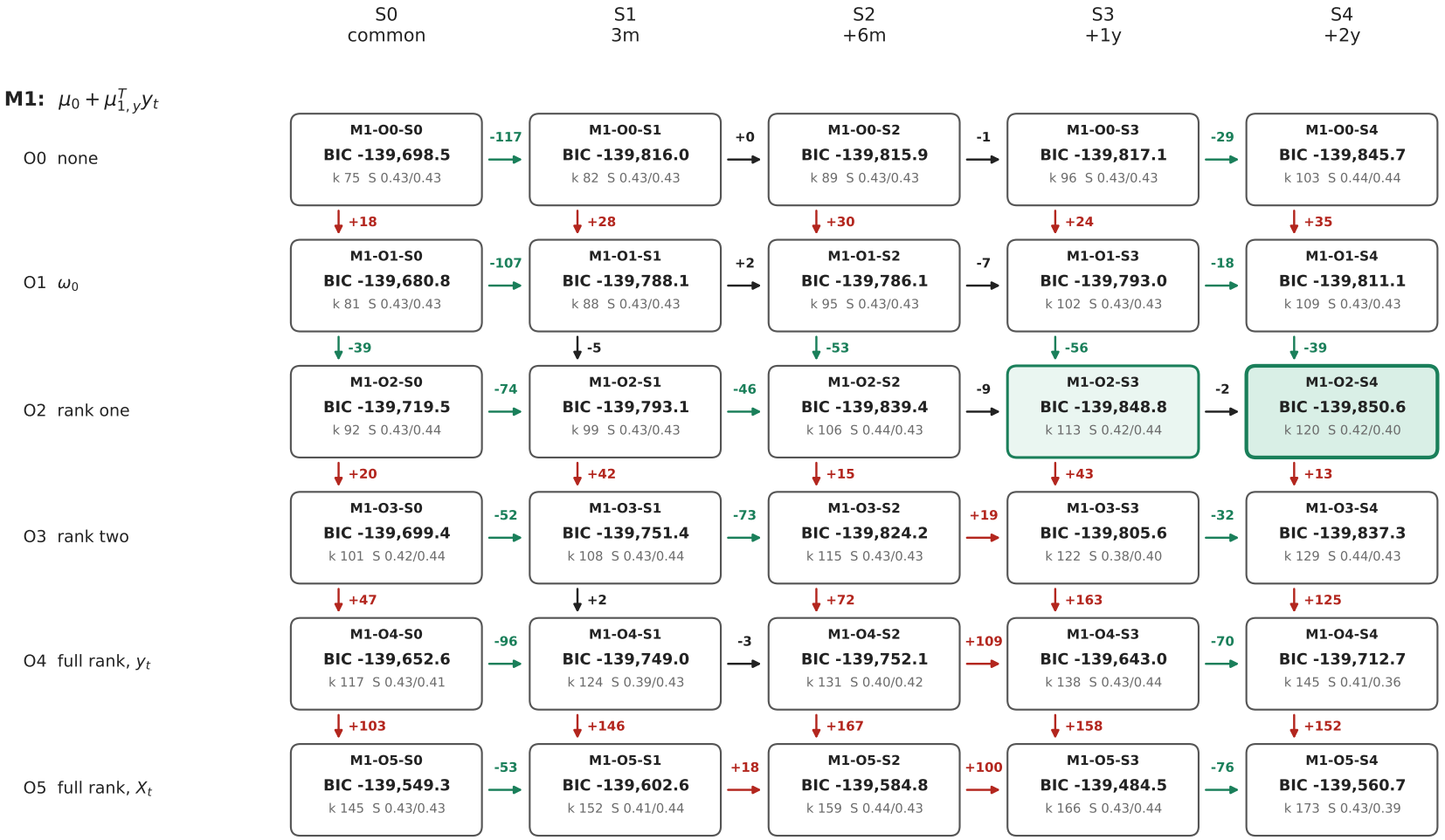


Figure G.51: Wedge model comparison by Sharpe ratio constrained maximum-likelihood BIC: no drift loading on debt or growth states (specification M1 from Table 4). Every box is estimated subject to a bound of 0.433 per quarter on the sample average of the maximum conditional Sharpe ratio. The line labelled S in each box reports this average for the corporate and the Treasury kernel. Each box is one specification, with the maturity-specific wedge S0 to S4 in the columns (the common wedge, then the Treasury loadings freed at the 3-month, 6-month, 1-year and 2-year maturities in turn) and the exposure O0 to O5 in the rows. A box reports the BIC and the number of free parameters k. Each arrow gives the change in BIC from adding a maturity-specific wedge (horizontal, to the next column) or a richer exposure (vertical, to the next row), that is, the BIC of the box at the head of the arrow minus the BIC of the box at its tail. Green arrows lower the BIC by 10 or more and red arrows raise it by 10 or more. The shaded box has the lowest constrained BIC of all M1 and M2 specifications.

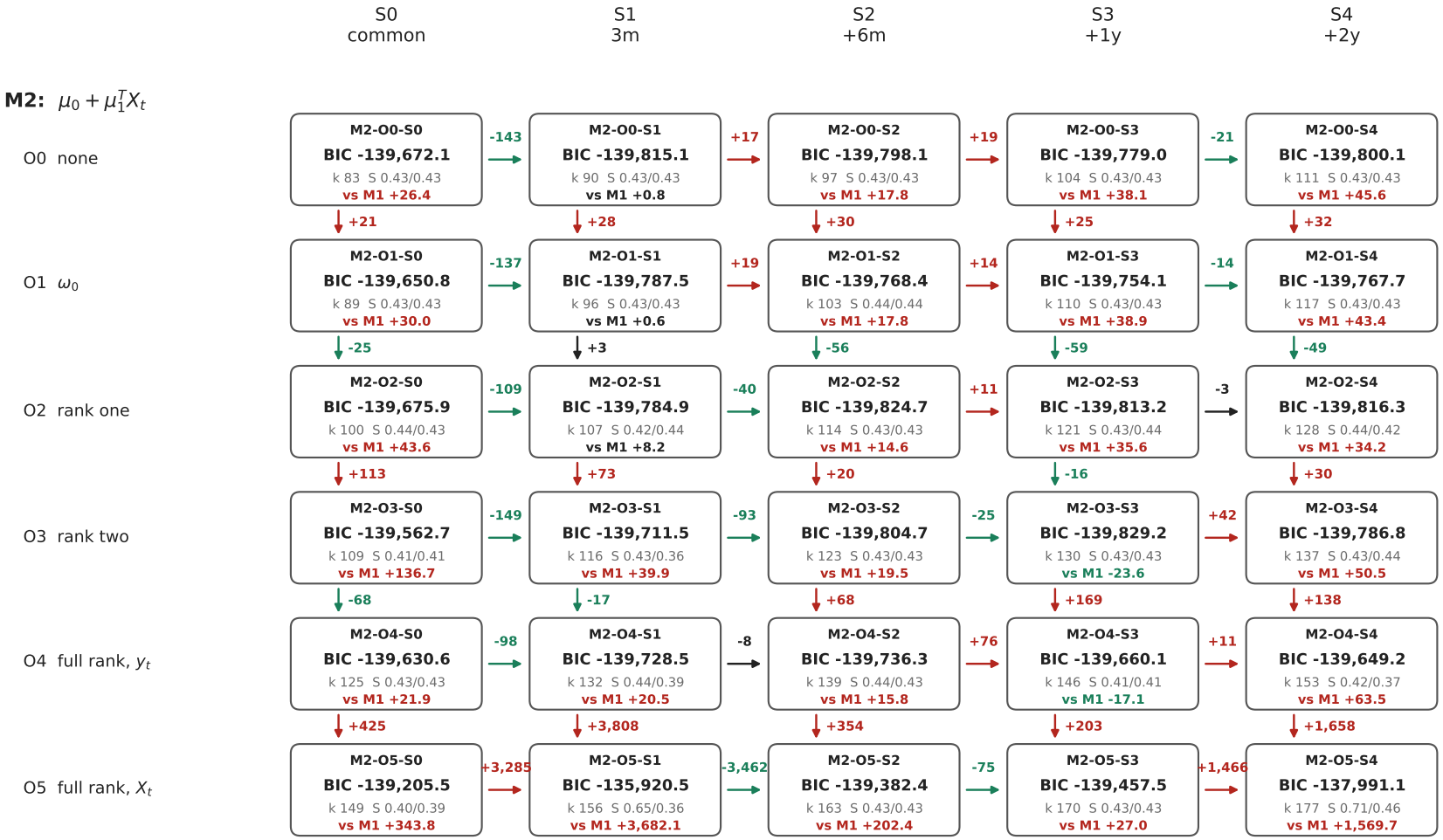


Figure G.52: Wedge model comparison by Sharpe ratio constrained maximum-likelihood BIC: with drift loading on debt and growth states (specification M2 from Table 4). Every box is estimated subject to a bound of 0.433 per quarter on the sample average of the maximum conditional Sharpe ratio. The line labelled S in each box reports this average for the corporate and the Treasury kernel. Each box is one specification, with the maturity-specific wedge S0 to S4 in the columns (the common wedge, then the Treasury loadings freed at the 3-month, 6-month, 1-year and 2-year maturities in turn) and the exposure O0 to O5 in the rows. A box reports the BIC and the number of free parameters k. Each arrow gives the change in BIC from adding a maturity-specific wedge (horizontal, to the next column) or a richer exposure (vertical, to the next row), that is, the BIC of the box at the head of the arrow minus the BIC of the box at its tail. Green arrows lower the BIC by 10 or more and red arrows raise it by 10 or more. The shaded box has the lowest constrained BIC of all M1 and M2 specifications.

responds on impact only to ε_g, issuance does not respond to yield-curve shocks, and a tilt shock leaves total issuance unchanged, since the impact on total debt of shock k is $\mathbf{1}'\Delta V^{n,(k)} = -(\mathbf{1}'\ell)c_{gk} + c_{Ik}$ and $c_{Ik} = 0$ for the tilts.

Inside the yield block the Cholesky order is arbitrary since replacing C_{ss} by $C_{ss}Q$ with Q orthogonal leaves Σ and the likelihood unchanged. Let $u_s = C_{ss}\varepsilon_s$ be the part of V^s that the first four shocks do not explain, with covariance $\Omega_{ss} = C_{ss}C_{ss}'$. Each identification chooses an invertible $T \in \mathbb{R}^{6\times 6}$ whose leading rows are the named directions and whose remaining rows Q_2' are an orthonormal basis of their orthogonal complement, and sets $\varepsilon_s^* = \Pi^{-1}Tu_s$ and $C_{ss}^* = T^{-1}\Pi$ with $\Pi = \mathrm{chol}(T\Omega_{ss}T')$. This changes only the yield block of $\mathcal{S}$, so the responses to the first four shocks do not depend on the identification. In identification I, $T = (b_{\mathrm{lev}},\, b_{\mathrm{slope}},\, Q_2)'$, where $b_{\mathrm{lev}}'s_t$ is the funding advantage averaged across maturities (the combination of the funding-advantage states whose loading curve is closest to flat) and $b_{\mathrm{slope}}'s_t$ is the fitted 15-year minus 2-year funding advantage. The first shock is the standardized innovation to the level, $b_{\mathrm{lev}}'u_s/\sqrt{b_{\mathrm{lev}}'\Omega_{ss}b_{\mathrm{lev}}}$, and the second is the part of the slope innovation that the level shock does not explain. In identification II, $T = (a,\, Q_2)'$ with $a = -\bar{\Lambda}_s$, where $\bar{\Lambda}_s = \Lambda_{s,0} + \Lambda_{s,1}\bar{X}$ is the risk price at the sample-mean state. The first shock is the SDF shock, $\xi_t = -\bar{\Lambda}_s'u_{y,t}/\bar{\theta}$ with $\bar{\theta} = \sqrt{\bar{\Lambda}_s'\Omega_{ss}\bar{\Lambda}_s}$, and its impact on the yield block is $-\Omega_{ss}\bar{\Lambda}_s/\bar{\theta}$.